

Mechanics

Chapter 1: Oscillation

Simple Harmonic Motion (SHM): The periodic to and fro motion in which acceleration of an oscillating body is directly proportional to the displacement from mean position and is always directed towards the mean position is called SHM.

i.e. acceleration $\propto$ displacement

or, $a \propto x$

or, $a = -\omega^2 x$

where ω^2 is proportionality constant and –ve sign indicates that acceleration is directed towards mean position.

Differential Equation of SHM: If x be the displacement of a particle from mean position due to a restoring force F then from Hooke's law, $F = -kx$(i) where k is force constant.

If m be the mass of the particle then from Newton's Second law, $F = ma$(ii)

Now from equations (i) and (ii)

$$ma = -kx$$

$$\text{or, } m \frac{d^2x}{dt^2} = -kx$$

$$\text{or, } \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

$$\text{or, } \frac{d^2x}{dt^2} + \omega^2 x = 0 \text{.....(iii) where } \omega = \sqrt{\frac{k}{m}}$$

Here, equation (iii) represents the differential equation of SHM whose angular frequency is given by

$$\omega = \sqrt{\frac{k}{m}}.$$

The solution of equation (iii) can be written as $x = A \cos(\omega t \pm \phi)$ where A is amplitude and ϕ is phase constant.

Characteristics of SHM:

1) Displacement: It is the distance covered by the oscillating body from mean position and is given by $x = A \cos(\omega t \pm \phi)$

2) Velocity: It is the rate of change displacement.

$$\text{i.e. velocity } v = \frac{dx}{dt} = -\omega A \sin(\omega t \pm \phi) = \omega A \cos(\omega t \pm \phi + \pi/2)$$

This shows that velocity leads the displacement by phase angle $\pi/2$

$$\text{Also, } v = -\omega A \sin(\omega t \pm \phi) = \pm \omega A \sqrt{1 - \cos^2(\omega t \pm \phi)} = \pm \omega \sqrt{A^2 - A^2 \cos^2(\omega t \pm \phi)}$$

$$\text{or, } v = \pm \omega \sqrt{A^2 - x^2}$$

This gives the velocity of a particle executing SHM.

$$\therefore v_{\max} = \pm \omega A \quad \text{at } x = 0 \text{ i.e. at mean position}$$

$$\text{and } v_{\min} = 0 \quad \text{at } x = \pm A \text{ i.e. at extreme positions.}$$

3) Acceleration: It is the rate of change of velocity.

$$\text{i.e. acceleration } a = \frac{dv}{dt} = \frac{d}{dt} \{-\omega A \sin(\omega t \pm \phi)\} = -\omega^2 A \cos(\omega t \pm \phi)$$

$$\text{or, } a = -\omega^2 x$$

This gives the acceleration of a particle executing SHM.

$$\therefore a_{\max} = \pm \omega^2 A \quad \text{at } x = \pm A \text{ i.e. at extreme positions}$$

$$\text{and } a_{\min} = 0 \quad \text{at } x = 0 \text{ i.e. at mean position.}$$

4) Time period: It is the time taken by the oscillating body for one complete oscillation.

$$\text{i.e. time period, } T = \frac{\text{angular distance covered in a complete oscillation}}{\text{angular velocity}} = \frac{2\pi}{\omega}$$

5) Frequency: It is the number of complete oscillations made by a particle in one second and is given by $f = \frac{1}{T}$

6) Phase: The phase of an oscillating body gives the position and direction of motion of the oscillating body since it last passed through its mean position.

In the displacement equation $x = A \cos(\omega t \pm \phi)$, $(\omega t \pm \phi)$ is called phase angle and ϕ is phase constant or initial phase.

Energy of SHM

A particle executing SHM has two types of energy. Due to displacement from mean position, it has potential energy and due to velocity, it has kinetic energy. The sum of kinetic energy and potential energy give total energy. Since there are no any resistive force acting in the system, the total energy is always constant.

$$\text{Now, Kinetic energy, K.E.} = \frac{1}{2} m v^2 = \frac{1}{2} m \omega^2 (A^2 - x^2) \quad (\because v = \pm \omega \sqrt{A^2 - x^2})$$

$$\text{and potential energy P.E.} = \frac{1}{2} k x^2 = \frac{1}{2} m \omega^2 x^2 \quad (\because \omega = \sqrt{\frac{k}{m}})$$

$$\therefore \text{Total mechanical energy } E = \text{K.E.} + \text{P.E.} = \frac{1}{2} m \omega^2 (A^2 - x^2) + \frac{1}{2} m \omega^2 x^2$$

$$\text{or, } E = \frac{1}{2} m \omega^2 A^2$$

This shows that total mechanical energy is conserved.

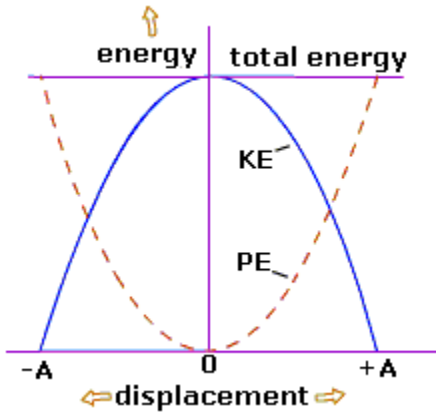
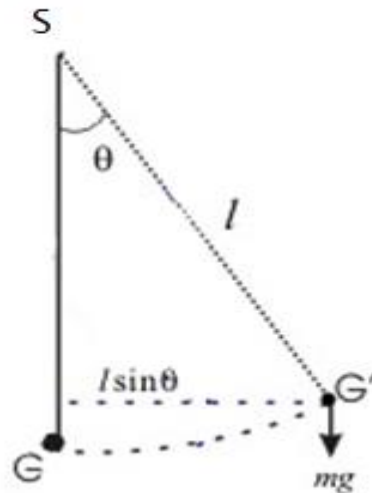


Fig: Variation of energy with displacement

Examples of SHM

- 1) **Compound Pendulum/Physical pendulum:** A compound pendulum consists of a rigid body of any shape which is capable of oscillating about horizontal axis passing through it. The distance between point of suspension and centre of gravity is called length of pendulum.



Here, $SG = SG' = l$, the length of pendulum

Consider a rigid body of mass m is suspended through a rigid support about a point S . If the body is displaced by small angle θ ($\theta < 4^\circ$) and released then it oscillates in to and fro motion due to restoring torque $\tau = -mgl \sin \theta$

Where -ve sign indicates that restoring torque is directed towards mean position.

or, $\tau = -mgl \theta$ (i) ($\because \sin \theta \approx \theta$ for small angle)

If I be the moment of inertia of the rigid body about an axis passing through S , then from

Newton's second law, $\tau = I\alpha = I \frac{d^2 \theta}{dt^2}$ (ii)

Now from equations (i) & (ii)

$$I \frac{d^2 \theta}{dt^2} = -mgl \theta$$

$$\text{or, } \frac{d^2 \theta}{dt^2} + \frac{mgl}{I} \theta = 0$$

$$\text{or, } \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0 \text{(iii) where, } \omega = \sqrt{\frac{mgl}{I}}$$

Here, equation (iii) shows that the motion of compound pendulum is simple harmonic in nature

whose angular frequency is given by $\omega = \sqrt{\frac{mgl}{I}}$

$$\therefore \text{Frequency, } f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{mgl}{I}}$$

$$\text{Hence, time period, } T = \frac{1}{f} = 2\pi \sqrt{\frac{I}{mgl}}$$

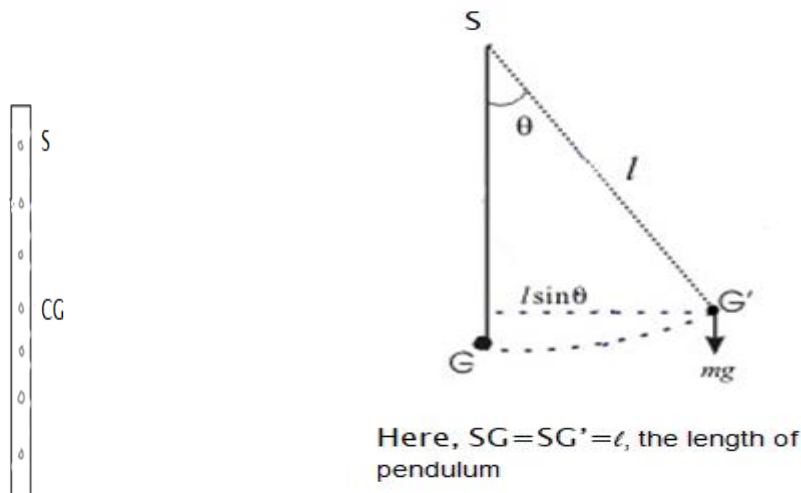
We have from Parallel axis theorem, $I = I_{CG} + ml^2 = mk^2 + ml^2$ where k is radius of gyration.

$$\therefore T = 2\pi \sqrt{\frac{mk^2 + ml^2}{mgl}} = 2\pi \sqrt{\frac{l^2 + k^2}{gl}} = 2\pi \sqrt{\frac{l + k^2/l}{g}}$$

$$\text{or, } T = 2\pi \sqrt{\frac{L}{g}}$$

where, $L = l + k^2/l$ is equivalent length of simple pendulum. ($L > l$)

- 2) Bar Pendulum:** It is the special case of compound pendulum which consists of uniform and symmetrical metal bar having finite numbers of holes at equal distance on either side of C.G. The time period of bar pendulum is determined by suspending the pendulum at each holes except C.G.



Consider a bar pendulum of mass m is suspended through a rigid support about a point S . If the body is displaced by small angle θ ($\theta < 4^\circ$) and released then it oscillates in to and fro motion due to restoring torque $\tau = -mgl \sin \theta$

Where $-ve$ sign indicates that restoring torque is directed towards mean position.

or, $\tau = -mgl \theta$ (i) ($\therefore \sin \theta \approx \theta$ for small angle)

If I be the moment of inertia of the rigid body about an axis passing through S , then from

Newton's second law, $\tau = I\alpha = I \frac{d^2 \theta}{dt^2}$ (ii)

Now from equations (i) & (ii)

$$I \frac{d^2 \theta}{dt^2} = -mgl \theta$$

$$\text{or, } \frac{d^2 \theta}{dt^2} + \frac{mgl}{I} \theta = 0$$

$$\text{or, } \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0 \text{(iii) where, } \omega = \sqrt{\frac{mgl}{I}}$$

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$$\text{Hence, time period, } T = \frac{1}{f} = 2\pi \sqrt{\frac{I}{mgl}}$$

We have from Parallel axis theorem, $I = I_{CG} + ml^2 = mk^2 + ml^2$ where k is radius of gyration.

$$\therefore T = 2\pi \sqrt{\frac{mk^2 + ml^2}{mgl}} = 2\pi \sqrt{\frac{l^2 + k^2}{gl}} = 2\pi \sqrt{\frac{l + k^2/l}{g}}$$

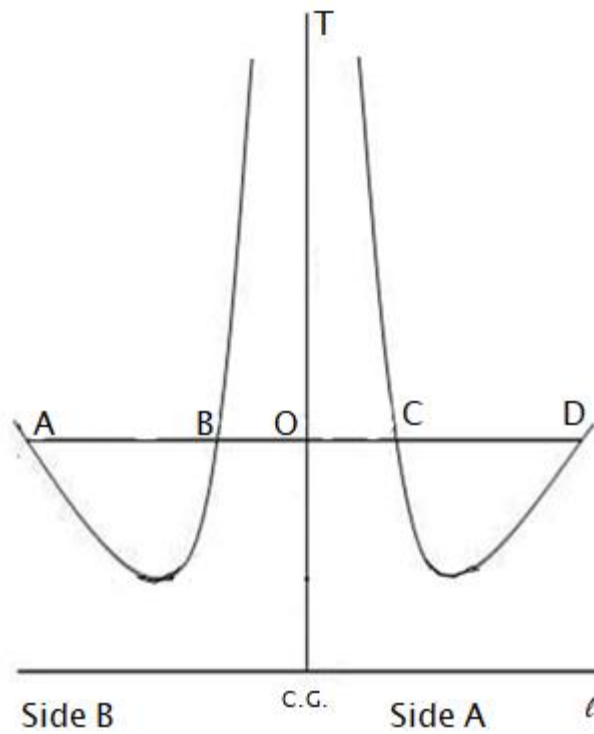
$$\text{or, } T = 2\pi \sqrt{\frac{L}{g}}$$

where, $L = l + k^2/l$ is equivalent length of simple pendulum. ($L > l$)

Question 1: Show that there exist four collinear points in a bar pendulum/compound pendulum which have same time period.

We have time period of bar pendulum is $T = 2\pi \sqrt{\frac{l^2 + k^2}{gl}}$ (a)

If we plot the graph of $T \sim l$, the graph is as shown in figure.



From figure it is seen that there exist four collinear points (A,B,C&D) having same time period at O.

On squaring equation (a) we have,

$$T^2 = \frac{4\pi^2}{lg} (l^2 + K^2)$$

$$\text{or, } 4\pi^2 l^2 + (-gT^2)l + 4\pi^2 k^2 = 0$$

This is quadratic in l so it possesses two roots. If l_1 and l_2 be the two roots then,

$$l_1 + l_2 = -\frac{b}{a} = \frac{gT^2}{4\pi^2}$$

$$\therefore g = \frac{4\pi^2}{T^2} (l_1 + l_2) = 4\pi^2 \frac{L}{T^2} \quad (\because L = l_1 + l_2)$$

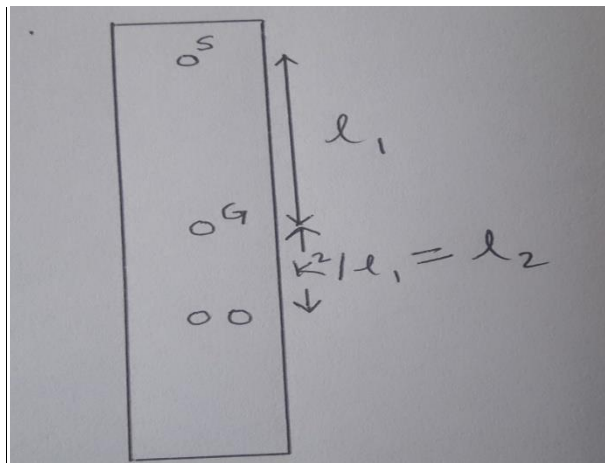
$$\text{Again, } l_1 l_2 = \frac{c}{a} = k^2$$

$$\therefore k = \sqrt{l_1 l_2}$$

From above results, we see that there are two lengths of pendulum having same time period, on either side of C.G. Hence there are four different collinear points having same time period.

Question 2: Show that point of suspension and point of oscillation are interchangeable.

Point of oscillation: Any point which lies at a distance of k^2/l from C.G. on the opposite side of point of suspension is called point of oscillation.



Consider S be point of suspension, G be C.G. and O be the point of oscillation. When the pendulum is suspended from S then we have $SG = l_1$ and $OG = k^2/l_1$.

In this case time period is $T_1 = 2\pi \sqrt{\frac{l_1 + k^2/l_1}{g}}$

When the pendulum is suspended from O then $OG = l_2$ and $SG = \frac{k^2}{l_2}$.

In this case, time period is $T_2 = 2\pi \sqrt{\frac{l_2 + k^2/l_2}{g}}$

We have, $k = \sqrt{l_1 l_2}$

$$\therefore k^2 = l_1 l_2$$

$$\text{or, } l_1 = \frac{k^2}{l_2} \text{ and } l_2 = \frac{k^2}{l_1}$$

$$\text{Now, } T_1 = 2\pi \sqrt{\frac{l_1 + k^2/l_1}{g}} = 2\pi \sqrt{\frac{\frac{k^2}{l_2} + l_2}{g}} = 2\pi \sqrt{\frac{l_2 + k^2/l_2}{g}} = T_2$$

This shows that point of suspension and point of oscillation are interchangeable.

Question 3: Show that time period is minimum when length of pendulum is equal to radius of gyration.

or

Show that time period is minimum when point of suspension and point of oscillation are equidistant from C.G.

We have time period of compound pendulum is $T = 2\pi \sqrt{\frac{l_1 + k^2/l_1}{g}} = 2\pi \sqrt{\frac{L}{g}}$

Where, $L = l_1 + k^2/l_1$

Here, l_1 is the distance of point of suspension from C.G. or length of pendulum and k is radius of gyration.

For minimum time period, L must be minimum.

For minimum L , $\frac{dL}{dl_1} = 0$

$$\text{or, } \frac{d}{dl_1} \left(l_1 + \frac{k^2}{l_1} \right) = 0$$

$$\text{or, } 1 - \frac{k^2}{l_1^2} = 0$$

$$\text{or, } k^2 = l_1^2 \dots \dots \dots (a)$$

$$\text{or, } k = l_1$$

This shows that time period is minimum when length of pendulum is equal to radius of gyration.

If l_2 be the distance of point of oscillation from C.G. then

$$k = \sqrt{l_1 l_2}$$

$$\text{or, } k^2 = l_1 l_2$$

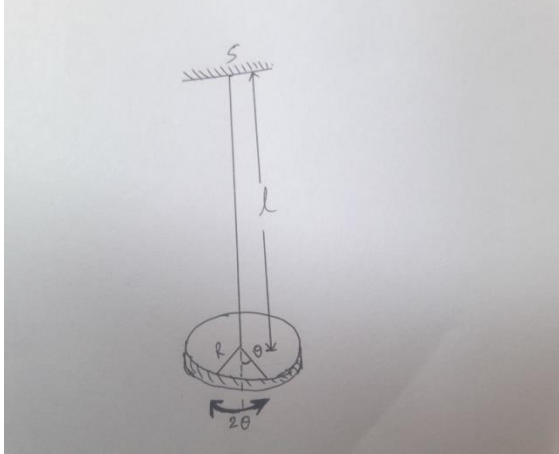
$$\text{or, } l_1^2 = l_1 l_2 \quad \because \text{ using equation (a)}$$

$$\text{or, } l_1 = l_2$$

Hence time period is minimum when distance of point of suspension and distance of point of oscillation are equidistant from C.G.

Note: If $l = 0$ then time period is maximum ($T = \infty$)

3) Torsion Pendulum: A torsion pendulum consists of a circular disc which is suspended through a rigid support by means of a metallic wire passing through C.G. of the disc.



Consider a circular disc of mass m and radius R is suspended through a rigid support by means of a wire of length l and radius r . If the disk is turned by θ and released then it oscillates about an angle 2θ due to restoring couple

$$\tau = -C\theta \dots\dots\dots(i)$$

Where $-ve$ sign indicates that restoring couple is directed towards the mean position and C is torsional constant whose value is $C = \frac{\pi\eta r^4}{2l}$ where, η is modulus of rigidity of the wire.

If I be the moment of inertia of the disc about an axis passing through C.G., then from Newton's second law,

$$\tau = I\alpha = I \frac{d^2 \theta}{dt^2} \dots\dots\dots(ii)$$

Now from equations (i) and (ii)

$$I \frac{d^2 \theta}{dt^2} = -C\theta$$

$$\text{or, } \frac{d^2 \theta}{dt^2} + \frac{C}{I} \theta = 0$$

$$\text{or, } \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0 \dots\dots\dots(iii) \quad \text{where, } \omega = \sqrt{\frac{C}{I}}$$

Here, equation (iii) shows that the motion of torsion pendulum is angular simple harmonic in nature whose angular frequency is given by

$$\omega = \sqrt{\frac{C}{I}}$$

$$\therefore \text{Frequency, } f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{C}{I}}$$

$$\text{Hence, time period, } T = \frac{1}{f} = 2\pi \sqrt{\frac{I}{C}}$$

This gives the time period of torsion pendulum. This shows that time period is not affected even if the amplitude of oscillation is large, provided elastic limit of the wire is not exceeded.

Determination of moment of Inertia of an irregular body using torsion pendulum

We have the time period of torsion pendulum is $T = 2\pi \sqrt{\frac{I}{C}}$ (i)

When a regular body of known moment of inertia I_1 is placed over the disc, then time period is

$$T_1 = 2\pi \sqrt{\frac{I+I_1}{C}} \text{(ii)}$$

When an irregular body of unknown moment of inertia I_2 is placed over the disc, then time period is

$$T_2 = 2\pi \sqrt{\frac{I+I_2}{C}} \text{(iii)}$$

From equations (i) and (ii), we have

$$\frac{T_1^2}{T^2} = \frac{I + I_1}{I} = 1 + \frac{I_1}{I}$$

$$\text{or, } \frac{I_1}{I} = \frac{T_1^2}{T^2} - 1 = \frac{T_1^2 - T^2}{T^2} \text{(iv)}$$

Again, from equations (i) and (iii), we have

$$\frac{T_2^2}{T^2} = \frac{I + I_2}{I} = 1 + \frac{I_2}{I}$$

$$\text{or, } \frac{I_2}{I} = \frac{T_2^2}{T^2} - 1 = \frac{T_2^2 - T^2}{T^2} \text{(v)}$$

Now, dividing equation (v) by (iv) we get

$$\frac{I_2}{I_1} = \frac{T_2^2 - T^2}{T_1^2 - T^2}$$

$$\text{or, } I_2 = \left(\frac{T_2^2 - T^2}{T_1^2 - T^2} \right) I_1$$

knowing the values of T_1, T_2, T and I_1 we can determine I_2 .

Determination of modulus of rigidity of a wire using torsion pendulum

We have the time period of torsion pendulum is $T = 2\pi \sqrt{\frac{I}{C}}$ (i)

When a circular ring of moment of inertia I_1 is placed over the circular disc then the time period is

$$T_1 = 2\pi \sqrt{\frac{I+I_1}{C}} \text{(ii)}$$

Here, $I_1 = \frac{mR_1^2 + mR_2^2}{2}$ where, m is mass and R_1 & R_2 are internal and external radii of circular ring.

Now from equations (i) and (ii)

$$T_1^2 - T^2 = \frac{4\pi^2}{C} I_1$$

$$\text{or, } T_1^2 - T^2 = \frac{4\pi^2}{\frac{\pi\eta r^4}{2l}} I_1 = \frac{8\pi^2 I_1 l}{\pi\eta r^4}$$

$$\text{or, } \eta = \frac{8\pi I_1 l}{(T_1^2 - T^2) r^4}$$

This gives the modulus of rigidity of metallic wire.

Also we can write

$$\frac{T_1^2 - T^2}{T^2} = \frac{I_1}{I}$$

$$\text{or, } I = \left(\frac{T^2}{T_1^2 - T^2} \right) I_1$$

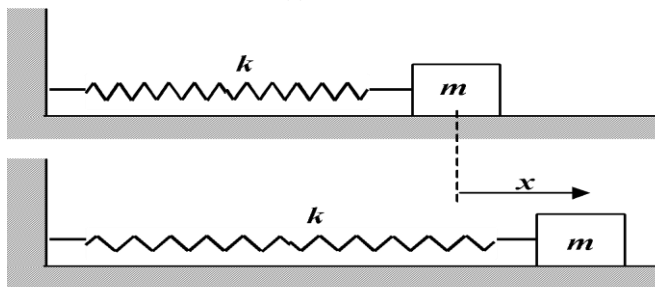
This gives the moment of inertia of circular disc.

4) Mass Spring System:

If the oscillation of a system is due to the restoring force stored in a spring, then it is called mass spring system.

Consider a spring of spring constant k whose one end is fixed to a horizontal frictionless surface and other end is attached to a block of mass m . If the mass is displaced by x and is released then it oscillates in to and fro motion due to restoring force

$$F = -kx \dots\dots\dots(i)$$



From Newton's Second law, we have

$$F = ma = m \frac{d^2 x}{dt^2} \dots\dots\dots (ii)$$

From equations (i) and (ii), we have

$$m \frac{d^2 x}{dt^2} = -kx$$

$$\text{or, } \frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$$

$$\text{or, } \frac{d^2 x}{dt^2} + \omega^2 x = 0 \dots\dots\dots (iii) \quad \text{where, } \omega = \sqrt{\frac{k}{m}}$$

Here, equation (iii) shows that the motion of mass spring system is simple harmonic in nature

whose angular frequency is given by $\omega = \sqrt{\frac{k}{m}}$

$$\therefore \text{Frequency, } f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$\text{Hence, time period, } T = \frac{1}{f} = 2\pi \sqrt{\frac{m}{k}}$$

Free Oscillation: If the oscillation of a system is only due to the effect of restoring force, then it is called free oscillation. In this oscillation, the amplitude remains constant with increase in time but it is rare in actual practice.

Damping Force: The force which opposes the motion of a system due to which the amplitude goes on decreasing with increase in time is called damping force. For example, air resistance, frictional force, viscous force etc.

The damping force present in a system is directly proportional to the velocity of the system.

i.e. Damping force, $F_d \propto \text{velocity, } v$

$$\text{or, } F_d \propto v$$

$$\text{or, } F_d = -bv$$

where, b is damping coefficient and –ve sign indicates that damping force acts in the opposite direction of velocity.

Damped Oscillation:

If we consider the effect of damping force in a system, then it is called damped oscillation. The forces acting on this system are

- i) Restoring force, $F_r = -kx$ and
- ii) Damping force, $F_d = -bv$

Therefore net force acting on this system is

$$F = F_r + F_d$$

$$\text{Or, } F = -kx - bv$$

$$\text{Or, } ma = -kx - bv$$

$$\text{Or, } m \frac{d^2x}{dt^2} = -kx - b \frac{dx}{dt}$$

$$\text{Or, } m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

$$\text{Or, } \frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

$$\text{Or, } \frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \omega_0^2 x = 0 \dots \dots \dots (i)$$

$$\text{where, } \omega_0 = \sqrt{\frac{k}{m}}$$

Here equation (i) represents the differential equation of damped oscillation. The solution of this equation can be written as

$$x = R e^{\frac{-bt}{2m}} \cos(\beta t \pm \phi) \dots \dots \dots (ii)$$

Where, $R e^{\frac{-bt}{2m}} = A(t)$ is amplitude and $\beta = \sqrt{\omega_0^2 - \frac{b^2}{4m^2}}$ is angular frequency of damped oscillation.

Here, R is amplitude at $t=0$ called initial amplitude.

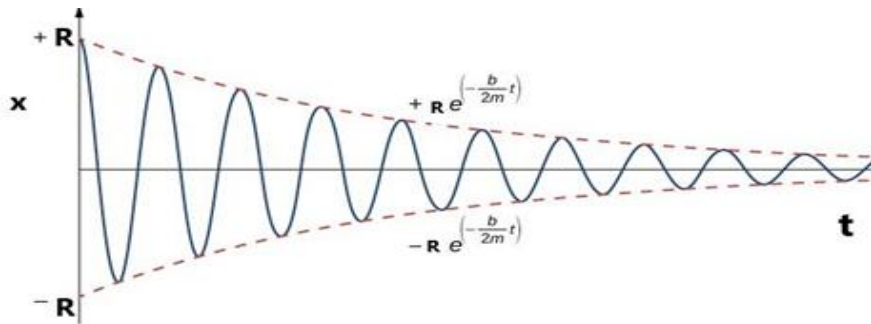


Fig: Variation of amplitude in damped oscillation

Note:

- 1) If $\omega_0 > \frac{b}{2m}$ then the system is oscillating called under damped case.
- 2) If $\omega_0 < \frac{b}{2m}$ then the system is non oscillating called over damped case.
- 3) If $\omega_0 = \frac{b}{2m}$ then the system is non oscillating called critical damped case.

Energy Loss Per Cycle in Damped Oscillation

We have the energy of damped oscillator at any instant of time is

$$E(t) = \frac{1}{2} m \omega_0^2 \{A(t)\}^2 \text{ where, } A(t) = R e^{\frac{-bt}{2m}} \text{ is amplitude of damped oscillator.}$$

$$\text{Or, } E(t) = \frac{1}{2} m \omega_0^2 R^2 e^{\frac{-bt}{m}}$$

$$\text{or, } E(t) = E_0 e^{\frac{-bt}{m}} \text{ where, } E_0 = \frac{1}{2} m \omega_0^2 R^2 \text{ is the initial energy.}$$

Now energy of the oscillator after one cycle becomes

$$E(t + T) = E_0 e^{\frac{-b(t+T)}{m}} \text{ where T is time period.}$$

Therefore, energy loss per cycle is

$$\Delta E = E(t) - E(t + T)$$

$$\text{Or, } \Delta E = E_0 e^{\frac{-bt}{m}} - E_0 e^{\frac{-b(t+T)}{m}} = E_0 e^{\frac{-bt}{m}} \left(1 - e^{\frac{-bT}{m}}\right) = E(t) \left(1 - e^{\frac{-bT}{m}}\right)$$

We have, $e^{\frac{-bT}{m}} = 1 - \frac{bT}{m} + \frac{1}{2!} \left(\frac{bT}{m}\right)^2 - \dots \approx 1 - \frac{bT}{m}$ (Since, b is very small so b^2 and higher powers can be neglected)

$$\text{Therefore, } 1 - e^{\frac{-bT}{m}} = \frac{bT}{m}$$

$$\text{Now, } \Delta E = \frac{EbT}{m}$$

This gives the energy loss per cycle.

Quality Factor: It is defined as 2π times the ratio between energy stored to the energy loss per cycle. It is denoted by Q . Higher the quality factor, higher will be the efficiency and vice versa.

$$\text{Mathematically, } Q = 2\pi \frac{\text{Energy stored}}{\text{Energy loss per cycle}}$$

$$\text{Or, } Q = 2\pi \frac{E}{E b T} m = \frac{2\pi}{T} \frac{m}{b} = \beta \frac{m}{b}$$

$$\text{In case of low damping, } \beta = \sqrt{\omega_0^2 - \frac{b^2}{4m^2}} \approx \omega_0$$

$$\text{Therefore, } Q = \omega_0 \frac{m}{b}$$

$$\text{This shows that } Q \propto \frac{1}{b}$$

$$\text{Also, } Q = \omega_0 \tau \text{ where } \tau = \frac{m}{b} \text{ is relaxation time.}$$

Relaxation time: The time in which energy of the damped oscillator becomes $1/e$ times the initial energy is called relaxation time. It is denoted by τ .

We have energy of oscillator at any instant of time is

$$E(t) = E_0 e^{\frac{-bt}{m}}$$

At $t = \tau$, the energy becomes

$$E(\tau) = E_0 e^{\frac{-b\tau}{m}}$$

$$\text{Or, } \frac{1}{e} E_0 = E_0 e^{\frac{-b\tau}{m}}$$

$$\text{Or, } e^{-1} = e^{\frac{-b\tau}{m}}$$

$$\text{Or, } \frac{b\tau}{m} = 1$$

$$\text{Or, } \tau = \frac{m}{b}$$

Note:

$$x = Re^{\frac{-bt}{2m}} \cos(\beta t \pm \phi) = Re^{\frac{-t}{2\tau}} \cos(\beta t \pm \phi)$$

$$A(t) = Re^{\frac{-bt}{2m}} = Re^{\frac{-t}{2\tau}}$$

$$\text{And } E(t) = E_0 e^{\frac{-bt}{m}} = E_0 e^{\frac{-t}{\tau}}$$

Forced Oscillation (Driven Oscillation)

If the oscillation of a system is due to the continuous application of external force, then the system is called forced oscillation. In this system, the frequency of oscillation is the frequency of applied force. The forces acting on this system are,

- i) Restoring force, $F_r = -kx$
- ii) Damping force, $F_d = -bv$ and
- iii) External force, $F_{ext} = F_m \sin \omega t$

Where, F_m is the amplitude and ω is the angular frequency of applied force.

Therefore net force acting on this system is

$$F = F_r + F_d + F_{ext}$$

$$\text{Or, } F = -kx - bv + F_m \sin \omega t$$

$$\text{Or, } ma = -kx - bv + F_m \sin \omega t$$

$$\text{Or, } m \frac{d^2 x}{dt^2} = -kx - b \frac{dx}{dt} + F_m \sin \omega t$$

$$\text{Or, } m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = F_m \sin \omega t$$

$$\text{Or, } \frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = \frac{F_m}{m} \sin \omega t$$

$$\text{Or, } \frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \omega_0^2 x = \frac{F_m}{m} \sin \omega t \dots \dots \dots (i)$$

where, $\omega_0 = \sqrt{\frac{k}{m}}$ is natural frequency of the system.

Here equation (i) represents the differential equation of forced oscillation. The solution of this equation can be written as

$$x = A \sin(\omega t + \varphi)$$

Where, A is the amplitude of forced oscillation whose value is

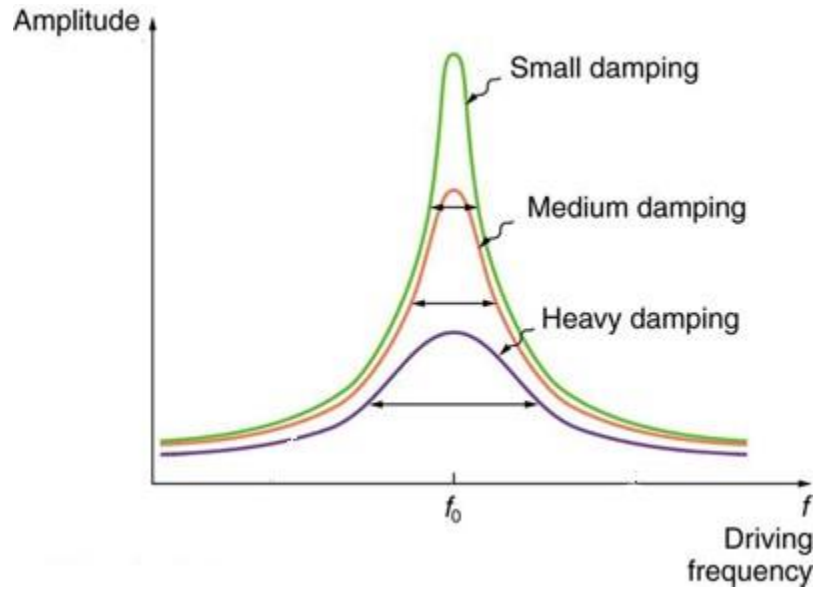
$$A = \frac{F_m}{\sqrt{m^2(\omega^2 - \omega_0^2)^2 + b^2 \omega^2}} \text{ and}$$

$$\varphi = \tan^{-1} \left\{ \frac{m(\omega^2 - \omega_0^2)}{b\omega} \right\} \text{ is phase constant.}$$

Resonance: when $\omega = \omega_0$ then the amplitude of forced oscillation becomes the maximum. This condition is called resonance and the maximum amplitude is given by

$$A_{max} = \frac{F_m}{b\omega_0} = \frac{F_m}{b} \sqrt{\frac{m}{k}}$$

$$\therefore A_{max} \propto \frac{1}{b}$$

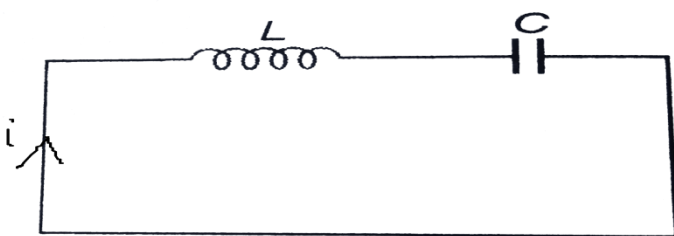


The amplitude decrease from this maximum value on either side of resonant frequency. This rate of fall in amplitude is called sharpness of resonance.

Electromagnetic Oscillations

1) LC oscillation

Consider a fully charged capacitor having capacitance C is connected to an inductor of inductance L as shown in figure. Let i be the instantaneous current through the circuit and q be the instantaneous charge through the capacitor.



Now the energy stored in the capacitor and inductor are

$$U_E = \frac{q^2}{2C} \quad \text{and} \quad U_B = \frac{1}{2} Li^2 \quad \text{respectively.}$$

Then, total energy in the circuit is

$$U = U_E + U_B = \frac{q^2}{2C} + \frac{1}{2} Li^2$$

$$\text{or, } \frac{dU}{dt} = \frac{1}{2C} 2q \frac{dq}{dt} + \frac{L}{2} 2i \frac{di}{dt}$$

$$\text{or, } 0 = \frac{q}{C} \frac{dq}{dt} + Li \frac{di}{dt} \quad (\text{Since, } U \text{ is constant})$$

$$\text{or, } \frac{q}{C} i + Li \frac{di}{dt} = 0 \quad (\text{Since, } \frac{dq}{dt} = i)$$

$$\text{or, } L \frac{di}{dt} + \frac{q}{C} = 0$$

$$\text{or, } \frac{d}{dt} \left(\frac{dq}{dt} \right) + \frac{1}{LC} q = 0$$

$$\text{or, } \frac{d^2 q}{dt^2} + \omega^2 q = 0 \dots\dots\dots(i) \quad \text{where, } \omega = \frac{1}{\sqrt{LC}}$$

This is the differential equation of LC Oscillation. This shows that the charge variation in the LC circuit is an analogy of SHM whose angular frequency is given by

$$\omega = \frac{1}{\sqrt{LC}}$$

$$\therefore \text{The frequency of LC oscillation is } f = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

Now the solution of equation (i) is

$$q = q_m \cos(\omega t \pm \varphi) \text{ where, } q_m \text{ is the maximum charge in the capacitor.}$$

$$\therefore \text{current, } i = \frac{dq}{dt} = -\omega q_m \sin(\omega t \pm \varphi)$$

$$\text{Now, electrical energy, } U_E = \frac{q^2}{2C} = \frac{q_m^2}{2C} \cos^2(\omega t \pm \varphi)$$

$$\therefore (U_E)_{\max.} = \frac{q_m^2}{2C} \quad \text{and} \quad (U_E)_{\min.} = 0$$

$$\begin{aligned} \text{And magnetic energy, } U_B &= \frac{1}{2} Li^2 = \frac{1}{2} L \omega^2 q_m^2 \sin^2(\omega t \pm \varphi) = \frac{1}{2} L \frac{1}{LC} q_m^2 \sin^2(\omega t \pm \varphi) \\ &= \frac{q_m^2}{2C} \sin^2(\omega t \pm \varphi) \end{aligned}$$

$$\therefore (U_B)_{\max.} = \frac{q_m^2}{2C} \quad \text{and} \quad (U_B)_{\min.} = 0$$

Now total electromagnetic energy is

$$U = U_E + U_B = \frac{q_m^2}{2C} \cos^2(\omega t \pm \varphi) + \frac{q_m^2}{2C} \sin^2(\omega t \pm \varphi) = \frac{q_m^2}{2C}$$

From above results we see that

$$(U_E)_{\max.} = (U_B)_{\max.} = U = \frac{q_m^2}{2C}$$

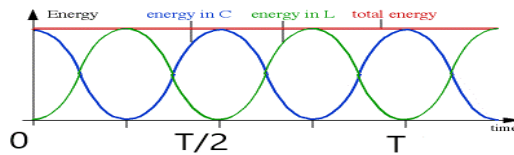


Fig: variation of energy with time

2. Electromagnetic Damped Oscillation (LCR Damped Oscillation)

Consider a fully charged capacitor having capacitance C is connected in series with a inductor of inductance L and a resistor of resistance R as shown in figure. Let i be the instantaneous current through the circuit and q be the instantaneous charge through the capacitor.

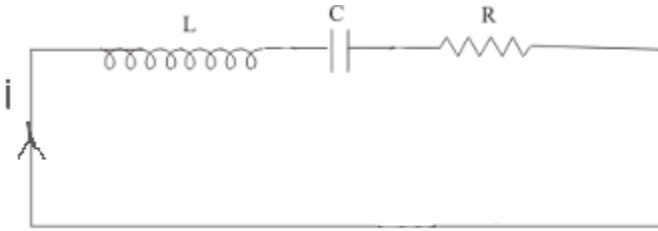


Figure Circuit containing inductor capacitor and an resistor in series

Now the energy stored in the capacitor and inductor are

$$U_E = \frac{q^2}{2C} \quad \text{and} \quad U_B = \frac{1}{2} Li^2 \quad \text{respectively.}$$

Then, total energy in the circuit is

$$U = U_E + U_B = \frac{q^2}{2C} + \frac{1}{2} Li^2$$

$$\text{or, } \frac{dU}{dt} = \frac{1}{2C} 2q \frac{dq}{dt} + \frac{L}{2} 2i \frac{di}{dt}$$

$$\text{or, } -i^2 R = \frac{q}{C} \frac{dq}{dt} + Li \frac{di}{dt} \quad (-ve \text{ sign indicates that energy decreases with increase in time})$$

$$\text{or, } \frac{q}{C} i + Li \frac{di}{dt} + i^2 R = 0 \quad (\text{Since, } \frac{dq}{dt} = i)$$

$$\text{or, } L \frac{di}{dt} + iR + \frac{q}{C} = 0$$

$$\text{or, } L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = 0 \quad \dots\dots\dots(i)$$

$$\text{or, } \frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC} q = 0$$

$$\text{or, } \frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \omega_0^2 q = 0 \dots\dots\dots(ii) \quad \text{where, } \omega_0 = \frac{1}{\sqrt{LC}}$$

This is the differential equation of LCR damped Oscillation. Here, equation (i) can be compared to mechanical damped oscillation as,

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

Where, $L \leftrightarrow m$, $R \leftrightarrow b$ and $\frac{1}{C} \leftrightarrow k$

The solution of equation (ii) can be written as

$$q = q_0 e^{\frac{-Rt}{2L}} \cos(\beta t \pm \varphi)$$

where, $q_0 e^{\frac{-Rt}{2L}} = q_m(t)$ is charge amplitude and $\beta = \sqrt{\omega_0^2 - \frac{R^2}{4L^2}}$ is angular frequency of damped LCR oscillation. Here, q_0 is charge amplitude at $t=0$.

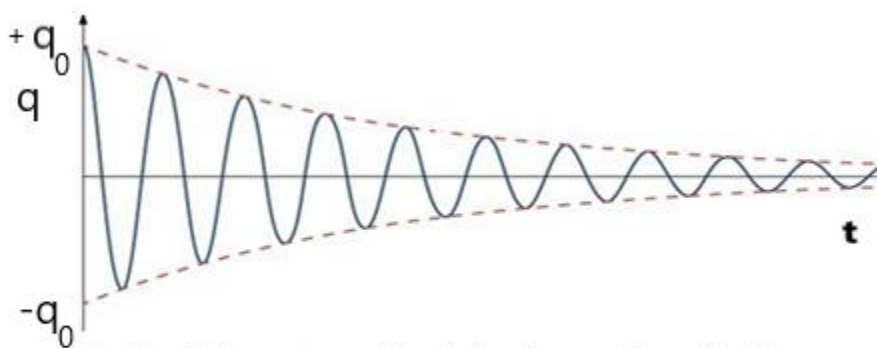


Fig: Variation of amplitude in damped oscillation

3. Forced electromagnetic Oscillation (LCR Forced oscillation)

Consider a capacitor having capacitance C is connected in series with a inductor of inductance L , a resistor of resistance R and an external ac source $E = E_0 \sin \omega t$ as shown in figure. Let i be the instantaneous current through the circuit and q be the instantaneous charge through the capacitor.

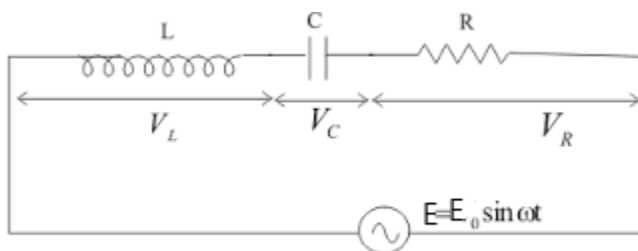


Figure 12(a). A.C. Circuit containing inductor capacitor and an resistor in series

Using Kirchhoff's voltage law,

$$E = V_L + V_C + V_R$$

$$\text{or, } L \frac{di}{dt} + \frac{q}{C} + iR = E_0 \sin \omega t$$

$$\text{or, } \frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC} q = \frac{E_0}{L} \sin \omega t$$

$$\text{or, } \frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \omega_0^2 q = \frac{E_0}{L} \sin \omega t \dots \dots \dots (i) \quad \text{where, } \omega_0 = \frac{1}{\sqrt{LC}} \text{ is natural frequency.}$$

This is differential equation of forced e-m oscillation. The solution of this equation can be written as

$$q = q_0 \sin(\omega t + \varphi)$$

Where, $q_0 = \frac{E_0}{\sqrt{L^2(\omega^2 - \omega_0^2)^2 + R^2\omega^2}}$ is charge amplitude and $\varphi = \tan^{-1} \left\{ \frac{L(\omega^2 - \omega_0^2)}{R\omega} \right\}$ is phase difference.

Now, current, $i = \frac{dq}{dt} = \omega q_0 \cos(\omega t + \varphi) = i_0 \cos(\omega t + \varphi)$

Where, $i_0 = \omega q_0$ is amplitude of current.

$$\text{Now, } i_0 = \omega q_0 = \frac{\omega E_0}{\sqrt{L^2(\omega^2 - \omega_0^2)^2 + R^2\omega^2}} = \frac{E_0}{\sqrt{\left(L\omega - \frac{1}{c\omega}\right)^2 + R^2}} = \frac{E_0}{Z}$$

Where, $Z = \sqrt{\left(L\omega - \frac{1}{c\omega}\right)^2 + R^2}$ is impedance of LCR series circuit.

Resonance: for the maximum amplitude of current,

$$L\omega = \frac{1}{c\omega}$$

$$\text{or, } \omega^2 = \frac{1}{LC}$$

$$\text{or, } \omega = \frac{1}{\sqrt{LC}}$$

$$\therefore f_r = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

This gives the resonance frequency.

Quality factor: it is defined as the ratio of p.d. across capacitor or inductor to the p.d. across resistor at resonance.

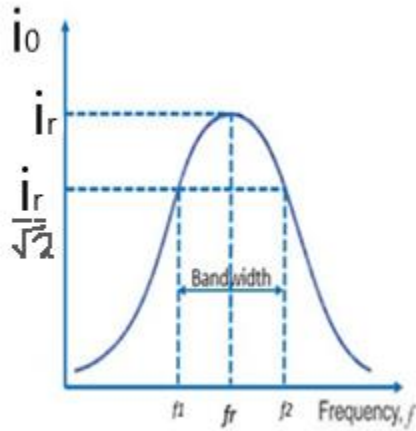
$$\text{i.e. quality factor, } Q = \frac{\text{P.d. across } C \text{ or } L}{\text{P.d. across } R} \text{ at resonance}$$

$$\text{or, } Q = \frac{1}{C\omega R} = \frac{L\omega}{R}$$

Frequency band Width: It is the difference in two frequencies in the resonant curve when the maximum power in the circuit is reduced to half.

$$\text{We have, } P_{\max.} = i_{\max.}^2 R = i_r^2 R$$

$$\therefore \frac{P_{max.}}{2} = \frac{i_r^2 R}{2} = \left(\frac{i_r}{\sqrt{2}}\right)^2 R$$



Hence, for maximum power reduced to half, the current in the circuit is $\frac{i_r}{\sqrt{2}}$ and let the corresponding frequencies are f_1 and f_2 .

$\therefore$ frequency band width, $\Delta f = f_2 - f_1$

$$\text{Also, } Q = \frac{2\pi f_r}{f_2 - f_1}$$

Numerical

Moment of inertia of some regular bodies about C.G.

a) Solid sphere, $I_{C.G.} = \frac{2}{5}mR^2$

b) Hollow Sphere, $I_{C.G.} = \frac{2}{3}mR^2$

c) Circular disc, $I_{C.G.} = \frac{1}{2}mR^2$

d) Circular ring, $I_{C.G.} = mR^2$

e) Thin rod or bar, $I_{C.G.} = \frac{ml'^2}{12}$

Also, $I_{C.G.} = \frac{m(l'^2 + b^2)}{12}$ if breadth is given

1. An oscillatory motion of a body is represented by $y = ae^{i\omega t}$, where y is displacement in time t and ω is angular velocity. Show that the motion is simple harmonic.

→ Here the displacement equation is $y = ae^{i\omega t}$

Now, $\frac{dy}{dt} = i\omega ae^{i\omega t}$

$$\therefore \frac{d^2y}{dt^2} = (i\omega)^2 ae^{i\omega t} = -\omega^2 y$$

$$\text{or, } \frac{d^2y}{dt^2} + \omega^2 y = 0$$

This is the differential equation of SHM. So the motion is simple Harmonic.

- 2. Show that the velocity of simple harmonic oscillator at any instant leads the displacement by a phase angle of $\pi/2$. Hence find the general expression for the velocity of a simple harmonic oscillator**

→ Refer theory

- 3. At $t=0$, the displacement $x(0)$ of the block in linear oscillator is -8.50cm. The block's velocity $v(0)$ then is -0.920m/s and its acceleration $a(0)$ is 47m/s². Find (a) angular frequency (b) Phase constant.**

→ Here, $x(0) = -8.50\text{cm} = -0.085\text{m}$, $v(0) = -0.920\text{ms}^{-1}$, and $a(0) = 47\text{ms}^{-2}$

a) we have, $a(0) = -\omega^2 x(0)$

$$\therefore \omega = \sqrt{-\frac{a(0)}{x(0)}} = ?$$

b) we have, $x = A \cos(\omega t + \phi) \therefore x(0) = A \cos \phi \dots \dots \dots (i)$

$v = -A\omega \sin(\omega t + \phi) \therefore v(0) = -A\omega \sin \phi \dots \dots \dots (ii)$

Now, dividing equation (ii) by (i) we get

$$\phi = \tan^{-1}\left(-\frac{v(0)}{\omega x(0)}\right) = ?$$

- 4. A 0.12kg body undergoes simple harmonic motion of amplitude 8.5cm and period 0.20sec. a) What is the magnitude of the maximum force acting on it? b) If the oscillations are produced by a spring, what is the spring constant?**

→ Here, $m = 0.12\text{kg}$, $A = 8.5\text{cm} = 0.085\text{m}$, $T = 0.20\text{s}$

a) Force, $F = m\omega^2 A = m \times \left(\frac{2\pi}{T}\right)^2 \times A = ?$

b) we have, $F = kA$

$$\therefore k = \frac{F}{A} = ?$$

- 5. A block on a horizontal surface that is moving back and forth horizontally with SHM of frequency 2Hz. The coefficient of static friction between block and surface is 0.5. How great can the amplitude of SHM be if the block is not to slip along the surface?**

→ Here, for the block not to slip along the surface,

restoring force < frictional force

$$\text{or, } m\omega^2 x < \mu mg$$

$$\text{or, } x < \frac{\mu g}{\omega^2} < \frac{\mu g}{4\pi^2 f^2} = ?$$

6. The motion of the particle in simple harmonic motion is given by $x = A \sin \omega t$. If it has a speed u when the displacement is x_1 and speed v when the displacement is x_2 . Show that the amplitude A of the motion is

$$A = \sqrt{\frac{v^2 x_1^2 - u^2 x_2^2}{v^2 - u^2}}$$

$$\rightarrow \text{Here, } u = \omega \sqrt{A^2 - x_1^2} \dots \dots \dots (i)$$

$$\text{And } v = \omega \sqrt{A^2 - x_2^2} \dots \dots \dots (ii)$$

$$\therefore v^2 - u^2 = \omega^2 (x_1^2 - x_2^2)$$

$$\text{or, } \omega = \sqrt{\frac{v^2 - u^2}{x_1^2 - x_2^2}}$$

Now from equation (i)

$$u^2 = \omega^2 (A^2 - x_1^2) = \frac{v^2 - u^2}{x_1^2 - x_2^2} (A^2 - x_1^2)$$

Now solve for A .

7. When the displacement is half the amplitude, what fraction of the total energy is K.E. and what fraction is P.E. in SHM? At what displacement is the energy half K.E. and half P.E.

$\rightarrow$ if x be the displacement and A be the amplitude then by question, $x = A/2$

$$\text{we have, } K.E. = \frac{1}{2} m \omega^2 (A^2 - x^2) = \frac{1}{2} m \omega^2 \left(A^2 - \frac{A^2}{4} \right) = \frac{1}{2} m \omega^2 \frac{3}{4} A^2 = \frac{3}{4} E$$

$$\therefore \frac{K.E.}{E} = \frac{3}{4}$$

$$\text{Now, } \frac{P.E.}{E} = 1 - \frac{3}{4} = 1/4$$

For second part,

$$K.E. = P.E.$$

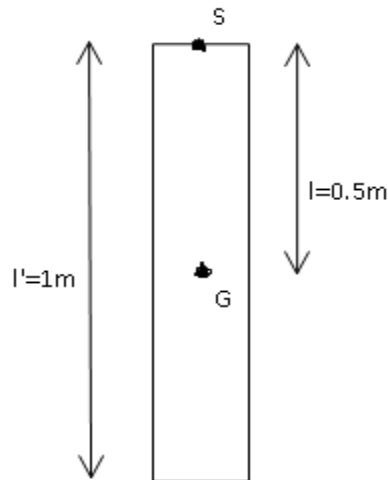
$$\text{or, } \frac{1}{2} m \omega^2 (A^2 - x^2) = \frac{1}{2} m \omega^2 x^2$$

$$\text{or, } A^2 - x^2 = x^2$$

$$\text{or, } x = A/\sqrt{2}$$

8. A meter stick swings about a pivot point at one of its end. What is the time period of oscillation? What is the distance of the pivot point from the centre of oscillation of the stick?

$$\rightarrow \text{Here, } l' = 1m, \quad \text{and } l = 0.5m$$



We have, $T = 2\pi \sqrt{\frac{l'^2 + k^2}{gl}} = 2\pi \sqrt{\frac{l + k^2/l}{g}}$

Since, $I_{C.G.} = mk^2 = \frac{ml'^2}{12}$

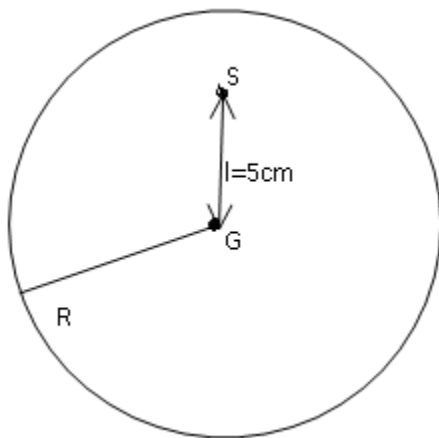
$\therefore k^2 = \frac{l'^2}{12} = ?$

$\therefore T = 2\pi \sqrt{\frac{l + k^2/l}{g}} = ?$

Again, the distance of pivot point from center of oscillation is $L = l + \frac{k^2}{l} = ?$

9. A uniform circular disc of diameter 20cm vibrates about a horizontal axis perpendicular plane and at a distance of 5cm from the centre. Calculate, the time period of oscillation and the equivalent length of the simple pendulum.

→ Here, $R = 10\text{cm} = 0.1\text{m}$



We have, $T = 2\pi \sqrt{\frac{l^2 + k^2}{gl}} = 2\pi \sqrt{\frac{l + k^2/l}{g}}$

Since, $I_{C.G.} = mk^2 = \frac{mR^2}{2}$

$\therefore k^2 = \frac{R^2}{2} = ?$

$\therefore T = 2\pi \sqrt{\frac{l + k^2/l}{g}} = ?$

Again, the equivalent length of simple pendulum is $L = l + \frac{k^2}{l} = ?$

- 10. A thin, straight, uniform rod of length $l=1\text{m}$ and mass 160gm hangs from a pivot at one end. What is the period for small oscillations? What is the length of simple pendulum that will have the same time period.**

→ Same as Q.N. 8.

- 11. A stick of length 1.85m oscillates as a physical pendulum. a) What is the length of pendulum that gives the least time period? b) What is the minimum time period?**

→ Here, $l' = 1.85\text{m}$

a) for minimum time period, $l = k$

We have, $I_{C.G.} = mk^2 = \frac{ml'^2}{12}$

$\therefore k = \frac{l'}{\sqrt{12}} = ?$

$\therefore l = k = ?$

b) we have, $T = 2\pi \sqrt{\frac{l + \frac{k^2}{l}}{g}}$

$\therefore T_{min.} = 2\pi \sqrt{\frac{2k}{g}} = ?$

- 12. A heavy circular ring of radius R oscillates in a vertical plane about a horizontal axis at a distance x from the centre, Show that the time period is minimum when $x=R$.**

→ Here, length of pendulum $l = x$

we have, time period is $T = 2\pi \sqrt{\frac{l + \frac{k^2}{l}}{g}} = 2\pi \sqrt{\frac{x + \frac{k^2}{x}}{g}} = 2\pi \sqrt{\frac{L}{g}}$, where $L = x + \frac{k^2}{x}$

For minimum time period, L must be minimum.

For minimum L , $\frac{dL}{dx} = 0$

or, $\frac{d}{dx} \left(x + \frac{k^2}{x} \right) = 0$

or, $1 - \frac{k^2}{x^2} = 0$

$$\text{or, } k^2 = x^2$$

$$\text{or, } x = k$$

we have, for ring

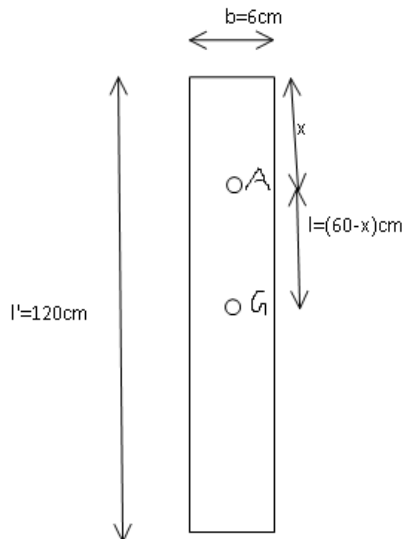
$$I_{C.G.} = mk^2 = mR^2$$

$$\therefore k = R$$

Hence T is minimum when $x = R$

13. A uniform thin rod of length 120cm and width 6cm is swinging in a vertical plane on pendulum about a point A at some distance from one end. If the time of swing is minimum, find the distance A from the end of the rod.

→let x be the distance of A from the end of the rod.



Here, length of pendulum, $l = (60 - x)$

Since, time period is minimum so $l = k$

$$\text{or, } 60 - x = k \dots \dots \dots (i)$$

We have,

$$I_{C.G.} = mk^2 = \frac{m(l'^2 + b^2)}{12}$$

$$\therefore k = \sqrt{\frac{l'^2 + b^2}{12}} = ?$$

Now from equation (i), $x = 60 - k = ?$

14. A simple pendulum of length l and mass m is suspended in a car that is travelling with a constant speed v around a circle of radius R . If the pendulum undergoes small oscillations about its equilibrium position x , prove that the frequency of oscillation is $f = \frac{1}{2\pi} \sqrt{\frac{gx}{lR}}$.

→ Here, the necessary centripetal force for the pendulum to move in circular path is provided by the restoring force.

i.e. centripetal force = restoring force

$$\text{or, } m\omega^2 R = mg \sin \theta$$

$$\text{or, } \omega^2 R = g \frac{x}{l}$$

$$\text{or, } \omega = \sqrt{\frac{gx}{lR}}$$

$$\therefore \text{frequency, } f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{gx}{lR}}$$

15. A simple pendulum of length 20cm and mass 5gm is suspended in a car travelling with constant speed 70m/s around a circle of radius 50m. If the pendulum undergoes small oscillations in a radial direction about its equilibrium position, what is the frequency of oscillation?

→ Here, effective value of acceleration due to gravity is

$$g_{eff.} = \sqrt{g^2 + \left(\frac{v^2}{R}\right)^2} = ?$$

$$\therefore T = 2\pi \sqrt{\frac{l}{g_{eff.}}}$$

$$\text{or, frequency, } f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{g_{eff.}}{l}} = ?$$

16. A simple pendulum of 1m length is suspended in an elevator. If the elevator is ascending at an acceleration of 3 m/s², what is the period of oscillation of the pendulum?

→ Here, effective value of acceleration due to gravity is

$$g_{eff.} = g + a$$

$$\therefore T = 2\pi \sqrt{\frac{l}{g_{eff.}}} = 2\pi \sqrt{\frac{l}{g+a}} = ?$$

17. A metallic disc of mass 0.5kg and radius 0.1m is suspended by a wire of length 40 cm and radius 1mm. The torsion oscillations of the disc are found to have a period of 2.5sec. Find the modulus of rigidity of the wire.

→ We have, Time period of Torsion Pendulum is

$$T = 2\pi \sqrt{\frac{I}{C}} = 2\pi \sqrt{\frac{\frac{mR^2}{2}}{\frac{\pi\eta r^4}{2l}}} = 2\pi \sqrt{\frac{mR^2}{2} \times \frac{2l}{\pi\eta r^4}}$$

$$\therefore \eta = ?$$

18. A wire has a torsional constant of 2Nm/rad. A disc of radius 5cm and mass 100gm is suspended at its centre. What is the frequency of torsional oscillations?

→ We have, Time period of Torsion Pendulum is

$$T = 2\pi \sqrt{\frac{I}{C}} = 2\pi \sqrt{\frac{mR^2}{2C}}$$

$$\therefore f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{2C}{mR^2}} = ?$$

19. A solid sphere of mass 2kg and diameter 0.30m is suspended on a wire. Find the period of angular oscillation for small displacements if the torque constant of the wire is 6×10^{-3} Nm/rad.

→ We have, Time period of Torsion Pendulum is

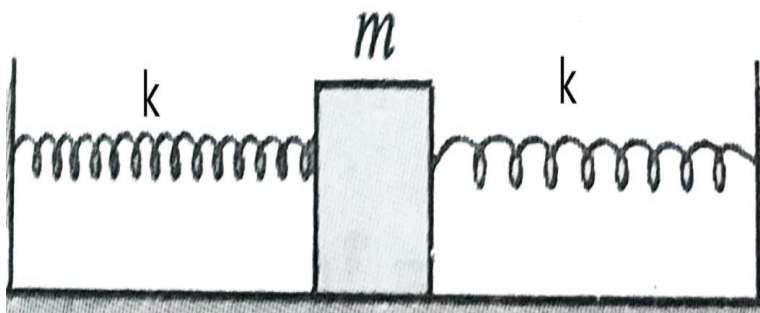
$$T = 2\pi \sqrt{\frac{I}{C}} = 2\pi \sqrt{\frac{2mR^2}{5C}} = ?$$

20. Two identical springs are attached to a block of mass m horizontally. Show that the frequency of

oscillation on the frictionless surface is given by $f = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$

→ If x be the extension in first spring, then restoring force $F_1 = -kx$

The compression in second spring is also x so restoring force $F_2 = -kx$



$$\therefore \text{net force, } F = F_1 + F_2 = -2kx$$

$$\text{or, } ma = -2kx$$

$$\text{or, } m \frac{d^2 x}{dt^2} = -2kx$$

$$\text{or, } \frac{d^2 x}{dt^2} + \frac{2k}{m}x = 0$$

$$\text{or, } \frac{d^2 x}{dt^2} + \omega^2 x = 0 \dots\dots\dots (i) \quad \text{where, } \omega = \sqrt{\frac{2k}{m}}$$

Here, equation (i) shows that the motion of mass spring system is simple harmonic in nature whose angular frequency is given by $\omega = \sqrt{\frac{2k}{m}}$

$$\therefore \text{Frequency, } f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$$

- 21. A 2kg mass hangs from a spring. A 300gm body hung below the mass stretches the spring 2cm further. If the 300gm body is removed and the mass is set in oscillation, find the period of motion.**

$$\rightarrow \text{Here, } F = mg = kx$$

$$\therefore k = \frac{mg}{x} = \frac{0.3 \times 10}{0.02} = ?$$

$$\text{Now time period, } T = 2\pi \sqrt{\frac{M}{k}} = 2\pi \sqrt{\frac{2}{k}} = ?$$

- 22. A mass of 1 kg is suspended from a spring of spring constant 25N/m. If the undamped frequency is $2/\sqrt{3}$ times the damped frequency, what will be the damping factor?**

$$\rightarrow \text{Here, } f = \frac{2}{\sqrt{3}} f'$$

$$\text{or, } \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{2}{\sqrt{3}} \times \frac{1}{2\pi} \times \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

$$\text{or, } \frac{k}{m} = \frac{4}{3} \times \left(\frac{k}{m} - \frac{b^2}{4m^2} \right)$$

$$\therefore b = ?$$

23. A 750gm block oscillates one end of a spring whose force constant is 56N/m. The mass moves in a fluid which offers a resistive force $F = -bv$, where $b = 0.162 \text{Ns/m}$. What the period of motion.

$$\rightarrow \text{we have, } \beta = \sqrt{\omega_0^2 - \frac{b^2}{4m^2}} = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} = ?$$

$$\therefore T = \frac{2\pi}{\beta} = ?$$

24. The energy of guitar string of frequency 256 Hz reduces to half its initial value in 2 seconds. What is the Q value of the string?

$$\rightarrow \text{Here, } f' = 256 \text{Hz,}$$

$$\text{We have, } E(t) = E_0 e^{\frac{-bt}{m}} = E_0 e^{\frac{-t}{\tau}}$$

$$\text{or, } \frac{1}{2} E_0 = E_0 e^{\frac{-2}{\tau}}$$

$$\text{or, } \ln\left(\frac{1}{2}\right) = \frac{-2}{\tau}$$

$$\text{or, } \tau = \frac{-2}{\ln\left(\frac{1}{2}\right)} = ?$$

$$\therefore \text{Quality factor, } Q = \beta\tau = 2\pi f' \times \tau = ?$$

25. The quality factor of a piano wire of frequency 500 Hz is 5000. In what time will its energy reduce to $1/e$ of its value because of damping?

$$\rightarrow \text{Here, } f' = 500 \text{Hz, } Q = 5000$$

$$\text{we have, Quality factor, } Q = \beta\tau = 2\pi f' \times \tau$$

$$\therefore \tau = \frac{Q}{2\pi f'} = ?$$

$$\text{Again, we have, } E(t) = E_0 e^{\frac{-bt}{m}} = E_0 e^{\frac{-t}{\tau}}$$

$$\text{or, } \frac{1}{e} E_0 = E_0 e^{\frac{-t}{\tau}}$$

$$\text{or, } e^{-1} = e^{\frac{-t}{\tau}}$$

$$\text{or, } -1 = \frac{-t}{\tau}$$

$$\therefore t = \tau = ?$$

26. A simple pendulum has a period of 1 sec and amplitude 10° . After 10 complete oscillations its amplitude is reduced to 5° . What is the relaxation time of the pendulum and the quality factor.

$$\rightarrow \text{Here, } T = 1\text{s, } \theta_0 = 10^\circ, \quad t = 10\text{s, } \theta(t) = 5^\circ$$

$$\text{we have } A(t) = R e^{\frac{-bt}{2m}} = R e^{\frac{-t}{2\tau}}$$

$$\therefore \theta(t) = \theta_0 e^{\frac{-t}{2\tau}}$$

$$\text{or, } 5^\circ = 10^\circ e^{\frac{-10}{2\tau}}$$

$$\text{or, } \frac{1}{2} = e^{\frac{-10}{2\tau}}$$

$$\text{or, } \ln\left(\frac{1}{2}\right) = \frac{-10}{2\tau}$$

$\therefore \tau = ?$

Again, quality factor, $Q = \beta\tau = \frac{2\pi}{T} \times \tau = ?$

27. Show that damping reduces the frequency of oscillator by $\frac{12.5}{Q^2} \%$.

→ We have the angular frequency of damped oscillation is

$$\beta = \sqrt{\omega_0^2 - \frac{b^2}{4m^2}} = \omega_0 \sqrt{1 - \frac{b^2}{4\omega_0^2 m^2}} = \omega_0 \sqrt{1 - \frac{1}{4\omega_0^2 \tau^2}} = \omega_0 \sqrt{1 - \frac{1}{4Q^2}}$$

($\because Q = \beta\tau \approx \omega_0\tau$ for low damping)

$$\text{or, } \frac{\beta}{\omega_0} = \left(1 - \frac{1}{4Q^2}\right)^{\frac{1}{2}} \approx 1 - \frac{1}{2} \times \frac{1}{4Q^2}$$

$$\text{or, } \frac{2\pi f'}{2\pi f} = 1 - \frac{1}{8Q^2}$$

$$\text{or, } \frac{f'}{f} = 1 - \frac{1}{8Q^2}$$

$$\text{or, } 1 - \frac{f'}{f} = \frac{1}{8Q^2}$$

$$\text{or, } \frac{f-f'}{f} = \frac{1}{8Q^2} \text{ This gives the fractional reduced in frequency due to the presence of damping.}$$

$$\therefore \frac{f-f'}{f} \times 100\% = \frac{1}{8Q^2} \times 100\% = \frac{12.5}{Q^2} \%$$

28. The quality factor of a tuning fork is 50000. Calculate the number of vibrations after which its energy becomes $1/10^{\text{th}}$ of the initial value.

→ We have, $Q = \beta\tau = \frac{2\pi}{T} \times \tau$

$$\therefore T = \frac{2\pi\tau}{Q} = \frac{2\pi\tau}{50000}$$

Again, we have, $E(t) = E_0 e^{\frac{-t}{\tau}}$

$$\text{or, } \frac{1}{10} E_0 = E_0 e^{\frac{-t}{\tau}}$$

$$\text{or, } \ln\left(\frac{1}{10}\right) = \frac{-t}{\tau}$$

$$\therefore t = -\tau \times \ln\left(\frac{1}{10}\right)$$

$$\text{Now, number of vibration, } n = \frac{t}{T} = \frac{-\tau \times \ln\left(\frac{1}{10}\right)}{2\pi\tau} \times 50000 = ?$$

29. If the relaxation time of the damped harmonic oscillator is 50sec, find the time period for which
a) the amplitude falls to $1/e$ times the initial value b) energy falls to $1/e^4$ times the initial value c) the energy falls to $1/e$ times the initial value.

→ a) We have,

$$A(t) = R e^{\frac{-bt}{2m}} = R e^{\frac{-t}{2\tau}}$$

$$\text{or, } \frac{1}{e} R = R e^{\frac{-t}{2\tau}}$$

$$\text{or, } e^{-1} = e^{\frac{-t}{2\tau}}$$

$$\text{or, } -1 = \frac{-t}{2\tau}$$

$$\therefore t = ?$$

$$\text{b) We have, } E(t) = E_0 e^{\frac{-bt}{m}} = E_0 e^{\frac{-t}{\tau}}$$

$$\text{or, } \frac{1}{e^4} E_0 = E_0 e^{\frac{-t}{\tau}}$$

$$\text{or, } e^{-4} = e^{\frac{-t}{\tau}}$$

$$\text{or, } -4 = \frac{-t}{\tau}$$

$$\therefore t = ?$$

c) Same as (b)

30. For damped oscillator, $m=250\text{gm}$, $k=85\text{N/m}$, and $b=70\text{g/sec}$. a) what is the period of motion? b) how long does it take for the amplitude of the damped oscillations to drop to half its initial value? C) How long does it take for the mechanical energy to drop to one half its initial value?

$$\rightarrow \text{a) } T = \frac{2\pi}{\beta} = \frac{2\pi}{\sqrt{\omega_0^2 - \frac{b^2}{4m^2}}} = \frac{2\pi}{\sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}} = ?$$

$$\text{b) } A(t) = R e^{\frac{-bt}{2m}}$$

$$\text{or, } \frac{1}{2} R = R e^{\frac{-bt}{2m}}$$

$$\therefore t = ?$$

$$\text{c) } E(t) = E_0 e^{\frac{-bt}{m}}$$

$$\text{or, } \frac{1}{2} E_0 = E_0 e^{\frac{-bt}{m}}$$

$$\therefore t = ?$$

31. The amplitude of a lightly damped oscillator decreases by 3% during each cycle. What percent of the mechanical energy of the oscillator is lost in each cycle?

$$\rightarrow \text{If } R \text{ be the initial amplitude, the initial energy is } E_0 = \frac{1}{2} m \omega_0^2 R^2$$

The amplitude after one cycle becomes $0.97R$. So the energy after one cycle becomes

$$E = \frac{1}{2} m \omega_0^2 (0.97R)^2 = (0.97)^2 \times E_0$$

$$\text{or, } \frac{E}{E_0} = (0.97)^2$$

$$\text{Now, \% of energy lost per cycle} = \frac{E_0 - E}{E_0} \times 100\% = \left(1 - \frac{E}{E_0}\right) \times 100\%$$

$$= [1 - (0.97)^2] \times 100\% \approx 6\%$$

32. If Q of an oscillator be 40, calculate the ratio of the amplitude of forced vibration and the amplitude at resonance when the frequency of the applied force is 99% of the natural frequency in the absence of damping.

→ Here, $Q=40$, $\omega = 0.99\omega_0$

We have, amplitude of forced oscillation is $A = \frac{F_m}{\sqrt{m^2(\omega^2 - \omega_0^2)^2 + b^2\omega^2}}$

$$\therefore \text{amplitude at resonance, } A_{max.} = \frac{F_m}{b\omega_0}$$

$$\text{Now, } \frac{A}{A_{max.}} = \frac{b\omega_0}{\sqrt{m^2(\omega^2 - \omega_0^2)^2 + b^2\omega^2}} = \frac{1}{\sqrt{\frac{m^2}{b^2\omega_0^2}(\omega^2 - \omega_0^2)^2 + \frac{\omega^2}{\omega_0^2}}} = \frac{1}{\sqrt{\frac{m^2\omega_0^4}{b^2\omega_0^2}\left(\frac{\omega^2}{\omega_0^2} - 1\right)^2 + \frac{\omega^2}{\omega_0^2}}}$$

$$\text{or, } \frac{A}{A_{max.}} = \frac{1}{\sqrt{Q^2\left(\frac{\omega^2}{\omega_0^2} - 1\right)^2 + \frac{\omega^2}{\omega_0^2}}} = ?$$

ELECTROMAGNETIC OSCILLATION

33. A circuit has $L=10\text{mH}$ and $C=1\mu\text{F}$. How much resistance must be inserted in the circuit to reduce the resonant frequency by 0.01%?

→ Here, $L=10\text{mH}$ and $C=1\mu\text{F}$

$$\frac{f - f'}{f} \times 100\% = 0.01\%$$

$$\text{or, } 1 - \frac{f'}{f} = 10^{-4}$$

$$\text{or, } \frac{f'}{f} = 1 - 10^{-4}$$

$$\text{or, } \frac{\frac{1}{2\pi\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}}}{\frac{1}{2\pi\sqrt{\frac{1}{LC}}}} = 1 - 10^{-4}$$

$$\text{or, } \sqrt{1 - \frac{R^2 C}{4L}} = 1 - 10^{-4}$$

$$\therefore R = ?$$

34. A circuit has $L=10\text{mH}$ and $C=1\mu\text{F}$. How much resistance should be added to circuit so that the frequency of oscillation will be 1% less than that of the frequency of free LC oscillation.

→ Here, $L=10\text{mH}$ and $C=1\mu\text{F}$

$$f' = f - 1\% \text{ of } f = 0.99f$$

$$\text{or, } \frac{f'}{f} = 0.99$$

$$\text{or, } \frac{\frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}}{\frac{1}{2\pi} \sqrt{\frac{1}{LC}}} = 0.99$$

$$\text{or, } \sqrt{1 - \frac{R^2 C}{4L}} = 0.99$$

$$\therefore R = ?$$

35. At $t=0$, a 40mH inductor is placed in series with a resistance $R=3\Omega$ and a charged capacitor $C=4.8\mu\text{F}$. Determine the frequency of oscillation. What is the time required for the charge amplitude to drop to half its starting value?

$$\rightarrow f' = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = ?$$

Again, we have, charge amplitude $q_m(t) = q_0 e^{\frac{-Rt}{2L}}$

$$\text{or, } \frac{1}{2} q_0 = q_0 e^{\frac{-Rt}{2L}}$$

$$\text{or, } \ln\left(\frac{1}{2}\right) = \frac{-Rt}{2L}$$

$$\therefore t = ?$$

36. You are given 10mH inductor and two capacitors of $5\mu\text{F}$ and $2\mu\text{F}$ capacitances. What resonant frequencies can be obtained by connecting these elements in various ways?

$\rightarrow$ Here, $C_1=5\mu\text{F}$, $C_2=2\mu\text{F}$, $L=10\text{mH}=10 \times 10^{-3}\text{H}$

When, C_1 and C_2 are in series, $C = \frac{C_1 C_2}{C_1 + C_2} = ?$

$$\therefore f_1 = \frac{1}{2\pi\sqrt{LC}} = ?$$

Again, when C_1 and C_2 are in parallel, $C' = C_1 + C_2 = ?$

$$\therefore f_2 = \frac{1}{2\pi\sqrt{LC'}} = ?$$

- 37. A 0.5kg body oscillates in SHM on a spring that when extended 2mm from its equilibrium position, has 8N restoring force. What are the angular frequency of oscillation and the capacitance of LC circuit with same period if inductor is 5H?**

$\rightarrow$ we have, $F = kx$

$$\therefore k = \frac{F}{x} = \frac{8}{2 \times 10^{-3}} = ?$$

Now, angular frequency, $\omega = \sqrt{\frac{k}{m}} = ?$

Again, for LC oscillation, $\omega = \frac{1}{\sqrt{LC}}$

$$\therefore C = ?$$

38. A $1\mu\text{F}$ capacitor, a 0.2H inductor and a 800Ω resistor are connected in series. Is the circuit oscillatory?

→ Here, $\omega_0 = \frac{1}{\sqrt{LC}} = ?$

and $\frac{R}{2L} = ?$

For the system to be oscillatory, $\omega_0 > \frac{R}{2L}$ otherwise not.

39. A 1200pF capacitor is charged by a 500V battery. It is disconnected from the battery and is connected to a 75mH inductor at t=0. Determine a) the initial charge on the capacitor, b) the maximum current, c) the frequency and period of oscillation d) the total energy oscillating in the system. Ans. 0.6μC, 63mA, 12kHz, 6x10⁻⁵sec, 1.5x10⁻⁴J

→ a) $q_m = CV = ?$

b) We have, $q = q_m \cos(\omega t \pm \varphi)$

∴ Current, $i = \frac{dq}{dt} = -\omega q_m \sin(\omega t \pm \varphi)$

∴ $i_{max.} = \omega q_m = \frac{1}{\sqrt{LC}} q_m = ?$

c) $f = \frac{1}{2\pi\sqrt{LC}} = ?$ And $T = \frac{1}{f} = ?$

d) Total Energy, $U = \frac{q_m^2}{2C} = ?$

Chapter: Wave Motion

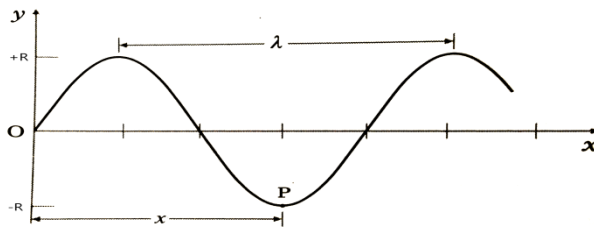
Progressive Wave:

If a wave is propagating along positive x-direction then it is called progressive wave.

The equation of progressive wave is

$$y = R\sin(\omega t - kx)$$

Where R is amplitude of wave and $k = \frac{2\pi}{\lambda}$ is wave number.



Simple harmonic progressive wave

Note:

$$\text{Particle velocity, } v_p = \frac{dy}{dt} = R\omega \cos(\omega t - kx) = \pm\omega\sqrt{R^2 - y^2}$$

$$\text{Wave velocity, } v = \left| \frac{dx}{dt} \right| = \left| \frac{dy}{dt} \frac{dx}{dy} \right| = \frac{R\omega \cos(\omega t - kx)}{Rk \cos(\omega t - kx)} = \frac{\omega}{k} = \frac{2\pi f}{2\pi} \times \lambda = \lambda f$$

$$\text{Particle acceleration, } a_p = \frac{dv_p}{dt} = -R\omega^2 \sin(\omega t - kx) = -\omega^2 y$$

Relation between Particle velocity and Wave velocity

We have displacement of the particles in a medium due to the propagation of progressive wave is

$$y = R\sin(\omega t - kx)$$

$$\text{Now, Particle velocity } v_p = \frac{dy}{dt} = R\omega \cos(\omega t - kx) \dots \dots \dots (i)$$

$$\text{Again, slope at any point is } m = \frac{dy}{dx} = -Rk \cos(\omega t - kx) \dots \dots \dots (ii)$$

Dividing equation (i) by (ii), we get

$$\frac{v_p}{m} = -\frac{\omega}{k} = -\lambda f = -v, \text{ where } v = \lambda f \text{ is wave velocity.}$$

$$\therefore \frac{v_p}{v} = -m$$

i.e. the ratio of particle to the wave velocity is numerically equal to the slope at any point.

$$\text{Also, } v_p = (-m) \times v$$

Differential Equation of wave motion

We have displacement of the particles in a medium due to the propagation of progressive wave is

$$y = R \sin(\omega t - kx) \dots \dots \dots (i)$$

$$\Rightarrow y = f(x, t)$$

Now differentiating equation (i) with respect to t twice, we get

$$\frac{dy}{dt} = R\omega \cos(\omega t - kx)$$

$$\therefore \frac{d^2y}{dt^2} = -R\omega^2 \sin(\omega t - kx) = -\omega^2 y$$

$$\text{or, } y = -\frac{1}{\omega^2} \frac{d^2y}{dt^2} \dots \dots \dots (ii)$$

Again, differentiating equation (i) with respect to x twice, we get

$$\frac{dy}{dx} = -Rk \cos(\omega t - kx)$$

$$\therefore \frac{d^2y}{dx^2} = -Rk^2 \sin(\omega t - kx) = -k^2 y = -k^2 \times -\frac{1}{\omega^2} \frac{d^2y}{dt^2} \quad (\text{since, using equation (ii)})$$

$$\text{or, } \frac{d^2y}{dx^2} = \frac{k^2}{\omega^2} \frac{d^2y}{dt^2}$$

$$\text{or, } \frac{d^2y}{dx^2} = \frac{1}{v^2} \frac{d^2y}{dt^2}$$

$$(\text{since, } \frac{\omega}{k} = v \text{ is wave velocity})$$

This is the differential equation of wave motion.

$$\text{Also, we can write } \frac{d^2y}{dt^2} = v^2 \times \frac{d^2y}{dx^2}$$

$$\therefore \text{Particle acceleration} = (\text{wave velocity})^2 \times \text{curvature at a point}$$

Energy, Power and Intensity of Progressive Wave

Small workdone dW to displace the particles by dy from mean position is

$$dW = Fdy = ma_p dy = m\omega^2 y dy$$

If ρ be the density of medium then small workdone per unit volume is

$$dw = \rho\omega^2 y dy$$

Now, total workdone per unit volume to displace the particle from 0 to y is

$$\begin{aligned} w &= \int dw = \int_0^y \rho\omega^2 y dy = \rho\omega^2 \int_0^y y dy = \frac{1}{2}\rho\omega^2 y^2 \\ &= \frac{1}{2}\rho\omega^2 R^2 \sin^2(\omega t - kx) \quad (\because y = R\sin(\omega t - kx)) \end{aligned}$$

This workdone per unit volume is stored as potential energy per unit volume (u).

$$\begin{aligned} \therefore u &= w = \frac{1}{2}\rho\omega^2 R^2 \sin^2(\omega t - kx) = 2\pi^2 f^2 R^2 \rho \sin^2(\omega t - kx) \\ \Rightarrow u_{avg.} &= \pi^2 f^2 R^2 \rho \end{aligned}$$

Again, kinetic energy per unit volume is

$$\begin{aligned} k &= \frac{1}{2}\rho v_p^2 = \frac{1}{2}\rho\omega^2 R^2 \cos^2(\omega t - kx) = 2\pi^2 f^2 R^2 \rho \cos^2(\omega t - kx) \\ \Rightarrow k_{avg.} &= \pi^2 f^2 R^2 \rho \end{aligned}$$

Now total energy per unit volume is

$$\begin{aligned} e &= u + k = 2\pi^2 f^2 R^2 \rho \sin^2(\omega t - kx) + 2\pi^2 f^2 R^2 \rho \cos^2(\omega t - kx) \\ &= 2\pi^2 f^2 R^2 \rho \end{aligned}$$

$\therefore$ Average P.E. per unit volume and average K.E. per unit volume are equal and each is equal to half of total energy per unit volume.

Also, Potential energy per unit volume and kinetic energy per unit volume both vary with time but their averages remain constant.

Power:

We have, total energy of progressive wave is

$$E = e \times \text{volume} = e \times Al$$

Where, l is the length and A is the area of cross-section of the medium through which wave is propagating.

$$\text{or, } E = 2\pi^2 f^2 R^2 \rho \times Al$$

$$\therefore \text{Power, } P = \frac{E}{t} = 2\pi^2 f^2 R^2 \rho \frac{l}{t} A = 2\pi^2 f^2 R^2 \rho v A$$

Where, $v = \frac{l}{t}$ is the speed of wave.

Now the intensity of wave is given by

$$I = \frac{P}{A} = 2\pi^2 f^2 R^2 \rho v$$

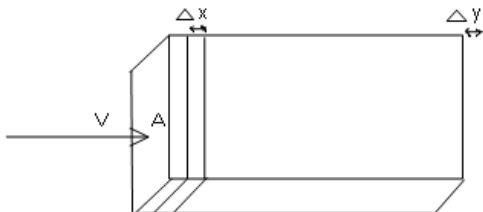
Note: If r be the distance of an observer from a speaker then

$$\text{Power, } P = 2\pi^2 f^2 R^2 \rho v A = 2\pi^2 f^2 R^2 \rho v \times 4\pi r^2 = \frac{1}{2} \omega^2 R^2 \rho v \times 4\pi r^2 = \omega^2 R^2 \rho v \times 2\pi r^2$$

$$\therefore \text{the amplitude of wave is } R = \frac{1}{r} \sqrt{\frac{P}{2\pi \omega^2 \rho v}}$$

Chapter: Acoustics

Excess Pressure in a medium due to the propagation of longitudinal wave



Consider a longitudinal wave is propagating in a medium with speed v as shown in figure. Let us divide the medium into infinite number of small layer of cross-section area A and thickness Δx . Let Δy be the displacement in each layer. If B be the bulk modulus of the medium then,

$$B = \frac{\text{Normal stress}}{\text{volumetric strain}} = \frac{\frac{F}{A}}{-\frac{\Delta V}{V}} = -\frac{P}{\frac{\Delta V}{V}}$$

Where, -ve sign indicates that there is inverse relation between pressure and volume.

$$\therefore P = -B \frac{\Delta V}{V} = -B \frac{A \Delta y}{A \Delta x} = -B \frac{\Delta y}{\Delta x}$$

Taking limit $\Delta x \rightarrow 0$ on both sides,

$$P = -B \frac{dy}{dx}$$

We have, $y = R \sin(\omega t - kx)$

$$\therefore \frac{dy}{dx} = -kR \cos(\omega t - kx)$$

$$\text{So, } P = BkR \cos(\omega t - kx)$$

$$\text{Or, } P = P_m \cos(\omega t - kx)$$

Where, $P_m = BkR$ is Pressure amplitude.

If ρ be the density of medium then the velocity of sound in a medium is $v = \sqrt{\frac{B}{\rho}}$

$$\therefore B = v^2 \rho$$

$$\text{Now, } P_m = BkR = v^2 \rho \times \frac{2\pi}{\lambda} \times R = 2\pi \frac{v}{\lambda} R \rho v = 2\pi f R \rho v$$

We have intensity of wave in a medium is

$$I = 2\pi^2 f^2 R^2 \rho v = \frac{(2\pi f R \rho v)^2}{2\rho v} = \frac{P_m^2}{2\rho v}$$

Threshold of hearing:

The minimum amount of sound energy which is just received by human's ear is called threshold of hearing and corresponding intensity of sound wave is threshold intensity.

Loudness:

The amount of sound energy which is greater than threshold of hearing is called loudness. The loudness of sound wave is measured in terms of intensity level. The intensity level of sound wave having intensity I is given by

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right), \text{ where } I_0 \text{ is threshold intensity. The unit of intensity level is dB(decibel)}$$

Reverberation

The phenomenon of intermixing between original and reflected sound wave in a medium due to which the sound wave received by the observer is somewhat extended is called reverberation.

Reverberation Time

The time in which the intensity of sound wave becomes one millionth of the initial intensity is reverberation time.

Absorption Coefficient

The ratio of perfectly sound absorbing area to the total available area is called absorption coefficient.

$$\text{i.e. absorption coefficient, } \alpha = \frac{\text{perfectly absorbing area}}{\text{total available area}}$$

If $\alpha_1, \alpha_2, \dots, \alpha_n$ be the absorption coefficient for surfaces having areas $S_1, S_2, \dots, S_n$ then the average absorption coefficient is given by

$$\alpha = \frac{\alpha_1 S_1 + \alpha_2 S_2 + \dots + \alpha_n S_n}{S_1 + S_2 + \dots + S_n}$$

Sabine's Relation

Let us consider a hall having volume V and total surface area S . Let α be the average absorption coefficient of the hall and I be the instantaneous intensity of sound wave. If ΔI be the change in intensity in time Δt , then according to Sabine's assumption,

$$\Delta I = -I\alpha n\Delta t$$

Where n is number of reflection per second and $-ve$ sign indicates that intensity goes on decreasing with increase in time.

$$\therefore \frac{\Delta I}{\Delta t} = -I\alpha n$$

Taking limit $\Delta t \rightarrow 0$ on both sides, we have

$$\frac{dI}{dt} = -I\alpha n$$

$$\text{or, } \frac{dI}{I} = -\alpha n dt$$

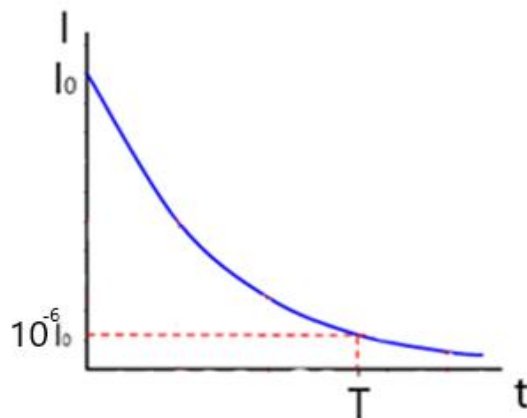
Integrating both sides from time 0 to t

$$\int_{I_0}^I \frac{dI}{I} = -\alpha n \int_0^t dt \quad \text{Here, } I_0 \text{ is initial intensity.}$$

$$\text{or, } \ln\left(\frac{I}{I_0}\right) = -\alpha n t$$

$$\text{or, } \frac{I}{I_0} = e^{-\alpha n t}$$

$$\text{or, } I = I_0 e^{-\alpha n t} \dots \dots \dots (i)$$



We have the mean free path is $\sigma = \frac{4V}{S}$

If v be the velocity of sound in air then time taken by the sound wave for two successive reflections is

$$T' = \frac{\sigma}{v} = \frac{4V}{Sv}$$

$$\therefore \text{number of reflections per second, } n = \frac{1}{T'} = \frac{Sv}{4V}$$

Now from equation (i)

$$I = I_0 e^{-\alpha n t} = I_0 e^{-\alpha \frac{Sv}{4V} t}$$

When $I = 10^{-6} I_0$, then $t = T$; the reverberation time

$$\therefore 10^{-6} I_0 = I_0 e^{-\alpha \frac{Sv}{4V} T}$$

$$\text{or, } 10^{-6} = e^{-\alpha \frac{Sv}{4V} T}$$

$$\text{or, } 10^6 = e^{\alpha \frac{Sv}{4V} T}$$

$$\text{or, } \alpha \frac{Sv}{4V} T = \ln(10^6)$$

$$\text{or, } \alpha \frac{Sv}{4V} T = 13.8$$

$$\text{or, } T = \frac{4 \times 13.8V}{\alpha S \times v}$$

We have the velocity of sound in air is $v \approx 340 \text{ m/s}$

$$\therefore T = \frac{4 \times 13.8V}{\alpha S \times 340}$$

$$\text{or, } T = \frac{0.16V}{\alpha S} \text{ This is Sabine's relation in SI}$$

In FPS system, this becomes

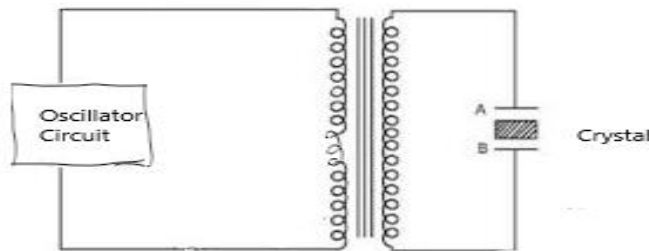
$$T = \frac{0.05V}{\alpha S}$$

Ultrasound

If the frequency of sound wave is greater than 20 kHz, then it is called ultrasound.

Production of Ultrasound (Piezoelectric Generator)

If mechanical stress is applied on certain crystal such as quartz, tourmaline etc., an electric field is developed surrounding the crystal. Conversely, if an electric field is applied on the crystal, mechanical stress is developed on it. This effect is called piezoelectric effect and is used to produce the ultrasound.



An ac field produced by an oscillator circuit is applied between two metallic plates A and B with the help of a transformer as shown in figure.

When a crystal is placed between two metallic plates as a dielectric, the crystal starts to vibrate. When the frequency of applied ac field is equal to natural frequency of the crystal, the crystal starts to vibrate with maximum amplitude. During that period, the frequency of wave surrounding the crystal is greater than 20 kHz which is ultrasound.

Uses of ultrasound

1. Medical uses
 - a) It is used to identify the presence of tumors.
 - b) It is used to identify the fracture of bones.
 - c) It is used in bloodless surgery.
 - d) It is used to investigate internal organs.
 - e) It is used to cure the muscular pain.
2. Directional signaling: Due to very high frequency (low wavelength), its angular spread is too small. So it can be used as directional signaling.
3. It is used in cutting, welding and drilling purposes.
4. It is used to identify the depth of sea.

Numerical

Speed of a wave on a stretched string $v = \sqrt{\frac{T}{\mu}}$, where T is tension and μ is mass per unit length.

Velocity of longitudinal wave in a fluid $v = \sqrt{\frac{B}{\rho}}$, Where B is Bulk modulus and ρ is density.

Pressure amplitude $P_m = 2\pi f R \rho v$

1. The displacement equation of a transverse plane wave at any instant is given by $y = 0.03 \sin(3\pi t - 0.03\pi x)$ where x and t are in meter and sec. Calculate the wavelength, frequency and velocity of the wave. Also calculate phase difference between two particles 0.05m apart at same instant.

→ Here the given wave equation is $y = 0.03 \sin(3\pi t - 0.03\pi x)$

Comparing this equation with $y = R \sin(\omega t - kx)$ we get

Amplitude, $R = 0.03\text{m}$, angular frequency, $\omega = 3\pi \text{ rad/s}$, wave number, $k = 0.03\pi \text{ m}^{-1}$

Path difference, $x = 0.05\text{m}$

Now, wavelength, $\lambda = \frac{2\pi}{k} = ?$

Frequency, $f = \frac{\omega}{2\pi} = ?$

Wave velocity, $v = \lambda f = ?$

Phase difference, $\phi = \frac{2\pi}{\lambda} \times x = ?$

2. The function $y = (15\text{cm}) \cos(\pi x - 15\pi t)$, with x in meter and t in sec., describes the wave on a taut string. What is the transverse speed for a point on the string at an instant when the point has the displacement $y = 12\text{cm}$?

→ Here the given wave equation is $y = (15\text{cm}) \cos(\pi x - 15\pi t)$

Comparing this equation with $y = R \cos(kx - \omega t)$ we get

$R = 15\text{cm} = 0.15\text{m}$, $k = \pi \text{ m}^{-1}$, $\omega = 15\pi \text{ rad/s}$,

Transverse speed, $v_p = \omega \sqrt{R^2 - y^2} = ?$ take $y = 0.12\text{m}$

3. Here are three equations of a wave $y = 2 \sin(4x - 2t)$, $y = \sin(3x - 4t)$ and $y = 2 \sin(3x - 3t)$. Rank the waves according to their a) wave speed and b) maximum transverse speed, greatest first.

→ Comparing given equations with $y = R \sin(kx - \omega t)$ we get

$$R_1 = 2m, \quad R_2 = 1m, \quad R_3 = 2m$$

$$k_1 = 4m^{-1}, \quad k_2 = 3m^{-1}, \quad k_3 = 3m^{-1} \text{ and}$$

$$\omega_1 = 2\text{rads}^{-1}, \quad \omega_2 = 4\text{rads}^{-1}, \quad \omega_3 = 3\text{rads}^{-1}$$

Now, wave speeds,

$$v_1 = \frac{\omega_1}{k_1} = \frac{2}{4} = 0.5\text{m/s}, \quad v_2 = \frac{\omega_2}{k_2} = \frac{4}{3} = 1.33\text{m/s}, \quad v_3 = \frac{\omega_3}{k_3} = \frac{3}{3} = 1\text{m/s}$$

$$\therefore v_2 > v_3 > v_1$$

And maximum transverse speeds,

$$(v_{p1})_{\max} = R_1 \omega_1 = 2 \times 2 = 4\text{m/s}, \quad (v_{p2})_{\max} = R_2 \omega_2 = 1 \times 4 = 4\text{m/s}$$

$$(v_{p3})_{\max} = R_3 \omega_3 = 2 \times 3 = 6\text{m/s}$$

$$\therefore (v_{p3})_{\max} > (v_{p1})_{\max} = (v_{p2})_{\max}$$

Now, Rank according to wave speed

$$1) y = \sin(3x-4t) \quad (2) y = 2\sin(3x-3t) \quad (3) y = 2\sin(4x-2t)$$

and Rank according to maximum transverse speed

$$(1) y = 2\sin(3x-3t) \quad (2) y = 2\sin(4x-2t) \quad (3) y = \sin(3x-4t)$$

4. A sinusoidal wave travels along a string. The time for a particular point to move from maximum displacement to zero is 0.170 sec. What are the a) period b) frequency c) the wave speed if the wavelength is 1.40m.

$$\rightarrow \text{Here, } \frac{T}{4} = 0.170\text{s}, \quad \text{wavelength, } \lambda = 1.40\text{m}$$

$$\text{a) Period, } T = ? \quad \text{b) frequency, } f = \frac{1}{T} = ? \quad \text{c) wave speed, } v = \lambda f = ?$$

5. **Prove that if the transverse wave is travelling along a string, then the slope at any point of the string is numerically equal to ratio of the particle speed to the wave speed at that point.**

→ Refer to theory

6. Determine the amplitude of resultant wave when two sinusoidal waves having the same frequency and travelling in the same direction are combined, if their amplitudes are 3cm and 4cm and they differ in phase by $\frac{\pi}{2}$ rad.

$$\rightarrow \text{Here, } a_1 = 3\text{cm}, a_2 = 4\text{cm} \quad \text{and } \varphi = \frac{\pi}{2}\text{rad}$$

$$\text{Now, amplitude of resultant wave, } R = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\varphi} = ?$$

7. **Show that the function $y = A \sin kx \cos \omega t$ satisfies the wave equation.**

$$\rightarrow \text{Here, the given function is } y = A \sin kx \cos \omega t \dots\dots\dots(i)$$

Now differentiating equation (i) with respect to t twice, we get

$$\frac{dy}{dt} = -\omega A \sin kx \sin \omega t$$

$$\therefore \frac{d^2 y}{dt^2} = -\omega^2 A \sin kx \cos \omega t = -\omega^2 y$$

$$\text{or, } y = -\frac{1}{\omega^2} \frac{d^2 y}{dt^2} \dots \dots \dots (ii)$$

Again, differentiating equation (i) with respect to x twice, we get

$$\frac{dy}{dx} = k A \cos kx \cos \omega t$$

$$\therefore \frac{d^2 y}{dx^2} = -k^2 A \sin kx \cos \omega t = -k^2 y = -k^2 \times -\frac{1}{\omega^2} \frac{d^2 y}{dt^2} \quad (\text{since, using equation (ii)})$$

$$\text{or, } \frac{d^2 y}{dx^2} = \frac{k^2}{\omega^2} \frac{d^2 y}{dt^2}$$

$$\text{or, } \frac{d^2 y}{dx^2} = \frac{1}{v^2} \frac{d^2 y}{dt^2} \quad (\text{since, } \frac{\omega}{k} = v \text{ is wave velocity})$$

This is the differential equation of wave motion. So the given function satisfies the wave equation.

- 8. A source of sound has a frequency of 256 Hz and amplitude of 0.50 cm. Calculate the energy flow across a square cm per sec. The velocity of sound in air is 330 m/s and density of air is 1.29 kg/m³.**

→ Here, frequency, $f = 256 \text{ Hz}$, amplitude, $R = 0.50 \text{ cm} = 0.005 \text{ m}$, velocity, $v = 330 \text{ m/s}$

Density $\rho = 1.29 \text{ kg/m}^3$, Intensity, $I = ?$

$$\text{We have, } I = \frac{P}{A} = 2\pi^2 f^2 R^2 \rho v = \dots \dots \dots \text{Jm}^{-2}\text{s}^{-1} = \dots \dots \dots \times 10^{-4} \text{Jcm}^{-2}\text{s}^{-1}$$

- 9. A string has linear density 525 g/m and tension 45 N. We send a sinusoidal wave with frequency 120 Hz and amplitude 8.5 mm along the string. At what average rate does the wave transport energy?**

→ Here, linear mass density, $\mu = 525 \text{ gm/m} = 0.525 \text{ kg/m}$, Tension, $T = 45 \text{ N}$,
 $f = 120 \text{ Hz}$, $R = 8.5 \text{ mm} = 8.5 \times 10^{-3} \text{ m}$, Power, $P = ?$

$$\text{we have, Power, } P = I \times A = 2\pi^2 f^2 R^2 \rho v \times A = 2\pi^2 f^2 R^2 \mu v = 2\pi^2 f^2 R^2 \mu \sqrt{\frac{T}{\mu}} = 2\pi^2 f^2 R^2 \sqrt{T\mu} = ?$$

- 10. A piano wire with mass 3 gm and length 80 cm is stretched with tension of 25 N. A wave with frequency 120 Hz and amplitude 1.6 mm travels along the wire. a) Calculate the average power carried by the wave b) what happens to the average power if the wave amplitude is halved?**

→ Here, $\mu = \frac{3 \times 10^{-3}}{80 \times 10^{-2}} \text{ kg/m}$, $T = 25 \text{ N}$, $f = 120 \text{ Hz}$, $R = 1.6 \text{ mm} = 1.6 \times 10^{-3} \text{ m}$

$$\begin{aligned}
 \text{a) we have, Power, } P &= I \times A = 2\pi^2 f^2 R^2 \rho v \times A = 2\pi^2 f^2 R^2 \mu v = 2\pi^2 f^2 R^2 \mu \sqrt{\frac{T}{\mu}} \\
 &= 2\pi^2 f^2 R^2 \sqrt{T\mu} = ?
 \end{aligned}$$

b) When amplitude is halved, the average power becomes, $P' = \frac{P}{4} = ?$

11. Calculate the average amplitude of a sinusoidal wave in air of a frequency of 1.5kHz and average intensity 10^{-5}W/cm^2 , where density of air is 1.29kg/m^3 .

$$\rightarrow R = ?, f = 1.5 \text{kHz} = 1500 \text{Hz}, I = 10^{-5} \text{W/cm}^2 = 0.1 \text{W/m}^2, \rho = 1.29 \text{kg/m}^3$$

$$\text{we have, } I = \frac{P}{A} = 2\pi^2 f^2 R^2 \rho v \quad \text{where } v = 330 \text{ m/s}$$

$$\therefore R = ?$$

12. The speed of transverse wave on a string is 170m/s when the string tension is 120N. To what value must the tension be changed to raise the wave speed to 180m/s?

$$\rightarrow \text{Here, } v_1 = 170 \text{m/s}, T_1 = 120 \text{N}, \quad v_2 = 180 \text{m/s}, \quad T_2 = ?$$

$$\text{We have, } v = \sqrt{\frac{T}{\mu}}$$

$$\therefore \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$$

$$\text{or, } T_2 = \frac{v_2^2}{v_1^2} \times T_1 = ?$$

13. Two strings are made of same material. String 1 has twice the diameter of string 2, but is under half the tension. Find $\frac{v_2}{v_1}$.

$$\rightarrow \text{Here, } \rho_1 = \rho_2, \quad d_1 = 2d_2, \quad T_1 = \frac{1}{2}T_2, \quad \frac{v_2}{v_1} = ?$$

$$\text{We have, } v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{A\rho}} = \sqrt{\frac{4 \times T}{\pi d^2 \rho}}$$

$$\therefore \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1} \times \frac{d_1^2}{d_2^2} \times \frac{\rho_1}{\rho_2}} = ?$$

14. If a wave of frequency 500Hz is travelling with a velocity of 200m/s, find the change in phase at a given point in space between a time interval of 1ms. Also find the path difference between two points which differ in phase by $\frac{\pi}{2}$ radian.

$$\rightarrow \text{Here, } f = 500 \text{Hz}, \quad v = 200 \text{m/s}, \quad \Delta t = 1 \text{ms} = 10^{-3} \text{s}, \quad \text{Phase difference, } \phi = \frac{\pi}{2}$$

$$\text{we have, the phase angle is } \theta = \omega t - kx$$

$$\therefore \text{change in phase at a given point is } \theta_2 - \theta_1 = \omega(t_2 - t_1) = 2\pi f \times \Delta t = ?$$

$$\text{Again we have, phase difference, } \phi = \frac{2\pi}{\lambda} \times \text{path difference, } x$$

$$\therefore x = \frac{\lambda}{2\pi} \times \varphi = \frac{v}{2\pi f} \times \varphi = ?$$

15. One end of a 14 m long rubber tube with total mass 0.8kg is fastened to a fixed support. A cord attached to the other end passes over a pulley and supports an object with a mass of 7.5kg. The tube struck a transverse blow at one end. Find the time required for the pulse to reach the other end.

→ Here, length of tube, $l = 14\text{m}$, mass of tube = 0.8kg $\therefore \mu = \frac{0.8}{14}\text{kg/m}$

$$\text{Tension } T = mg = 7.5 \times 10 = 75\text{N} \quad \therefore v = \sqrt{\frac{T}{\mu}} = ?$$

Now time required to reach other end, $t = \frac{l}{v} = ?$

16. Calculate the reverberation time for a hall of volume 1400m^3 , which has seating capacity of 110 persons with full capacity of audience and when audience are occupying only cushioned seats. Relevant data can be taken from the following table

Surface	area(m^2)	coefficient of absorption
Plastered wall	98	0.03
Plastered ceiling	144	0.04
Wooden door	15	0.06
Cushioned chairs	88	1.00
Audience	150	4.70

→ Here, total absorption coefficient is

$$\alpha S = \alpha_1 S_1 + \alpha_2 S_2 + \alpha_3 S_3 + \alpha_4 S_4 + \alpha_5 S_5$$

$$= 0.03 \times 98 + 0.04 \times 144 + 0.06 \times 15 + 1 \times 88 + 4.7 \times 110 = ?$$

$$\text{Now, reverberation time, } T = \frac{0.16V}{\alpha S} = \frac{0.16 \times 1400}{\alpha S} = ?$$

17. The volume of a room is 980m^3 . The wall area of the room is 150m^2 , ceiling area 95m^2 and floor area is 90m^2 . The average sound absorption coefficient a) for wall is 0.03, b) for ceiling is 0.80 and c) for the floor is 0.06, calculate the average sound absorption coefficient and reverberation time.

$$\rightarrow \text{We have average absorption coefficient, } \alpha = \frac{\alpha_1 S_1 + \alpha_2 S_2 + \alpha_3 S_3}{S_1 + S_2 + S_3} = ?$$

$$\text{And reverberation time, } T = \frac{0.16V}{\alpha S} = ? \quad \text{Here, } S = S_1 + S_2 + S_3$$

18. How much acoustic power enters the window of area 1.58m^2 , via the sound wave (standard intensity level = 10^{-16}W/cm^2). The window opens on a street where the street noise results in an intensity level at the window of 60dB.

→ Here, Power, $P = ?$, area, $A = 1.58\text{m}^2$, standard intensity level, $I_0 = 10^{-16}\text{W/cm}^2 = 10^{-12}\text{W/m}^2$

Intensity level, $L = 60\text{dB}$

$$\text{We have, } L = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

$$\text{or, } 60 = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

$$\text{or, } 6 = \log_{10} \left(\frac{I}{I_0} \right)$$

$$\text{or, } \frac{I}{I_0} = 10^6$$

$$\text{or, } I = 10^6 I_0 = 10^6 \times 10^{-12} = 10^{-6} \text{ W/m}^2$$

Now power, $P = I \times A = ?$

19. A room has dimensions of 6x4x5 m. Find a) the mean free path of the sound wave in the room. b) no. of reflections made per second wave with the walls of the room. Given velocity of sound in air = 350 m/s

$$\rightarrow \text{Here, (a) mean free path, } \sigma = \frac{4V}{S} = \frac{l \times b \times h}{2(lb + bh + lh)} = ?$$

$$\text{(b) no. of reflections per sec., } n = \frac{Sv}{4V} = ?$$

20. The time of reverberation of an empty hall without and with 500 audiences is 1.5 sec and 1.4 sec respectively. Find the reverberation time with 800 audiences in the hall.

$\rightarrow$ Here, reverberation time of empty hall, $T_0 = 1.5 \text{ s}$

reverberation time with 500 audience, $T_{500} = 1.4 \text{ sec}$

reverberation time with 800 audience, $T_{800} = ?$

Here, decrease in reverberation time due to 500 audiences = $1.5 - 1.4 = 0.1 \text{ s}$

$$\therefore \text{decrease in reverberation time due to 800 audiences} = \frac{0.1}{500} \times 800 = 0.16 \text{ s}$$

$$\text{hence the reverberation time with 800 audiences } T_{800} = 1.5 - 0.16 = 1.34 \text{ sec}$$

21. The time of reverberation of an empty hall is 1.5 sec. With 500 audience present in the hall, the reverberation time falls down to 1.4 sec. Find the no. of persons present in the hall if the reverberation time falls down to 1.312 sec.

$\rightarrow$ Here, reverberation time of empty hall, $T_0 = 1.5 \text{ s}$

reverberation time with 500 audience, $T_{500} = 1.4 \text{ sec}$

reverberation time with n audience, $T_n = 1.312 \text{ s}$

Here, decrease in reverberation time due to 500 audiences = $1.5 - 1.4 = 0.1s$

$$\therefore \text{decrease in reverberation time due to } n \text{ audiences} = \frac{0.1}{500} \times n$$

$$\text{hence the reverberation time with } n \text{ audiences, } T_n = 1.5 - \frac{0.1}{500} \times n$$

$$\text{or, } 1.312 = 1.5 - \frac{0.1}{500} \times n$$

$$\text{or, } \frac{0.1}{500} \times n = 1.5 - 1.312 = 0.188$$

$$\text{or, } n = \frac{0.188 \times 500}{0.1} = 940$$

22. A lecture hall of volume $12 \times 10^4 \text{m}^3$ has a total absorption of 13200m^2 of open window unit. Entry of students in the hall rises the coefficient by another 13200m^2 of open window unit. Find the change in reverberation time.

$$\rightarrow \text{here, reverberation time for empty hall, } T_0 = \frac{0.16V}{\alpha S} = \frac{0.16 \times 12 \times 10^4}{13200} = ?$$

$$\text{Reverberation time with students, } T_s = \frac{0.16 \times 12 \times 10^4}{13200 + 13200} = ?$$

$$\therefore \text{Change in reverberation time} = T_0 - T_s = ?$$

23. Entry of people in an auditorium hall of volume $1.6 \times 10^5 \text{ft}^3$ and total absorption of 1000ft^2 of open window raises the absorption by 600ft^2 . Find the change in reverberation time.

$$\rightarrow \text{here, reverberation time for empty hall, } T_0 = \frac{0.05V}{\alpha S} = \frac{0.05 \times 1.6 \times 10^5}{1000} = ?$$

$$\text{Reverberation tie with People, } T_p = \frac{0.05 \times 1.6 \times 10^5}{1000 + 600} = ?$$

$$\therefore \text{Change in reverberation time} = T_0 - T_p = ?$$

24. A baby's mouth is 30 cm from his father's ear and 1.5m from his mother's ear. What is the difference between sound intensity levels heard by the father and mother when the baby is crying?
 $\rightarrow$ let r_1 and r_2 be the distance of baby's mouth from his father and mother respectively.

$$\text{here, } r_1 = 30 \text{cm} \quad \text{and} \quad r_2 = 1.5 \text{m} = 150 \text{cm}$$

$$\text{we have ratio of intensity, } \frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} = ?$$

$$\text{Now difference in intensity level, } \Delta L = L_1 - L_2 = 10 \log_{10} \left(\frac{I_1}{I_2} \right) = ?$$

25. An engineer set a record for the loudest speech program in a hall, the sound level 46m in front of the speaker system was 120dB. What is the ratio of the intensity at that spot to the intensity of traffic sound level of 92 dB?

→ Here, $L_1 = 120dB$, $L_2 = 92dB$

we have $L_1 - L_2 = 10 \log_{10} \left(\frac{I_1}{I_2} \right)$

or, $120 - 92 = 10 \log_{10} \left(\frac{I_1}{I_2} \right)$

or, $2.8 = \log_{10} \left(\frac{I_1}{I_2} \right)$

or, $\frac{I_1}{I_2} = 10^{2.8} = ?$

26. A loudspeaker produces sound level of 8dB above reference level at a point 40m from it. Find the intensity level at a point 30m from the source.

→ Here, $L_1 = 8dB$ $r_1 = 40m$ $L_2 = ?$ $r_2 = 30m$

$\therefore \frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} = ?$

We have, $L_1 - L_2 = 10 \log_{10} \left(\frac{I_1}{I_2} \right)$

$\therefore L_2 = ?$

27. A source of sound emits energy in all direction at a rate of 0.638J/s. Find the intensity level at a distance 11.5m from the source. (take $I_0 = 1.2 \times 10^{-12} \text{ W/m}^2$)

→ here, $P = 0.638J/s$, $r = 11.5m$, $I_0 = 1.2 \times 10^{-12} \text{ W/m}^2$ $L = ?$

we have intensity $I = \frac{P}{A} = \frac{P}{4\pi r^2} = ?$

now intensity level, $L = 10 \log_{10} \left(\frac{I}{I_0} \right) = ?$

28. The maximum pressure variation that the ear can tolerate in loud sound is about 10 Nm^{-2} . Normal atmospheric pressure is 10^5 N/m^2 . Find the corresponding maximum displacement for a sound wave in air having a frequency of 1000Hz. Given, density of air 1.293 kg/m^3 and velocity of sound in air is 343 m/s

→ we have, pressure amplitude, $P_m = 2\pi f R \rho v$

$\therefore R = ?$ here, $P_m = 10 \text{ Nm}^{-2}$

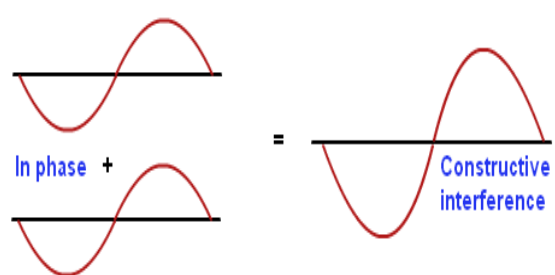
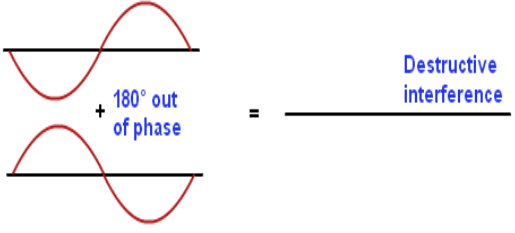
Optics

Chapter: Interference

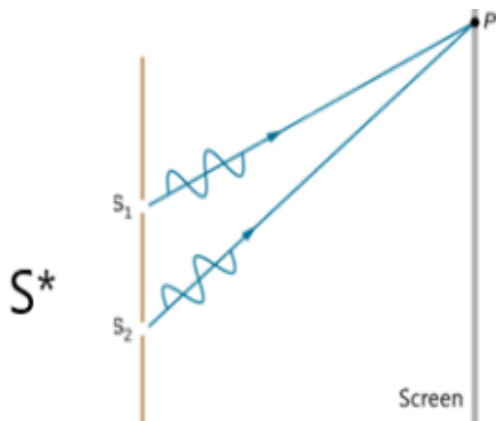
Coherent sources: Two sources are said to be coherent if they emit light waves having same wavelength (or frequency), nearly the same amplitude which are either in phase with each other or have constant phase difference.

Interference: The phenomenon of non-uniform distribution of light energy in a medium due to the superposition of light waves coming from two coherent sources is called interference.

Types of interference

Constructive Interference	Destructive interference
If the superposition of two light waves is such that crest/trough of one wave falls over the crest/trough of another wave, as a result of this, the resultant amplitude increases and hence the intensity distribution becomes maximum. This type of interference is called constructive interference.	If the superposition of two light waves is such that crest/trough of one wave falls over the trough/crest of another wave, as a result of this, the resultant amplitude decreases and hence the intensity distribution becomes minimum. This type of interference is called destructive interference.
 The diagram shows two sine waves in phase, labeled 'In phase +'. An equals sign follows, leading to a single wave with double the amplitude, labeled 'Constructive interference'.	 The diagram shows two sine waves 180 degrees out of phase, labeled '+ 180° out of phase'. An equals sign follows, leading to a flat line, labeled 'Destructive interference'.

Mathematical Interpretation of Interference (Analytical treatment of Interference)



Here, S is monochromatic source, S_1 and S_2 are two pin holes act as two coherent sources.

Consider a point P on the screen at which superposition of two light waves coming from S_1 and S_2 takes place.

The light wave coming from S_1 and S_2 are

$$y_1 = a_1 \sin \omega t \quad \text{and} \quad y_2 = a_2 \sin(\omega t + \varphi) \quad \text{respectively.}$$

Here, φ is the phase difference between y_1 and y_2 .

Now the resultant wave at P becomes

$$y = y_1 + y_2$$

$$\text{or, } y = a_1 \sin \omega t + a_2 \sin(\omega t + \varphi) = a_1 \sin \omega t + a_2 \sin \omega t \cos \varphi + a_2 \cos \omega t \sin \varphi$$

$$\text{or, } y = (a_1 + a_2 \cos \varphi) \sin \omega t + a_2 \sin \varphi \cos \omega t$$

$$\text{Put, } a_1 + a_2 \cos \varphi = R \cos \delta \dots \dots \dots (i)$$

$$\text{and } a_2 \sin \varphi = R \sin \delta \dots \dots \dots (ii)$$

$$\therefore y = R(\sin \omega t \cos \delta + \cos \omega t \sin \delta)$$

$$\text{or, } y = R \sin(\omega t + \delta)$$

This shows that resultant wave is simple harmonic in nature where, R is resultant amplitude and $\delta = \tan^{-1}\left(\frac{a_2 \sin \varphi}{a_1 + a_2 \cos \varphi}\right)$ is phase constant.

Now squaring and adding equations (i) and (ii) we get

$$R^2 = (a_1 + a_2 \cos \varphi)^2 + (a_2 \sin \varphi)^2$$

$$\text{or, } R^2 = a_1^2 + 2a_1a_2 \cos \varphi + a_2^2 \cos^2 \varphi + a_2^2 \sin^2 \varphi = a_1^2 + a_2^2 + 2a_1a_2 \cos \varphi$$

$$\therefore R = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \varphi}$$

This gives the resultant amplitude.

Now the resultant intensity at P is

$$I = R^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos \varphi$$

$$\text{or, } I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \varphi$$

Case i: When phase difference, $\varphi = 0(2\pi), 1(2\pi), 2(2\pi), \dots$ or

Path difference, $x = n\lambda$, $n = 0, 1, 2, \dots$ then the intensity becomes maximum and is given by

$$I_{\max} = (a_1 + a_2)^2 = (\sqrt{I_1} + \sqrt{I_2})^2$$

This gives the condition for construction interference.

Case ii: When phase difference, $\varphi = \pi, 3\pi, 5\pi, \dots$ or

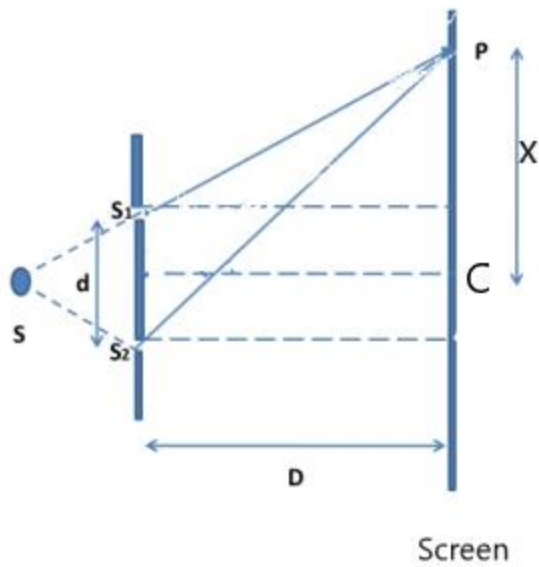
Path difference, $x = (2n - 1)\frac{\lambda}{2}$, $n = 1, 2, 3, \dots$ then the intensity becomes minimum and is

$$\text{given by } I_{\min} = (a_1 - a_2)^2 = (\sqrt{I_1} - \sqrt{I_2})^2$$

This gives the condition for destruction interference.

Young's Double slit Experiment

Consider a monochromatic source S and two pin holes S_1 and S_2 equidistant from S separated by a distance d. Let a screen be placed at a distance D from the slits S_1 and S_2 . Point C on the screen is the position of maximum intensity, as it is equidistant from S_1 and S_2 . Consider a point P on the screen at a distance x from C which may be the position of maximum or minimum intensity depending up on path difference



From figure we have, $(S_2P)^2 = D^2 + \left(x + \frac{d}{2}\right)^2$

And $(S_1P)^2 = D^2 + \left(x - \frac{d}{2}\right)^2$

$$\therefore (S_2P)^2 - (S_1P)^2 = 2xd$$

$$\text{or, } (S_2P + S_1P)(S_2P - S_1P) = 2xd$$

$$\text{or, } S_2P - S_1P = \frac{2xd}{S_2P + S_1P}$$

When point P is very close to C then $S_2P \approx S_1P \approx D$

$$\therefore S_2P - S_1P = \frac{2xd}{2D}$$

$$\therefore \text{Path difference} = \frac{xd}{D} \dots \dots \dots (i)$$

Case i: Constructive interference (bright fringe)

For this, path difference = $n\lambda$, $n = 0, 1, 2, \dots$

$$\therefore \frac{xd}{D} = n\lambda$$

$$\text{or, } x_n = \frac{n\lambda D}{d}$$

This gives the distance of n^{th} bright fringe from the center C on the screen.

Now fringe width $\beta = x_{n+1} - x_n = \frac{\lambda D}{d}$

Case ii: Destructive interference (dark fringe)

For this, path difference = $(2n - 1) \frac{\lambda}{2}$, $n = 1, 2, 3, \dots$

$$\therefore \frac{x d}{D} = (2n - 1) \frac{\lambda}{2}$$

$$\text{or, } x_n = (2n - 1) \frac{\lambda D}{2d}$$

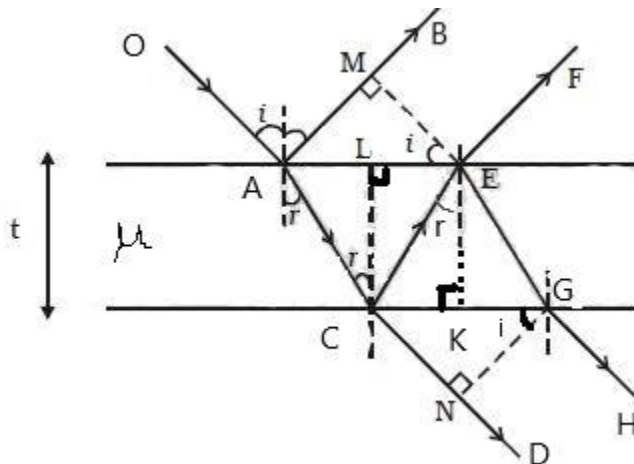
This gives the distance of n^{th} dark fringe from the center C on the screen.

Now fringe width $\beta = x_{n+1} - x_n = \frac{\lambda D}{d}$

From above results we see that bright fringe and dark fringe are of same width in Young's double slit experiment.

Interference due to thin film

Consider a thin film of thickness t having refractive index μ .



(a) Interference due to reflected system

Interference pattern can be observed due to the superposition of two reflected rays AB and EF .

Let EM perpendicular to AB and CL perpendicular to AE .

The optical path difference between two reflected rays AB and EF is

$$x = \mu(AC + CE) - AM \dots \dots \dots (i)$$

$$\text{Now from } \triangle ACL, \cos r = \frac{CL}{AC} = \frac{t}{AC}$$

$$\therefore AC = \frac{t}{\cos r} = CE$$

$$\text{and } \tan r = \frac{AL}{CL} = \frac{AL}{t}$$

$$\therefore AL = t \tan r = EL$$

$$\text{Again from } \triangle AME, \quad \sin i = \frac{AM}{AE} = \frac{AM}{2AL} = \frac{AM}{2t \tan r}$$

$$\therefore AM = 2t \sin i \tan r$$

Now from equation (i)

$$\begin{aligned} x &= \mu \left(\frac{t}{\cos r} + \frac{t}{\cos r} \right) - 2t \sin i \tan r = \frac{2\mu t}{\cos r} - 2 \frac{\sin i}{\sin r} t \frac{\sin^2 r}{\cos r} \\ &= \frac{2\mu t}{\cos r} - \frac{2\mu t}{\cos r} \sin^2 r = \frac{2\mu t}{\cos r} (1 - \sin^2 r) = \frac{2\mu t}{\cos r} \cos^2 r \end{aligned}$$

$$\therefore x = 2\mu t \cos r$$

This represents the apparent path difference.

Since the ray AB is reflected from optically denser medium, so it suffers additional π phase difference ($\frac{\lambda}{2}$ path difference).

$\therefore$ actual path difference between two reflected rays AB and EF is

$$x' = 2\mu t \cos r - \frac{\lambda}{2}$$

Case i: Constructive interference

For this, path difference = $n\lambda$

$$\therefore 2\mu t \cos r - \frac{\lambda}{2} = n\lambda$$

$$\text{or, } 2\mu t \cos r = (2n + 1) \frac{\lambda}{2}$$

$$\text{or, } 2\mu t \cos r = \text{odd integer} \times \frac{\lambda}{2}$$

Case ii: Destructive Interference

For this, path difference = $(2n - 1) \frac{\lambda}{2}$

$$\therefore 2\mu t \cos r - \frac{\lambda}{2} = (2n - 1) \frac{\lambda}{2}$$

$$\text{or, } 2\mu t \cos r = n\lambda$$

(b) Interference due to transmitted system

Interference pattern can be observed due to the superposition of two transmitted rays CD and GH.

Let GN perpendicular to CD and EK perpendicular to CG.

The optical path difference between two transmitted rays CD and GH is

$$x = \mu(CE + EG) - CN$$

$$\begin{aligned} x &= \mu \left(\frac{t}{\cos r} + \frac{t}{\cos r} \right) - 2t \sin i \tan r = \frac{2\mu t}{\cos r} - 2 \frac{\sin i}{\sin r} t \frac{\sin^2 r}{\cos r} \\ &= \frac{2\mu t}{\cos r} - \frac{2\mu t}{\cos r} \sin^2 r = \frac{2\mu t}{\cos r} (1 - \sin^2 r) = \frac{2\mu t}{\cos r} \cos^2 r \end{aligned}$$

$$\therefore x = 2\mu t \cos r$$

Case i: Constructive Interference

For this, path difference = $n\lambda$

$$\therefore 2\mu t \cos r = n\lambda$$

Case ii: Destructive Interference

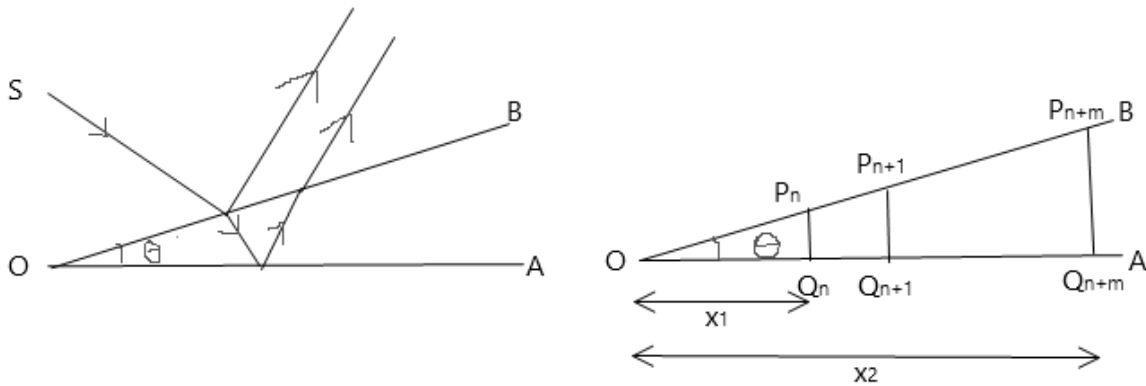
For this, path difference = $(2n - 1) \frac{\lambda}{2}$

$$\therefore 2\mu t \cos r = (2n - 1) \frac{\lambda}{2}$$

$$\text{or, } 2\mu t \cos r = \text{odd integer} \times \frac{\lambda}{2}$$

From above results we see that formation of interference patterns in the case of reflected and transmitted system is opposite to each other.

Interference due to Wedge Shaped thin film



Consider two thin glass plates OA and OB inclined at an angle θ so that a thin film of air is enclosed between two surfaces in wedge shaped. When a ray of light from a monochromatic source is incident on the upper glass plate, then interference patterns can be observed due to reflected rays as shown in figure.

Consider n^{th} order bright fringe occurs at P_n . The thickness of air film at P_n is $t = P_n Q_n$. Applying the condition for bright fringe in reflected system,

$$2\mu t \cos r = (2n - 1) \frac{\lambda}{2}$$

For normal incidence, $\cos r = 1$ and for air, $\mu = 1$

$$\therefore 2P_n Q_n = (2n - 1) \frac{\lambda}{2}$$

$$\text{or, } P_n Q_n = (2n - 1) \frac{\lambda}{4}$$

Similarly, for $(n+1)^{\text{th}}$ bright fringe

$$P_{n+1}Q_{n+1} = [2(n+1) - 1] \frac{\lambda}{4} = (2n+1) \frac{\lambda}{4}$$

Now, increase in thickness of air film for two consecutive bright fringes is

$$P_{n+1}Q_{n+1} - P_nQ_n = \frac{\lambda}{2}$$

$\therefore$ Increase in thickness of air film for two consecutive bright fringes is equal to half of the wavelength of light.

Again, we can write

$$P_{n+m}Q_{n+m} - P_nQ_n = \frac{m\lambda}{2} \dots \dots \dots (i)$$

Let $OQ_n = x_1$ and $OQ_{n+m} = x_2$

Then $Q_nQ_{n+m} = x_2 - x_1 = x$ (say)

We have, $\theta = \frac{P_nQ_n}{x_1} = \frac{P_{n+m}Q_{n+m}}{x_2}$

$$\therefore x_1 = \frac{P_nQ_n}{\theta} \text{ and } x_2 = \frac{P_{n+m}Q_{n+m}}{\theta}$$

$$\text{Now, } x = x_2 - x_1 = \frac{P_{n+m}Q_{n+m} - P_nQ_n}{\theta} = \frac{m\lambda}{2\theta} \quad (\text{Since, using equation (i)})$$

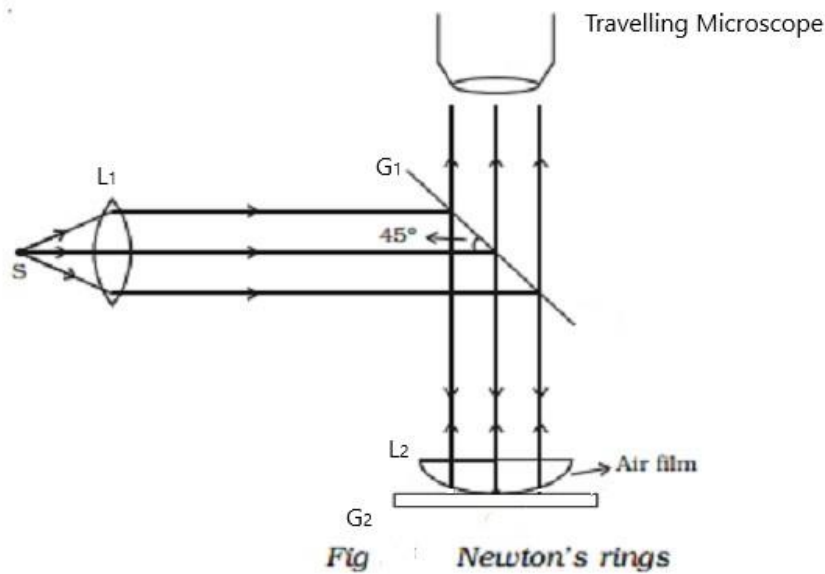
$$\therefore \text{fringe width, } \beta = \frac{x}{m} = \frac{\lambda}{2\theta} \text{ for air}$$

For a liquid having refractive index μ

$$\beta = \frac{\lambda}{2\mu\theta}$$

Newton's Ring

When a planoconvex lens of large radius of curvature is placed over the plane glass plate, then a thin film of air is enclosed between two surfaces in circular shape. When a parallel beam of monochromatic light is incident over the planoconvex lens, then interference patterns can be observed. The patterns are the concentric circles known as Newton's ring.



Consider a parallel beam of monochromatic light is incident normally over a planoconvex lens L_2 of radius of curvature R with the help of a glass plate G_1 inclined at an angle 45° with the horizon.

a) Newton's rings by reflected system

Newton's rings can be observed due to the superposition of two reflected rays, one from lower surface of lens L_2 and the other from upper surface of glass plate G_2 .

We have in reflected system,

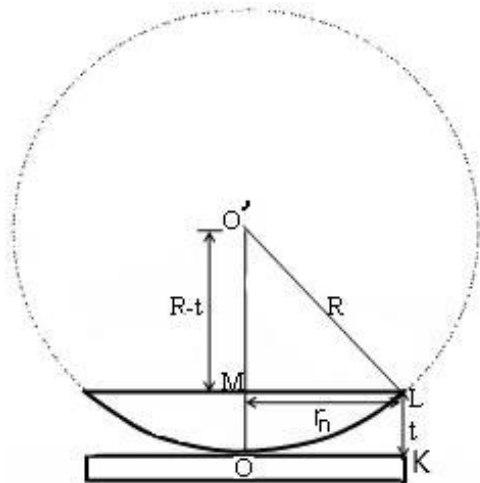
$$2\mu t \cos r = (2n - 1) \frac{\lambda}{2} \quad \text{for bright}$$

$$\text{and } 2\mu t \cos r = n\lambda \quad \text{for dark}$$

But for normal incidence, $\cos r = 1$ and for air, $\mu = 1$

$$\therefore 2t = (2n - 1) \frac{\lambda}{2} \dots \dots \dots (i) \quad \text{for bright ring}$$

$$\text{and } 2t = n\lambda \dots \dots \dots (ii) \quad \text{for dark ring}$$



Here, $O'L=OO'=R$, $OM=KL=t$ and $ML=r_n$

We have from $\Delta O'ML$, $O'L^2 = O'M^2 + ML^2$

$$\text{or, } R^2 = (R - t)^2 + r_n^2$$

$$\text{or, } R^2 = R^2 - 2Rt + t^2 + r_n^2$$

$$\text{or, } r_n^2 = 2Rt - t^2 \approx 2Rt \quad (\text{Since, } R \gg t)$$

$$\therefore 2t = \frac{r_n^2}{R} \dots \dots \dots (iii)$$

Now from equations (i) and (iii)

$$\frac{r_n^2}{R} = (2n - 1) \frac{\lambda}{2}$$

$$\text{or, } r_n = \sqrt{\frac{(2n-1)\lambda R}{2}}, n=1,2,3,\dots$$

This gives the radius of nth bright ring.

$$\Rightarrow r_n \propto \sqrt{2n - 1}$$

Again, from equations (ii) and (iii)

$$\frac{r_n^2}{R} = n\lambda$$

$$\text{or, } r_n = \sqrt{n\lambda R}, n=0,1,2,3,\dots$$

This gives the radius of nth dark ring.

$$\Rightarrow r_n \propto \sqrt{n}$$

From above results, we see that central ring is dark in the case of reflected system.

Also, it is seen that the rings become thinner and thinner as one goes from center to outside.

b) Newton's rings by transmitted system

Newton's rings can be observed due to the superposition of two transmitted rays, one from lower surface of lens L_2 and the other from upper surface of glass plate G_2 .

We have in transmitted system,

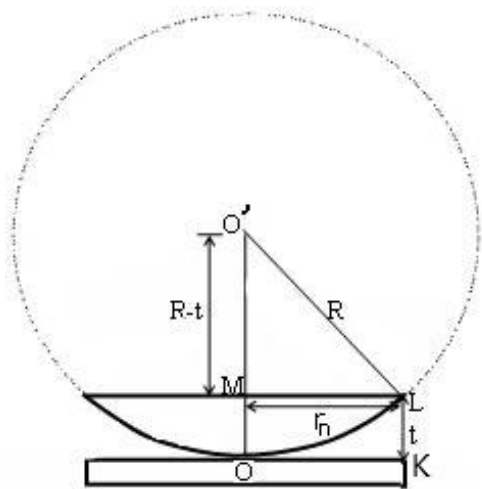
$$2\mu t \cos r = n\lambda \quad \text{for bright}$$

$$\text{and } 2\mu t \cos r = (2n - 1) \frac{\lambda}{2} \quad \text{for dark}$$

But for normal incidence, $\cos r = 1$ and for air, $\mu = 1$

$$\therefore 2t = n\lambda \dots \dots \dots (i) \quad \text{for bright ring}$$

$$\text{and } 2t = (2n - 1) \frac{\lambda}{2} \dots \dots \dots (ii) \quad \text{for dark ring}$$



Here, $O'L = OO' = R$, $OM = KL = t$ and $ML = r_n$

We have from $\Delta O'ML$, $O'L^2 = O'M^2 + ML^2$

$$\text{or, } R^2 = (R - t)^2 + r_n^2$$

$$\text{or, } R^2 = R^2 - 2Rt + t^2 + r_n^2$$

$$\text{or, } r_n^2 = 2Rt - t^2 \approx 2Rt \quad (\text{Since, } R \gg t)$$

$$\therefore 2t = \frac{r_n^2}{R} \dots \dots \dots (iii)$$

Now from equations (i) and (iii)

$$\frac{r_n^2}{R} = n\lambda$$

$$\text{or, } r_n = \sqrt{n\lambda R}, n=0,1,2,3,\dots$$

This gives the radius of nth bright ring.

$$\Rightarrow r_n \propto \sqrt{n}$$

Again, from equations (ii) and (iii)

$$\frac{r_n^2}{R} = (2n-1) \frac{\lambda}{2}$$

$$\text{or, } r_n = \sqrt{\frac{(2n-1)\lambda R}{2}}, n=1,2,3,\dots$$

This gives the radius of nth dark ring.

$$\Rightarrow r_n \propto \sqrt{2n-1}$$

From above results, we see that central ring is bright in the case of transmitted system.

Also, it is seen that the rings become thinner and thinner as one goes from center to outside.

c) Determination of wavelength of monochromatic light using Newton's ring method

We have the radius of nth dark ring in reflected system is

$$r_n = \sqrt{n\lambda R}$$

$$\therefore \text{diameter of } n^{\text{th}} \text{ dark ring is } D_n = 2r_n = 2\sqrt{n\lambda R} = \sqrt{4n\lambda R}$$

$$\therefore D_n^2 = 4n\lambda R$$

$$\text{Similarly, for } m^{\text{th}} \text{ dark ring, } D_m^2 = 4m\lambda R$$

$$\text{Now, } D_n^2 - D_m^2 = 4(n-m)\lambda R$$

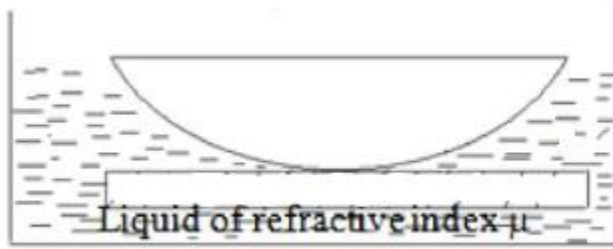
$$\text{or, } \lambda = \frac{D_n^2 - D_m^2}{4(n-m)R}$$

This gives the wavelength of monochromatic light.

d) Determination of refractive index of liquid using Newton's ring

When air is enclosed between planoconvex lens and glass plate we have

$$D_n^2 - D_m^2 = 4(n-m)\lambda R \dots \dots \dots (a)$$



When a liquid having refractive index μ is inserted between planoconvex lens and glass plate,

then for n^{th} ring, $D_n'^2 = \frac{4n\lambda R}{\mu}$ and for m^{th} dark ring, $D_m'^2 = \frac{4m\lambda R}{\mu}$

$$\therefore D_n'^2 - D_m'^2 = \frac{4(n - m)\lambda R}{\mu} \dots \dots \dots (b)$$

Now dividing equation (a) by (b) we get

$$\mu = \frac{D_n^2 - D_m^2}{D_n'^2 - D_m'^2}$$

This gives the refractive index of liquid.

Also we can write, $\mu = \frac{D_n^2}{D_n'^2}$

Note 1: Newton's rings with white light

When white light is used, the diameter of the rings of different colors will be different and colored rings are observed.

Note 2: Newton's rings with bright center due to reflected light

When a transparent liquid of refractive index μ is enclosed between lens of refractive index μ_1 and glass plate with refractive index μ_2 such that $\mu_1 > \mu > \mu_2$ (or $\mu_1 < \mu < \mu_2$), then the

central ring will be bright. In this case the radius of n^{th} bright ring is $r_n = \sqrt{\frac{n\lambda R}{\mu}}$

Numerical

$$\text{Visible Range: } 4000\text{\AA} - 7000\text{\AA} = 4 \times 10^{-7}\text{m} - 7 \times 10^{-7}\text{m}$$

$$= 4 \times 10^{-5}\text{cm} - 7 \times 10^{-5}\text{cm} = 4 \times 10^{-4}\text{mm} - 7 \times 10^{-4}\text{mm}$$

$$\text{Angular Separation for } n^{\text{th}} \text{ fringe, } \theta_n = \frac{n\lambda}{d}$$

$$\text{Angular Separation for a fringe, } \theta = \frac{\lambda}{d}$$

1. The ratio between max. and min. intensities is 36:1 in an interference pattern. Find the ratio between the amplitudes and intensities

$$\Rightarrow \text{Here, } \frac{I_{\max}}{I_{\min}} = \frac{36}{1}, \quad \frac{a_1}{a_2} = ? \quad \frac{I_1}{I_2} = ?$$

$$\text{We have, } \frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \left\{ \frac{\frac{a_1}{a_2} + 1}{\frac{a_1}{a_2} - 1} \right\}^2$$

$$\text{or, } \frac{36}{1} = \left\{ \frac{\frac{a_1}{a_2} + 1}{\frac{a_1}{a_2} - 1} \right\}^2$$

$$\text{or, } \left\{ \frac{6}{1} \right\}^2 = \left\{ \frac{\frac{a_1}{a_2} + 1}{\frac{a_1}{a_2} - 1} \right\}^2$$

$$\text{or, } \frac{6}{1} = \frac{\frac{a_1}{a_2} + 1}{\frac{a_1}{a_2} - 1}$$

$$\therefore \frac{a_1}{a_2} = ?$$

$$\text{And } \frac{I_1}{I_2} = \left\{ \frac{a_1}{a_2} \right\}^2 = ?$$

2. The two slits have width in the ratio 4:9. Find the ratio of the intensity at max, and min. in the interference pattern.

$$\Rightarrow \text{Here, } \frac{w_1}{w_2} = \frac{I_1}{I_2} = \frac{4}{9}, \quad \frac{I_{\max}}{I_{\min}} = ?$$

$$\text{we have, } \frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \left\{ \frac{\sqrt{\frac{I_1}{I_2}} + 1}{\sqrt{\frac{I_1}{I_2}} - 1} \right\}^2 = ?$$

3. The path difference between two interfering waves at a point on the screen is $\frac{1}{8}^{\text{th}}$ of a wavelength. Find the ratio of intensity at this point to that at the center of bright fringe.

$$\Rightarrow \text{Here, Path difference, } x = \frac{\lambda}{8}$$

$$\therefore \text{Phase difference, } \varphi = \frac{2\pi}{\lambda} \times x = \frac{\pi}{4}$$

$$\text{We have, } I = a_1^2 + a_2^2 + 2a_1a_2\cos\varphi = 4a^2\cos^2\frac{\varphi}{2} \text{ for } a_1 = a_2 = a$$

$$\therefore \text{Intensity at the center is } I_0 = 4a^2$$

$$\text{Now, } \frac{I}{I_0} = \cos^2\frac{\varphi}{2} = ?$$

4. Two sources of intensities $4I$ and I are used in an interference exp. Obtain the intensity at points where waves from the two sources superimposed with phase difference π and $\pi/2$.

$$\Rightarrow \text{Here, } I_1 = 4I, \quad I_2 = I$$

$$\text{We have, } I' = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\varphi$$

$$\text{When } \varphi = \pi, \quad I' = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\varphi = ?$$

$$\text{When } \varphi = \frac{\pi}{2}, \quad I' = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\varphi = ?$$

5. Two waves of same frequency have amplitude 3 units and 4 units. They interfere at a point where their phase difference is 60° . What is the resultant amplitude?

$$\Rightarrow \text{Here, } a_1 = 3\text{units}, \quad a_2 = 4\text{units}, \quad \varphi = 60^\circ, \quad R = ?$$

$$\text{We have, } R = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\varphi} = ?$$

6. A screen containing two slits 0.1mm apart is 1m from the viewing screen. Light of wavelength 500nm falls on the slits from a distant source. Approximately how far apart will the bright interference fringes be seen on the screen?

$$\Rightarrow \text{Here, slit separation } d = 0.1\text{mm} = 0.1 \times 10^{-3}\text{m}, \text{ distance between slits and screen } D = 1\text{m}, \text{ Wavelength } \lambda = 500\text{nm} = 500 \times 10^{-9}\text{m}, \text{ Fringe width } \beta = ?$$

$$\text{We have, } \beta = \frac{\lambda D}{d} = ?$$

7. In a double slit exp., the distance between slits is 5mm and the slits are 1m from the screen. Two interference patterns can be seen on the screen, one due to light of wavelength 480nm, and other due to light of wavelength 600nm. What is the separation on the screen between the third order bright fringes of two interference patterns?

$$\Rightarrow \text{Here, } d = 5\text{mm} = 5 \times 10^{-3}\text{m}, D = 1\text{m}, \lambda_1 = 480\text{nm} = 480 \times 10^{-9}\text{m}, \lambda_2 = 600\text{nm} = 600 \times 10^{-9}\text{m}$$

$$\text{And } n = 3$$

$$\text{We have distance of } n^{\text{th}} \text{ bright fringe from central maxima is } x_n = \frac{n\lambda D}{d}$$

$$\therefore x_3 - x'_3 = \frac{n\lambda_2 D}{d} - \frac{n\lambda_1 D}{d} = \frac{nD}{d}(\lambda_2 - \lambda_1) = ?$$

8. In a double slit pattern, the distance between the fourth bright fringes on either side of the center of the screen is 7cm. If the slit separation is 0.2mm and the distance to the screen is 3m, what is the wavelength of light?

⇒ Here, $n=4$, $x_4=3.5\text{cm}=3.5 \times 10^{-2}\text{m}$, $d = 0.2\text{mm}=0.2 \times 10^{-3}\text{m}$, $D=3\text{m}$, $\lambda=?$

We have, distance of n^{th} bright fringe from central maxima is $x_n = \frac{n\lambda D}{d}$

$$\therefore x_4 = \frac{4\lambda D}{d}$$

$$\therefore \lambda = ?$$

9. If the distance between the first and tenth minima of a double slit pattern is 18mm and the slits are separated by 0.15mm with the screen 50cm from the slits, what is the wavelength if light used.

⇒ Here, $x_{10} - x_1 = 18\text{mm} = 18 \times 10^{-3}\text{m}$, $d=0.15\text{mm}=0.15 \times 10^{-3}\text{m}$, $D = 0.5\text{m}$, $\lambda = ?$

We have, distance of n^{th} dark fringe from central maxima is $x_n = (2n - 1) \frac{\lambda D}{2d}$

$$\therefore x_{10} - x_1 = \frac{19\lambda D}{2d} - \frac{\lambda D}{2d}$$

$$\text{or, } 18 \times 10^{-3} = \frac{9\lambda D}{2d}$$

$$\therefore \lambda = ?$$

10. In Young's double slit experiment, a source of light of wavelength $4.2 \times 10^{-7}\text{m}$ is used to obtain interference fringes of width $6.4 \times 10^{-3}\text{m}$. What should be the wavelength of light to obtain fringes $4.6 \times 10^{-3}\text{m}$ wide, if the distance between screen and the slit is reduced to half the initial value.

⇒ Here, $\lambda_1 = 4.2 \times 10^{-7}\text{m}$, $\beta_1 = 6.4 \times 10^{-3}\text{m}$, $D_1 = D$

$\lambda_2 = ?$, $\beta_2 = 4.6 \times 10^{-3}\text{m}$, $D_2 = D/2$

We have, $\beta = \frac{\lambda D}{d}$

$$\therefore \frac{\beta_2}{\beta_1} = \frac{\lambda_2 D_2}{\lambda_1 D_1}$$

or, $\lambda_2 = ?$

11. Monochromatic green light, of wavelength 550nm, illuminates two parallel narrow slits $7.7\mu\text{m}$ apart. Calculate the angular deviation of the third order bright fringe a) in radian b) in degrees.

⇒ Here, $\lambda = 550\text{nm} = 550 \times 10^{-9}\text{m}$, $d = 7.7\mu\text{m} = 7.7 \times 10^{-6}\text{m}$, $n = 3$, angular deviation $\theta = ?$

We have angular deviation for n^{th} order, $\theta_n = \frac{n\lambda}{d}$

$$(a) \therefore \theta_3 = \frac{3\lambda}{d} = \dots\dots\dots \text{radian}$$

$$(b) \theta_3 = \dots\dots\dots \times \left(\frac{180}{\pi}\right)^\circ = ?$$

12. A double slit arrangement produces interference fringes for sodium light ($\lambda=589\text{nm}$) that have an angular separation of $3.5 \times 10^{-3}\text{rad}$. For what wavelength would the angular separation be 10% greater?

⇒ Here, $\lambda_1 = 589\text{nm}$ $\theta_1 = 3.5 \times 10^{-3}\text{rad}$ $\lambda_2 = ?$ $\theta_2 = 1.1\theta_1$

We have angular separation $\theta = \frac{\lambda}{d}$

$$\therefore \frac{\theta_2}{\theta_1} = \frac{\lambda_2}{\lambda_1}$$

or, $\lambda_2 = ?$

13. A double slit arrangement produces interference fringes for sodium light ($\lambda=589\text{nm}$) that are 0.2° apart. What is the angular fringes separation if the entire arrangement is immersed in water ($\mu=1.33$)?

$\Rightarrow$ Here, $\lambda=589\text{nm}$, $\theta = 0.2^\circ$ $\mu=1.33$

We have angular separation $\theta = \frac{\lambda}{d}$ (for air)

When immersed in water, angular separation $\theta' = \frac{\lambda}{\mu d} = \frac{\theta}{\mu} = ?$

14. Light of wavelength 624nm is incident perpendicularly on a soap film ($\mu=1.33$) suspended in air. What are the a) least and b) second least thickness of the film for which the reflections from the film undergo fully constructive interference?

$\Rightarrow$ Here, $\lambda=624\text{nm}=624 \times 10^{-9}\text{m}$, $r = 0^\circ$, $\mu = 1.33$,

We have, $2\mu t \cos r = (2n - 1) \frac{\lambda}{2}$

$$\therefore t = \frac{(2n - 1)\lambda}{4\mu \cos r}$$

a) For least thickness, $n=1$

$$\therefore t = \frac{(2 \times 1 - 1)\lambda}{4\mu \cos r} = ?$$

b) For second least thickness, $n=2$

$$\therefore t = \frac{(2 \times 2 - 1)\lambda}{4\mu \cos r} = ?$$

15. White light falls normally on a film of soapy water whose thickness is $5 \times 10^{-5}\text{cm}$ and $\mu=1.33$. Which wavelength in the visible region will be reflected most strongly?

$\Rightarrow$ Here, $r = 0^\circ$, $t = 5 \times 10^{-5}\text{cm}$, $\mu = 1.33$,

We have, $2\mu t \cos r = (2n - 1) \frac{\lambda}{2}$

$$\text{or, } \lambda = \frac{4\mu t \cos r}{2n - 1}$$

$$\text{for } n = 1, \lambda_1 = \frac{4 \times 1.33 \times 5 \times 10^{-5} \times \cos 0^\circ}{2 \times 1 - 1} = 26.6 \times 10^{-5}\text{cm (invisible)}$$

$$\text{for } n = 2, \lambda_2 = \frac{4 \times 1.33 \times 5 \times 10^{-5} \times \cos 0^\circ}{2 \times 2 - 1} = 8.86 \times 10^{-5}\text{cm (invisible)}$$

$$\text{for } n = 3, \lambda_3 = \frac{4 \times 1.33 \times 5 \times 10^{-5} \times \cos 0^\circ}{2 \times 3 - 1} = 5.32 \times 10^{-5}\text{cm (visible)}$$

$$\text{for } n = 4, \lambda_4 = \frac{4 \times 1.33 \times 5 \times 10^{-5} \times \cos 0^\circ}{2 \times 4 - 1} = 3.8 \times 10^{-5}\text{cm (invisible)}$$

Hence, light of wavelength $5.32 \times 10^{-5}\text{cm}$ will be reflected most strongly.

16. A soap film $5 \times 10^{-5}\text{cm}$ is viewed at an angle of 35° to the normal. Find the wavelengths of light in the visible spectrum which will be absent from the reflected light if μ of soap film is 1.33.

$\Rightarrow$ Here, $t = 5 \times 10^{-5}\text{cm}$, $i = 35^\circ$ $\mu = 1.33$,

We have, $\mu = \frac{\sin i}{\sin r}$

$$\therefore r = \sin^{-1} \left(\frac{\sin i}{\mu} \right) = 25.55^\circ$$

Again, we have $2\mu t \cos r = n\lambda$

$$\text{or, } \lambda = \frac{2\mu t \cos r}{n}$$

$$\text{for } n = 1, \lambda_1 = \frac{2 \times 1.33 \times 5 \times 10^{-5} \times \cos 25.55^\circ}{1} = 12 \times 10^{-5} \text{ cm (invisible)}$$

$$\text{for } n = 2, \lambda_2 = \frac{2 \times 1.33 \times 5 \times 10^{-5} \times \cos 25.55^\circ}{2} = 6 \times 10^{-5} \text{ cm (visible)}$$

$$\text{for } n = 3, \lambda_3 = \frac{2 \times 1.33 \times 5 \times 10^{-5} \times \cos 25.55^\circ}{3} = 4 \times 10^{-5} \text{ cm (visible)}$$

$$\text{for } n = 4, \lambda_4 = \frac{2 \times 1.33 \times 5 \times 10^{-5} \times \cos 25.55^\circ}{4} = 3 \times 10^{-5} \text{ cm (invisible)}$$

Hence the required wavelengths are $6 \times 10^{-5} \text{ cm}$ and $4 \times 10^{-5} \text{ cm}$

17. A thin film suspended in air is $0.41 \mu\text{m}$ thick and is illuminated with white light incident perpendicularly on its surface. The r.i. of the film is 1.5. At what wavelength will visible light that is reflected from the two surfaces of the film undergo fully constructive interference?
 $\Rightarrow$ Same as question number 15.

18. A 600 nm thick film ($\mu=1.4$) in air is illuminated with white light in a direction perpendicular to the film. For how many different wavelengths in the 300 to 700 nm range is there a) fully constructive interference and b) fully destructive interference in the reflected light?
 $\Rightarrow$ Same as question number 15, 16, 17.

19. A beam of parallel rays is incident at an angle of 30° with the normal on a plane parallel film of thickness $4 \times 10^{-5} \text{ cm}$ and r.i. 1.50. Show that the reflected light whose wavelength is $7.539 \times 10^{-5} \text{ cm}$, will be strengthened by reinforcement.

$$\Rightarrow \text{Here, } i = 30^\circ, \quad t = 4 \times 10^{-5} \text{ cm}, \quad \mu = 1.50, \quad \lambda = 7.539 \times 10^{-5} \text{ cm}$$

We have, $\mu = \frac{\sin i}{\sin r}$

$$\therefore r = \sin^{-1} \left(\frac{\sin i}{\mu} \right) = ?$$

We have for constructive interference in reflected system,

$$2\mu t \cos r = (2n - 1) \frac{\lambda}{2}$$

$$\text{or, } 2\mu t \cos r = n' \frac{\lambda}{2}$$

Here you need to show that n' is odd.

20. White light is incident on a soap film at an angle $\sin^{-1}(4/5)$ and the reflected light on examination by a spectrometer shows a dark bands. The consecutive dark bands corresponds to wavelength $6.1 \times 10^{-5} \text{ cm}$ and $6 \times 10^{-5} \text{ cm}$ overlap. If $\mu=1.333$ for the film, calculate its thickness.

$\Rightarrow$ Here, $i = \sin^{-1}(4/5)$, $\lambda_1 = 6.1 \times 10^{-5} \text{ cm}$, $\lambda_2 = 6 \times 10^{-5} \text{ cm}$ $\mu = 1.333$, $t = ?$

We have, $\mu = \frac{\sin i}{\sin r}$

$$\therefore r = \sin^{-1} \left(\frac{\sin i}{\mu} \right) = \sin^{-1} \left(\frac{4}{5 \times 1.333} \right) = ?$$

Again, by question

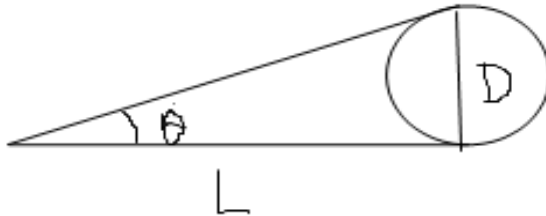
$$2\mu t \cos r = n\lambda_1 = (n+1)\lambda_2$$

$$\therefore n = ?$$

$$\text{Now, } t = \frac{n\lambda_1}{2\mu \cos r} = ?$$

- 21. A wedge shaped film of air is produced by placing a fine wire of diameter D between the ends of two flat glass plates of length $L=20\text{cm}$. When the film is illuminated with light of wavelength 550nm , there are 12 dark fringes per cm. Find D .**

$\Rightarrow$ Here, $L=20\text{cm}$, $\lambda=550\text{nm}=550 \times 10^{-7} \text{ cm}$, $\beta = \frac{1}{12} \text{ cm}$, $D = ?$



$$\text{We have, } \beta = \frac{\lambda}{2\theta} = \frac{\lambda L}{2D}$$

$$\therefore D = ?$$

- 22. Two glass plate enclosing a wedge shaped air film touching at one edge are separated by a wire of 0.03mm diameter at a distance of 15cm from the edge. Monochromatic light of $\lambda=600\text{nm}$ from a broad source falls normally on the film. Calculate the fringe width.**

$\Rightarrow$ Same as 21.

- 23. Interference fringes are produced with monochromatic light falling normally on a wedge shaped film of cellophane with $\mu = 1.4$. The angle of the wedge is 40 seconds of an arc and the distance between the successive fringes is 0.125cm . Calculate the wavelength of light.**

$\Rightarrow$ Here, $\mu = 1.4$, $\theta = 40'' = \frac{40}{60 \times 60} \times \frac{\pi}{180} \text{ radian}$, $\beta = 0.125\text{cm}$, $\lambda = ?$

$$\text{We have, } \beta = \frac{\lambda}{2\mu\theta}$$

$$\therefore \lambda = ?$$

24. Light of wavelength $6000\text{\AA}$ falls normally on a thin wedge shaped film of refractive index 1.4 forming fringes that are 2mm apart. Find the angle of wedge.

$$\Rightarrow \text{We have, } \beta = \frac{\lambda}{2\mu\theta}$$

$$\therefore \theta = ?$$

25. In Newton's ring exp., the radius of curvature R of the lens is 5m and the lens diameter is 20mm. a) How many bright rings are produced? Assume that $\lambda=589\text{nm}$. B) How many bright fringes would be produced if the arrangement were immersed in water($n=1.33$)

$$\Rightarrow \text{Here, } R=5\text{m, } r_n = 10\text{mm} = 10 \times 10^{-3}\text{m, } \lambda = 589\text{nm} = 589 \times 10^{-9}\text{m, } \mu = 1.33$$

- a) We have the radius of bright ring in reflected system is

$$r_n = \sqrt{\frac{(2n-1)\lambda R}{2}}$$

$$\therefore n = ?$$

- b) When immersed in water,

$$r_{n'} = \sqrt{\frac{(2n'-1)\lambda R}{2\mu}} \quad (\text{here, } r_{n'} = r_n)$$

$$\therefore n' = ?$$

26. A plano-convex lens of radius 300cm is placed on an optically flat glass plate and is illuminated by monochromatic light. The diameter of 8th dark ring in the transmitted system is 0.72cm. Calculate the wavelength of light used.

$\Rightarrow$ We have diameter of nth dark ring in transmitted system is

$$D_n = 2r_n = 2\sqrt{\frac{(2n-1)\lambda R}{2}}$$

$$\therefore D_8 = 2\sqrt{\frac{(2 \times 8 - 1)\lambda R}{2}}$$

$$\text{or, } \lambda = ?$$

27. Newton's rings formed by sodium light viewed normally. What is the order of the dark ring which will have double the diameter of 40th ring.

$\Rightarrow$ Let the diameter of nth dark ring is double the diameter of 40th ring.

$$\text{i.e. } D_n = 2D_{40}$$

$$\text{or, } \sqrt{4n\lambda R} = 2 \times \sqrt{4 \times 40 \times \lambda R}$$

$$\text{or, } n = ?$$

28. In Newton's ring experiment the diameter of 5th dark ring is reduced to half of its value after introducing a liquid below the convex surface. Calculate the refractive index of the liquid.

$$\Rightarrow \text{Here, } D_5 = x \quad D'_5 = \frac{x}{2}$$

$$\therefore \mu = \frac{D_n^2}{D_n'^2} = \frac{D_5^2}{D_5'^2} = ?$$

29. When oil is introduced into the space between the lens and the plate in Newton's arrangement, the radius of the 8th dark ring decreases from 1.80cm to 1.64cm. What is the refractive index of the oil?

$$\Rightarrow \text{Here, } r_8 = 1.8\text{cm} \quad r'_8 = 1.64\text{cm}$$

$$\therefore \mu = \frac{D_n^2}{D_n'^2} = \frac{r_n^2}{r_n'^2} = \frac{r_8^2}{r_8'^2} = ?$$

30. A Newton's rings apparatus is to be used to determine the radius of curvature of a lens. The radii of n^{th} and $(n+20)^{\text{th}}$ bright rings are measured and found to be 0.162 and 0.368 cm respectively, in light of $\lambda=546\text{nm}$. Calculate the radius of curvature of the lower surface of the lens.

$$\Rightarrow \text{Here, } D_n = 2 \times 0.162\text{cm}, \quad D_{n+20} = 2 \times 0.368\text{cm}, \quad \lambda=546\text{nm}=546 \times 10^{-7}\text{cm}, \quad R=?$$

$$\text{We have, } \lambda = \frac{D_n^2 - D_m^2}{4(n-m)R} = \frac{D_{n+20}^2 - D_n^2}{4(n+20-n)R}$$

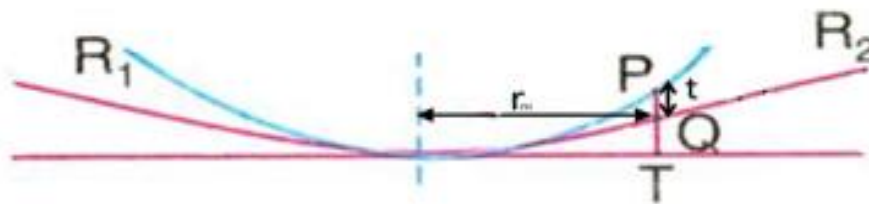
$$\therefore R=?$$

31. Show that the diameters of the Newton's rings when two surfaces of radii R_1 and R_2 are placed in contact are related by the relation $(1/R_1) - (1/R_2) = (4m\lambda/d^2_m)$, where n is the integer number of the & fringes.

$$\Rightarrow \text{We have } 2\mu t \cos r = m\lambda$$

$$\text{But for air, } \mu = 1 \text{ and for normal incidence, } \cos r = 1$$

$$\therefore 2t = m\lambda \dots \dots \dots (i)$$



$$\text{Here, the thickness of air film is } t = PT - QT = \frac{r_m^2}{2R_1} - \frac{r_m^2}{2R_2} = \frac{r_m^2}{2} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{d_m^2}{8} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Now from (i)

$$2 \times \frac{d_m^2}{8} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = m\lambda$$

$$\text{or, } \frac{1}{R_1} - \frac{1}{R_2} = \frac{4m\lambda}{d_m^2}$$

32. In Newton's ring arrangement a source emitting two wavelengths λ_1 and λ_2 is used. It is found that n^{th} dark ring due to one wavelength coincides with $(n+1)^{\text{th}}$ dark ring due to other. Find the radius of the n^{th} dark ring if radius of curvature of lens is R .

$\Rightarrow$ We have the radius of n^{th} dark ring due to wavelength λ_1 is $r_n = \sqrt{n\lambda_1 R}$

And the radius of $(n+1)^{\text{th}}$ dark ring due to wavelength λ_2 is $r_{n+1} = \sqrt{(n+1)\lambda_2 R}$

By question, $r_n = r_{n+1}$

$$\text{or, } \sqrt{n\lambda_1 R} = \sqrt{(n+1)\lambda_2 R}$$

$$\text{or, } n = \frac{\lambda_2}{\lambda_1 - \lambda_2}$$

Now the radius of n^{th} dark ring is

$$r_n = \sqrt{n\lambda_1 R} = \sqrt{\frac{\lambda_2 \lambda_1 R}{\lambda_1 - \lambda_2}}$$

Also diameter of n^{th} dark ring is $D_n = 2r_n = 2\sqrt{\frac{\lambda_2 \lambda_1 R}{\lambda_1 - \lambda_2}}$

33. In Newton's ring arrangement a source emitting two wavelengths $6 \times 10^{-7} \text{ m}$ and $5.9 \times 10^{-7} \text{ m}$ is used. It is found that n^{th} dark ring due to one wavelength coincides with $(n+1)^{\text{th}}$ dark ring due to other. Find the diameter of the n^{th} dark ring if radius of curvature of lens is 0.9 m

$\Rightarrow$ Same as 32.

34. In a Newton's ring experiment, the diameter of the 15^{th} ring was found to be 0.590 cm and that of the 5^{th} ring was 0.336 cm . if the radius of the plano-convex lens is 100 cm , calculate the wavelength of light used.

$\Rightarrow$ Here, $D_{15} = 0.590 \text{ cm}$, $D_5 = 0.336 \text{ cm}$, $R = 100 \text{ cm}$, $\lambda = ?$

$$\text{We have, } \lambda = \frac{D_n^2 - D_m^2}{4(n-m)R} = \frac{D_{15}^2 - D_5^2}{4(15-5)R} = ?$$

Chapter: Diffraction

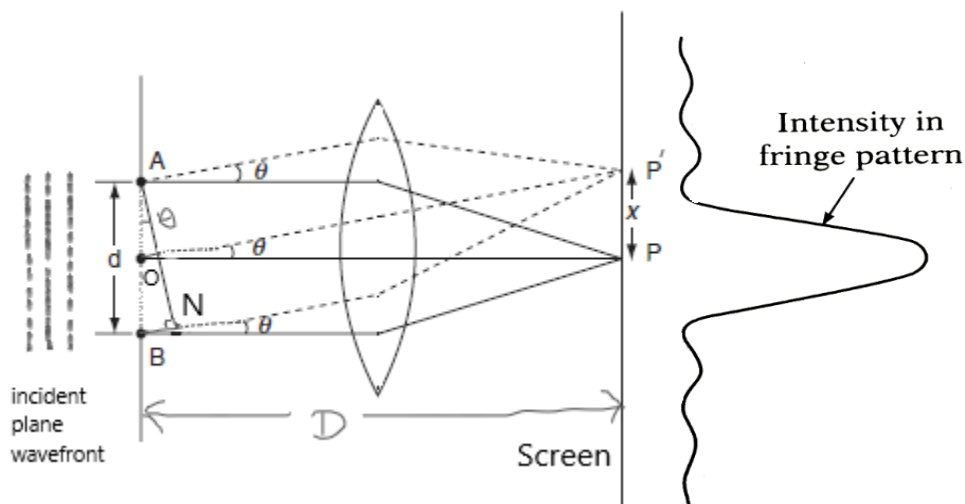
The phenomenon of bending of light round the corners of an obstacle (or small openings) and spreading into the regions of geometrical shadow is called diffraction. For the diffraction of light waves, size of the obstacle must be comparable to wavelength of light.

Types of Diffraction

Fresnel's Diffraction	Fraunhofer's Diffraction
1. In this diffraction, source and screen are at finite distance from the obstacle. 2. The incident wavefront is either spherical or cylindrical. 3. No lens is to be used in order to study the diffraction pattern.	1. In this diffraction, source and screen are at infinite distance from the obstacle. 2. The incident wavefront is plane. 3. Lens is to be used in order to study the diffraction pattern.

Fraunhofer's diffraction through a single slit

Consider a plane wavefront is incident on the slit AB of width d and each point on this wavefront is a source of secondary disturbance. The secondary waves travelling in the direction parallel to OP comes to focus at P which is the position of central maxima. Now consider the secondary waves travelling in the direction inclined at an angle θ and reach at point P' on the screen which may be the position of maximum or minimum intensity depending upon the path difference between the secondary waves originating from the corresponding points of the wavefront.



The path between the secondary waves originating from extreme points A and B is $BN = d \sin \theta$

If we divide the slit AB into two equal zones of width $d/2$, then for minima, path difference for a zone should be equal to $\lambda/2$ (according to Fresnel)

$$i.e. \frac{d}{2} \sin \theta = \frac{\lambda}{2}$$

$$or, d \sin \theta = \lambda \quad (\text{First order secondary minima})$$

Similarly, if the slit is divided into four equal zones of width $d/4$ then for a zone,

$$\frac{d}{4} \sin \theta = \frac{\lambda}{2}$$

$$or, d \sin \theta = 2\lambda \quad (\text{Second order secondary minima})$$

Now, to obtain n^{th} minima, we divide the slit into $2n$ number of equal zones of width $d/2n$ then,

$$\frac{d}{2n} \sin \theta = \frac{\lambda}{2}$$

$$or, d \sin \theta = n\lambda \quad n=1,2,3,\dots\dots\dots (n^{\text{th}} \text{ secondary minima})$$

In between these secondary minima, secondary maxima can be found for which path difference is odd number multiple of $\frac{\lambda}{2}$.

$$i.e. d \sin \theta = (2n + 1) \frac{\lambda}{2}, \quad n = 1,2,3, \dots\dots\dots (\neq 0) \quad (n^{\text{th}} \text{ secondary maxima})$$

Let x be the distance of point P' from center P on the screen and D be the distance between slit and screen.

$$\text{We have, } \tan \theta = \frac{x}{D}$$

For small angle, $\tan \theta \approx \sin \theta$

$$\therefore \sin \theta = \frac{x}{D}$$

$$or, d \sin \theta = \frac{xd}{D}$$

$$or, n\lambda = \frac{xd}{D}$$

$$or, x_n = \frac{n\lambda D}{d}, \quad n = 1,2,3 \dots\dots$$

This gives the distance of n^{th} minima from central maxima.

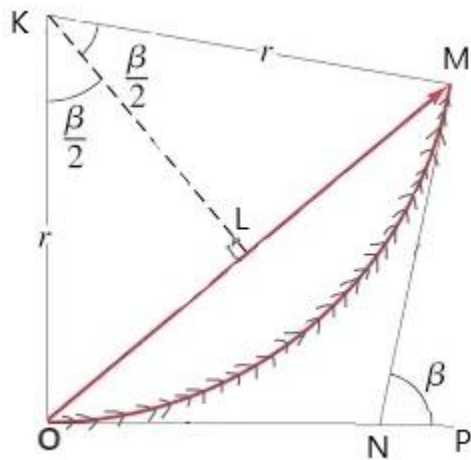
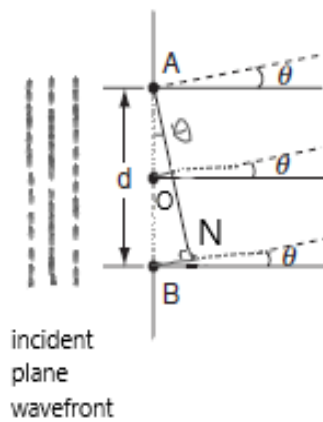
If $n=1$ then $x_1 = \frac{\lambda D}{d}$

This gives the half width of central maximum.

$\therefore$ The width of central maximum is $2x_1 = \frac{2\lambda D}{d}$

Also, the angular width of central maximum is 2θ .

Intensity distribution in diffraction pattern through a single slit



Here, θ is angle of diffraction and d is width of single slit.

The incident plane wavefront on the slit AB can be divided into large number of small circular strips. The path difference between secondary wavefronts originating from extreme points A and B is $d \sin \theta$ and let the corresponding phase difference is β .

Here, the resultant amplitude is obtained by vector polygon method. Let the vibration polygon coincide with a circular arc OM of radius r with center at K.

$$\text{Now from } \triangle OKL, \quad \sin \frac{\beta}{2} = \frac{OL}{r}$$

$$\therefore OL = r \sin \frac{\beta}{2}$$

$$\therefore \text{Chord } OM = 2 OL = 2r \sin \frac{\beta}{2} \dots \dots \dots (i)$$

Here, the length of arc OM is proportional to width of slit d .

$$\text{i.e. arc } OM = md$$

Where, m is proportionality constant.

$$\text{Now from figure, } \beta = \frac{\text{arc } OM}{\text{radius}} = \frac{md}{r}$$

$$\text{or, } r = \frac{md}{\beta}$$

$\therefore$ From equation (i),

$$\text{Chord } OM = 2r \sin \frac{\beta}{2} = 2 \frac{md}{\beta} \sin \frac{\beta}{2} = md \frac{\sin \frac{\beta}{2}}{\beta/2}$$

$$\text{or, } A = A_0 \frac{\sin \frac{\beta}{2}}{\beta/2}$$

Where, Chord OM=A is the resultant amplitude and $md = A_0$

Now, the resultant intensity at a point on the screen is given by

$$I = A^2 = A_0^2 \left\{ \frac{\sin \frac{\beta}{2}}{\beta/2} \right\}^2$$

$$\text{or, } I = I_0 \left\{ \frac{\sin \frac{\beta}{2}}{\beta/2} \right\}^2$$

Where, $I_0 = A_0^2$

$\therefore$ The resultant intensity at a point on the screen is proportional to $\left\{ \frac{\sin \frac{\beta}{2}}{\beta/2} \right\}^2$

Now, the phase difference β is given by

$$\beta = \frac{2\pi}{\lambda} \times d \sin \theta$$

Case i: For central maxima

For central maxima, we have $\theta = 0^\circ$ so that $\beta = 0$

$$\therefore I = \lim_{\frac{\beta}{2} \rightarrow 0} I_0 \left\{ \frac{\sin \frac{\beta}{2}}{\beta/2} \right\}^2 = I_0$$

Case ii: For secondary maxima,

For secondary maxima, we have $d \sin \theta = (2n + 1) \frac{\lambda}{2}$, $n = 1, 2, 3, \dots$

(a) If $n = 1$ (i.e. for first order secondary maxima)

$$\text{Then, } d \sin \theta = \frac{3\lambda}{2}$$

$$\therefore \beta = \frac{2\pi}{\lambda} \times \frac{3\lambda}{2} = 3\pi$$

$$\text{Hence, } I = I_0 \left\{ \frac{\sin \frac{\beta}{2}}{\beta/2} \right\}^2 = I_0 \left\{ \frac{\sin \frac{3\pi}{2}}{3\pi/2} \right\}^2$$

$$\text{or, } I = \left\{ \frac{2}{3\pi} \right\}^2 I_0 = \frac{4}{9\pi^2} I_0 = 4.5\% \text{ of } I_0 = \frac{1}{22} I_0$$

(b) If $n = 2$ (i.e. for second order secondary maxima)

$$d \sin \theta = \frac{5\lambda}{2}$$

$$\therefore \beta = \frac{2\pi}{\lambda} \times \frac{5\lambda}{2} = 5\pi$$

$$\text{Hence, } I = I_0 \left\{ \frac{\sin \frac{\beta}{2}}{\beta/2} \right\}^2 = I_0 \left\{ \frac{\sin \frac{5\pi}{2}}{5\pi/2} \right\}^2$$

$$\text{or, } I = \left\{ \frac{2}{5\pi} \right\}^2 I_0 = \frac{4}{25\pi^2} I_0 = 1.62\% \text{ of } I_0 = \frac{1}{62} I_0$$

And so on.

Case iii: For secondary minima

For this we have $d \sin \theta = n\lambda$

$$\therefore \beta = \frac{2\pi}{\lambda} \times n\lambda = 2n\pi$$

$$\text{Hence, } I = I_0 \left\{ \frac{\sin n\pi}{n\pi} \right\}^2 = 0 \text{ for all values of } n$$

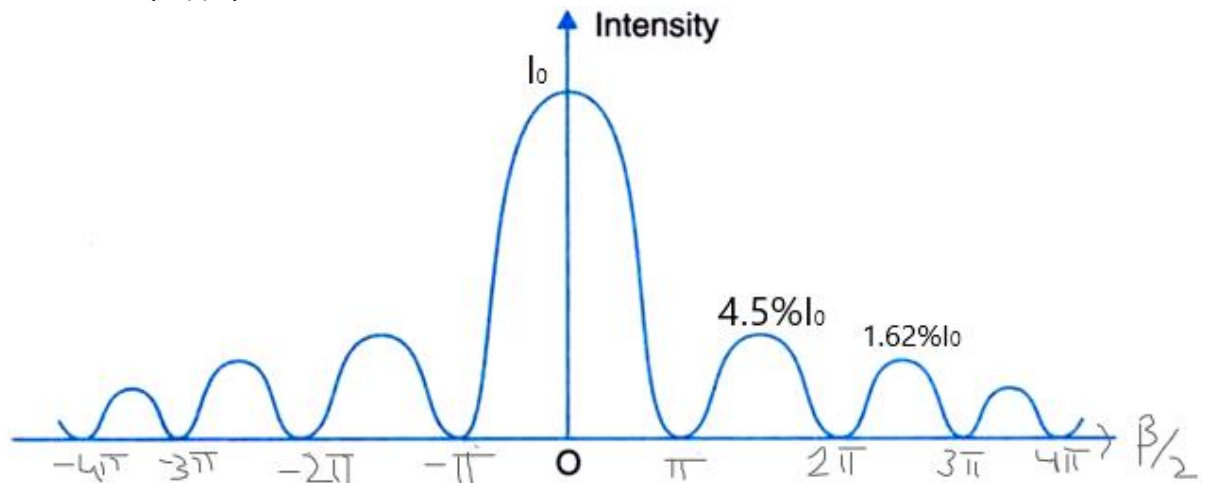


Fig: Variation of intensity in diffraction through a single slit

Diffraction Grating

It is an optical device which is used to study the diffraction of light. It consists of thousands of very fine equally spaced parallel slits made by drawing parallel lines in a plane glass plate by using diamond tip. The coated portion is opaque for light. The width of each transparent portion (a) is same and the width of each opaque portion (b) is also same. The sum of the width of transparent portion and opaque portion (a+b) is called grating element or grating spacing or grating constant.

The direction of n th order primary maxima is given by

$$(a + b) \sin \theta = n\lambda$$

Where, $a + b = \frac{1}{N}$ (N is number of lines per unit width of grating)

θ is angle of diffraction

n is number of order

And λ is wavelength of light used.

Dispersive power and Resolving power of grating

Dispersive power: The ability of a grating to disperse the spectral line is called dispersive power. It is given by the ratio of the difference in angle of diffraction of any two neighboring spectral lines to the difference in wavelength between these lines.

We have, the direction of n th order primary maxima is given by

$$(a + b) \sin \theta = n\lambda$$

Differentiating both sides with respect to λ we get

$$(a + b) \cos \theta \frac{d\theta}{d\lambda} = n$$

$$\text{or, } \frac{d\theta}{d\lambda} = \frac{n}{(a + b) \cos \theta}$$

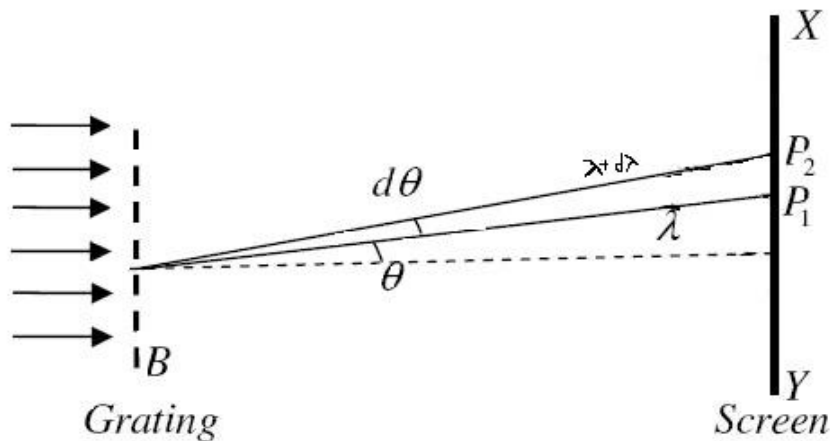
$$\text{or, } \frac{d\theta}{d\lambda} = \frac{nN}{\cos \theta}$$

This gives the dispersive power of grating.

Note1: If $a + b = d$, then $\frac{d\theta}{d\lambda} = \frac{n}{(a+b) \cos \theta} = \frac{n}{d \cos \theta} = \frac{n}{\sqrt{d^2 - n^2 \lambda^2}} = \frac{1}{\sqrt{\frac{d^2}{n^2} - \lambda^2}}$

Note2: $\frac{d\theta}{d\lambda} = \frac{n}{(a+b) \cos \theta} = \frac{n}{d \cos \theta} = \frac{d \sin \theta}{\lambda} \times \frac{1}{d \cos \theta} = \frac{\tan \theta}{\lambda}$

Resolving power: It is defined as the ratio of the wavelength of any spectral line to the difference in wavelength between this line and its neighboring line such that two lines appear to be just resolved.



Consider P_1 and P_2 be the positions of n^{th} order primary maxima for wavelengths λ and $\lambda + d\lambda$ with corresponding angle of diffractions θ and $\theta + d\theta$ respectively.

For wavelength λ we have

$$(a + b) \sin \theta = n\lambda$$

And for wavelength $\lambda + d\lambda$ we have

$$(a + b) \sin(\theta + d\theta) = n(\lambda + d\lambda) \dots \dots \dots (i)$$

These two lines can be resolved only if P_2 also corresponds to the first minimum after P_1 .

This is possible only if the extra path difference introduced is $\frac{\lambda}{N'}$. Here, N' is minimum number of lines that a grating should have in order to resolve the two spectral lines.

$$\therefore (a + b) \sin(\theta + d\theta) = n\lambda + \frac{\lambda}{N'} \dots \dots \dots (ii)$$

From equations (i) and (ii) we have

$$n(\lambda + d\lambda) = n\lambda + \frac{\lambda}{N'}$$

$$\text{or, } n\lambda + nd\lambda = n\lambda + \frac{\lambda}{N'}$$

$$\text{or, } nd\lambda = \frac{\lambda}{N'}$$

$$\text{or, } \frac{\lambda}{d\lambda} = nN'$$

This gives the resolving power of grating.

Relation between Dispersive power and Resolving power

We have dispersive power is $\frac{d\theta}{d\lambda} = \frac{nN}{\cos \theta}$

If w be the width of grating then we have $N' = w \times N$

$$\therefore \frac{d\theta}{d\lambda} = \frac{nN'}{w \cos \theta}$$

$$\text{or, } \frac{d\theta}{d\lambda} = \frac{\lambda}{d\lambda} \times \frac{1}{w \cos \theta}$$

Physical significance: Higher the disperse power higher will be the separation between two spectral lines where as high resolving power refers to ability of the instrument to show two nearby spectral lines distinctly.

Bragg's diffraction Law $2d \sin \theta = n\lambda$ **Where, d = interplanar spacing**

$$\text{Also } d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad \text{Where, } a \text{ is lattice constant and } (hkl) \text{ are Miller indices.}$$

DIFFRACTION

1. Light of wavelength 600nm is incident normally on a slit of width 0.1mm a) what is the angular position of the first minimum? b) What is the position of the second order minimum on a screen 3m from the slit?

→ Here, wavelength $\lambda = 600\text{nm} = 600 \times 10^{-9}\text{m}$, width of slit $d = 0.1\text{mm} = 0.1 \times 10^{-3}\text{m}$, distance between slit and screen $D = 3\text{m}$

- a) We have for minima, $d \sin \theta = n\lambda$
for first minimum, $n = 1$
 $\therefore d \sin \theta = \lambda$
or, $\theta = \sin^{-1}\left(\frac{\lambda}{d}\right) = ?$

- b) We have the distance of n th minima from central maxima is $x_n = \frac{n\lambda D}{d}$
 $\therefore x_2 = \frac{2\lambda D}{d} = ?$

2. Light of wavelength 750nm passes through a slit of width $1 \times 10^{-3}\text{mm}$. Find the angular and linear width of central maximum on a screen at 20cm from the slit.

→ Here, wavelength $\lambda = 750\text{nm} = 750 \times 10^{-9}\text{m}$, width of slit $d = 1 \times 10^{-3}\text{mm} = 1 \times 10^{-6}\text{m}$, distance between slit and screen $D = 20\text{cm} = 0.2\text{m}$

We have, for minima, $d \sin \theta = n\lambda$

For central maximum, $n=1$

$$\therefore d \sin \theta = \lambda$$

$$\text{or, } \theta = \sin^{-1}\left(\frac{\lambda}{d}\right) = \sin^{-1}\left(\frac{750 \times 10^{-9}}{1 \times 10^{-6}}\right) = 49^\circ$$

$$\therefore \text{Width of central maximum} = 2\theta = 98^\circ$$

If x be the distance of first minima from central maxima then we have

$$\tan \theta = \frac{x}{D}$$

$$\text{or, } x = D \tan \theta = ?$$

$$\therefore \text{Linear width of central maxima} = 2x = ?$$

3. The distance between the first and fifth minima of a single slit diffraction pattern is 0.35mm with the screen 40cm away from the slit, when light of wavelength 550nm is used. a) Find the slit width. b) Calculate the angle of the first diffraction minimum.

$$\rightarrow \text{Here, } x_5 - x_1 = 0.35 \text{ mm} = 0.35 \times 10^{-3} \text{ m, } D = 40 \text{ cm} = 0.4 \text{ m, } \lambda = 550 \text{ nm} = 550 \times 10^{-9} \text{ m}$$

$$\text{a) We have the distance of } n\text{th minima from central maxima is } x_n = \frac{n\lambda D}{d}$$

$$\therefore x_5 - x_1 = \frac{5\lambda D}{d} - \frac{\lambda D}{d}$$

$$\text{or, } 0.35 \times 10^{-3} = \frac{4\lambda D}{d}$$

$$\text{or, } d = ?$$

$$\text{b) We have for minima, } d \sin \theta = n\lambda$$

$$\text{for first minimum, } n = 1$$

$$\therefore d \sin \theta = \lambda$$

$$\text{or, } \theta = \sin^{-1}\left(\frac{\lambda}{d}\right) = ?$$

4. A plane wave of wavelength 590nm is incident on a slit with a width of 0.4mm. A thin converging lens of focal length 70cm is placed between the slit and a viewing screen focuses the light on the screen. a) How far is the screen from the lens? b) What is the distance on the screen from the center of the diffraction pattern to the first minimum?

$$\rightarrow \text{Here, } \lambda = 590 \text{ nm} = 590 \times 10^{-9} \text{ m, } d = 0.4 \text{ mm} = 0.4 \times 10^{-3} \text{ m, } f = 70 \text{ cm} = 0.7 \text{ m}$$

$$\text{a) } D = f = 0.7 \text{ m}$$

$$\text{b) } x_1 = \frac{\lambda D}{d} = ?$$

5. Diffraction pattern of a single slit of width 0.5cm is formed by a lens of focal length 40cm. Calculate the distance between the first dark and the next bright fringe from the axis. Wavelength = 4890Å

$$\rightarrow \text{Here, } d = 0.5 \text{ cm} = 0.005 \text{ m, } f = D = 40 \text{ cm} = 0.4 \text{ m, } \lambda = 4890 \text{ Å} = 4890 \times 10^{-10} \text{ m}$$

$$\text{We have the distance of } n\text{th minima from central maxima is } x_n = \frac{n\lambda D}{d}$$

$$\therefore x_1 = \frac{\lambda D}{d}$$

And the distance of n th maxima from central maxima is $x'_n = \frac{(2n+1)\lambda D}{2d}$

$$\therefore x'_1 = \frac{3\lambda D}{2d}$$

Now, $x'_1 - x_1 = ?$

6. What must be ratio of the slit width to wavelength for a single slit to have 1st minimum at 45°.

→ We have for first minimum, $d \sin \theta = \lambda$

$$\therefore \frac{d}{\lambda} = \frac{1}{\sin \theta} = ?$$

7. **Light of wavelength 600nm is incident normally on a slit of width 0.1mm. What is the intensity at $\theta=0.2^\circ$.**

→ Here, $\lambda = 600\text{nm} = 600 \times 10^{-9}\text{m}$, $d = 0.1\text{mm} = 0.1 \times 10^{-3}\text{m}$, angle of diffraction, $\theta=0.2^\circ$, $I=?$

$$\text{We have, } I = I_0 \left\{ \frac{\sin \frac{\beta}{2}}{\beta/2} \right\}^2$$

Where, $\beta = \frac{2\pi}{\lambda} \times d \sin \theta = ?$

$$\therefore I = ?$$

8. Hydrogen gas in a discharge tube emits a red line at 656.2nm. The light passes through a single slit of width 0.08mm. a) at what angle is the 1st minimum? b) What is the intensity at half the angle formed in a?

→ Same as 7.

9. Monochromatic light with wavelength 538nm is incident on a slit with width 0.025mm. the distance from the slit to a screen is 3.5m. Consider a point on a screen 1.1cm from the central maximum. Calculate a) θ for that point b) α c) the ratio of the intensity at that point to the intensity at the central maximum.

→ Here, $\lambda = 538\text{nm} = 538 \times 10^{-9}\text{m}$, $d = 0.025\text{mm} = 0.025 \times 10^{-3}\text{m}$, $D = 3.5\text{m}$, $x = 1.1\text{cm} = 0.011\text{m}$

a) We have $\tan \theta = \frac{x}{D}$

$$\therefore \theta = ?$$

b) $\alpha = \frac{\beta}{2} = ?$

c) $\frac{I}{I_0} = \left\{ \frac{\sin \frac{\beta}{2}}{\beta/2} \right\}^2 = \left\{ \frac{\sin \alpha}{\alpha} \right\}^2 = ?$

10. Calculate the first and second order angles for light of wavelength 400nm and 700nm, if the grating contains 10,000lines/cm.

→ Here, $\lambda_1 = 400\text{nm} = 400 \times 10^{-7}\text{cm}$, $\lambda_2 = 700\text{nm} = 700 \times 10^{-7}\text{cm}$

$N=10000\text{lines/cm}$

We have $(a + b) \sin \theta = n\lambda$

For wavelength λ_1 ,

$$\theta_1 = \sin^{-1} \left\{ \frac{\lambda_1}{a + b} \right\} = \sin^{-1}(N\lambda_1) = ?$$

$$\theta_2 = \sin^{-1} \left\{ \frac{2\lambda_1}{a + b} \right\} = \sin^{-1}(2N\lambda_1) = ?$$

and for wavelength λ_2

$$\theta'_1 = \sin^{-1} \left\{ \frac{\lambda_2}{a+b} \right\} = \sin^{-1}(N\lambda_2) = ?$$

$$\theta'_2 = \sin^{-1} \left\{ \frac{2\lambda_2}{a+b} \right\} = \sin^{-1}(2N\lambda_2) = ? \text{ (Not Possible)}$$

11. A diffraction grating 20mm wide has 6000 rulings. Light of wavelength 589nm is incident perpendicularly on a grating. What are the a) largest, b) second largest and c) third largest values of θ at which maxima appear on a distance viewing screen?

$$\rightarrow \text{Here, } N = \frac{6000}{2} = 3000 \text{ lines/cm, } \lambda = 589\text{nm} = 589 \times 10^{-7} \text{ cm}$$

$$\text{We have } (a+b) \sin \theta = n\lambda$$

$$\text{or, } n = \frac{(a+b) \sin \theta}{\lambda} = \frac{\sin \theta}{N\lambda}$$

$$\text{for maximum } n, \sin \theta = 1$$

$$\therefore n = \frac{1}{N\lambda} = 5.66$$

- a) For largest value of $\theta, n = 5$

$$\therefore \theta_5 = \sin^{-1}(5N\lambda) = ?$$

- b) For second largest value of $\theta, n = 4$

$$\therefore \theta_4 = \sin^{-1}(4N\lambda) = ?$$

- c) For third largest value of $\theta, n = 3$

$$\therefore \theta_3 = \sin^{-1}(3N\lambda) = ?$$

12. A grating places a spectral line of wavelength 460nm at 11° in 1st order. At what angle is a line of $\lambda=490\text{nm}$ observed in 2nd order.

$$\rightarrow \text{Here, } \lambda_1 = 460\text{nm, } \theta_1 = 11^\circ, n_1 = 1, \lambda_2 = 490\text{nm, } \theta_2 = ?, n_2 = 2$$

$$\text{We have } (a+b) \sin \theta_1 = n_1 \lambda_1$$

$$\text{And } (a+b) \sin \theta_2 = n_2 \lambda_2$$

$$\therefore \frac{\sin \theta_2}{\sin \theta_1} = \frac{n_2 \lambda_2}{n_1 \lambda_1}$$

$$\text{or, } \theta_2 = ?$$

13. A diffraction grating used at normal incidence gives a line $6000\text{\AA}$ in a certain order superimposed on another line $4500\text{\AA}$ of the next higher order. If the angle of diffraction is 30° , how many lines are there in a cm in the grating?

$$\rightarrow \text{Here, } \lambda_1 = 6000\text{\AA} = 6000 \times 10^{-8} \text{ cm, } \lambda_2 = 4500\text{\AA} = 4500 \times 10^{-8} \text{ cm, } \theta = 30^\circ, N = ?$$

$$\text{We have by question, } (a+b) \sin \theta = n\lambda_1 = (n+1)\lambda_2$$

$$\therefore n = ?$$

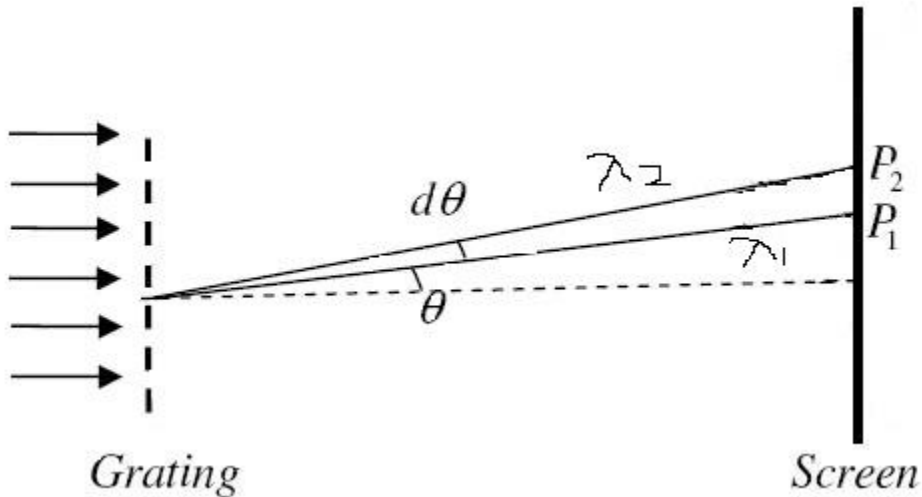
$$\text{Again, } (a+b) \sin \theta = n\lambda_1$$

$$\text{or, } a+b = \frac{n\lambda_1}{\sin \theta}$$

$$\text{or, } N = \frac{1}{a+b} = \frac{\sin \theta}{n\lambda_1} = ?$$

14. Assume that the limits of the visible spectrum are arbitrarily chosen as 430 and 680nm. Calculate the number of rulings per mm of a grating that will spread the first order spectrum through an angle of 20° .

→ Here, $\lambda_1 = 430\text{nm} = 430 \times 10^{-6}\text{mm}$, $\lambda_2 = 680\text{nm} = 680 \times 10^{-6}\text{mm}$, $d\theta = 20^\circ$, $n = 1$, $N = ?$



We have $(a + b) \sin \theta = n\lambda$

For wavelength λ_1

$$D \sin \theta = \lambda_1 \quad \text{Where, } D = a + b$$

And for wavelength λ_2

$$D \sin(\theta + d\theta) = \lambda_2$$

$$\text{or, } D \sin \theta \cos d\theta + D \cos \theta \sin d\theta = \lambda_2$$

$$\text{or, } \lambda_1 \cos d\theta + \sqrt{D^2 - \lambda_1^2} \sin d\theta = \lambda_2$$

$$D = \sqrt{\left\{ \frac{\lambda_2 - \lambda_1 \cos d\theta}{\sin d\theta} \right\}^2 + \lambda_1^2} = ?$$

$$\therefore N = \frac{1}{D} = ?$$

15. How many complete orders are formed for the visible range by a grating that has 6000 lines /cm.

→ Here, $N=6000 \text{ lines /cm}$, $n=?$

$$(a + b) \sin \theta = n\lambda$$

$$\therefore \lambda = \frac{a+b}{n} \text{ for } \sin \theta = 1$$

$$\text{or, } \lambda = \frac{1}{nN}$$

$$\text{For } n=1, \lambda_1 = \frac{1}{N} = \frac{1}{6000} = 16.66 \times 10^{-5} \text{ cm (invisible)}$$

$$\text{For } n=2, \lambda_2 = \frac{1}{2N} = \frac{1}{2 \times 6000} = 8.33 \times 10^{-5} \text{ cm (invisible)}$$

$$\text{For } n=3, \lambda_3 = \frac{1}{3N} = \frac{1}{3 \times 6000} = 5.55 \times 10^{-5} \text{ cm (visible)}$$

$$\text{For } n=4, \lambda_4 = \frac{1}{4N} = \frac{1}{4 \times 6000} = 4.16 \times 10^{-5} \text{ cm (visible)}$$

$$\text{For } n=5, \lambda_5 = \frac{1}{5N} = \frac{1}{5 \times 6000} = 3.33 \times 10^{-5} \text{ cm (invisible)}$$

Hence the number of complete order is 2.

16. A diffraction grating has 8900 slits across 1.2cm. If light with a wavelength of 500nm is sent through it, how many orders (maxima) lie on one side of central maxima?

$$\rightarrow \text{Here, } N = \frac{8900}{1.2} \text{ lines/cm, } \lambda = 500 \text{ nm} = 500 \times 10^{-7} \text{ cm, } n = ?$$

$$\text{We have, } (a+b) \sin \theta = n\lambda$$

$$\therefore n = \frac{a+b}{\lambda} \text{ for } \sin \theta = 1$$

$$\text{or, } n = \frac{1}{N\lambda} = ?$$

17. How many orders will be visible if the wavelength of incident light is $5 \times 10^{-7} \text{ m}$ and no. of lines /inch on the grating is 2620?

$$\rightarrow \text{Here, } N = 2620 \text{ lines/inch} = \frac{2620}{2.54} \text{ lines/cm, } \lambda = 5 \times 10^{-7} \text{ m} = 5 \times 10^{-5} \text{ cm, } n = ?$$

Same as 16.

18. A diffraction grating 3cm wide produces the second order at 33° with light of wavelength 600nm. What is the total no. of lines on the grating?

$$\rightarrow \text{Here, width } w = 3 \text{ cm, } n=2, \theta = 33^\circ, \lambda = 600 \text{ nm} = 600 \times 10^{-7} \text{ cm}$$

$$\text{We have, } (a+b) \sin \theta = n\lambda$$

$$\therefore a+b = \frac{n\lambda}{\sin \theta}$$

$$\text{or, } N = \frac{1}{a+b} = \frac{\sin \theta}{n\lambda} = ?$$

Now, total number of lines in the grating = $N \times w = ?$

19. A grating has 600 rulings/mm and is 5mm width. a) What is the smallest wavelength interval it can resolve in the 3rd order at $\lambda = 500\text{nm}$? b) What is the resolving power of grating? c) How many higher orders of maxima can be seen?

→ Here, $N = 600\text{rulings/mm}$, width $w = 5\text{mm}$, $n = 3$, $\lambda = 500\text{nm} = 500 \times 10^{-7}\text{cm}$

$$\therefore N' = N \times w = 3000\text{rulings}$$

a) We have, $\frac{\lambda}{d\lambda} = nN'$
 or, $d\lambda = \frac{\lambda}{nN'} = ?$

b) Resolving power = $nN' = ?$

c) Same as 16.

20. a) What is the resolving power required to resolve the two sodium lines at 589nm and 589.6nm? b) If a grating is 2cm wide, how many lines/mm are needed to resolve these wavelengths in 3rd order?

→ Here, $\lambda_1 = 589\text{nm}$, $\lambda_2 = 589.6\text{nm}$, $w = 2\text{cm} = 20\text{mm}$, $n = 3$

a) Resolving Power = $\frac{\lambda}{d\lambda} = \frac{589.3}{0.6} = ?$

b) We have, $\frac{\lambda}{d\lambda} = nN'$
 $\therefore N' = \frac{1}{n} \times \frac{\lambda}{d\lambda} = ?$

Now, $N = \frac{N'}{w} = ?$

21. The sodium light has wavelengths 589.0nm and 589.6nm. If these lines are to be resolved what is the required width of grating with 300 lines/mm.

→ Here, $\lambda_1 = 589\text{nm}$, $\lambda_2 = 589.6\text{nm}$, $N = 300\text{lines/mm}$, $w = ?$

We have, $\frac{\lambda}{d\lambda} = nN'$

$\therefore N' = \frac{1}{n} \times \frac{\lambda}{d\lambda} = ?$ Take $n = 1$

Now, $w = \frac{N'}{N} = ?$

22. Calculate the minimum no. of lines/cm in a 2.5cm wide grating which will just resolve the sodium lines in second order.

→ Here, $\lambda_1 = 589\text{nm}$, $\lambda_2 = 589.6\text{nm}$, $w = 2.5\text{cm}$, $n = 2$, $N = ?$

We have, $\frac{\lambda}{d\lambda} = nN'$

$\therefore N' = \frac{1}{n} \times \frac{\lambda}{d\lambda} = ?$

Now, $N = \frac{N'}{w} = ?$

23. What should be the minimum no of lines in a grating which will just resolve D lines of sodium in 2nd order?

→ Here, $\lambda_1 = 589\text{nm}$, $\lambda_2 = 589.6\text{nm}$, $n = 2$, $N' = ?$

Same as 22.

24. How many lines per cm must a 4cm wide grating contains if it can just resolve two wavelengths 4187.23 and 4187.41A° in 1st order?

→ Same as 22.

25. A diffraction grating is just able to resolve two lines of wavelength 5140.34A° and 5140.85A° in first order. Will it resolve lines 8037.20A° and 8037.50A° in 2nd order?

→ Here, $\lambda_1 = 5140.34A^0$, $\lambda_2 = 5140.85A^0$, $n = 1$

We have, $\frac{\lambda}{d\lambda} = nN'$

$$\therefore N' = \frac{1}{n} \times \frac{\lambda}{d\lambda} = \frac{1}{1} \times \frac{5140.595}{0.51} = 10079 \quad (\text{This is the total number of lines in the grating.})$$

Again, for second pair of wavelengths,

We have, $\frac{\lambda'}{d\lambda'} = n'N''$

$$\text{or, } N'' = \frac{1}{n'} \times \frac{\lambda'}{d\lambda'} = \frac{1}{2} \times \frac{8037.35}{0.30} = 13395$$

Since, $N' < N''$ so the second pair cannot be resolved in 2nd order.

26. Light is incident normally on a grating 0.5cm wide with 2500lines. Find the angles of diffraction for the principal maxima of the two sodium lines in the first order spectrum. ($\lambda_1=5890A^0$ and $\lambda_2=5896A^0$). Also calculate the angular separation between the lines. Are the two lines resolved?

→ Here, $N = \frac{2500}{0.5} \text{ lines/cm} = 5000 \text{ lines/cm}$, $n=1$, $\lambda_1 = 5890A^0 = 5890 \times 10^{-8} \text{ cm}$,

$$\lambda_2 = 5896A^0 = 5890 \times 10^{-8} \text{ cm}$$

We have, $(a + b) \sin \theta = n\lambda$

For wavelength λ_1

$$(a + b) \sin \theta_1 = \lambda_1$$

$$\therefore \theta_1 = \sin^{-1} \left\{ \frac{\lambda_1}{a+b} \right\} = \sin^{-1} \{ N\lambda_1 \} = ?$$

And for wavelength λ_2

$$(a + b) \sin \theta_2 = \lambda_2$$

$$\therefore \theta_2 = \sin^{-1} \left\{ \frac{\lambda_2}{a+b} \right\} = \sin^{-1} \{ N\lambda_2 \} = ?$$

Now, angular separation = $\theta_2 - \theta_1 = ?$

Again we have resolving power, $\frac{\lambda}{d\lambda} = nN'$

$$\therefore N' = \frac{1}{n} \times \frac{\lambda}{d\lambda} = \frac{1}{1} \times \frac{5893}{6} = 982$$

Since, $982 < 2500$ so the lines will be well resolved.

27. Light which is a mixture of two wavelengths $5000\text{\AA}$ and $5200\text{\AA}$ is incident normally on a plane transmission grating having 10000lines/cm . A lens of focal length 150cm is used to observe the spectrum on a screen. Calculate the separation in cm of the two lines in the first order spectrum.

$$\rightarrow \text{Here, } \lambda_1 = 5000\text{\AA} = 5000 \times 10^{-8}\text{cm}, \lambda_2 = 5200\text{\AA} = 5200 \times 10^{-8}\text{cm}$$

$$N = 10000\text{lines/cm}, n = 1$$

$$\text{We have, } (a + b) \sin \theta = n\lambda$$

For wavelength λ_1

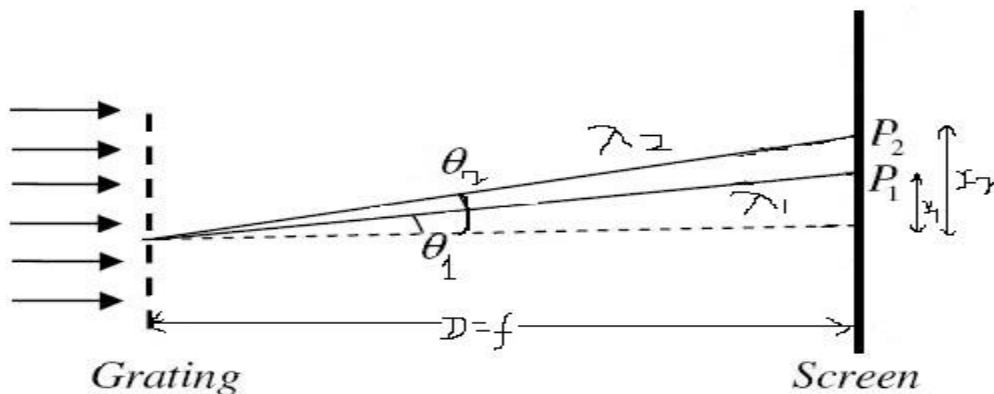
$$(a + b) \sin \theta_1 = \lambda_1$$

$$\therefore \theta_1 = \sin^{-1} \left\{ \frac{\lambda_1}{a+b} \right\} = \sin^{-1} \{ N\lambda_1 \} = ?$$

And for wavelength λ_2

$$(a + b) \sin \theta_2 = \lambda_2$$

$$\therefore \theta_2 = \sin^{-1} \left\{ \frac{\lambda_2}{a+b} \right\} = \sin^{-1} \{ N\lambda_2 \} = ?$$



$$\text{Now, } \tan \theta_1 = \frac{x_1}{D} \text{ and } \tan \theta_2 = \frac{x_2}{D}$$

$$\therefore x_1 = D \tan \theta_1 \text{ and } x_2 = D \tan \theta_2$$

$$\text{Hence, } x_2 - x_1 = ?$$

28. Yellow sodium light consists of two wavelengths (589nm and 589.59nm), falls on a 7500 line/cm diffraction grating. Determine a) Maximum order that will be present for sodium light b) the angular

dispersion of the grating c) the width of grating necessary to resolve the two sodium lines d) the angular width of each sodium line.

→ Here, $\lambda_1 = 589\text{nm}$, $\lambda_2 = 589.6\text{nm}$, $N = 7500\text{lines/cm}$

a) Here, maximum order will be present for $\lambda_1 = 589\text{nm}$

We have $(a + b) \sin \theta = n\lambda_1$

$$\therefore n = \frac{a+b}{\lambda_1} \quad \text{for } \sin \theta = 1$$

$$\text{or, } n = \frac{1}{N\lambda_1} = \frac{1}{7500 \times 589 \times 10^{-7}} \approx 2$$

b) We have $\sin \theta = \frac{n\lambda_1}{a+b} = nN\lambda_1$

$$\text{or, } \theta = \sin^{-1}(nN\lambda_1) = 62.37^\circ$$

$$\therefore \text{Angular dispersion, } \frac{d\theta}{d\lambda} = \frac{nN}{\cos \theta} = \frac{2 \times 7500}{\cos 62.37^\circ} = 32344 \text{ rad/cm}$$

c) We have, $\frac{\lambda}{d\lambda} = nN'$

$$\therefore N' = \frac{1}{n} \times \frac{\lambda}{d\lambda} = \frac{1}{2} \times \frac{589.295}{0.59} = 499$$

$$\text{Hence, the width of grating } w = \frac{N'}{N} = \frac{499}{7500} = 0.066\text{cm}$$

d) Angular width $d\theta = \frac{nN}{\cos \theta} \times d\lambda = 32344 \times 0.59 \times 10^{-7} = 0.0019\text{rad}$

29. Two spectral lines have wavelengths λ and $\lambda + d\lambda$ respectively where $d\lambda \ll \lambda$. Show that their angular separation is $d\theta = \frac{d\lambda}{\sqrt{\left(\frac{d}{m}\right)^2 - \lambda^2}}$, where d and m are grating elements and no. of order respectively.

→ Refer to theory.

30. In the second order spectrum of a plane diffraction grating, a certain spectrum lines appear at an angle 10° , while another line of wavelength $5 \times 10^{-9}\text{cm}$ appears at an angle $3''$ greater. Find the wavelength of the lines and maximum grating width to resolve them.

$$\rightarrow \text{Here, } n=2, \theta = 10^\circ, d\lambda = 5 \times 10^{-9}\text{cm}, d\theta = 3'' = \frac{3}{60 \times 60} \times \frac{\pi}{180} \text{radian}$$

$$\text{We have, } \frac{d\theta}{d\lambda} = \frac{nN}{\cos \theta}$$

$$\therefore N = ?$$

$$\text{Now, } (a + b) \sin \theta = n\lambda$$

$$\therefore \lambda = \frac{(a+b) \sin \theta}{n} = \frac{\sin \theta}{nN} = ?$$

Hence the wavelengths are

$$\lambda_1 = \lambda = ? \text{ and } \lambda_2 = \lambda + d\lambda = ?$$

$$\text{Again, we have } \frac{\lambda}{d\lambda} = nN'$$

$$\therefore N' = \frac{1}{n} \times \frac{\lambda}{d\lambda} = ?$$

$$\therefore \text{Width of grating } w = \frac{N'}{N} = ?$$

31. A plane transmission grating has 40000 lines with grating element $12.5 \times 10^{-5} \text{ cm}$. Calculate maximum resolving power for which it can be used in wavelength range 500 nm.

$$\rightarrow \text{Here, } N' = 40000, a+b = 12.5 \times 10^{-5} \text{ cm}, \lambda = 500 \text{ nm} = 500 \times 10^{-7} \text{ cm}$$

$$\text{We have, resolving power } \frac{\lambda}{d\lambda} = nN'$$

$$\text{Since, } (a+b) \sin \theta = n\lambda$$

$$\therefore n = \frac{a+b}{\lambda} = ? \quad (\text{Since, } \sin \theta = 1 \text{ for maximum } n)$$

$$\therefore \frac{\lambda}{d\lambda} = nN' = ?$$

32. A diffraction grating has 4000 lines/cm and is used at normal incidence. Calculate the dispersive power of grating in the third order for wavelength $5000 \text{ \AA}$.

$$\rightarrow \text{Here, } N = 4000 \text{ lines/cm}, n = 3, \lambda = 5000 \text{ \AA} = 5000 \times 10^{-8} \text{ cm}$$

$$\text{We have, } (a+b) \sin \theta = n\lambda$$

$$\therefore \theta = \sin^{-1} \left\{ \frac{n\lambda}{a+b} \right\} = \sin^{-1} \{ Nn\lambda \} = ?$$

$$\text{Now dispersive power, } \frac{d\theta}{d\lambda} = \frac{nN}{\cos \theta} = ?$$

33. In first order reflection occurs in a crystal at Bragg angle 3.4° , at what angle does 2^{nd} order reflection occur from the same family of reflecting planes.

$$\rightarrow \text{Here, } n_1 = 1, \theta_1 = 3.4^\circ, n_2 = 2, \theta_2 = ?$$

$$\text{We have, } 2d \sin \theta = n\lambda$$

$$\therefore \frac{\sin \theta_2}{\sin \theta_1} = \frac{n_2}{n_1} \quad (\text{Since, for same family, } d \text{ and } \lambda \text{ are same})$$

$$\text{or, } \theta_2 = ?$$

34. What is the smallest Bragg angle for x rays of wavelength 30 pm to reflecting planes spaced 0.3 nm apart in a calcite crystal?

$$\rightarrow \text{We have, } 2d \sin \theta = n\lambda$$

$$\text{For smallest Bragg angle, } n = 1$$

$$\therefore \theta = \sin^{-1} \left(\frac{\lambda}{2d} \right) = ?$$

35. X-rays with wavelength $0.58 \text{ \AA}$ are used for calculating d_{200} in nickel. The angle of reflection is 9.5° . What is the size of unit cell?

→ Here, $\lambda = 0.58 \text{Å}$, $(hkl) = (200)$, $\theta = 9.5^\circ$, $a = ?$

We have, $2d_{200}\sin\theta = n\lambda$

$$\therefore d_{200} = \frac{\lambda}{2\sin\theta} = ? \text{ for } n = 1$$

$$\text{Again we have } d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$\therefore d_{200} = \frac{a}{\sqrt{2^2 + 0^2 + 0^2}} = \frac{a}{2}$$

$$\text{or, } a = 2 \times d_{200} = ?$$

36. For a NaCl crystal, molar mass is 58.5g, density 2.17g/cm^3 , calculate the a) crystal spacing b) angle of glancing for first order and c) Miller indices of the reflecting plane if lattice parameter is 2.81 and incident wavelength of x-rays is 0.98Å .

→ Here, $M=58.5 \text{g}$, $\rho = 2.17 \text{g/cm}^3$, $a=2.81 \text{Å}$, $\lambda = 0.98 \text{Å}$

$$\text{(a) crystal spacing, } d = \left\{ \frac{M}{2N_A \rho} \right\}^{\frac{1}{3}} = \left\{ \frac{58.5}{2 \times 6.02 \times 10^{23} \times 2.17} \right\}^{\frac{1}{3}} = 2.81 \times 10^{-8} \text{cm} = 2.81 \text{Å}$$

(b) We have, $2d\sin\theta = n\lambda$

$$\therefore \theta = \sin^{-1} \left\{ \frac{n\lambda}{2d} \right\} = \sin^{-1} \left\{ \frac{1 \times 0.98}{2 \times 2.81} \right\} = 11.8^\circ$$

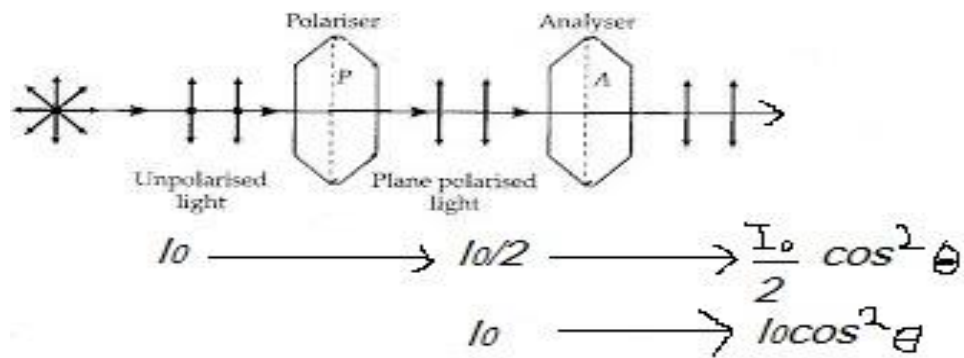
$$\text{(c) we have } d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$\text{or, } \sqrt{h^2 + k^2 + l^2} = \frac{a}{d_{hkl}} = \frac{2.81}{2.81} = 1$$

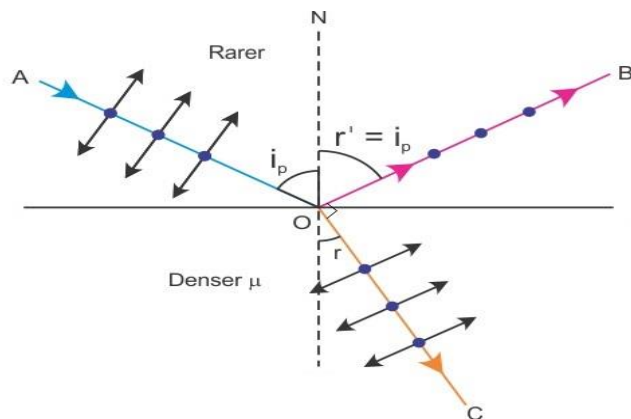
$$\therefore (hkl) = (100)$$

Chapter: Polarization

The phenomenon due to which vibration of light wave is restricted in a particular plane is called polarization and such a light is called plane polarized light.



Polarizing angle (i_p): The angle of incidence for which the reflected ray is completely plane polarized is called polarizing angle.



Here, $i_p + r = 90^\circ$

Brewster's Law: $\mu = \tan i_p$

Double Refraction

The phenomenon due to which refracted ray splits into two components whose vibrations are perpendicular to each other is called double refraction.

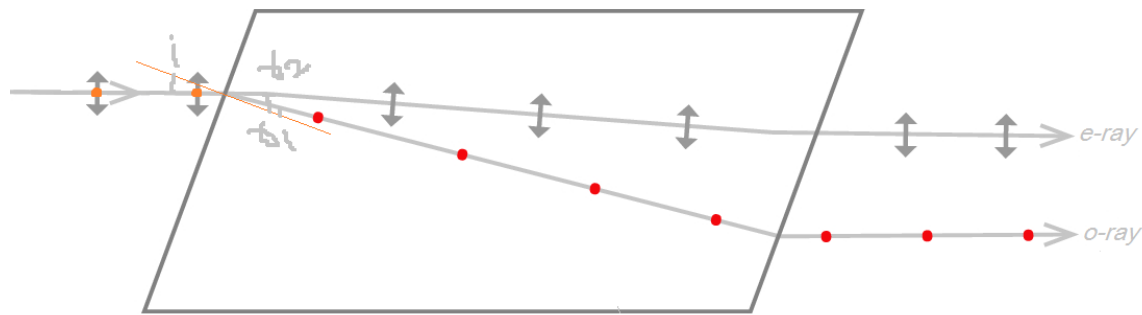


Fig: Double refraction through calcite crystal

Consider an unpolarized light is incident on a calcite crystal then the refracted ray splits into o and e components.

$$\text{For o-ray, } \mu_o = \frac{\sin i}{\sin r_1}$$

$$\text{And for e-ray, } \mu_e = \frac{\sin i}{\sin r_2}$$

$$\text{As } r_1 < r_2 \text{ so } \mu_o > \mu_e$$

$$\text{Since, } \mu = \frac{c}{v} \text{ so } v_o < v_e$$

It is seen that the velocity of o-ray is less than that of e-ray. Such crystals in which velocity of o-ray is less than the velocity of e-ray are called negative uniaxial crystal. For example, calcite, tourmaline, ruby etc.

Those crystals in which velocity of o-ray is greater than the velocity of e-ray are called positive uniaxial crystal. For example, quartz.

Nicol Prism

It is an optical device which is used to produce and analyze the plane polarized light.

Construction: It is made by cutting a calcite crystal along a diagonal as shown in figure. The cutting pieces are then joined back together by using a substance called Canada balsam. Canada balsam is a transparent substance whose refractive index lies between μ_o and μ_e .

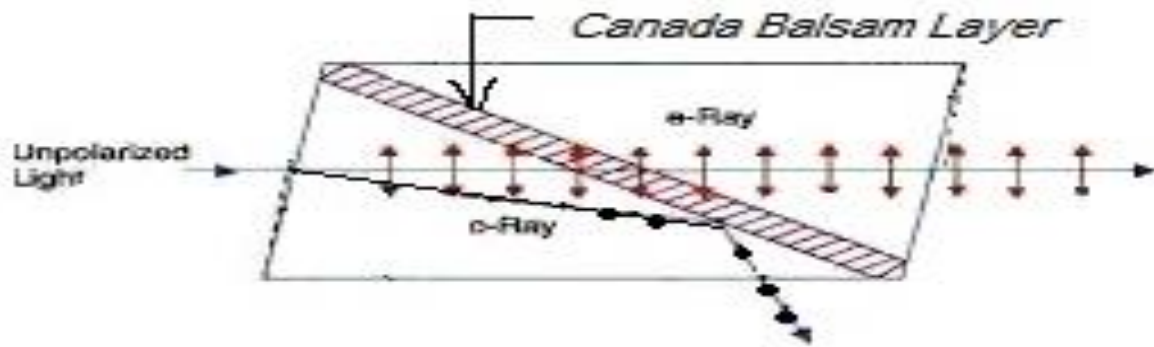
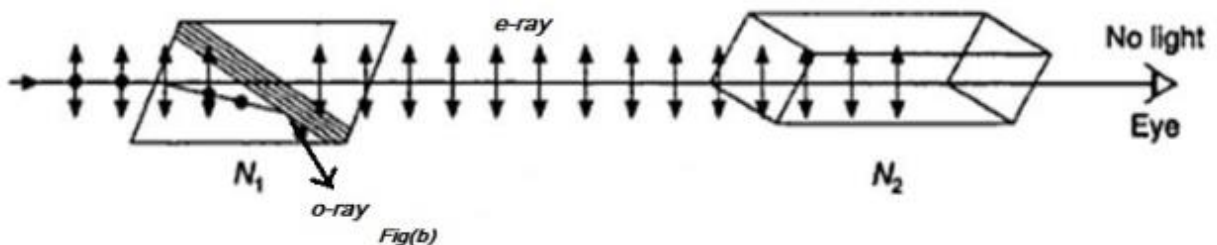
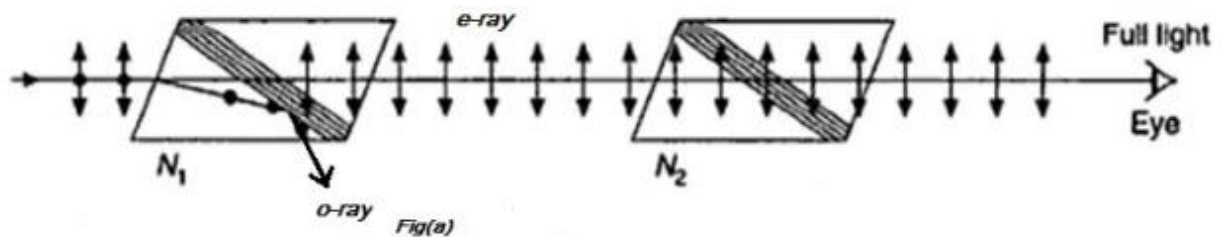


Fig: Nicol Prism

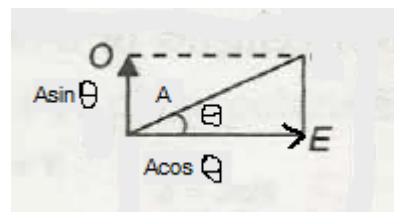
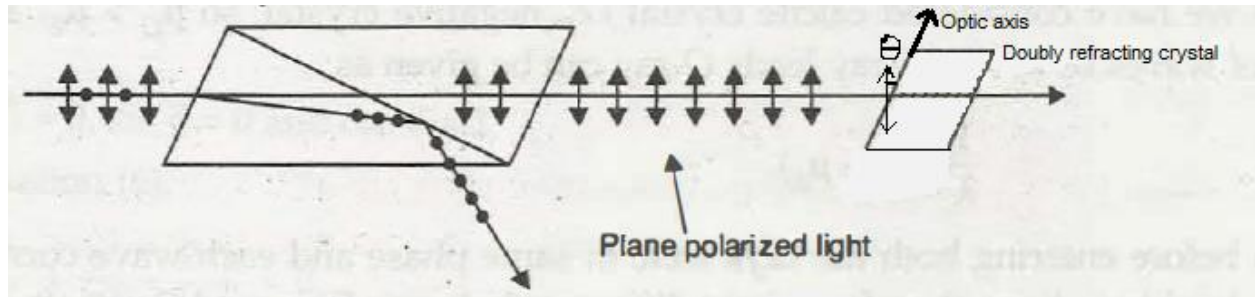
Working: When an unpolarized ray of light is incident on the first face of Nicol prism, the refracted ray splits into o and e components. For o-ray, the Canada balsam satisfies the condition of total internal reflection whereas e-ray passes on to emerge through the second face of Nicol prism. In this way, a Nicol prism can transmit only a plane polarized light.

Use: Nicol Prism can be used as polarizer and analyzer of plane polarized light. For this two prisms N_1 and N_2 are placed one after another as shown in figure (a).



The prism N_1 is used to produce the plane polarized light so is called as polarizer and the prism N_2 is used to analyze the plane polarized light so is called as analyzer. When N_1 and N_2 are parallel, the light from the polarizer passes on through the analyzer (Fig a). If we rotate the analyzer, the intensity of light goes on decreasing and no light can be found after the analyzer when it is completely turned by 90° . In this way Nicol prism can be used as polarizer and analyzer of plane polarized light.

Linearly, Circularly and Elliptically Polarized light



Consider a plane polarized light is incident normally on a doubly refracting crystal whose faces are cut parallel to the optic axis. The refracted ray splits into o and e components which move in same direction but with different velocities. Let A be the amplitude of incident plane polarized light and θ be the angle made by the plane of vibration with optic axis.

We have for o-ray, $y = A \sin \theta \sin \omega t$

And for e-ray, $x = A \cos \theta \sin(\omega t + \delta)$

Where, δ is phase difference between o and e rays.

Put $A \cos \theta = a$ and $A \sin \theta = b$, then

$$y = A \sin \theta \sin \omega t = b \sin \omega t$$

$$\therefore \sin \omega t = \frac{y}{b} \text{ and } \cos \omega t = \sqrt{1 - \frac{y^2}{b^2}}$$

$$\text{Now, } x = A \cos \theta \sin(\omega t + \delta) = a \sin(\omega t + \delta)$$

$$\text{or, } \frac{x}{a} = \sin(\omega t + \delta)$$

$$\text{or, } \frac{x}{a} = \sin \omega t \cos \delta + \cos \omega t \sin \delta$$

$$\text{or, } \frac{x}{a} = \frac{y}{b} \cos \delta + \sqrt{1 - \frac{y^2}{b^2}} \sin \delta$$

$$\text{or, } \frac{x}{a} - \frac{y}{b} \cos \delta = \sqrt{1 - \frac{y^2}{b^2}} \sin \delta$$

On squaring,

$$\frac{x^2}{a^2} - 2 \frac{xy}{ab} \cos \delta + \frac{y^2}{b^2} \cos^2 \delta = \left(1 - \frac{y^2}{b^2}\right) \sin^2 \delta$$

$$\text{or, } \frac{x^2}{a^2} - 2 \frac{xy}{ab} \cos \delta + \frac{y^2}{b^2} = \sin^2 \delta \dots \dots \dots (i)$$

This is the general equation of ellipse.

Case i: When $\delta = 0$ or π then equation (i) becomes

$$\frac{x^2}{a^2} \mp 2 \frac{xy}{ab} + \frac{y^2}{b^2} = 0$$

$$\text{or, } \left\{ \frac{x}{a} \mp \frac{y}{b} \right\}^2 = 0$$

$$\text{or, } \frac{x}{a} \mp \frac{y}{b} = 0$$

$$\text{or, } \frac{x}{a} = \pm \frac{y}{b}$$

$$\text{or, } y = \pm \frac{b}{a} x$$

This is the equation of straight line passing through origin. Hence the emergent light will be linearly polarized when phase difference between o and e rays is 0 or π . (fig a)

Case ii: When $\delta = \frac{\pi}{2}$ then equation (i) becomes

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

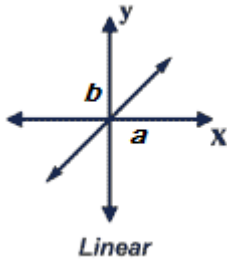
This is the equation of ellipse. Hence the emergent light will be elliptically polarized when phase difference between o and e rays is $\frac{\pi}{2}$. (fig b)

Case iii: When $\delta = \frac{\pi}{2}$ and $a = b$ then equation (i) becomes

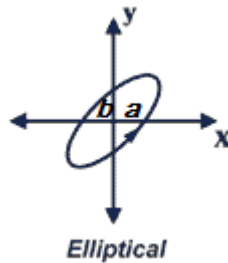
$$\frac{x^2}{a^2} + \frac{y^2}{a^2} = 1$$

$$\text{or, } x^2 + y^2 = a^2$$

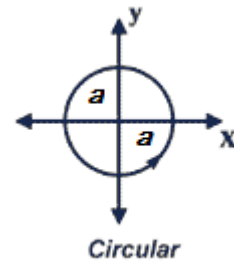
This is the equation of circle having center at origin. Hence the emergent light will be circularly polarized when phase difference between o and e rays is $\frac{\pi}{2}$ and $a = b$. (fig c)



Fig(a)



Fig(b)



Fig(c)

Retardation Plate

The plate which produces certain path difference between o-ray and e-ray is called retardation plate.

Types

1) Quarter-wave plate

The retardation plate which produces the path difference of $\frac{\lambda}{4}$ (or phase difference $\frac{\pi}{2}$) between o-ray and e-ray is called quarter-wave plate.

If μ_o and μ_e be the refractive indices of o-ray and e-ray on passing through a quarter-wave plate of thickness t then

$$(\mu_e - \mu_o)t = \frac{\lambda}{4} \quad (\text{For positive uniaxial crystal})$$

$$\text{and } (\mu_o - \mu_e)t = \frac{\lambda}{4} \quad (\text{For negative uniaxial crystal})$$

2) Half-wave plate

The retardation plate which produces the path difference of $\frac{\lambda}{2}$ (or phase difference π) between o-ray and e-ray is called half-wave plate.

If μ_o and μ_e be the refractive indices of o-ray and e-ray on passing through a half-wave plate of thickness t then

$$(\mu_e - \mu_o)t = \frac{\lambda}{2} \quad (\text{For positive uniaxial crystal})$$

$$\text{and } (\mu_o - \mu_e)t = \frac{\lambda}{2} \quad (\text{For negative uniaxial crystal})$$

Optical Activity

The phenomenon due to which rotation of the plane of vibration of plane polarized light takes place is called optical activity.

Specific Rotation

It is defined as the angle of rotation θ per unit concentration C (in gm/cc) per unit length of the medium l (in decimeter) at a given temperature and wavelength of light.

i.e. specific rotation, $S = \frac{\theta}{cl}$ Where, l is in decimeter.

or, $S = \frac{10\theta}{cl}$ Where, l is in centimeter.

Production and Detection of plane, circularly and elliptically polarized light

Production

- Plane polarized: The unpolarized light is incident on the Nicol prism so that the emergent ray is plane polarized.
- Elliptically polarized: When a plane polarized light is incident on a quarter-wave plate, than the emergent ray is elliptically polarized. In this case, the plane of vibration should make an angle other than 45° with the optic axis.
- Circularly polarized: When a plane polarized light is incident on a quarter-wave plate in such a way that the plane of vibration makes an angle 45° with the optic axis, than the emergent ray is circularly polarized.

Detection

- Plane polarized light: The given ray of light is incident on a Nicol prism. If the intensity of light vanishes twice in a complete rotation of the Nicol prism then the given ray of light is plane polarized, otherwise it is unpolarized.
- Elliptically polarized light: The given ray of light is incident on a Nicol prism. If the intensity of light through Nicol prism is non-uniform as it is rotated, then either it is elliptically polarized or the mixture of unpolarized and plane polarized. To distinguish this, the given ray of light is incident on the quarter-wave plate and on the Nicol prism. If the intensity of light vanishes twice in a complete rotation of Nicol prism, then the given ray of light is elliptically polarized otherwise the mixture of unpolarized and plane polarized.

Circularly polarized light: The given ray of light is incident on a Nicol prism. If the intensity of light through Nicol prism remains same as it is rotated, then either it is circularly polarized or unpolarized. To distinguish this, the given ray of light is incident on the quarter-wave plate and on the Nicol prism. If the intensity of light vanishes twice in a complete rotation of Nicol prism, then the given ray of light is circularly polarized otherwise unpolarized.

POLARIZATION

- If the polarizing angle of a piece of glass for green light is 60° , Find the angle of minimum deviation for a 60° prism made of same material of glass.

→ Here, polarizing angle $i_p = 60^\circ$, angle of prism $A=60^\circ$, minimum deviation $\delta_m = ?$

We have, $\mu = \tan i_p = ?$

$$\text{Again, } \mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$$

$$\text{or, } \delta_m = 2 \sin^{-1} \left(\mu \sin \frac{A}{2} \right) - A = ?$$

2. At a certain temperature, the critical angle of water for total internal reflection is 48° for certain wavelength. What is the polarizing angle and angle of refraction for light incident on water at polarizing angle.

→ Here, critical angle $C = 48^\circ$, $i_p = ?$, angle of refraction $r = ?$

$$\text{We have, } \mu = \frac{1}{\sin C} = ?$$

$$\therefore i_p = \tan^{-1} \mu = ?$$

$$\text{Also, } i_p + r = 90^\circ$$

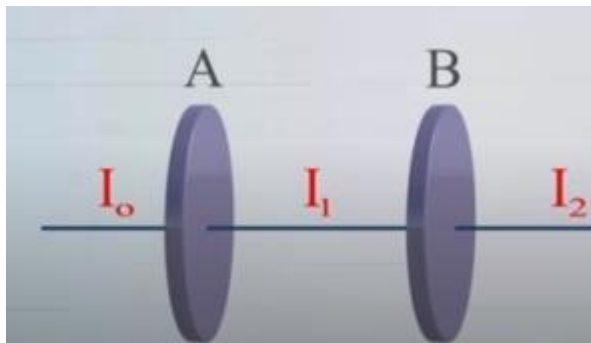
$$\therefore r = ?$$

3. Unpolarized light passes through two Polaroid; the axis of one is vertical and that of the other is at 60° to the vertical. What is the intensity of transmitted light?

→ Let I_0 be the initial intensity, I_1 be the intensity after first polaroid and I_2 be the intensity after second Polaroid.

$$\text{Now, } I_1 = I_0 / 2$$

$$\text{and } I_2 = I_1 \cos^2 60^\circ = \frac{I_0}{2} \times \frac{1}{4} = \frac{I_0}{8}$$



4. A horizontal beam of vertically polarized light of intensity 43 W/m^2 is sent through two polarizing sheets. The polarizing direction of first is at 70° to vertical and that of second is horizontal. What is the intensity of light transmitted by the pair of sheets?

→ Let I_0 be the initial intensity, I_1 be the intensity after first sheet and I_2 be the intensity after second sheet.

$$\text{Here, } I_0 = 43 \text{ W/m}^2$$

$$I_1 = I_0 \cos^2 70^\circ = ?$$

$$\text{and } I_2 = I_1 \cos^2(90^\circ - 70^\circ) = ?$$

5. A beam of polarized light is sent into a system of two polarizing sheets. Relative to the polarization direction of that incident light, the polarizing directions of the sheets are at angles θ for the first sheet and 90° for the second sheet. If 0.1 of the incident intensity is transmitted by the two sheets, what is θ .

→ Let I_0 be the initial intensity, I_1 be the intensity after first sheet and I_2 be the intensity after second sheet.

$$\text{We have, } I_1 = I_0 \cos^2 \theta$$

$$\text{And } I_2 = I_1 \cos^2(90^\circ - \theta) = I_0 \cos^2 \theta \sin^2 \theta$$

$$\text{or, } I_2 = \frac{I_0}{4} \sin^2 2\theta$$

$$\text{or, } 0.1 I_0 = \frac{I_0}{4} \sin^2 2\theta$$

$$\text{or, } \theta = ?$$

6. If the plane of vibration makes an angle 45° with the optic axis, compare the ratio of the intensities of o and e rays.

$$\rightarrow \text{Here, } \theta = 45^\circ$$

$$\text{We have, the intensity of e-ray is } I_E = A^2 \cos^2 \theta$$

$$\text{And the intensity of o-ray is } I_o = A^2 \sin^2 \theta$$

$$\text{Now, } \frac{I_o}{I_E} = \tan^2 \theta = \tan^2 45^\circ = ?$$

7. If the r.i. for o-ray and e-ray is 1.65 and 1.55 when incident on retardation plate. Calculate the thickness such that I) the emergent ray is plane polarized II) emergent ray is elliptically polarized. ($\lambda = 6000 \text{ \AA}$)

$$\rightarrow \text{Here, } \mu_o = 1.65, \mu_e = 1.55, \lambda = 6000 \text{ \AA}$$

- i) Since the emergent ray is plane polarized, so the phase difference between o and e rays is π i.e. path difference is $\frac{\lambda}{2}$.

$$\therefore (\mu_o - \mu_e)t = \frac{\lambda}{2}$$

$$\text{or, } t = ?$$

- ii) Since the emergent ray is elliptically polarized, so the phase difference between o and e rays is $\frac{\pi}{2}$ i.e. path difference is $\frac{\lambda}{4}$.

$$\therefore (\mu_o - \mu_e)t = \frac{\lambda}{4}$$

$$\text{or, } t = ?$$

8. The refractive index of Calcite crystal of thickness 0.01mm is 1.648 for ordinary ray and 1.486 for extraordinary ray. For what wavelengths in the visible region will this plate behaves as a) quarter wave plate b) a half wave plate?

$$\rightarrow \text{a) For quarter wave plate, } (\mu_o - \mu_e)t = \frac{\lambda}{4}$$

$$\therefore \lambda = ?$$

$$\text{b) For half wave plate, } (\mu_o - \mu_e)t = \frac{\lambda}{2}$$

$$\therefore \lambda = ?$$

9. A beam of plane polarized light is converted into circularly polarized light by passing it through a crystal slice of thickness $3 \times 10^{-5} \text{m}$. Calculate the difference in the r.i. of the two rays inside the crystal assuming the above thickness to be the minimum value required to produce the effect. ($\lambda = 600 \text{nm}$)

→ For circularly polarized light, path difference between o and e rays is $\frac{\lambda}{4}$.

$$\therefore (\mu_o - \mu_e)t = \frac{\lambda}{4}$$

$$\text{or, } \mu_o - \mu_e = ?$$

10. Plane polarized light is incident on a piece of quartz cut parallel to the axis. Find the least thickness for which the o-ray and e-ray combine to form a plane polarized light. Given $\mu_o = 1.5442$, $\mu_e = 1.5533$, $\lambda = 5 \times 10^{-5} \text{cm}$

→ For plane polarized light, path difference between o and e rays is $\frac{\lambda}{2}$.

$$\therefore (\mu_e - \mu_o)t = \frac{\lambda}{2}$$

$$\text{or, } t = ?$$

11. Calculate the thickness of i) a quarter wave plate and ii) a half wave plate given that $\mu_o = 1.544$, $\mu_e = 1.553$, $\lambda = 5 \times 10^{-5} \text{cm}$

→ Same as 8.

12. A quarter-wave plate is meant for $\lambda_0 = 5.893 \times 10^{-5} \text{cm}$. What phase retardation will show for $\lambda = 4.358 \times 10^{-5} \text{cm}$? (neglect changes of μ_o and μ_e with λ)

→ We have Phase retardation $\phi = \frac{2\pi}{\lambda} \times \text{path difference}$

$$\text{or, } \phi = \frac{2\pi}{\lambda} \times \frac{\lambda_0}{4} = ?$$

13. A sheet of cellophane is a half plate for light of wavelength $4000 \text{\AA}$. If the refractive index do not change with wavelength, explain how the sheet would behave with respect to light of wavelength $8000 \text{\AA}$.

→ For wavelength $\lambda_1 = 4000 \text{\AA}$, the sheet behaves as half wave plate.

$$\therefore (\mu_o - \mu_e)t = \frac{\lambda_1}{2}$$

For wavelength $\lambda_2 = 8000 \text{\AA}$

$$(\mu_o - \mu_e)t = \frac{\lambda_2}{n}$$

$$\therefore \frac{\lambda_1}{2} = \frac{\lambda_2}{n}$$

$$\text{or, } n = 2 \times \frac{\lambda_2}{\lambda_1} = 2 \times \frac{8000}{4000} = 4$$

$\therefore$ The sheet behaves as quarter-wave plate for $8000\text{\AA}$.

- 14. Find the specific rotation of a given sample of sugar solution if plane of polarization is turned through 26.4° . The length of the tube containing 20% sugar solution is 20cm.**

$\rightarrow$ Here, Specific rotation $S=?$, angle of rotation $\theta = 26.4^\circ$, Concentration $C=20\%=0.2\text{gm/cc}$, length of tube $l = 20\text{cm}$

$$\text{we have, } S = \frac{10\theta}{Cl} = ?$$

- 15. A 200mm long tube and containing 48cm^3 of sugar solution produces and optical rotation of 11° when placed on a sacchari meter. If the specific rotation of the sugar solution is 66° , calculate the quantity of sugar contained in the tube in the form of solution.**

$$\rightarrow \text{we have, } S = \frac{10\theta}{Cl}$$

$$\text{or, } C = \frac{10\theta}{Sl} = \dots \dots \dots \text{gm/cc}$$

$$\text{Now, quantity of sugar contained} = C \times V = \dots \dots \dots \text{gm}$$

- 16. A 20cm long tube having a sugar solution rotates the plane of polarization by 10° . If the specific rotation of sugar is 66° , calculate the strength of solution.**

$$\rightarrow C = ?$$

- 17. A sugar solution in a tube of length 20cm produces optical rotation of 13° . The solution is then diluted to $1/3$ of its previous concentration. Find the optical rotation produced by 30cm long tube containing the diluted solution.**

$$\rightarrow \text{Here, } l_1 = 20\text{cm}, \theta_1 = 13^\circ, C_1 = C, l_2 = 30\text{cm}, \theta_2 = ?, C_2 = C/3,$$

$$\text{We have, } S = \frac{10\theta_1}{C_1 l_1} = \frac{10\theta_2}{C_2 l_2}$$

$$\therefore \theta_2 = ?$$

- 18. A length of 25cm of solution containing 50gm of solute/liter causes a rotation of plane of polarization by 5° . Find the plane of polarization by a length of 75cm of similar solution containing 100gm solute/liter.**

$$\rightarrow \text{Here, } l_1 = 25\text{cm}, \theta_1 = 5^\circ, C_1 = 50 \frac{\text{gm}}{\text{liter}} = \frac{50}{1000} \frac{\text{gm}}{\text{cc}}, l_2 = 75\text{cm}, \theta_2 = ?, C_2 = 100 \frac{\text{gm}}{\text{liter}} = \frac{100}{1000} \frac{\text{gm}}{\text{cc}}$$

$$\text{We have, } S = \frac{10\theta_1}{C_1 l_1} = \frac{10\theta_2}{C_2 l_2}$$

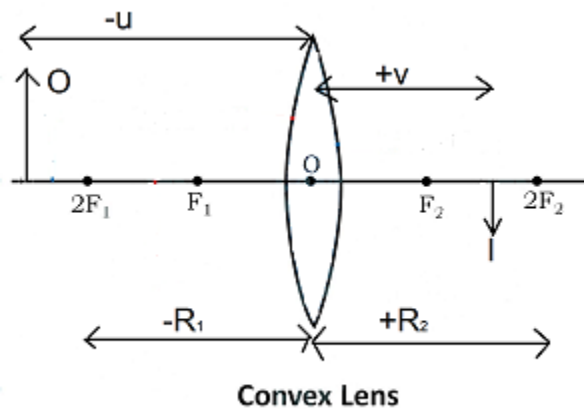
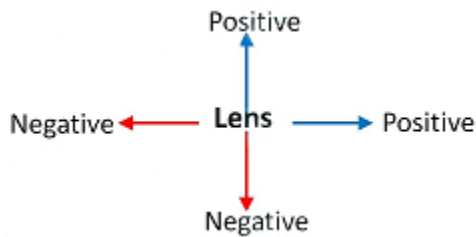
$$\therefore \theta_2 = ?$$

Chapter: Lens System

Sign Convention

- i) All the distances are measured from optical center.

- ii) Distances on the right of optical center can be taken as positive.
- iii) Distances on the left of optical center can be taken as negative.
- iv) Focal length of convex lens is always positive and focal length of concave lens is always negative.



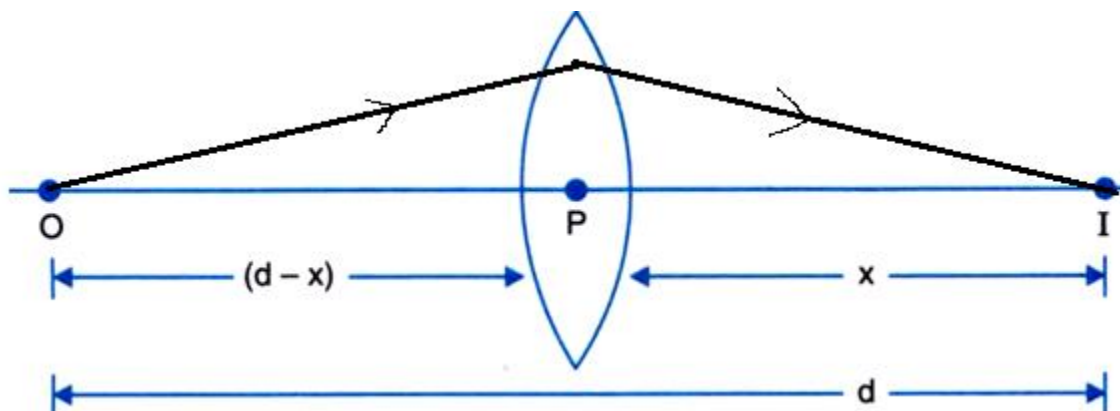
$$* \frac{1}{f} = -\frac{1}{u} + \frac{1}{v}$$

$$* \frac{1}{f} = (\mu - 1) \left(-\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$* \text{ Deviation angle produced by a lens, } \delta = \frac{h}{f}$$

Minimum Distance between an object and its real image in a convex lens

Consider a convex lens having focal length f . If an object is placed at O then its image will be formed at I.



Let $OI = d$ and $PI = x$ then

Object distance $u = -(d - x)$

Image distance $v = x$

We have $\frac{1}{f} = -\frac{1}{u} + \frac{1}{v}$

$$\text{or, } \frac{1}{f} = \frac{1}{d - x} + \frac{1}{x} = \frac{d}{x(d - x)}$$

$$\text{or, } xd - x^2 = fd$$

$$\text{or, } x^2 - dx + fd = 0$$

This is quadratic in x .

For image to be real, x should be real.

For real x , $d^2 - 4fd \geq 0$

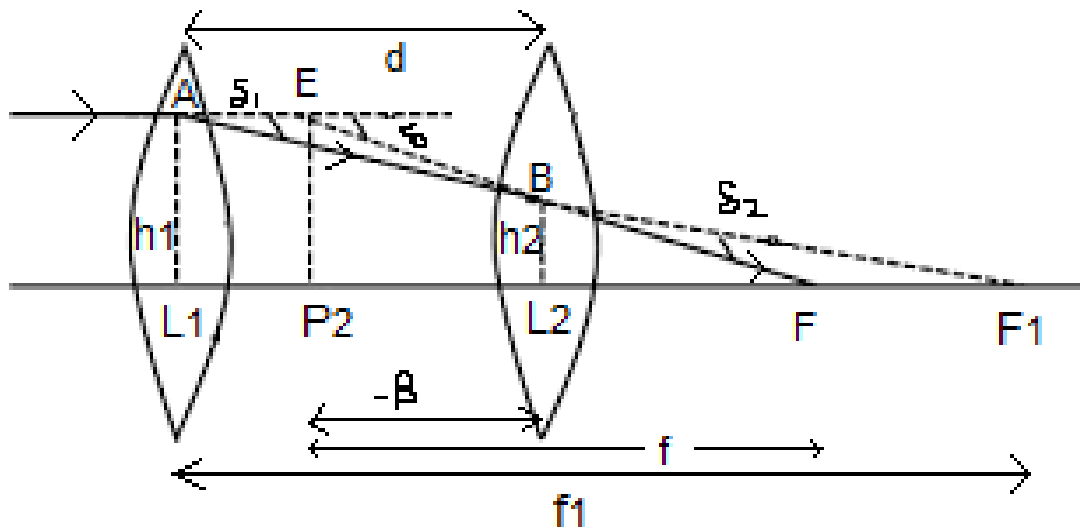
$$\text{or, } d^2 \geq 4fd$$

$$\text{or, } d \geq 4f$$

This shows that minimum distance between an object and its real image is four times the focal length of lens.

Equivalent focal length of two thin lenses separated by a distance

Consider two thin lenses L_1 and L_2 having focal lengths f_1 and f_2 are placed co-axially separated by a distance d . When a ray parallel to principal axis is incident on the lens L_1 then its final image will be formed at F.



Here, the deviations produced by the lenses L_1 and L_2 are $\delta_1 = \frac{h_1}{f_1}$ and $\delta_2 = \frac{h_2}{f_2}$ respectively.

If the final emergent ray BF is back produced then it meets original direction of incident ray at point E. So the lens placed in the position EP_2 is called equivalent lens. If f be the focal length of equivalent lens then deviation produced by it is given by, $\delta = \frac{h_1}{f}$.

We have net deviation $\delta = \delta_1 + \delta_2$

$$\text{or, } \frac{h_1}{f} = \frac{h_1}{f_1} + \frac{h_2}{f_2} \dots \dots \dots (i)$$

Since $\triangle AL_1F_1$ and $\triangle BL_2F_1$ are similar,

$$\therefore \frac{AL_1}{BL_2} = \frac{L_1F_1}{L_2F_1}$$

$$\text{or, } \frac{h_1}{h_2} = \frac{f_1}{f_1 - d} \dots \dots \dots (ii)$$

$$\text{or, } h_2 = \frac{f_1 - d}{f_1} \times h_1$$

Now from equation (i),

$$\frac{h_1}{f} = \frac{h_1}{f_1} + \frac{1}{f_2} \times \frac{f_1 - d}{f_1} \times h_1$$

$$\text{or, } \frac{1}{f} = \frac{1}{f_1} + \frac{f_1 - d}{f_1 f_2}$$

$$\text{or, } \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

This gives the equivalent focal length of two thin lenses separated by a distance d .

In terms of power, $P = P_1 + P_2 - dP_1P_2$

Also we can write, $f = \frac{f_1 f_2}{f_1 + f_2 - d}$

$$\text{or, } f = -\frac{f_1 f_2}{\Delta}$$

Where, $\Delta = d - (f_1 + f_2)$ is optical separation of lens system.

Let β be the position of second principle point (P_2) from the lens L_2 .

Since $\triangle EP_2F$ and $\triangle BL_2F$ are similar,

$$\therefore \frac{EP_2}{BL_2} = \frac{P_2F}{L_2F}$$

$$\text{or, } \frac{h_1}{h_2} = \frac{f}{f - (-\beta)} = \frac{f}{f + \beta} \dots \dots \dots (iii)$$

Now from equations (ii) and (iii) we get,

$$\frac{f_1}{f_1 - d} = \frac{f}{f + \beta}$$

$$\text{or, } f_1 f + f_1 \beta = f_1 f - f d$$

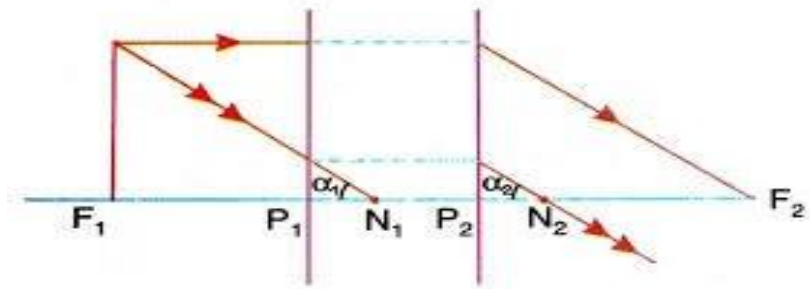
$$\text{or, } \beta = -\frac{f d}{f_1}$$

Similarly if α be the position of first principle point (P_1) from the lens L_1 then,

$$\alpha = \frac{f d}{f_2}$$

Cardinal Points

In the combination of two thin lenses, there exist three pairs of points in the principal axis which are considered as reference points to measure various distances, are called cardinal points. These are two principal points (P_1 & P_2), two focal points (F_1 & F_2) and two nodal points (N_1 & N_2). Nodal points are those points in the principal axis about which angular magnification is unity.

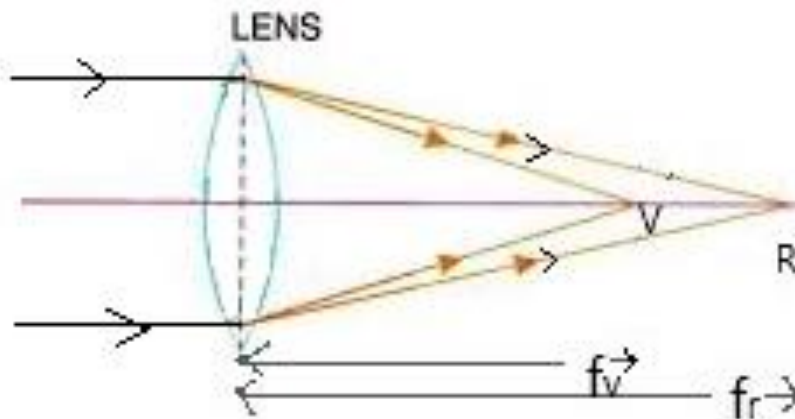


The planes which are perpendicular to the principal axis and pass through focal points are called focal planes. Similarly, the planes which are perpendicular to the principal axis and pass through principal points are called principal planes and the planes which are perpendicular to the principal axis and pass through nodal points are called nodal planes.

Chromatic aberration

The inability of a lens to focus all the colors of light at a single point after refraction is called chromatic aberration. It is given by

$$\text{Chromatic aberration} = f_r - f_v$$



Consider the rays of white light parallel to principal axis be incident on a convex lens of mean focal length f . After refraction, red and violet colored images will be formed at R and V respectively.

We have for mean color,

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_2} - \frac{1}{R_1} \right)$$

$$\text{or, } \frac{1}{R_2} - \frac{1}{R_1} = \frac{1}{f(\mu - 1)}$$

Now for red color,

$$\frac{1}{f_r} = (\mu_r - 1) \left(\frac{1}{R_2} - \frac{1}{R_1} \right) = \frac{\mu_r - 1}{f(\mu - 1)}$$

Similarly for violet color,

$$\frac{1}{f_v} = (\mu_v - 1) \left(\frac{1}{R_2} - \frac{1}{R_1} \right) = \frac{\mu_v - 1}{f(\mu - 1)}$$

$$\text{Now, } \frac{1}{f_v} - \frac{1}{f_r} = \frac{\mu_v - \mu_r}{f(\mu - 1)}$$

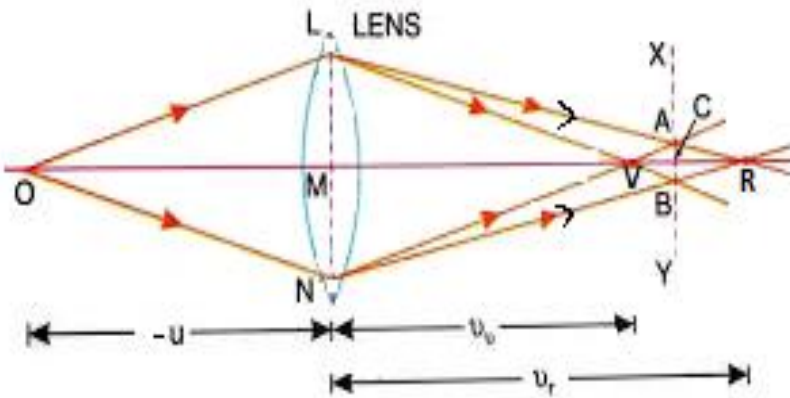
$$\text{or, } \frac{f_r - f_v}{f_r f_v} = \frac{\omega}{f} \quad (\text{Since, } \frac{\mu_v - \mu_r}{\mu - 1} = \omega \text{ is dispersive power of lens})$$

$$\text{or, } \frac{f_r - f_v}{f^2} = \frac{\omega}{f} \quad (\text{Since, } f_r f_v \approx f^2)$$

$$\text{or, } f_r - f_v = \omega f$$

This gives the longitudinal chromatic aberration.

Circle of least confusion



Consider the rays of white light from a point source O be incident on a convex lens of mean focal length f . After refraction through the lens, red and violet colored images will be formed at points R and V respectively. If a screen XY is placed between these images, then an image with least chromatic aberration is formed on the screen in the form of circle, called the circle of least confusion.

$$\text{We have, } \frac{1}{f_v} = \frac{1}{v_v} - \frac{1}{u} \quad \text{and} \quad \frac{1}{f_r} = \frac{1}{v_r} - \frac{1}{u}$$

$$\therefore \frac{1}{f_v} - \frac{1}{f_r} = \frac{1}{v_v} - \frac{1}{v_r}$$

$$\text{or, } \frac{f_r - f_v}{f_r f_v} = \frac{v_r - v_v}{v_r v_v}$$

$$\text{or, } \frac{f_r - f_v}{f^2} = \frac{v_r - v_v}{v^2} \text{ (Since } f_r f_v \approx f^2 \text{ and } v_r v_v \approx v^2 \text{)}$$

$$\text{or, } \frac{\omega f}{f^2} = \frac{v_r - v_v}{v^2} \text{ (Since } f_r - f_v = \omega f \text{)}$$

$$\therefore v_r - v_v = \frac{\omega v^2}{f} \dots \dots \dots (i)$$

Question: Show that longitudinal chromatic aberration is equal to

(i) ωf if object is at infinity

(ii) $\frac{\omega v^2}{f}$ if object is at finite distance.

Since $\triangle LNR$ and $\triangle ABR$ are similar,

$$\therefore \frac{MR}{LN} = \frac{CR}{AB} \dots \dots \dots (ii)$$

Again $\triangle LNV$ and $\triangle ABV$ are similar,

$$\therefore \frac{MV}{LN} = \frac{CV}{AB} \dots \dots \dots (iii)$$

Now adding equations (ii) and (iii) we get

$$\frac{MR + MV}{LN} = \frac{CR + CV}{AB}$$

$$\text{or, } \frac{v_r + v_v}{LN} = \frac{v_r - v_v}{AB}$$

$$\text{or, } \frac{v_r + v_v}{D} = \frac{v_r - v_v}{d}$$

Where $LN=D$ is the aperture of the lens and $AB=d$ is the diameter of the circle of least confusion.

$$\text{or, } \frac{v_r + v_v}{D} = \frac{\omega v^2}{fd} \text{ (Since, using equation (i))}$$

$$\text{or, } \frac{2v}{D} = \frac{\omega v^2}{fd} \text{ (Since, } v_r \approx v_v \approx v \text{)}$$

$$\text{or, } d = \frac{1}{2} \omega D \frac{v}{f}$$

If the object is at infinity, then $v = f$

$$\therefore d = \frac{1}{2} \omega D$$

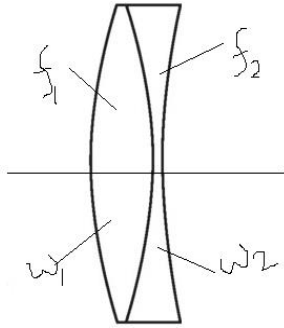
This shows that diameter of the circle of least confusion is independent of the focal length of lens.

Achromatism (Chromatic doublet)

The process of removal of chromatic aberration is called achromatism.

- (i) Achromatism by using two lenses placed in contact

Consider a convex lens having focal length f_1 and dispersive power ω_1 is placed in contact with a concave lens having focal length f_2 and dispersive power ω_2 as shown in figure.



We have the equivalent focal length is

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

on differentiating,

$$-\frac{1}{f^2} df = -\frac{1}{f_1^2} df_1 - \frac{1}{f_2^2} df_2$$

But for achromatism, $df = 0$

$$\therefore \left(-\frac{df_1}{f_1}\right) \frac{1}{f_1} + \left(-\frac{df_2}{f_2}\right) \frac{1}{f_2} = 0$$

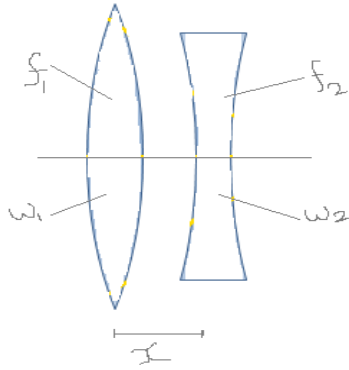
$$\text{or, } \frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$$

Where, $-\frac{df_1}{f_1} = \omega_1$ and $-\frac{df_2}{f_2} = \omega_2$

This gives the condition of achromatism. This shows that two lenses of same nature cannot be used for this purpose.

- (ii) Achromatism by using two lenses separated by a distance

Consider a convex lens having focal length f_1 and dispersive power ω_1 is separated by a distance x with a concave lens having focal length f_2 and dispersive power ω_2 as shown in figure.



We have the equivalent focal length is

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{x}{f_1 f_2}$$

on differentiating,

$$-\frac{1}{f^2} df = -\frac{1}{f_1^2} df_1 - \frac{1}{f_2^2} df_2 - x \left(-\frac{1}{f_1 f_2^2} df_2 - \frac{1}{f_2 f_1^2} df_1 \right)$$

But for achromatism, $df = 0$

$$\therefore \left(-\frac{df_1}{f_1} \right) \frac{1}{f_1} + \left(-\frac{df_2}{f_2} \right) \frac{1}{f_2} - \frac{x}{f_1 f_2} \left\{ \left(-\frac{df_1}{f_1} \right) + \left(-\frac{df_2}{f_2} \right) \right\} = 0$$

$$\text{or, } \frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} - \frac{x}{f_1 f_2} (\omega_1 + \omega_2) = 0$$

$$\text{Where, } -\frac{df_1}{f_1} = \omega_1 \quad \text{and} \quad -\frac{df_2}{f_2} = \omega_2$$

If $\omega_1 = \omega_2 = \omega$ then

$$\frac{\omega}{f_1} + \frac{\omega}{f_2} - \frac{x}{f_1 f_2} (\omega + \omega) = 0$$

$$\text{or, } \frac{1}{f_1} + \frac{1}{f_2} - \frac{2x}{f_1 f_2} = 0$$

$$\text{or, } 2x = f_1 + f_2$$

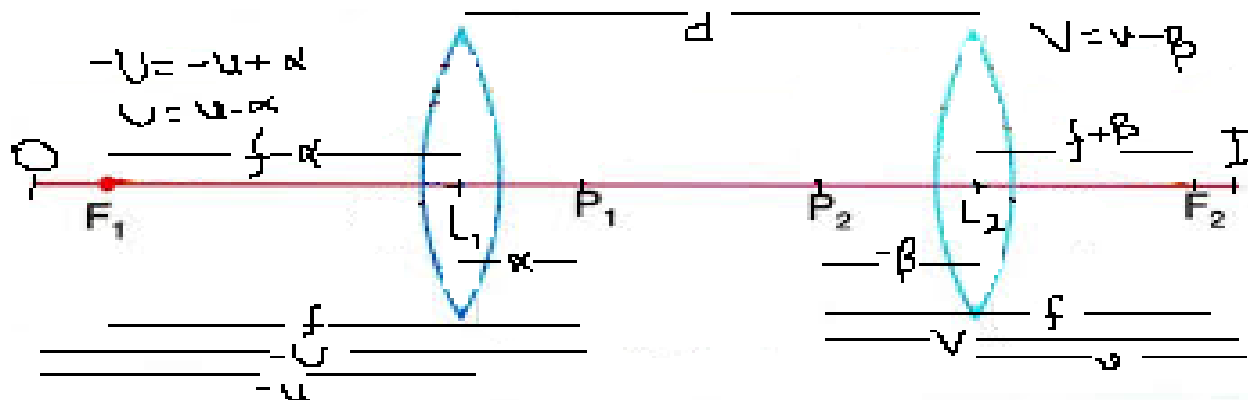
$$\text{or, } x = \frac{f_1 + f_2}{2}$$

If $f_1 = f_2 = f'$ then

$$x = f'$$

Problem Solving Technique

Given: f_1, f_2, d and u or v



1. Equivalent focal length $f = \frac{f_1 f_2}{f_1 + f_2 - d}$
2. Position of first principal point from $L_1 = \frac{f d}{f_2}$
3. Position of second principal point from $L_2 = \beta = -\frac{f d}{f_1}$
4. Position of first focal point F_1 from $L_1 = f - \alpha$
5. Position of Second focal point F_2 from $L_2 = f + \beta$
6. Object distance for equivalent lens $U = u - \alpha$
7. Image distance for equivalent lens $V = v - \beta$
8. For equivalent lens, $\frac{1}{f} = -\frac{1}{U} + \frac{1}{V}$
9. If object is at ∞ then, $u = \infty$ so $U = u - \alpha = \infty$
 $\therefore \frac{1}{f} = -\frac{1}{U} + \frac{1}{V}$
 $\therefore V = f$
10. If image is formed at ∞ then, $v = \infty$ so $V = v - \beta = \infty$
 $\therefore \frac{1}{f} = -\frac{1}{U} + \frac{1}{V}$
 $\therefore U = -f$

Numerical

1. A thin lens of refractive index n has focal length f in air is now immersed into a liquid of refractive index n_1 . Prove that new focal length of the lens in liquid is $f' = \left[\frac{n_1(n-1)}{n-n_1} \right] f$

→ Here, refractive index of lens $a\mu_g = n$, Focal length in air $f_a = f$, refractive index of liquid $a\mu_l = n_1$

When the lens is in air, $\frac{1}{f_a} = (a\mu_g - 1) \left(\frac{1}{R_2} - \frac{1}{R_1} \right)$

$$\text{or, } \frac{1}{f} = (n - 1) \left(\frac{1}{R_2} - \frac{1}{R_1} \right) \dots \dots \dots (i)$$

$$\text{When the lens is in liquid, } \frac{1}{f_l} = (l\mu_g - 1) \left(\frac{1}{R_2} - \frac{1}{R_1} \right) = \left(\frac{a\mu_g}{a\mu_l} - 1 \right) \left(\frac{1}{R_2} - \frac{1}{R_1} \right)$$

$$\text{or, } \frac{1}{f'} = \left(\frac{n}{n_1} - 1 \right) \left(\frac{1}{R_2} - \frac{1}{R_1} \right) = \left(\frac{n - n_1}{n_1} \right) \left(\frac{1}{R_2} - \frac{1}{R_1} \right) \dots \dots \dots (ii)$$

Now equation (i)/(ii) give $\frac{f'}{f} = \frac{n_1(n-1)}{n-n_1}$

$$\text{or, } f' = \left[\frac{n_1(n-1)}{n-n_1} \right] f$$

2. The focal length of the lens in air is 10cm. What will be the focal length if the air is replaced by water? Refractive index of water=4/3 and that of glass=3/2.

→ Same as 1.

3. Both surface of a double convex lens have radii of 28cm. If the focal length is 26.2cm, what is the index of refraction of the lens material?

→ Here, $R_2 = 28\text{cm}$, $R_1 = -28\text{cm}$, $f = 26.2\text{cm}$, $\mu = ?$

$$\text{We have, } \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_2} - \frac{1}{R_1} \right)$$

$$\therefore \mu = ?$$

4. Find the focal length of a plano-convex lens, the radius of surface being 40cm and $\mu = 1.50$

→ Here, $R_2 = 40\text{cm}$, $R_1 = \infty$, $f = ?$, $\mu = 1.5$

Same as 3.

5. **Two thin convex lenses having focal lengths 10cm and 4cm are coaxially separated by a distance of 5cm. Find the equivalent focal length of the combination. Determine also the position of the principle points.**

→ Here, $f_1 = 10\text{cm}$, $f_2 = 4\text{cm}$, $d = 5\text{cm}$

$$\text{We have equivalent focal length } f = \frac{f_1 f_2}{f_1 + f_2 - d} = ?$$

$$\text{Position of first principal point } \alpha = \frac{f d}{f_2} = ?$$

$$\text{And Position of second principal point } \beta = -\frac{f d}{f_1} = ?$$

6. **Two thin lenses of focal lengths 8cm each are identical and coaxially separated by 4cm. Determine the equivalent focal length of this lens combination and illustrate the principle points. If the image is formed at infinity at a particular position of the object, find the object distance.**

→ Here, $f_1 = 8\text{cm}$, $f_2 = 8\text{cm}$, $d = 4\text{cm}$, $v = \infty$, $u = ?$

$$\text{We have equivalent focal length } f = \frac{f_1 f_2}{f_1 + f_2 - d} = ?$$

$$\text{Position of first principal point } \alpha = \frac{f d}{f_2} = ?$$

$$\text{And Position of second principal point } \beta = -\frac{f d}{f_1} = ?$$

$$\text{Again image distance for equivalent lens } V = v - \beta = \infty$$

We have $\frac{1}{f} = -\frac{1}{u} + \frac{1}{v} = -\frac{1}{u}$

$\therefore U = -f = ?$

Now $u = U + \alpha = ?$

7. **A coaxial lens system placed in air has two lenses of equal focal length f are separated by a distance $2f/3$. Find the positions of the cardinal points.**

$\rightarrow$ Here, $f_1 = f, f_2 = f, d = \frac{2f}{3}$

We have equivalent focal length $f' = \frac{f_1 f_2}{f_1 + f_2 - d} = ?$

Position of first principal point $\alpha = \frac{f' d}{f_2} = ?$

And Position of second principal point $\beta = -\frac{f' d}{f_1} = ?$

Position of first focal point $= f' - \alpha = ?$

Position of second focal point $= f' + \beta = ?$

8. **Two thin converging lenses of focal length 20cm each are set coaxially 5cm apart. An image of an upright pole 200m distant and 10m high is formed by the combination. Find the position of the principal point, focal planes and the image. Also find the size of image.**

$\rightarrow$ Here, $f_1 = 20\text{cm}, f_2 = 20\text{cm}, d = 5\text{cm}, u = -200\text{m} = -20000\text{cm}, \text{ height of object } O = 10\text{m} = 1000\text{cm},$

We have equivalent focal length $f = \frac{f_1 f_2}{f_1 + f_2 - d} = ?$

Position of first principal point $\alpha = \frac{f d}{f_2} = ?$

And Position of second principal point $\beta = -\frac{f d}{f_1} = ?$

Position of first focal point $= f - \alpha = ?$

Position of second focal point $= f + \beta = ?$

Now, object distance for equivalent lens $U = u - \alpha = ?$

We have $\frac{1}{f} = -\frac{1}{u} + \frac{1}{v}$

$\therefore V = ?$

Now, $v = V + \beta = ?$

Again, magnification $m = \frac{\text{image distance}}{\text{object distance}} = \frac{\text{Height of image}}{\text{Height of object}}$

$$\text{or, } m = \frac{v}{u} = \frac{I}{O}$$

$$\therefore I = ?$$

9. Two thin converging lenses of focal length 3cm and 4cm respectively are placed coaxially in air and separated by a distance of 2cm. An object is placed 4cm in front of the first lens. Find the position, nature of the image and its lateral magnification.

$$\rightarrow \text{Here, } f_1 = 3\text{cm}, f_2 = 4\text{cm}, d = 2\text{cm}, u = -4\text{cm}$$

$$\text{We have equivalent focal length } f = \frac{f_1 f_2}{f_1 + f_2 - d} = ?$$

$$\text{Position of first principal point } \alpha = \frac{f d}{f_2} = ?$$

$$\text{And Position of second principal point } \beta = -\frac{f d}{f_1} = ?$$

$$\text{Now, object distance for equivalent lens } U = u - \alpha = ?$$

$$\text{We have } \frac{1}{f} = -\frac{1}{U} + \frac{1}{V}$$

$$\therefore V = ?$$

Now, $v = V + \beta = ?$ (if v is +ve, image is real and inverted, if v is -ve, image is virtual and erect)

$$\text{Again lateral magnification } m = \frac{v}{u} = ?$$

10. A convex lens A of focal length 20cm and concave lens B of focal length 5cm are kept along the same axis with distance d between them. If a parallel beam of light falling on A leaves B as parallel beam, what will be the value of d ?

$$\rightarrow \text{Here, } f_1 = 20\text{cm}, f_2 = -5\text{cm}, u = \infty, v = \infty$$

$$\text{Here, } U = u - \alpha = \infty \text{ and } V = v - \beta = \infty$$

$$\text{We have } \frac{1}{f} = -\frac{1}{U} + \frac{1}{V}$$

$$\therefore f = \infty$$

$$\text{Again, } f = \frac{f_1 f_2}{f_1 + f_2 - d}$$

$$\text{or, } \frac{1}{f} = \frac{f_1 + f_2 - d}{f_1 f_2}$$

$$\text{or, } 0 = \frac{f_1 + f_2 - d}{f_1 f_2}$$

$$\text{or, } f_1 + f_2 - d = 0$$

$$\therefore d = f_1 + f_2 = ?$$

- 11. A thin convex and thin concave lens, each of focal length 50cm are coaxially situated and separated by 10cm. Find the position and nature of the final image formed if an object placed 20cm from the convex lens.**

$$\rightarrow \text{Here, } f_1 = 50\text{cm}, f_2 = -50\text{cm}, d = 10\text{cm}, u = -20\text{cm}$$

$$\text{We have equivalent focal length } f = \frac{f_1 f_2}{f_1 + f_2 - d} = ?$$

$$\text{Position of first principal point } \alpha = \frac{f d}{f_2} = ?$$

$$\text{And Position of second principal point } \beta = -\frac{f d}{f_1} = ?$$

$$\text{Now, object distance for equivalent lens } U = u - \alpha = ?$$

$$\text{We have } \frac{1}{f} = -\frac{1}{U} + \frac{1}{v}$$

$$\therefore V = ?$$

$$\text{Now, } v = V + \beta = ? \text{ (if } v \text{ is +ve, image is real and inverted, if } v \text{ is -ve, image is virtual and erect)}$$

- 12. What is the power of combination of lenses of focal lengths 50cm and 10cm which are separated by a distance 10cm.**

$$\rightarrow \text{Here, } f_1 = 50\text{cm}, f_2 = 10\text{cm}, d = 10\text{cm},$$

$$\text{We have equivalent focal length } f = \frac{f_1 f_2}{f_1 + f_2 - d} = \dots \dots \text{cm} = \dots \dots \dots m$$

$$\text{Now power } P = \frac{1}{f} = ?$$

- 13. A lens made of crown glass has focal length 50cm. Calculate the longitudinal chromatic aberration. Given refractive indices of crown and flint glasses are 1.5164 and 1.5249.**

$$\rightarrow \text{Here, equivalent focal length } f = 50\text{cm}, \mu_1 = 1.5164, \mu_2 = 1.5249$$

$$\text{We have } \omega = \frac{\mu_2 - \mu_1}{\mu - 1} = ? \text{ (here, } \mu = \frac{\mu_2 + \mu_1}{2} \text{)}$$

$$\text{Now, longitudinal chromatic aberration} = \omega f = ?$$

- 14. The focal lengths of a thin convex lens are 1m and 0.968m for red and blue rays respectively. Calculate the chromatic aberration and dispersive power of the material of the lens.**

$$\rightarrow \text{Here, } f_r = 1\text{m}, f_b = 0.968\text{m}$$

$$\text{Chromatic aberration} = f_r - f_b = ?$$

$$\text{And dispersive power, } \omega = \frac{f_r - f_b}{f} = \frac{f_r - f_b}{\sqrt{f_r f_b}} = ?$$

15. Dispersive powers of two different glass types are 0.0163 and 0.0243 respectively. How can you design an achromatic contact doublet of focal length 50cm.

→ Here, $\omega_1 = 0.0163$, $\omega_2 = 0.0243$, equivalent focal length $f = 50\text{cm}$

We have, $\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$

$$\text{or, } \frac{f_2}{f_1} = -\frac{\omega_2}{\omega_1} = -\frac{0.0243}{0.0163} = -1.49$$

$$\text{or, } f_2 = -1.49f_1$$

$$\text{Again, } \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\text{or, } \frac{1}{50} = \frac{1}{f_1} - \frac{1}{1.49f_1}$$

$$\therefore f_1 = ? \text{ And } f_2 = -1.49f_1 = ?$$

16. An achromatic doublet of focal length 90cm is to be made of two lenses, the material of one having 1.5 times the dispersive power of the other. Find the focal length of each lenses.

→ Same as 15.

17. It is required to form a converging achromatic system of focal length 0.30m consisting of two lenses made of the same material. If one of the lens be convex and of focal length 0.2m, find a) the focal of the other, b) the distance between lenses.

→ Here, equivalent focal length = 0.30m, $f_1 = 0.2\text{m}$, $f_2 = ?$ $d = ?$

$$\text{We have, } d = \frac{f_1 + f_2}{2} = \frac{0.2 + f_2}{2} \dots \dots \dots (i)$$

$$\text{Also, } f = \frac{f_1 f_2}{f_1 + f_2 - d}$$

$$\text{or, } 0.30 = \frac{0.2 \times f_2}{0.2 + f_2 - (\frac{0.2 + f_2}{2})}$$

$$\text{or, } f_2 = ?$$

$$\text{Now from (i), } d = \frac{0.2 + f_2}{2} = ?$$

18. Two lenses of focal lengths +4.5cm and +1.5cm are separated by a finite distance. Find the position of principle points if the combination makes an achromatic combination.

→ Here, $f_1 = 4.5\text{cm}$, $f_2 = 1.5\text{cm}$

$$\text{Since, the combination makes achromatic combination, so } d = \frac{f_1 + f_2}{2} = ?$$

Now, equivalent focal length $f = \frac{f_1 f_2}{f_1 + f_2 - d} = ?$

Position of first principal point $\alpha = \frac{f d}{f_2} = ?$

And Position of second principal point $\beta = -\frac{f d}{f_1} = ?$

19. Two thin lenses of focal length f_1 and f_2 separated by a distance d have an equivalent focal length 50cm. The combination satisfies the conditions for no chromatic aberration and minimum spherical aberration. Find the values of f_1 , f_2 and d . Assume that both the lenses are of the same material.

→ Here, Equivalent focal length $f = 50\text{cm}$

For no chromatic aberration, $d = \frac{f_1 + f_2}{2} \dots \dots \dots (i)$

For minimum spherical aberration, $d = f_1 - f_2 \dots \dots \dots (ii)$

Solving (i) and (ii) we get

$$f_1 = \frac{3d}{2} \text{ and } f_2 = \frac{d}{2}$$

Again we have $f = \frac{f_1 f_2}{f_1 + f_2 - d}$

$$\text{or, } 50 = \frac{\frac{3d}{2} \times \frac{d}{2}}{\frac{3d}{2} + \frac{d}{2} - d}$$

or, $d = ?$

$$\therefore f_1 = \frac{3d}{2} = ? \text{ and } f_2 = \frac{d}{2} = ?$$

Chapter: LASER

The word LASER stands for Light Amplification by Stimulated Emission of Radiation. It is the process by which a parallel beam of monochromatic highly intense and coherence light is produced. The LASER light emerges out as a narrow beam which can travel long distance without much loss of intensity.

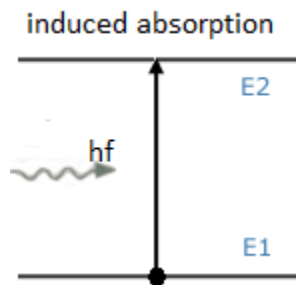
Properties

- The beam is monochromatic.
- The light is coherent with all the waves exactly in phase with each other.
- It is highly intense.
- The beam hardly diverges.

An atom can have energy in discrete value which is known as energy level of an atom. The transition of an electron from one energy level to another energy level is called quantum transition. There are three kinds of transition involving electromagnetic radiation between two energy levels E_1 and E_2 .

1. Induced absorption

Consider two energy levels of an atom with energies E_1 and E_2 as shown in figure. Consider an atom in E_1 level and a photon having energy $E = E_2 - E_1 = hf$ is incident on the atom at E_1 , the atom may absorb this incident photon and moves in the excited level E_2 . This is called induced absorption.

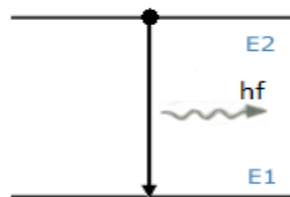


E_1 : Lower Energy State, E_2 : Higher Energy State

2. Spontaneous emission

The life time of an atom in the excited state is very short ($\approx 10^{-8} \text{sec}$). After finishing its life time, the atom moves in the lower state with emitting a photon of energy $E = E_2 - E_1 = hf$. This type of emission is called spontaneous emission.

Spontaneous Emission

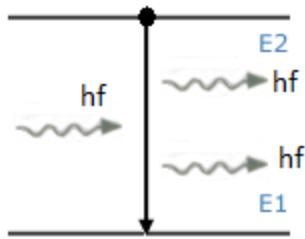


E_1 : Lower Energy State, E_2 : Higher Energy State

3. Stimulated emission

When a photon strikes on an atom on the excited state, the atom moves to the lower energy state with the emission of two photons of same frequency which are in phase and propagate in the same direction. This type of emission is called stimulated emission.

Stimulated Emission



E1: Lower Energy State, E2: Higher Energy State

Thus we get two coherent photons in this case. If these two photons are made to strike another two atoms in the excited state then four photons of same frequency are produced. If this process can be made go on, it may able to increase the intensity of radiation enormously. This is the basic operation of LASER.

The necessary condition for this type of amplification is that the number of atoms in the upper energy level must be sufficiently increased.

Active medium

The medium in which light gets amplified is called active medium. This medium may be solid, liquid or gas.

Population Inversion

The number of active atoms occupying an energy state is known as population of that state. At normal condition, the number of atoms in the lower state is greater than that on the excited state. The establishment of the situation, in which number of atoms in the excited state is greater than that on the lower state is called population inversion.

Metastable State

Normally, the life time of an atom in the excited state is very short ($\approx 10^{-8} \text{sec}$). Some energy states have comparatively longer life time (10^{-2}sec to 10^{-4}sec) and are known as metastable states.

Pumping

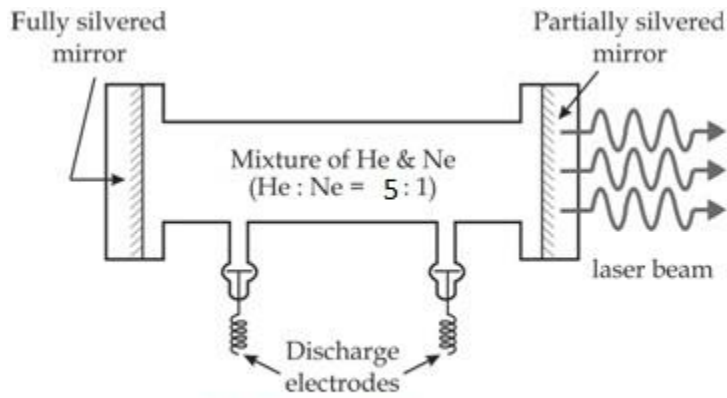
The process by which population inversion is achieved is called pumping.

Types of pumping

- Optical pumping
- Electrical pumping
- Chemical pumping
- Heating pumping

He-Ne LASER

The He-Ne LASER tube is about 5mm in diameter and 0.5m long. It contains helium and neon gas mixture in the ratio of 5:1 at a pressure of 1 mm of Hg. The tube has two parallel mirrors at two ends, one of them is partially reflecting.



He-Ne LASER

The He-Ne mixture contained in the discharge tube is ionized by passing a dc current through the gas. He atoms are excited very efficiently by electron impact in 2s level while Ne atoms are much less readily excited by the electrons. This excited 2s state of helium is relatively long lived. The energy of this level is almost the same as the energy of 5s level in Ne. Hence the energy of the He atoms is easily transferred to the Ne atoms when they collide. The 5s state of Ne results in a population inversion between 5s state and 3p state. The purpose of He atoms is thus for the population inversion in the Ne atoms. The transitions of atoms from 5s to 3p state produce photons of wavelength 632.8nm. Photons travelling parallel to the tube are reflected back and forth between the mirrors placed at the ends, and rapidly build up into an intense beam which escapes through the end without any losses.

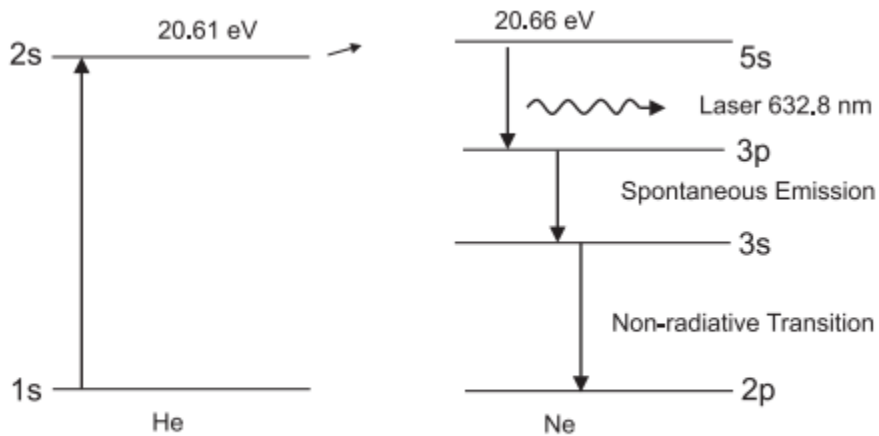


Figure 5.5: Helium-Neon energy level diagram

Application of LASER

1. Industrial Applications

- Welding purpose
- Cutting purpose
- Drilling purpose
- LASER heat treatment

2. Medical Application

- It is used in bloodless surgery
- It is used to cure ulcer, bone fracture, tumor.
- It is used in ophthalmology to cure retina.

3. Communication

LASER is used to establish communication between satellites, satellites and aircrafts and the satellite and the station in the ground. It is helpful to communicate between two sub marines in water.

4. LASER is used to check environmental pollutant. It is also used to forecast weather condition.

5. It is used in thermonuclear fusion.

6. It is used in isotopes separation.

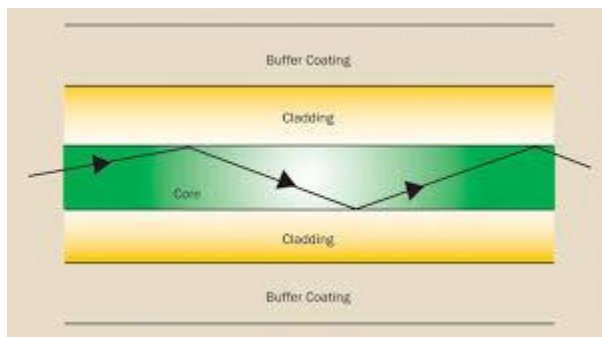
7. It is used in holography.

8. It is used in war.

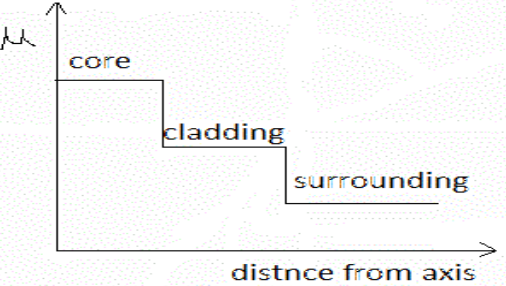
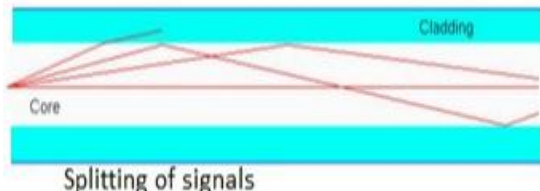
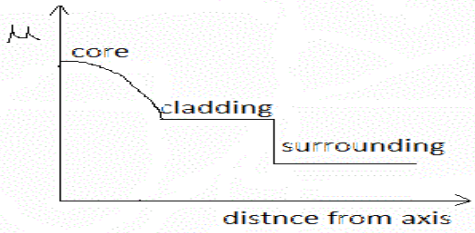
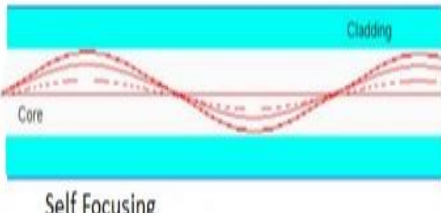
9. It is used to guide aircraft landing.

Chapter: Optical Fibers

The hair like structures which are made up of glass or plastic designed to transmit the signals from one end to other end is called optical fiber. It works on the principal of total internal reflection.



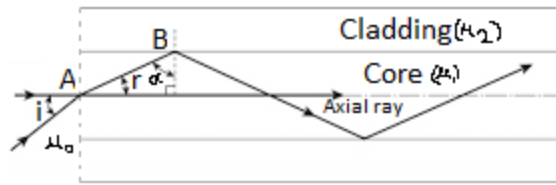
Types of optical fiber

Step index optical fiber	Graded index optical fiber
1. In this optical fiber, the refractive index of core (μ_1) is fixed and that of cladding (μ_2) is also fixed. 2. There is noticeable boundary between core and cladding.  3. There is high transmission loss due to splitting of the signals. 	1. In this optical fiber, the refractive index of cladding (μ_2) is fixed but that of core (μ_1) goes on decreasing from axis to core cladding interface. 2. There is no such noticeable boundary between core and cladding.  3. There is low transmission loss due to self-focusing. 

Acceptance angle and Numerical aperture of optical fiber

Acceptance angle: The angle of incidence for which refracted ray inside the core suffers total internal reflection is called acceptance angle.

Consider an optical fiber and light is sent from an end known as launching end. Let μ_1 and μ_2 be the refractive index of core and cladding respectively. Let μ_0 be the refractive index of surrounding of launching end.



Applying Snell's law at A,

$$\mu_0 \sin i = \mu_1 \sin r$$

$$\text{or, } \mu_0 \sin i = \mu_1 \sin(90 - \alpha) = \mu_1 \cos \alpha \dots \dots \dots (i)$$

Again using Snell's law at B when $\alpha = C$

$$\mu_1 \sin \alpha = \mu_2 \sin 90^\circ$$

$$\text{or, } \sin \alpha = \sin C = \frac{\mu_2}{\mu_1}$$

Now from (i)

$$\mu_0 \sin i = \mu_1 \sqrt{1 - \sin^2 \alpha}$$

$$\text{or, } \mu_0 \sin i = \mu_1 \sqrt{1 - \frac{\mu_2^2}{\mu_1^2}}$$

$$\text{or, } \mu_0 \sin i = \sqrt{\mu_1^2 - \mu_2^2}$$

$$\text{or, } \sin i = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0}$$

$$\text{or, } i = \sin^{-1} \left\{ \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0} \right\}$$

This gives the acceptance angle.

Numerical aperture (NA): The NA of optical fiber is defined as sine of maximum acceptance angle. It measures the light gathering power of optical fiber.

$$\text{Now, } NA = \sin i_{\max}$$

$$\text{or, } NA = \sqrt{\mu_1^2 - \mu_2^2} \quad (\text{Since, } \mu_0 = 1 \text{ for air})$$

$$\text{or, } NA = \sqrt{\left\{ \frac{\mu_1^2 - \mu_2^2}{2\mu_1^2} \right\} \times 2\mu_1^2}$$

$$\text{or, } NA = \sqrt{\Delta \times 2\mu_1^2}$$

$$\text{or, } NA = \mu_1 \sqrt{2\Delta}$$

Here, $\Delta = \frac{\mu_1^2 - \mu_2^2}{2\mu_1^2}$ is fractional refractive index change.

Optical fiber may be characterized by one more parameter called V-number or normalized frequency which is given by

$$V - \text{number} = \frac{2\pi a}{\lambda} \times NA = \frac{2\pi a}{\lambda} \times \mu_1 \sqrt{2\Delta}$$

Where, a is core radius and λ is wavelength of signal.

Numerical

1. A silica glass optical fiber has core of refractive index 1.50 and the cladding of refractive index 1.45. Calculate the critical angle for core cladding interface, the acceptance angle in air, corresponding angle of obliqueness, the numerical aperture of fiber and percentage of light collected by fiber w.r.t. the incident light.

→ Here, refractive index of core $\mu_1 = 1.50$, refractive index of cladding $\mu_2 = 1.45$

$$\text{We have } \sin C = \frac{\mu_2}{\mu_1}$$

$$\therefore C = \sin^{-1} \left(\frac{\mu_2}{\mu_1} \right) = ?$$

$$\text{Acceptance angle } i = \sin^{-1} \left\{ \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0} \right\} = \sin^{-1} \left\{ \sqrt{\mu_1^2 - \mu_2^2} \right\} = ? \quad (\mu_0 = 1 \text{ for air})$$

$$\text{Angle of obliqueness} = 90^\circ - C = ?$$

$$N.A. = \sqrt{\mu_1^2 - \mu_2^2} = ?$$

$$\% \text{ of light collected by optical fiber} = (N.A.)^2 \times 100\% = ?$$

2. Calculate the refractive indices of the core and cladding materials of a fiber from following data. $NA=0.22$ and fractional refractive index change $=0.012$.

→ Here, $N.A. = 0.22$, Fractional refractive index change $\Delta = 0.012$

We have, $NA = \mu_1 \sqrt{2\Delta}$

$$\therefore \mu_1 = \frac{NA}{\sqrt{2\Delta}} = ?$$

Again, $NA = \sqrt{\mu_1^2 - \mu_2^2}$

$$\therefore \mu_2 = ?$$

- 3. Calculate the numerical aperture and acceptance angle of fiber, given $\mu_{\text{core}}=1.62$ and $\mu_{\text{cladding}}=1.52$.**

→ Here, refractive index of core $\mu_1 = 1.62$, refractive index of cladding $\mu_2 = 1.52$

We have, $NA = \sqrt{\mu_1^2 - \mu_2^2} = ?$

and acceptance angle $i = \sin^{-1} \left\{ \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0} \right\} = \sin^{-1} \left\{ \sqrt{\mu_1^2 - \mu_2^2} \right\} = ?$ (assume $\mu_0 = 1$ for air)

- 4. The light ray enters from air to a fiber. The refractive index of air is 1.0 The fiber has refractive index of core equal to 1.5 and that of cladding equal to 1.48. Calculate the critical angle, the refractive index change, the acceptance angle and the numerical aperture.**

→ Here, $\mu_0 = 1$, $\mu_1 = 1.5$, $\mu_2 = 1.48$

We have, critical angle $C = \sin^{-1} \left(\frac{\mu_2}{\mu_1} \right) = ?$

refractive index change $\Delta = \frac{\mu_1^2 - \mu_2^2}{2\mu_1^2} = ?$

acceptance angle $i = \sin^{-1} \left\{ \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0} \right\} = ?$

$NA = \sqrt{\mu_1^2 - \mu_2^2} = ?$

- 5. An optical fiber has a numerical aperture of 0.2. The cladding has the r.i.1.48. Calculate the acceptance angle in water with r.i.4/3. Also calculate the critical angle at core cladding interface.**

→ Here, $NA = 0.2$, $\mu_2 = 1.48$, $\mu_0 = 4/3$

We have, acceptance angle $i = \sin^{-1} \left\{ \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0} \right\} = \sin^{-1} \left\{ \frac{NA}{\mu_0} \right\} = ?$

Again, $NA = \sqrt{\mu_1^2 - \mu_2^2}$

$$\therefore \mu_1 = ?$$

Now, critical angle $C = \sin^{-1} \left(\frac{\mu_2}{\mu_1} \right) = ?$

6. A glass clad fiber is made with a core glass of r.i.1.5 and the cladding is doped to give a fractional index change 5×10^{-4} . Determine a) cladding index b) critical internal reflection angle c) external critical acceptance angle and d) NA

→ Here, $\mu_1 = 1.5$, $\Delta = 5 \times 10^{-4}$

a) We have $\Delta = \frac{\mu_1^2 - \mu_2^2}{2\mu_1^2}$

$\therefore \mu_2 = ?$

b) Critical internal angle $C = \sin^{-1} \left(\frac{\mu_2}{\mu_1} \right) = ?$

c) External critical acceptance angle $i = \sin^{-1} \left\{ \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0} \right\} = ?$ (Take $\mu_0 = 1$)

d) $NA = \sqrt{\mu_1^2 - \mu_2^2} = ?$

7. Find the core diameter necessary for single mode operation at $450 \mu\text{m}$ in step index fiber with $n_1=1.48$ and $n_2=1.47$. What is NA and maximum acceptance angle of this fiber? Given V-number=2.405.

→ Here, Wavelength $\lambda = 450 \mu\text{m} = 450 \times 10^{-6} \text{m}$, $\mu_1 = 1.48$, $\mu_2 = 1.47$, V-number = 2.405

We have, $NA = \sqrt{\mu_1^2 - \mu_2^2} = ?$

maximum acceptance angle $i = \sin^{-1} \left\{ \sqrt{\mu_1^2 - \mu_2^2} \right\} = ?$

Again we have V-number $= \frac{2\pi a}{\lambda} \times NA$

or, $a = ?$

$\therefore$ diameter, $d = 2a = ?$

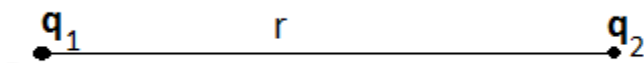
Electrostatics

Chapter: Electric Force and Field

Coulomb's law of electrostatics

Force of attraction or repulsion between two charges is

- directly proportional to the product of their charges and
- inversely proportional to the square of the distance between two charges



$$F \propto q_1 q_2 \quad \text{and} \quad F \propto \frac{1}{r^2}$$

On combining, we get

$$F \propto \frac{q_1 q_2}{r^2}$$

$$\text{or, } F = k \frac{q_1 q_2}{r^2}$$

Where $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$ in S.I. system and $k = 1$ in cgs system.

In vector form it can be written as

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r} \quad \text{Where } \hat{r} \text{ is a unit vector, } \hat{r} = \frac{\vec{r}}{|\vec{r}|}$$

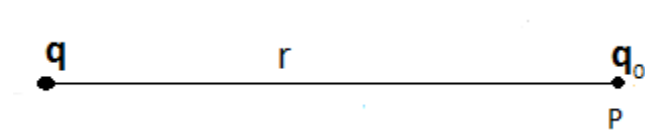
$$\text{Hence } \vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^3} \vec{r}$$

Electric Field

The space around a charge where another charge can experience a force due that charge is called electric field.

Electric Field Intensity

Electric Field Intensity at a point in side an electric field is defined as a force experienced by a unit positive test charge at that point. It is denoted by E. It is a vector quantity. It's unit is NC^{-1} or Vm^{-1} in S.I. system.



$$\text{We have} \quad F = \frac{1}{4\pi\epsilon_0} \frac{q q_0}{r^2}$$

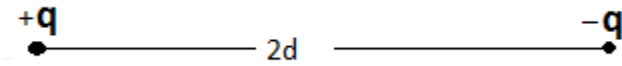
Where q_0 is a unit positive test charge.

$$\text{Now, } E = \frac{F}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Electric dipole

It is an arrangement of two equal and opposite charges separated by a certain distance so that net effect of electric field at a point due to the dipole is zero.

If two charges $+q$ and $-q$ are separated by a distance $2d$ as shown figure,



Then electric dipole moment is given by

$$P = q \times 2d$$

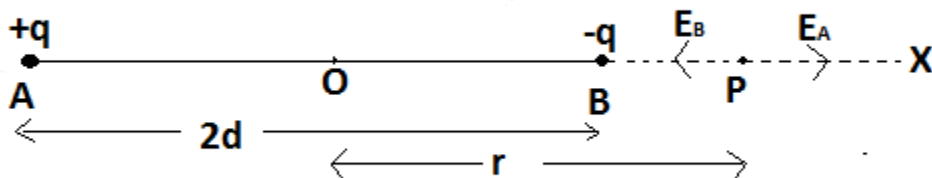
It is a vector quantity and its direction is always from negative charge to positive charge.

Its unit in SI system is C-m.

Electric Field Intensity due to an electric dipole

a) At a point on its axial line

Consider two equal and opposite charges $+q$ and $-q$ are located at points A and B are separated by a distance $2d$ as shown in figure. Suppose a point P lies on its axial line at a distance r from the center of the dipole.



Here electric field intensity at point P due to the charge $+q$ at A is given by

$$E_A = \frac{1}{4\pi\epsilon_0} \frac{q}{(r+d)^2} \quad (\text{Along PX})$$

Similarly, electric field intensity at point P due to the charge $-q$ at B is given by

$$E_B = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-d)^2} \quad (\text{Along PB})$$

Therefore, net electric field intensity at P due to the dipole is

$$E = E_B - E_A$$

$$\text{or, } E = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-d)^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{(r+d)^2}$$

$$\text{or, } E = \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{(r-d)^2} - \frac{1}{(r+d)^2} \right\}$$

$$\text{or, } E = \frac{q}{4\pi\epsilon_0} \left\{ \frac{(r+d)^2 - (r-d)^2}{(r^2 - d^2)^2} \right\}$$

$$\text{or, } E = \frac{q}{4\pi\epsilon_0} \left(\frac{r^2 + 2dr + d^2 - r^2 + 2dr - d^2}{(r^2 - d^2)^2} \right)$$

$$\text{or, } E = \frac{q}{4\pi\epsilon_0} \frac{4dr}{(r^2 - d^2)^2}$$

$$\text{or, } E = \frac{(q \times 2d)}{4\pi\epsilon_0} \frac{2r}{(r^2 - d^2)^2}$$

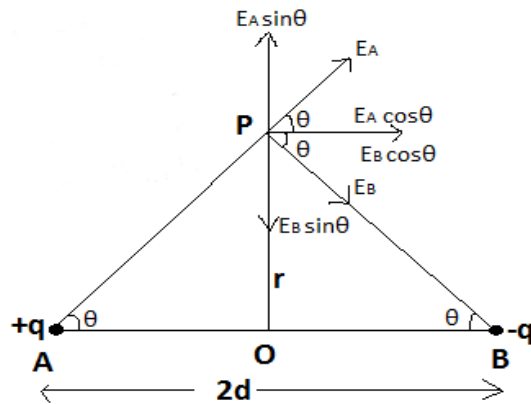
$$\therefore E = \frac{1}{4\pi\epsilon_0} \frac{2Pr}{(r^2 - d^2)^2} \quad \text{Where, } P = q \times 2d \text{ dipole moment}$$

For a very short dipole, $d \ll r$

$$\text{So } E = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$$

b) At a point on its equatorial line

Consider two equal and opposite charges $+q$ and $-q$ are located at points A and B are separated by a distance $2d$ as shown in figure. Suppose a point P lies on its equatorial line at a distance r from the center of the dipole.



$$\text{Here, } AP = BP = \sqrt{r^2 + d^2}$$

Now, electric field intensity at P due to a charge $+q$ at A is given by

$$E_A = \frac{1}{4\pi\epsilon_0} \frac{q}{(\sqrt{r^2 + d^2})^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2 + d^2}$$

Similarly, electric field intensity at p due to charge $-q$ at B is given by

$$E_B = \frac{1}{4\pi\epsilon_0} \frac{q}{(\sqrt{r^2 + d^2})^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2 + d^2}$$

Here, $E_A \sin \theta$ and $E_B \sin \theta$ are equal and opposite so they cancel each other.

$\therefore$ Net electric field intensity at p will be

$$E = E_A \cos \theta + E_B \cos \theta = 2E_A \cos \theta \quad (\text{Since, } E_A = E_B)$$

$$\text{or, } E = 2 \frac{1}{4\pi\epsilon_0} \frac{q}{r^2 + d^2} \frac{d}{\sqrt{r^2 + d^2}}$$

$$\therefore E = \frac{P}{4\pi\epsilon_0 (r^2 + d^2)^{3/2}}$$

Where, $P = q \times 2d$ dipole moment

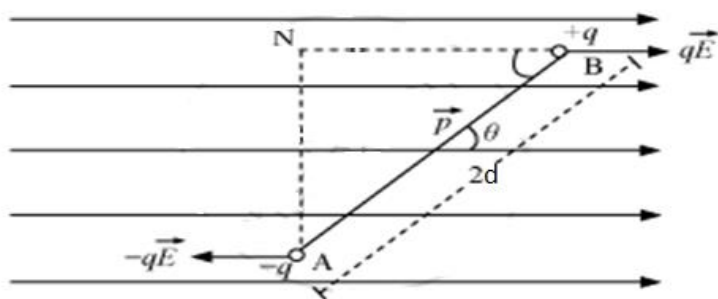
For a very short dipole, $d \ll r$

$$\text{So } E = \frac{1}{4\pi\epsilon_0} \frac{P}{r^3}$$

From above results we see that the electric field intensity at a point on the axial line is equal to twice that on the equatorial line for a very short dipole.

Torque and Potential Energy due to a Dipole in an electric field

Let a uniform electric field of strength E is represented by a number of line of force as shown in figure. Suppose an electric dipole AB having two equal and opposite charges $+q$ and $-q$ separated by a distance $2d$ is placed inside the electric field such that it makes an angle θ with the direction of E . Here the net force exerted on the dipole due to the field is zero. But these forces exert a net torque on the dipole about its center of mass.



We have, Torque $\tau = \text{Force} \times \text{perpendicular distance}$

$$\text{or, } \tau = qE \times 2d \sin \theta = PE \sin \theta$$

In vector form, $\vec{\tau} = \vec{P} \times \vec{E}$

When $\theta = 0^\circ$, $\tau_{\min} = 0$

When $\theta = 90^\circ$, $\tau_{\max} = PE$

When the dipole changes direction in a field, the electric torque does work on it.

The small work done dw by the torque τ for small displacement $d\theta$ is

$$dw = \tau d\theta$$

Since the torque is in the direction of decreasing θ , so $\tau = -PE \sin \theta$

$$\therefore dw = -PE \sin \theta d\theta$$

For finite displacement from θ_1 to θ_2 , total work done is

$$w = \int_{\theta_1}^{\theta_2} dw = \int_{\theta_1}^{\theta_2} -PE \sin \theta d\theta = -PE \int_{\theta_1}^{\theta_2} \sin \theta d\theta = PE(\cos \theta_2 - \cos \theta_1)$$

For $\theta_1 = \theta$ and $\theta_2 = 90^\circ$,

$$w = -PE \cos \theta$$

This work done is stored as potential energy (U) due to dipole.

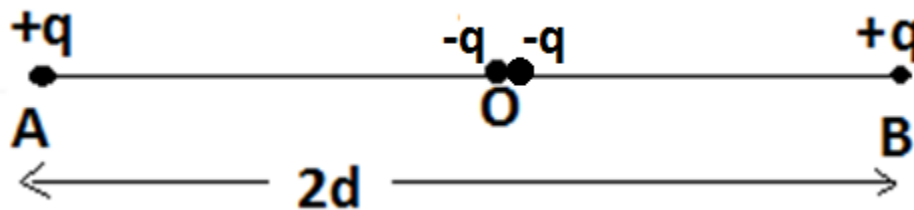
$$\therefore U = -PE \cos \theta$$

When $\theta = 0^\circ$, $U_{\max} = -PE$

When $\theta = 90^\circ$, $U_{\min} = 0$

Quadruple

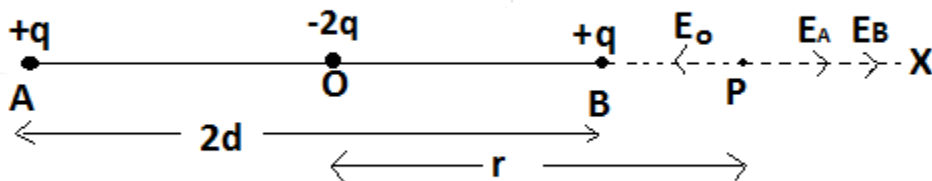
It is an arrangement of four equal and opposite charges or it is an arrangement of two dipoles.



The electric quadrupole moment is given by $Q = q \times 2d^2$. It is tensor quantity

Electric field due to a linear quadrupole

Let us consider an electric quadrupole separated by a distance $2d$. Consider a point at P at a distance r from the center of AB i.e. $OP=r$



Now electric field at P due to the charge $+q$ at A is

$$E_A = \frac{1}{4\pi\epsilon_0} \frac{q}{(r+d)^2} \text{ along PX}$$

Similarly, electric field at P due to +q at B is

$$E_B = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-d)^2} \text{ along PX}$$

And electric field at P due to charge -2q at O is

$$E_O = \frac{1}{4\pi\epsilon_0} \frac{2q}{r^2} \text{ along PO}$$

So total electric field at P due to the quadruple is

$$E = E_A + E_B - E_O$$

$$\begin{aligned} E &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(r+d)^2} + \frac{1}{(r-d)^2} - \frac{2}{r^2} \right] \\ &= \frac{q}{4\pi\epsilon_0} \left[\frac{(r-d)^2 + (r+d)^2}{(r^2 - d^2)^2} - \frac{2}{r^2} \right] \\ &= \frac{q}{4\pi\epsilon_0} \left[\frac{r^2(r^2 - 2dr + d^2 + r^2 + 2dr + d^2) - 2(r^4 - 2r^2d^2 + d^4)}{(r^2 - d^2)^2 r^2} \right] \\ &= \frac{q}{4\pi\epsilon_0} \left\{ \frac{2r^4 + 2d^2r^2 - 2r^4 + 4d^2r^2 - 2d^4}{(r^2 - d^2)^2 r^2} \right\} \\ &= \frac{q}{4\pi\epsilon_0} \left\{ \frac{6d^2r^2 - 2d^4}{(r^2 - d^2)^2 r^2} \right\} \\ &= \frac{q \times 2d^2}{4\pi\epsilon_0} \left\{ \frac{3r^2 - d^2}{(r^2 - d^2)^2 r^2} \right\} \\ \therefore E &= \frac{Q}{4\pi\epsilon_0} \left\{ \frac{3r^2 - d^2}{(r^2 - d^2)^2 r^2} \right\} \end{aligned}$$

Where, $Q = q \times 2d^2$ is quadruple moment.

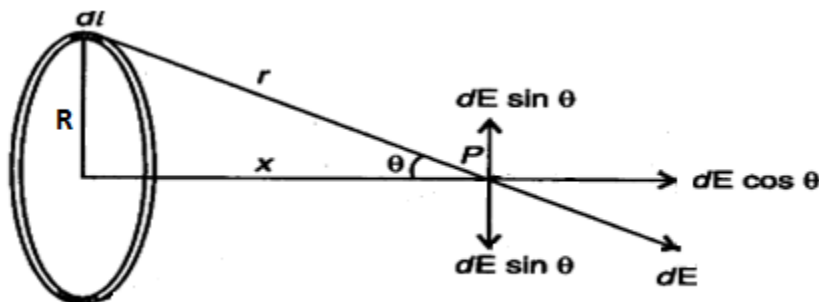
If $d \ll r$, then

$$E = \frac{3Q}{4\pi\epsilon_0 r^4}$$

This shows that electric field due to a quadruple varies inversely with fourth power of the distance from the center of quadruple.

Electric Field due to a charged ring

Let a circular ring of radius R and charge $+q$ is distributed uniformly on it with linear charge density λ . Consider a point P at a distance x from the center of ring on its axis.



$$\text{Here, } r = \sqrt{x^2 + R^2}$$

Now electric field at P due to the charge $dq = \lambda dl$ is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{(\sqrt{x^2 + R^2})^2}$$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda dl}{x^2 + R^2}$$

Here, $dE \sin \theta$ components of electric field cancel each other and only the effective components are $dE \cos \theta$.

$\therefore$ Net electric field intensity at P is

$$E = \int_0^{2\pi R} dE \cos \theta$$

$$E = \int_0^{2\pi R} \frac{1}{4\pi\epsilon_0} \frac{\lambda dl}{x^2 + R^2} \cos \theta$$

$$\text{or, } E = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi R} \frac{\lambda dl}{x^2 + R^2} \frac{x}{\sqrt{x^2 + R^2}}$$

$$\text{or, } E = \frac{\lambda}{4\pi\epsilon_0} \frac{x}{(x^2 + R^2)^{\frac{3}{2}}} \int_0^{2\pi R} dl$$

$$\text{or, } E = \frac{\lambda x}{4\pi\epsilon_0 (x^2 + R^2)^{\frac{3}{2}}} (2\pi R - 0)$$

$$\text{or, } E = \frac{(\lambda \times 2\pi R)x}{4\pi\epsilon_o (x^2 + R^2)^{\frac{3}{2}}}$$

$$\therefore E = \frac{qx}{4\pi\epsilon_o (x^2 + R^2)^{\frac{3}{2}}}$$

This is the electric field intensity due to the charged ring.

Note 1: Here electric field will be maximum when $\frac{dE}{dx} = 0$

$$\text{or, } \frac{d}{dx} \left\{ \frac{x}{(x^2 + R^2)^{\frac{3}{2}}} \right\} = 0$$

$$\text{or, } \frac{1}{(x^2 + R^2)^{\frac{3}{2}}} - \frac{3x^2}{(x^2 + R^2)^{\frac{5}{2}}} = 0$$

$$\text{or, } \frac{x^2 + R^2 - 3x^2}{(x^2 + R^2)^{\frac{5}{2}}} = 0$$

$$\text{or, } R^2 - 2x^2 = 0$$

$$\text{or, } x = \frac{R}{\sqrt{2}}$$

$$\therefore E \text{ is maximum at } x = \frac{R}{\sqrt{2}}$$

Note2: If $x \ll R$ then $E = \frac{qx}{4\pi\epsilon_o R^3}$

Now the force acting on a charge $-q'$ located near to the center of ring is

$$F = -q'E$$

$$\therefore F = -q' \frac{qx}{4\pi\epsilon_o R^3}$$

$$\text{or, } ma = -\left(q' \frac{q}{4\pi\epsilon_o R^3}\right) x$$

$$\text{or, } a = -\left(\frac{qq'}{4\pi\epsilon_o mR^3}\right) x$$

$$\text{or, } a = -\omega^2 x$$

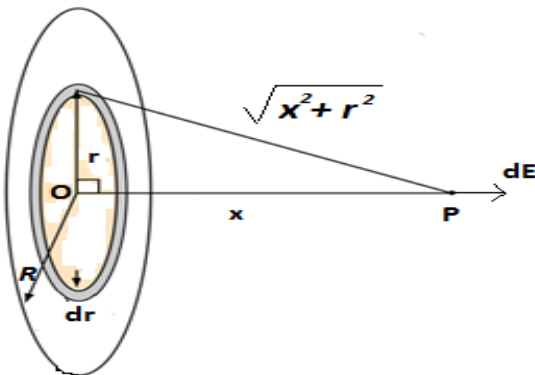
This shows that charge $-q'$ describes SHM whose angular frequency is

$$\omega = \sqrt{\frac{qq'}{4\pi\epsilon_0 mR^3}}$$

$$\therefore \text{frequency, } f = \frac{1}{2\pi} \sqrt{\frac{qq'}{4\pi\epsilon_0 mR^3}}$$

Electric field intensity due to a charged disc

Consider a charged disc of radius R with uniform surface charge density σ . Consider a point P at a distance x from the center O of the disc on its axis at which electric field intensity is to be determined.



As a disc consists of large number of circular rings, the electric field at P due to a circular ring of radius r and thickness dr is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dqx}{(x^2 + r^2)^{\frac{3}{2}}} = \frac{1}{4\pi\epsilon_0} \frac{(\sigma \times 2\pi r dr)x}{(x^2 + r^2)^{\frac{3}{2}}} = \frac{\sigma x}{4\epsilon_0} \frac{2r dr}{(x^2 + r^2)^{\frac{3}{2}}} \quad (\text{Since } dq = \sigma dA = \sigma 2\pi r dr)$$

Hence total electric field at point P due to the disc is

$$E = \int dE = \int_0^R \frac{\sigma x}{4\epsilon_0} \frac{2r dr}{(x^2 + r^2)^{\frac{3}{2}}} = \frac{\sigma x}{4\epsilon_0} \int_0^R \frac{2r dr}{(x^2 + r^2)^{\frac{3}{2}}}$$

$$\text{Put } x^2 + r^2 = y$$

$$\text{Then } 2r dr = dy$$

$$\therefore E = \frac{\sigma x}{4\epsilon_0} \int_0^R \frac{dy}{y^{\frac{3}{2}}} = \frac{\sigma x}{4\epsilon_0} \int_0^R y^{-\frac{3}{2}} dy = \frac{\sigma x}{4\epsilon_0} \left[\frac{y^{-1/2}}{-1/2} \right]_0^R = -\frac{\sigma x}{2\epsilon_0} \left[(x^2 + r^2)^{-1/2} \right]_0^R$$

$$\text{or, } E = \frac{\sigma x}{2\epsilon_0} \left[x^{-1} - (x^2 + R^2)^{-1/2} \right] = \frac{\sigma x}{2\epsilon_0} \left[\frac{1}{x} - \frac{1}{\sqrt{x^2 + R^2}} \right]$$

$$\therefore E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + R^2}} \right]$$

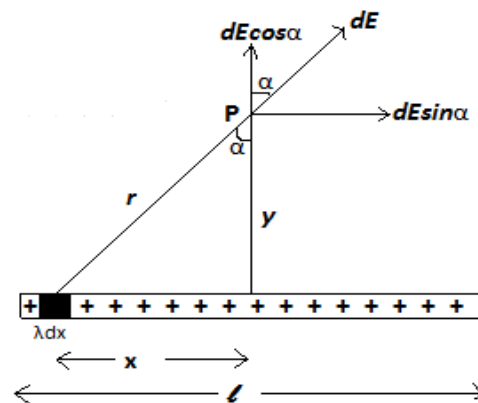
This is the electric field due to an insulating charge disc.

Note: If $R \rightarrow \infty$ with x finite, then $E = \frac{\sigma}{2\epsilon_0}$

This is the electric field produced by infinite sheet of uniform charge.

Electric Field due to a line charge

Consider a long thin rod with uniformly distributed charges q having linear charge density λ .



To find the electric field intensity due to the rod at point P located at perpendicular distance y from the middle of the rod, we consider an element of length dx at a distance x from its center.

Now, electric field intensity at P due to charge element $dq = \lambda dx$ is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{x^2 + y^2}$$

Here, $dE \sin \alpha$ components of electric field cancel each other and the effective components are $dE \cos \alpha$.

Now the resultant electric field intensity is

$$E = 2 \int dE \cos \alpha = 2 \int_0^{l/2} \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{x^2 + y^2} \frac{y}{\sqrt{x^2 + y^2}} = \frac{\lambda y}{2\pi\epsilon_0} \int_0^{l/2} \frac{dx}{(x^2 + y^2)^{3/2}}$$

Putting, $x = y \tan \theta$

We get $dx = y \sec^2 \theta d\theta$

$$\therefore E = \frac{\lambda y}{2\pi\epsilon_0} \int_0^{\frac{l}{2}} \frac{y \sec^2 \theta d\theta}{y^3 \sec^3 \theta}$$

$$= \frac{\lambda y}{2\pi\epsilon_0} \frac{1}{y^2} \int_0^{\frac{l}{2}} \cos \theta d\theta$$

$$= \frac{\lambda y}{2\pi\epsilon_0} \frac{1}{y^2} [\sin \theta]_0^{\frac{l}{2}}$$

$$= \frac{\lambda y}{2\pi\epsilon_0} \frac{1}{y^2} \left[\frac{x}{\sqrt{x^2 + y^2}} \right]_0^{l/2}$$

$$\therefore E = \frac{\lambda}{2\pi\epsilon_0 y} \frac{\frac{l}{2}}{\sqrt{(l/2)^2 + y^2}}$$

$$\text{or, } E = \frac{\lambda}{2\pi\epsilon_0 y} \frac{l}{\sqrt{l^2 + 4y^2}}$$

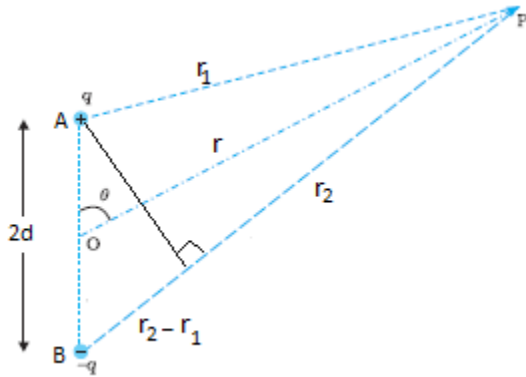
This is the electric field intensity due to finite line charge.

For infinite line of charge, $l \rightarrow \infty$ so

$$E = \frac{\lambda}{2\pi\epsilon_0 y} \frac{l}{\sqrt{l^2 + 4y^2}} = \frac{\lambda}{2\pi\epsilon_0 y} \frac{1}{\sqrt{1 + \frac{4y^2}{l^2}}} = \frac{\lambda}{2\pi\epsilon_0 y}$$

Electric Potential Due to a Dipole at any point

Consider an electric dipole AB with separation $2d$. Consider a point P at a distance r from the center O of dipole which is at a distance r_1 from $+q$ and r_2 from $-q$. Let $\angle POA = \theta$.



Now potential at P due to +q at A is

$$V_A = \frac{1}{4\pi\epsilon_0} \frac{q}{r_1}$$

And potential at P due to -q at B is

$$V_B = -\frac{1}{4\pi\epsilon_0} \frac{q}{r_2}$$

∴ Net potential at P is

$$V = V_A + V_B = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = \frac{q}{4\pi\epsilon_0} \frac{r_2 - r_1}{r_1 r_2}$$

For $r \gg d$, $r_2 - r_1 \approx 2d \cos \theta$ and $r_1 r_2 \approx r^2$

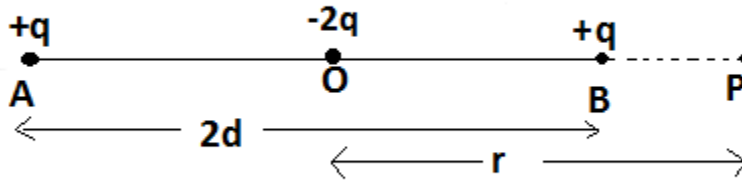
$$\therefore V = \frac{q}{4\pi\epsilon_0} \frac{2d \cos \theta}{r^2}$$

$$\text{or, } V = \frac{P \cos \theta}{4\pi\epsilon_0 r^2}$$

On the axis, $\theta = 0^\circ$ so $V = \frac{P}{4\pi\epsilon_0 r^2}$ and at equatorial point $\theta = 90^\circ$ so $V = 0$

Electric Potential due to a Quadruple

Consider a linear quadruple of length 2d. Consider a point P at a distance r from the center O of the quadruple at which electric potential is to be determined.



Now, electric potential at P due to charge +q at A is $V_A = \frac{1}{4\pi\epsilon_0} \frac{q}{r+d}$

Electric potential at P due to charge -2q at O is $V_O = -\frac{1}{4\pi\epsilon_0} \frac{2q}{r}$

And electric potential at P due to charge +q at B is $V_B = \frac{1}{4\pi\epsilon_0} \frac{q}{r-d}$

$\therefore$ Net potential at P is $V = V_A + V_O + V_B$

$$\text{or, } V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r+d} - \frac{2}{r} + \frac{1}{r-d} \right] = \frac{q}{4\pi\epsilon_0} \left[\frac{r^2 - rd - 2r^2 + 2d^2 + r^2 + rd}{r(r+d)(r-d)} \right]$$

$$\text{or, } V = \frac{q}{4\pi\epsilon_0} \frac{2d^2}{r(r^2 - d^2)}$$

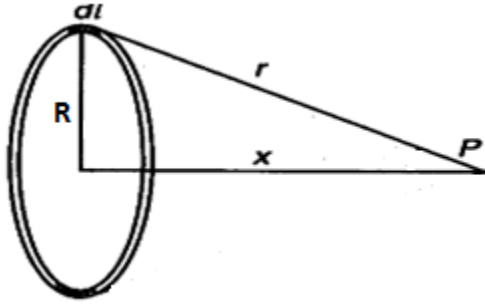
$$\therefore V = \frac{Q}{4\pi\epsilon_0 r(r^2 - d^2)}$$

Where, $Q = q \times 2d^2$ is electric quadrupole moment.

$$\text{For } r \gg d, V = \frac{Q}{4\pi\epsilon_0 r^3}$$

Electric potential due to ring of charge

Consider a charged ring of radius R having linear charge density λ . Consider a point P at a distance x from the center O of the ring at which electric potential is to be determined.



Electric potential at P due to elemental segment dl which contains charge dq is

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dl}{\sqrt{x^2 + R^2}}$$

$\therefore$ Net potential at P is

$$V = \int dV = \int_0^{2\pi R} \frac{1}{4\pi\epsilon_0} \frac{\lambda dl}{\sqrt{x^2 + R^2}} = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{\sqrt{x^2 + R^2}} \int_0^{2\pi R} dl = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{\sqrt{x^2 + R^2}} \times 2\pi R$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{x^2 + R^2}} \quad (\text{since, } q = \lambda \times 2\pi R)$$

Electric field and Potential Gradient

The electric field intensity at a point is equal to negative gradient of potential.

$$\text{i.e. } E = -\frac{\partial V}{\partial r}$$

$$\therefore V = -\int E dr$$

Gauss's law

It states that total electric flux through a closed surface enclosing a charge is equal to $\frac{1}{\epsilon_0}$ times the charge enclosed by the closed surface.

$$\text{i.e. } \oint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} q$$

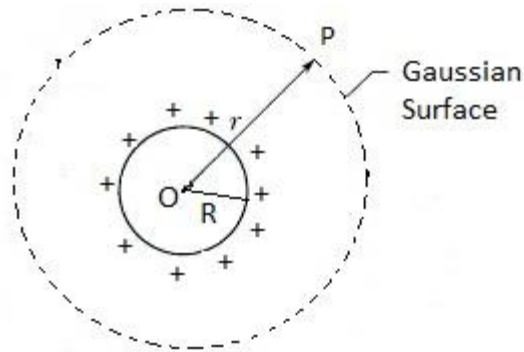
Applications of Gauss's Law

1. Electric field due to a charged conducting sphere

Consider a conducting sphere of radius R having charge q.

i) At a point outside the sphere

Consider a point P at a distance r from the center of sphere. To find the electric field at P, we draw a spherical Gaussian surface of radius r that encloses the charge q as shown in figure.



Using Gauss Law,

$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$

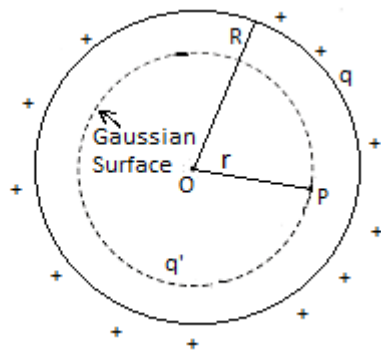
$$\text{or, } E \times 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$\therefore E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$\text{On the surface, } r=R \text{ so } E = \frac{q}{4\pi\epsilon_0 R^2}$$

ii) At a point inside the sphere

Consider a point P at a distance r from the center of sphere. To find the electric field at P, we draw a spherical Gaussian surface of radius r that encloses the charge q' as shown in figure.



Using Gauss Law,

$$\oint \vec{E} \cdot d\vec{a} = \frac{q'}{\epsilon_0}$$

$$\text{or, } E \times 4\pi r^2 = \frac{0}{\epsilon_0} \text{ (Since there are no charges inside the conducting sphere)}$$

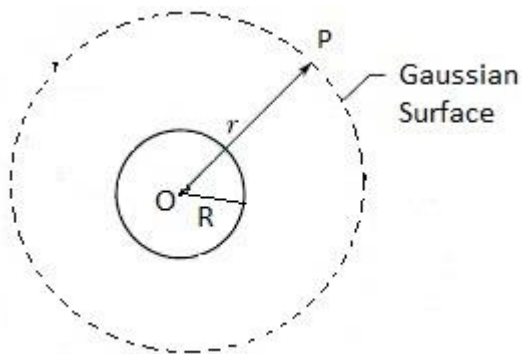
$$\therefore E = 0$$

2. Electric field due to a Non-Conducting Spherically Symmetric Charge Distribution

Consider a non-conducting sphere of radius R having uniformly distributed charge q.

i) At a point outside the sphere

Consider a point P at a distance r from the center of sphere. To find the electric field at P, we draw a spherical Gaussian surface of radius r that encloses the charge q as shown in figure.



Using Gauss Law,

$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$

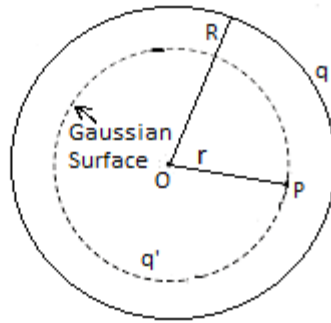
$$\text{or, } E \times 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$\therefore E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$\text{On the surface, } r=R \text{ so } E = \frac{q}{4\pi\epsilon_0 R^2}$$

ii) At a point inside the sphere

Consider a point P at a distance r from the center of sphere. To find the electric field at P, we draw a spherical Gaussian surface of radius r that encloses the charge q' as shown in figure.



Using Gauss Law,

$$\oint \vec{E} \cdot d\vec{a} = \frac{q'}{\epsilon_0}$$

$$\text{or, } E \times 4\pi r^2 = \frac{q'}{\epsilon_0}$$

$$\therefore E = \frac{q'}{4\pi\epsilon_0 r^2} \dots\dots\dots (i)$$

Since, volume charge density is same throughout the sphere, so

$$\rho = \frac{q}{\frac{4}{3}\pi R^3} = \frac{q'}{\frac{4}{3}\pi r^3}$$

$$\therefore q' = \frac{r^3}{R^3} q$$

Hence from equation (i),

$$E = \frac{q'}{4\pi\epsilon_0 r^2} = \frac{1}{4\pi\epsilon_0 r^2} \times \frac{r^3}{R^3} q$$

$$\text{or, } E = \frac{1}{4\pi\epsilon_0} \frac{qr}{R^3}$$

At the center, $r = 0$ so $E = 0$ and on the surface, $r=R$ so $E = \frac{q}{4\pi\epsilon_0 R^2}$

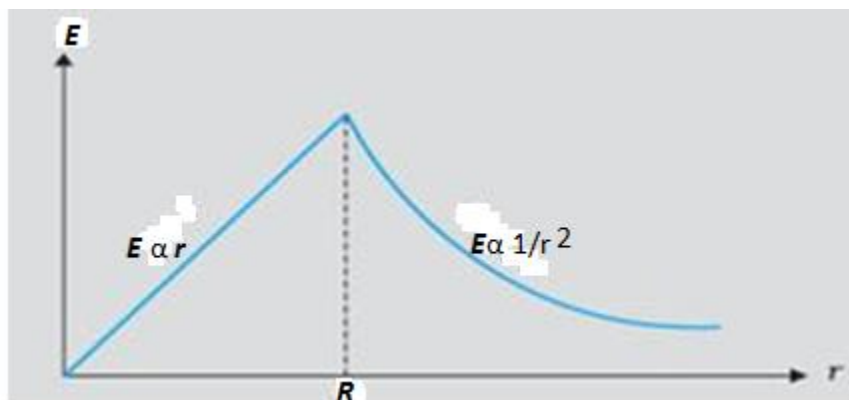


Fig: Variation of E with r

Numerical

1. Considering charge of one electron be equal to $-1.6 \times 10^{-19} \text{C}$, how many electrons must be removed from the body to give it a positive charge of $8 \times 10^{-19} \text{C}$?
 → Here, electronic charge, $e = 1.6 \times 10^{-19} \text{C}$, Total charge, $q = 8 \times 10^{-19} \text{C}$, no. of electrons=?
 We have, $q = ne$
 $\therefore n = \frac{q}{e} = ?$

2. If an atom of carbon has six electrons, what is the charge on the carbon nucleus?
 → Here, We have, $q = ne = ?$

3. If a copper coin has mass 3.11g, what is the number of atoms and total charge on the nucleus of the atoms in the coin? ($Z=29, M=63.5 \text{g/mol}$ for copper and $N_A=6.02 \times 10^{23} \text{ atom/mol}$)

→ Here, mass of copper coin, $m=3.11 \text{g}$, atomic number, $z=29$, molar mass, $M=63.5 \text{g}$,

Avogadro's number, $N_A = 6.02 \times 10^{23}$

We know, 63.5g of copper contains $6.02 \times 10^{23} \text{ atoms}$.

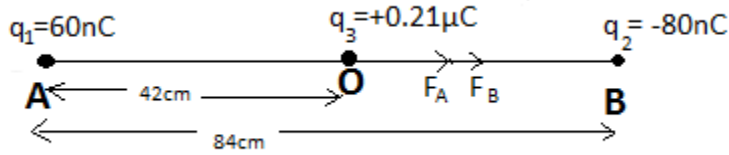
$$\therefore 3.11 \text{g of copper contains } \frac{6.02 \times 10^{23}}{63.5} \times 3.11 \text{ atoms}$$

$$\text{Hence the no of atoms, } N = \frac{6.02 \times 10^{23}}{63.5} \times 3.11 = ?$$

Since each atom contains 29 protons so total charge $q = ZNe = ?$

4. A charge of 60nC is 84cm from another charge of -80nC. What forces in air will these charges exert on a charge of +0.21μC placed halfway between them.

→ Here, $q_1 = 60\text{nC} = 60 \times 10^{-9}\text{C}$, $q_2 = -80\text{nC} = -80 \times 10^{-9}\text{C}$, $q_3 = +0.21\mu\text{C} = +0.21 \times 10^{-6}\text{C}$



Force on q_3 due to charge q_1 placed at A is $F_A = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{(0.42)^2}$ along OB

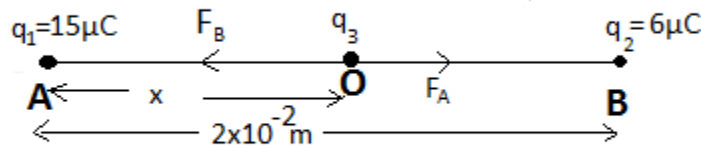
And force on q_3 due to charge q_2 placed at B is $F_B = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{(0.42)^2}$ along OB

∴ Net force on q_3 , $F = F_A + F_B = ?$

5. At what distance the resultant force will be zero if two charges 15 μC and 6 μC lie along the x-axis at a distance 2cm.

→ Here, $q_1 = 15\mu\text{C} = 15 \times 10^{-6}\text{C}$, $q_2 = 6\mu\text{C} = 6 \times 10^{-6}\text{C}$

Let the resultant force will be zero at a distance x m from 15μC.



Force on q_3 due to charge q_1 placed at A is $F_A = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{x^2}$ along OB

And force on q_3 due to charge q_2 placed at B is $F_B = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{(2 \times 10^{-2} - x)^2}$ along OA

By question, $F_A = F_B$

$$\text{or, } \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{(2 \times 10^{-2} - x)^2}$$

$$\text{or, } \frac{(2 \times 10^{-2} - x)^2}{x^2} = \frac{q_2}{q_1} = \frac{6}{15}$$

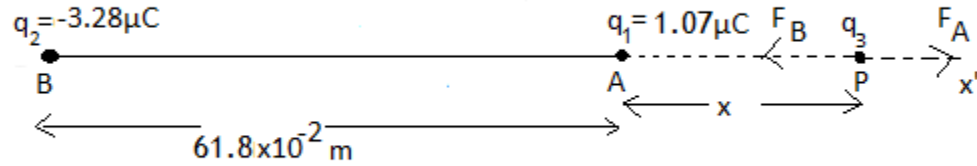
$$\text{or, } \frac{2 \times 10^{-2} - x}{x} = \sqrt{\frac{6}{15}}$$

∴ $x = ?$

6. Two fixed charges 1.07μC and -3.28 μC are 61.8cm apart. Where may a third charge be located so that no net force acts on it?

→ Here, $q_1 = 1.07\mu\text{C} = 1.07 \times 10^{-6}\text{C}$, $q_2 = -3.28\mu\text{C} = -3.28 \times 10^{-6}\text{C}$

Let the resultant force will be zero at a distance x m from $1.07\mu\text{C}$.



Force on q_3 due to charge q_1 placed at A is $F_A = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{x^2}$ along PX'

And force on q_3 due to charge q_2 placed at B is $F_B = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{(61.8 \times 10^{-2} + x)^2}$ along PA

By question, $F_A = F_B$

$$\text{or, } \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{(61.8 \times 10^{-2} + x)^2}$$

$$\text{or, } \frac{(61.8 \times 10^{-2} + x)^2}{x^2} = \frac{q_2}{q_1} = \frac{3.28}{1.07}$$

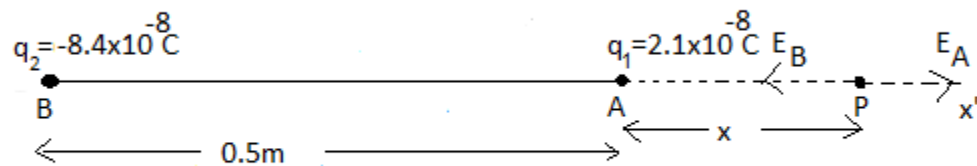
$$\text{or, } \frac{61.8 \times 10^{-2} + x}{x} = \sqrt{\frac{3.28}{1.07}}$$

$\therefore x = ?$

7. Two charges $2.1 \times 10^{-8}\text{C}$ and $-8.4 \times 10^{-8}\text{C}$ are placed at a separation of 50cm. Find the point along the straight line passing through the two charges at which electric field is zero.

→ Here, $q_1 = 2.1 \times 10^{-8}\text{C}$, $q_2 = -8.4 \times 10^{-8}\text{C}$

Let the resultant electric field will be zero at a distance x m from $2.1 \times 10^{-8}\text{C}$.



Electric field at P due to charge q_1 placed at A is $E_A = \frac{1}{4\pi\epsilon_0} \frac{q_1}{x^2}$ along PX'

And Electric field at P due to charge q_2 placed at B is $E_B = \frac{1}{4\pi\epsilon_0} \frac{q_2}{(0.5+x)^2}$ along PA

By question, $E_A = E_B$

$$\text{or, } \frac{1}{4\pi\epsilon_0} \frac{q_1}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{q_2}{(0.5+x)^2}$$

$$\text{or, } \frac{(0.5+x)^2}{x^2} = \frac{q_2}{q_1} = \frac{8.4}{2.1}$$

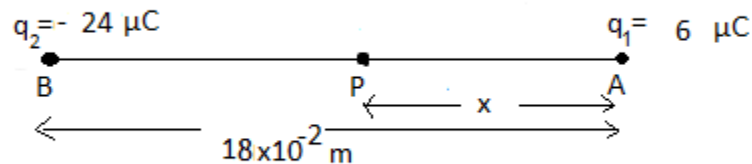
$$\text{or, } \frac{0.5+x}{x} = \sqrt{\frac{8.4}{2.1}}$$

$\therefore x = ?$

8. Two point charges $6 \mu\text{C}$ and $-24 \mu\text{C}$ are 18cm apart. Locate the position of zero potential on the line joining between them.

→ Here, $q_1 = 6\mu\text{C} = 6 \times 10^{-6}\text{C}$, $q_2 = -24\mu\text{C} = -24 \times 10^{-6}\text{C}$

Let the net potential will be zero at a distance $x \text{ m}$ from $6\mu\text{C}$.



$$\text{Electric potential at P due to charge placed at A, } V_A = \frac{1}{4\pi\epsilon_0} \frac{q_1}{x} = \frac{1}{4\pi\epsilon_0} \frac{6 \times 10^{-6}}{x}$$

$$\text{And Electric potential at P due to charge placed at B, } V_B = \frac{1}{4\pi\epsilon_0} \frac{q_2}{(18 \times 10^{-2} - x)} = -\frac{1}{4\pi\epsilon_0} \frac{24 \times 10^{-6}}{(18 \times 10^{-2} - x)}$$

By question, net potential, $V = V_A + V_B = 0$

$$\text{or, } \frac{1}{4\pi\epsilon_0} \frac{6 \times 10^{-6}}{x} - \frac{1}{4\pi\epsilon_0} \frac{24 \times 10^{-6}}{(18 \times 10^{-2} - x)} = 0$$

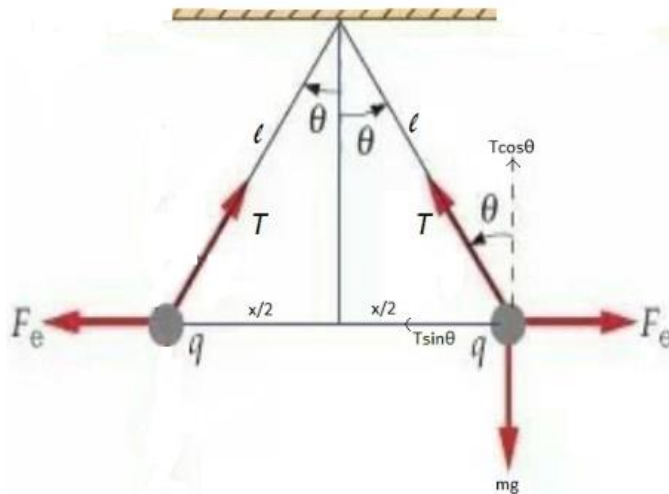
$$\text{or, } \frac{1}{4\pi\epsilon_0} \frac{6 \times 10^{-6}}{x} = \frac{1}{4\pi\epsilon_0} \frac{24 \times 10^{-6}}{(18 \times 10^{-2} - x)}$$

$\therefore x = ?$

9. Two similar balls of mass m are hung from silk threads of length l m and carry similar charges

q. Prove that the separation of the balls is $x = \sqrt[3]{\frac{q^2 l}{2\pi\epsilon_0 m g}}$.

→ Here,



Electrostatic force between two charges is $F_e = \frac{1}{4\pi\epsilon_0} \frac{q^2}{x^2}$

We have, from figure, $T \sin \theta = F_e$ (i)

And $T \cos \theta = mg$ (ii)

Dividing equation (i) by (ii) we get

$$\tan \theta = \frac{F_e}{mg}$$

For small angle, $\tan \theta \approx \sin \theta = \frac{x}{2l}$

$$\therefore \frac{x}{2l} = \frac{F_e}{mg}$$

$$\text{or, } \frac{x}{2l} = \frac{1}{mg} \frac{1}{4\pi\epsilon_0} \frac{q^2}{x^2}$$

$$\text{or, } x = \sqrt[3]{\frac{q^2 l}{2\pi\epsilon_0 mg}}$$

- 10. An electric dipole consists of charges +2e and -2e separated by 0.78nm. It is in electric field of strength $3.4 \times 10^6 \text{ N/C}$. Calculate the magnitude of the torque on the dipole when the dipole moment is (a) parallel, (b) at right angle and (c) opposite to the electric field.**

→ Here, charge, $q = 2e$, separation between two charges, $2d = 0.78 \text{ nm} = 0.78 \times 10^{-9} \text{ m}$,

Electric field intensity, $E = 3.4 \times 10^6 \text{ N/C}$

We have, torque due to a dipole is $\tau = PE \sin \theta$

Where dipole moment

$$P = q \times 2d = 2e \times 2d = 2 \times 1.6 \times 10^{-19} \times 0.78 \times 10^{-9} = \dots \dots \dots \text{C} - \text{m}$$

a) When the dipole moment is parallel to electric field, $\theta = 0^\circ$ so $\tau = PE \sin 0^\circ = 0$

b) When the dipole moment is at right angle to electric field, $\theta = 90^\circ$ so $\tau = PE \sin 90^\circ = ?$

c) When the dipole moment is opposite to electric field, $\theta = 180^\circ$ so $\tau = PE \sin 180^\circ = 0$

11. How much work is required to turn an electric dipole 180° in a uniform electric field of magnitude 46N/C if dipole moment is $3.02 \times 10^{-25}\text{Cm}$ and the initial angle is 64° ?

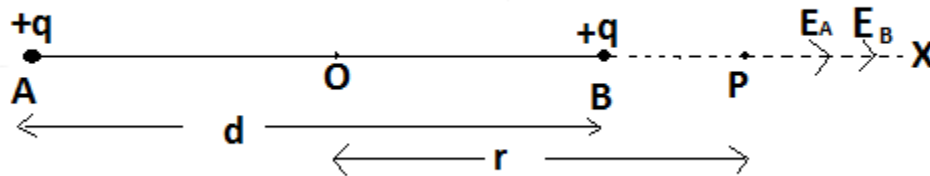
→ Here, Electric field intensity, $E=46\text{N/C}$, dipole moment $P= 3.02 \times 10^{-25}\text{C-m}$

We have Work done, $W = -PE\cos\theta$

∴ Required workdone to turn the dipole moment is $= -PE\cos 180^\circ - (-PE\cos 64^\circ) = ?$

12. For the system of two positive charges separated by a distance d , find an expression for an electric field along its axial line at a distance r and prove that $E = \frac{1}{4\pi\epsilon_0} \frac{2q}{r^2}$ for $r \gg d$.

→ Here,



Here electric field intensity at point P due to the charge $+q$ at A, $E_A = \frac{1}{4\pi\epsilon_0} \frac{q}{(r+\frac{d}{2})^2}$ (Along PX)

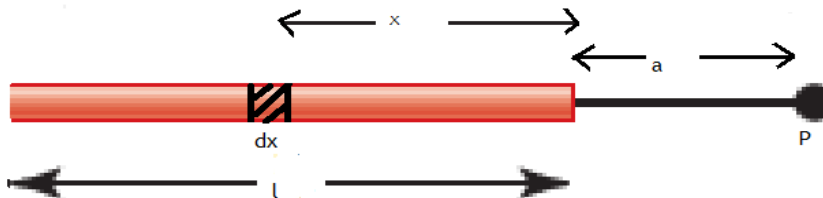
And electric field intensity at point P due to the charge $+q$ at B, $E_B = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-\frac{d}{2})^2}$ (Along PX)

$$\therefore \text{Net electric field intensity at P is } E = E_A + E_B = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(r+\frac{d}{2})^2} + \frac{1}{(r-\frac{d}{2})^2} \right] = \frac{q}{4\pi\epsilon_0} \left[\frac{2(r^2 + \frac{d^2}{4})}{(r^2 - \frac{d^2}{4})^2} \right]$$

$$\text{For } r \gg d, E = \frac{1}{4\pi\epsilon_0} \frac{2q}{r^2}$$

13. Calculate the electric field due to a uniformly charged rod of length l at a point along its long axis at a distance a from its nearest end.

→ Here,



The electric field intensity at P due to a small charge element $dq = \lambda dx$ is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{(x+a)^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{(x+a)^2}$$

∴ Net electric field at P is

$$E = \int dE = \int_0^l \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{(x+a)^2} = \frac{\lambda}{4\pi\epsilon_0} \int_0^l \frac{dx}{(x+a)^2} = \frac{\lambda}{4\pi\epsilon_0} \left[-\frac{1}{x+a} \right]_0^l = \frac{\lambda}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{l+a} \right] = \frac{\lambda l}{4\pi\epsilon_0 a(l+a)} = \frac{q}{4\pi\epsilon_0 a(l+a)}$$

14. **At what distance along the central axis of a uniform charged plastic disk of radius a is the electric field strength equal to half the value of the field at the center of the disk.**

→ Here, we have the electric field due to a charged disc of radius a at a distance x from its center is

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + a^2}} \right]$$

$$\text{At the center, } E_0 = \frac{\sigma}{2\epsilon_0}$$

$$\text{By question, } E = \frac{1}{2} E_0$$

$$\text{or, } \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + a^2}} \right] = \frac{1}{2} \frac{\sigma}{2\epsilon_0}$$

$$\text{or, } \frac{x}{\sqrt{x^2 + a^2}} = \frac{1}{2}$$

$$\text{or, } \frac{x^2}{x^2 + a^2} = \frac{1}{4}$$

$$\text{or, } 4x^2 = x^2 + a^2$$

$$\text{or, } x = \frac{a}{\sqrt{3}}$$

This is the required distance.

15. If a disk of radius 2.5cm had a surface charge density $5.5\mu\text{Cm}^{-2}$ on its upper surface, what is the electric field (i) at a point on the central axis at a distance 12cm from the disk and (ii) at the surface of the disk.

→ Here, radius of charged disk, $a=2.5\text{cm}=0.025\text{m}$, surface charge density, $\sigma = 5.5\mu\text{Cm}^{-2} = 5.5 \times 10^{-6}\text{Cm}^{-2}$

a) We have the electric field due to a charged disc of radius a at a distance x from its center is

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + a^2}} \right] = ?$$

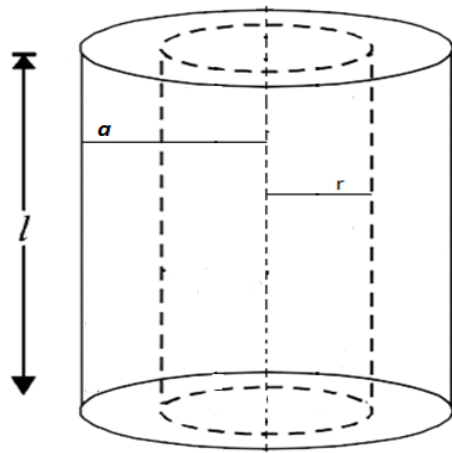
b) at the surface, $x=a$ so

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{a}{\sqrt{a^2 + a^2}} \right] = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{1}{\sqrt{2}} \right] = ?$$

16. **Charge is distributed uniformly throughout the volume of an infinitely charged cylinder of radius a and volume charge density ρ . Show that electric field at a distance r is given by**

$$E = \frac{\rho r}{2\epsilon_0} \text{ for } r < a \quad \text{and } E = \frac{\rho a^2}{2\epsilon_0 r} \text{ for } r > a$$

→ Here, for $r < a$



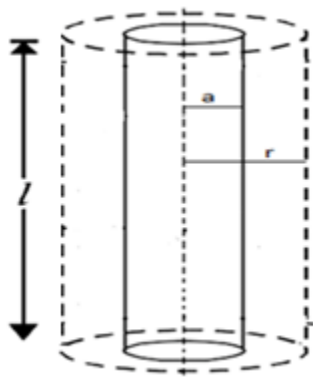
Using Gauss Law,

$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$

$$\text{or, } E \times 2\pi r l = \frac{\rho \times \pi r^2 l}{\epsilon_0}$$

$$\therefore E = \frac{\rho r}{2\epsilon_0}$$

Again for $r > a$



Using Gauss Law,

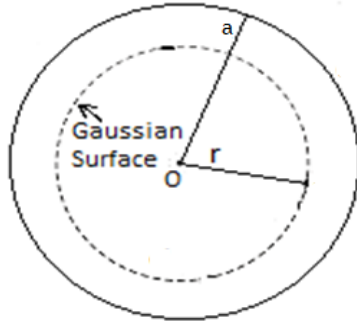
$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$

$$\text{or, } E \times 2\pi r l = \frac{\rho \times \pi a^2 l}{\epsilon_0}$$

$$\therefore E = \frac{\rho a^2}{2\epsilon_0 r}$$

17. A spherical charge distribution has volume charge density $\rho = Ar^n$ at $r < a$ and ρ_0 for $r > a$. Find the electric field in both case, where a is radius of the sphere.

→ Here, for $r < a$, $\rho = Ar^n$



Using Gauss law,

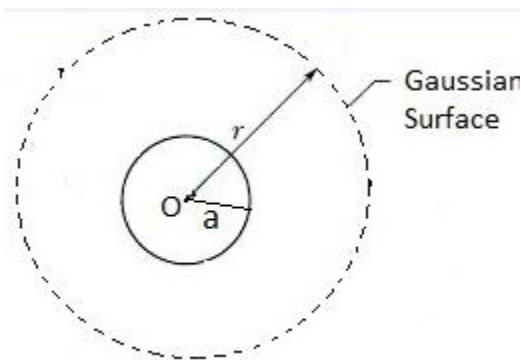
$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$

$$\text{or, } E \times 4\pi r^2 = \frac{\int dq}{\epsilon_0} = \frac{\int \rho dV}{\epsilon_0} = \frac{\int_0^r Ar^n 4\pi r^2 dr}{\epsilon_0} \quad \left(\text{Since, } V = \frac{4}{3}\pi r^3 \text{ so } dV = 4\pi r^2 dr \right)$$

$$\text{or, } E \times 4\pi r^2 = \frac{4\pi A}{\epsilon_0} \int_0^r r^{n+2} dr = \frac{4\pi A}{\epsilon_0} \frac{r^{n+3}}{n+3}$$

$$\text{or, } E = \frac{Ar^{n+1}}{\epsilon_0 (n+3)}$$

Again for $r > a$, $\rho = \rho_0$



Using Gauss law,

$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$

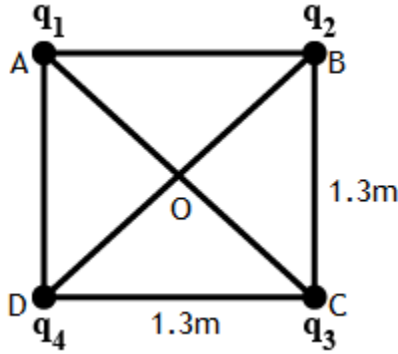
$$\text{or, } E \times 4\pi r^2 = \frac{\rho_0 \times \frac{4}{3}\pi a^3}{\epsilon_0}$$

$$\therefore E = \frac{\rho_0 a^3}{\epsilon_0 r^2}$$

18. Four charges 12nC, -24nC, 31nC and 17nC are placed at the corners of a square of side 1.3m.

Calculate the potential at the center of the square.

→ Here, $q_1 = 12nC = 12 \times 10^{-9}C$, $q_2 = -24nC = -24 \times 10^{-9}C$, $q_3 = 31nC = 31 \times 10^{-9}C$, $q_4 = 17nC = 17 \times 10^{-9}C$



Diagonal of square, $d = \sqrt{1.3^2 + 1.3^2} = ?$

$$\therefore r = OA = OB = OC = OD = \frac{d}{2} = ?$$

Now potential at the center is

$$V = V_A + V_B + V_C + V_D = \frac{1}{4\pi\epsilon_0 r} [q_1 + q_2 + q_3 + q_4] = \frac{9 \times 10^9}{r} [12 - 24 + 31 + 17] \times 10^{-9} = \dots \dots \dots \text{Volt}$$

19. If the electric field intensity of a charged disk of radius a at point P at a distance x along its axis from the center is $E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + a^2}} \right]$. Find the electric potential at point P .

→ Here, electric field due to a charged disc at a distance x is

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + a^2}} \right]$$

Now the electric potential at point P is

$$V = - \int E dx = - \frac{\sigma}{2\epsilon_0} \int \left[1 - \frac{x}{\sqrt{x^2 + a^2}} \right] dx = - \frac{\sigma}{2\epsilon_0} \left[x - \sqrt{x^2 + a^2} \right] = \frac{\sigma}{2\epsilon_0} \left[\sqrt{x^2 + a^2} - x \right]$$

20. Charge of uniform volume density $3.2\mu C m^{-3}$ fills a non conducting solid sphere of radius 5cm.

What is the magnitude of electric field at (a) 3.5cm and (b) 8cm from the sphere's center?

→ Here, volume charge density, $= 3.2\mu C m^{-3} = 3.2 \times 10^{-6} C m^{-3}$,

Radius of solid sphere, $R = 5cm = 0.05m$

Total charge on the sphere is $q = \rho \times \text{volume} = \rho \times \frac{4}{3}\pi R^3 = \dots \dots \dots C$

a) For $r = 3.5cm = 0.035m$

$$E = \frac{1}{4\pi\epsilon_0} \frac{qr}{R^3} = ?$$

b) For $r = 8cm = 0.08m$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = ?$$

21. Two large parallel plates are separated by a distance of 5cm. The plates have equal but opposite charges that create an electric field in the region between the plates. An alpha particle ($q=3.2 \times 10^{-19}$, $m=6.68 \times 10^{-27}$ kg) is released from the positively charged plate, and it strikes the negatively charged plate 2×10^{-6} s later. Assuming that the electric field between the plates is uniform and perpendicular to the plates, what is the strength of electric field?

→ Here, distance between two plates, $d=5\text{cm}=0.05\text{m}$, Charge, $q=3.2 \times 10^{-19}\text{C}$, mass, $m=6.68 \times 10^{-27}\text{kg}$, Time, $t=2 \times 10^{-6}\text{s}$

$$\text{We have, } S = d = ut + \frac{1}{2}at^2 = \frac{1}{2}at^2 \quad (\text{Since, } u = 0)$$

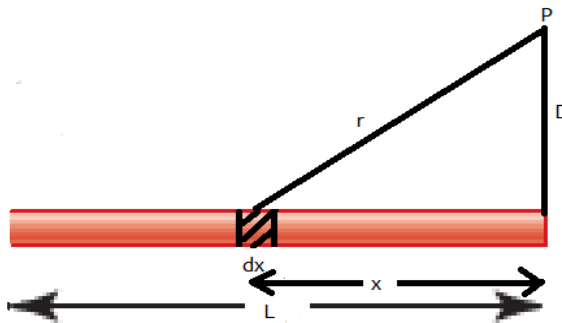
$$\therefore a = \frac{2d}{t^2}$$

$$\text{Again we have, } F = ma = qE$$

$$\therefore E = \frac{ma}{q} = \frac{m}{q} \frac{2d}{t^2} = ?$$

22. Calculate the electric potential at a point due to a uniform line of charge of length L at a distance D from its one end which lies in perpendicular line.

→ Here,



The electric potential due to a small charge element $dq = \lambda dx$ is

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{\sqrt{x^2 + D^2}}$$

$$\text{Hence the net potential at P is } V = \int dV = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{dx}{\sqrt{x^2 + D^2}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\ln(x + \sqrt{x^2 + D^2}) \right]_0^L$$

$$\text{or, } V = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(L + \sqrt{L^2 + D^2}) - \ln D \right] = \frac{q}{4\pi\epsilon_0 L} \ln \left(\frac{L + \sqrt{L^2 + D^2}}{D} \right)$$

- 23. A spherical drop of water carrying a charge of 30pC has a potential of 500V at its surface. (a) What is the radius of the drop? (b) if two such drops of the same charge and radius combine to form a single spherical drop, what is the potential at the surface of the new drop?**

→ Here, charge $q=30\text{pC}=30\times 10^{-12}\text{C}$, Potential, $v= 500\text{V}$

a) We have, $v = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$

$$\therefore r = \frac{1}{4\pi\epsilon_0} \frac{q}{v} = 9 \times 10^9 \times \frac{30 \times 10^{-12}}{500} = \dots \dots \dots m$$

- b) Let R and V be the radius and potential of bigger drop.

Charge on the bigger drop is $Q=2q$

Here, initial volume = final volume

$$\text{or, } 2 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$\text{or, } R = 2^{1/3} r$$

Now potential of bigger drop is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R} = \frac{1}{4\pi\epsilon_0} \frac{2q}{2^{1/3} r} = 2^{2/3} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right) = 2^{2/3} \times v = 2^{2/3} \times 500 = ?$$

- 24. What is the magnitude of electric field at point $(3\mathbf{i}-2\mathbf{j}+4\mathbf{k})$ m if the electric potential is given by $V=2xyz^2$, where V is in volts and x, y, and z are in meters.**

→ Here, potential, $V = 2xyz^2$

$$\text{Now, } x - \text{component of electric field, } E_x = -\frac{\partial V}{\partial x} = -2yz^2$$

$$y - \text{component of electric field, } E_y = -\frac{\partial V}{\partial y} = -2xz^2$$

$$\text{And } z - \text{component of electric field, } E_z = -\frac{\partial V}{\partial z} = -4xyz$$

$$\text{Hence the magnitude of electric field is } E = \sqrt{E_x^2 + E_y^2 + E_z^2} = \sqrt{4y^2z^4 + 4x^2z^4 + 16x^2y^2z^2}$$

Now at point $(3\mathbf{i}-2\mathbf{j}+4\mathbf{k})$

$$E = \sqrt{4y^2z^4 + 4x^2z^4 + 16x^2y^2z^2} = \dots \dots \dots V/m$$

- 25. The potential in a region between $x=0\text{m}$ and $x=6\text{m}$ is $V=a+bx^2$ where $a=10$ and $b = -7\text{V/m}$.**

Determine (i) the potentials at $x=0\text{m}$, 3m and 6m and (ii) the magnitude of electric fields at $x:=0\text{m}$, 3m and 6m .

→ Here, potential $V = a + bx^2$ where, $a = 10$ and $b = -7\text{V/m}$

i) At $x=0\text{m}$, $V = a + bx^2 = 10 + (-7) \times 0^2 = 10 \text{ Volt}$

At $x=3\text{m}$, $V = a + bx^2 = 10 + (-7) \times 3^2 = -53 \text{ Volt}$

At $x=6\text{m}$, $V = a + bx^2 = 10 + (-7) \times 6^2 = ?$

ii) For second part, $E = -\frac{dV}{dx} = -2bx$

∴ At $x=0\text{m}$, $E = -2bx = 0$

At $x=3\text{m}$, $E = -2bx = -2 \times -7 \times 3 = 42\text{V/m}$

At $x=6\text{m}$, $E = -2bx = ?$

26. What is the magnitude of the electric field at point (3,2) m if the electric potential is given by $V = 2x - 5xy + 3y^2$ volts. What acceleration does an electron experiences in the x-direction.

→ Here, potential $V = 2x - 5xy + 3y^2$

Now, x – component of electric field, $E_x = -\frac{\partial V}{\partial x} = -(2 - 5y)$

And y – component of electric field, $E_y = -\frac{\partial V}{\partial y} = -(-5x + 6y)$

Hence the magnitude of electric field is $E = \sqrt{E_x^2 + E_y^2} = \sqrt{(2 - 5y)^2 + (-5x + 6y)^2}$

Now at point (3,2)

$$E = \sqrt{(2 - 5y)^2 + (-5x + 6y)^2} = \dots \dots \dots V/m$$

Again we have $F = ma = eE_x$

$$\therefore a = \frac{eE_x}{m} \text{ Where, } E_x = -(2 - 5y) = 8V/m$$

$$\text{or, } a = \frac{1.6 \times 10^{-19} \times 8}{9.1 \times 10^{-31}} = ?$$

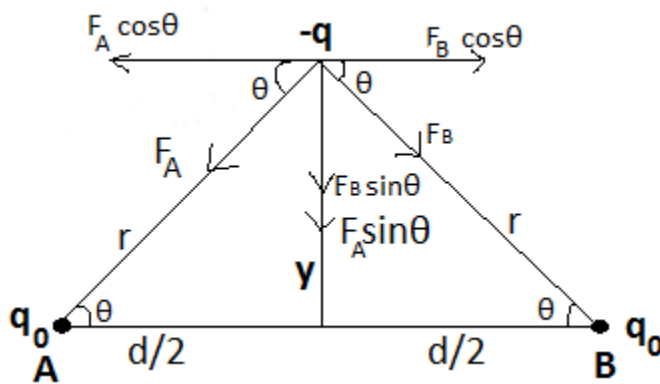
27. Over certain region of space, the electric potential is $V = 5x - 3x^2y + 2yz^2$. Find the expressions for the x, y and z components of the electric field over this region. What is the magnitude of the field at the point P that has coordinates (1,0,-2)m?

→ Here, Same as 21,22,23.

28. A particle of charge $-q$ and mass m is placed midway between two equal positive charges q_0 of separation d . If the negative charge $-q$ is displaced in perpendicular direction to the line

joining them and released, show that the particle describes a SHM with a period $T = \sqrt{\frac{\pi^3 d^3 m \epsilon_0}{qq_0}}$.

→ Here,



$$\text{Here, } r = \sqrt{y^2 + \frac{d^2}{4}}$$

$$\text{We have, } F_A = F_B = -\frac{1}{4\pi\epsilon_0} \frac{qq_0}{(y^2 + \frac{d^2}{4})} = F(\text{say})$$

$$\text{Net force is } F' = 2F \sin \theta$$

$$\text{or, } ma = -2 \frac{1}{4\pi\epsilon_0} \frac{qq_0}{(y^2 + \frac{d^2}{4})} \frac{y}{\sqrt{y^2 + \frac{d^2}{4}}}$$

$$\text{or, } ma = -\frac{2qq_0}{4\pi\epsilon_0} \frac{y}{(y^2 + \frac{d^2}{4})^{3/2}}$$

$$\text{For } y \ll d, (y^2 + \frac{d^2}{4})^{3/2} \approx \frac{d^3}{8}$$

$$\therefore ma = -\frac{4qq_0}{\pi\epsilon_0 d^3} y$$

$$\text{or, } a = -\frac{4qq_0}{\pi\epsilon_0 md^3} y$$

$$\text{or, } a = -\omega^2 y \quad \text{Where, } \omega = \sqrt{\frac{4qq_0}{\pi\epsilon_0 md^3}}$$

This shows that the motion is simple harmonic whose angular frequency is given by

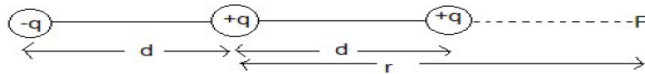
$$\omega = \sqrt{\frac{4qq_0}{\pi\epsilon_0 md^3}}$$

$$\therefore \text{frequency } f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{4qq_0}{\pi\epsilon_0 md^3}} = \sqrt{\frac{qq_0}{\pi^3\epsilon_0 md^3}}$$

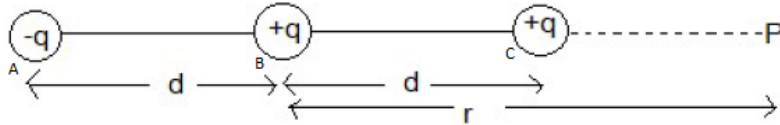
$$\text{and time period } T = \frac{1}{f} = \sqrt{\frac{\pi^3 d^3 m \epsilon_0}{qq_0}}$$

29. For the system of charges as shown in fig, prove that the potential at point p for $r \gg d$ is

$$V = \frac{q}{4\pi\epsilon_0} \left[1 + \frac{2d}{r} \right]$$



→ Here,



$$\text{Potential at P due to charge -q at A, } V_A = -\frac{1}{4\pi\epsilon_0} \frac{q}{r+d}$$

$$\text{Potential at P due to charge +q at B, } V_B = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

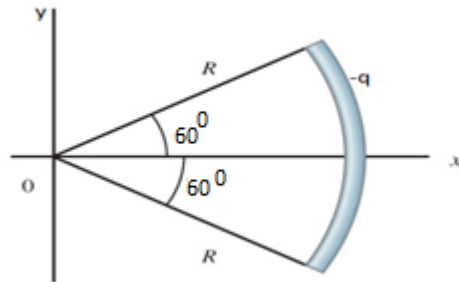
$$\text{Potential at P due to charge +q at C, } V_C = \frac{1}{4\pi\epsilon_0} \frac{q}{r-d}$$

$$\begin{aligned} \therefore \text{Total potential at P is } V &= V_A + V_B + V_C = \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{r+d} + \frac{1}{r} + \frac{1}{r-d} \right] \\ &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} + \frac{2d}{(r-d)(r+d)} \right] \end{aligned}$$

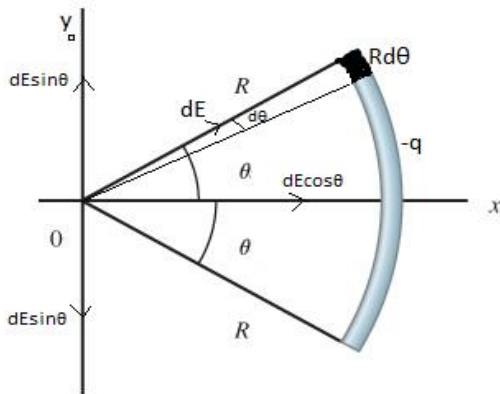
$$\text{or, } V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} + \frac{2d}{r^2 - d^2} \right]$$

$$\text{For } r \gg d, V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} + \frac{2d}{r^2} \right] = \frac{q}{4\pi\epsilon_0 r} \left[1 + \frac{2d}{r} \right]$$

30. A plastic rod contains uniformly distributed charge $-q$. the rod has been bent in 120° circular arc of radius R as in fig. Prove that the magnitude of electric field E produced at the center of the bent rod is $E = \frac{0.83q}{4\pi\epsilon_0 R^2}$.



→ Here, Consider the bent rod has linear charge density λ .



Electric field at the center due to a small charge element $dx = Rd\theta$ is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{R^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{R^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda R d\theta}{R^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda d\theta}{R}$$

Here, $dE \sin \theta$ components of electric field cancel out and the effective components are $dE \cos \theta$.

$$\therefore \text{net electric field at the center is, } E = \int dE \cos \theta = \int_{-\theta}^{\theta} \frac{1}{4\pi\epsilon_0} \frac{\lambda d\theta}{R} \cos \theta = \frac{\lambda}{4\pi\epsilon_0 R} \int_{-\theta}^{\theta} \cos \theta d\theta$$

$$\text{or, } E = \frac{\lambda}{4\pi\epsilon_0 R} [\sin \theta]_{-\theta}^{\theta} = \frac{\lambda}{4\pi\epsilon_0 R} [\sin \theta - \sin(-\theta)] = \frac{\lambda}{4\pi\epsilon_0 R} [\sin \theta + \sin \theta] = \frac{2\lambda \sin \theta}{4\pi\epsilon_0 R} = \frac{2\lambda \sin 60^\circ}{4\pi\epsilon_0 R}$$

Here, for 360° , the arc length $= 2\pi R$

$$\therefore \text{For } 120^\circ, \text{ the arc length} = \frac{120}{360} 2\pi R = \frac{2\pi R}{3}$$

$$\text{Hence, } \lambda = \frac{q}{\frac{2\pi R}{3}} = \frac{3q}{2\pi R}$$

$$\therefore E = \frac{2\lambda \sin 60^\circ}{4\pi\epsilon_0 R} = \frac{2 \sin 60^\circ}{4\pi\epsilon_0 R} \times \frac{3q}{2\pi R} = \frac{0.83q}{4\pi\epsilon_0 R^2}$$

Chapter: Capacitor

Capacitor: An electrical device which is used to store the electric charge is called capacitor.

Capacitance: The ability of a capacitor to store the electric charge is called its capacitance.

The amount of charge stored in a capacitor is directly proportional to the potential difference between the two plates of capacitor.

$$i.e. q \propto V$$

$$or, q = CV$$

Where, $C = \frac{q}{V}$ is capacitance of a capacitor. Its unit is Farad(F) in SI system.

Note: $1\mu F = 10^{-6}F$, $1nF = 10^{-9}F$, $1pF = 1\mu\mu F = 10^{-12}F$

Parallel Plate Capacitor

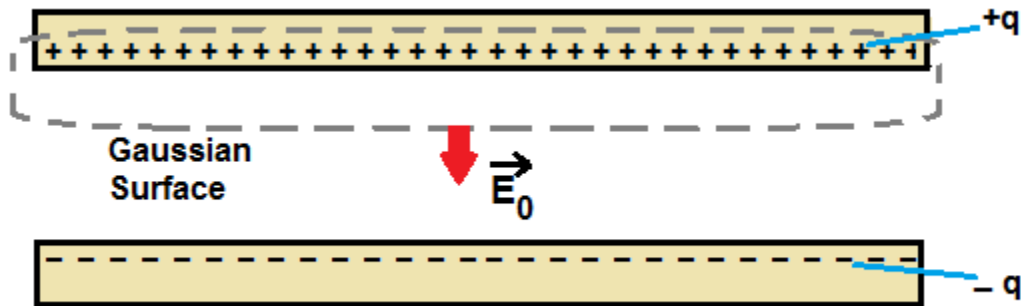


Fig: Parallel Plate Capacitor

Consider two parallel plates separated in space by d with plate area A . When $+q$ charge is given to first plate then inner side of other plate acquire $-q$ charge by induction. Due to this there is a uniform electric field between the plates. Let us construct a Gaussian surface that encloses the charge $+q$. Then according to Gauss Law,

$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$

$$\text{or, } EA = \frac{q}{\epsilon_0}$$

$$\text{or, } E = \frac{q}{\epsilon_0 A}$$

Now Potential difference between the plates of capacitor is

$$V = E \times d = \frac{qd}{\epsilon_0 A}$$

Hence the capacitance of the capacitor is

$$C = \frac{q}{V} = \frac{\epsilon_0 A}{d}$$

If a dielectric medium having dielectric constant K is inserted between the plates then

Capacitance becomes

$$C_m = \frac{\epsilon A}{d} = \frac{K\epsilon_0 A}{d} = KC$$

$$\therefore C_m > C$$

Spherical Capacitor

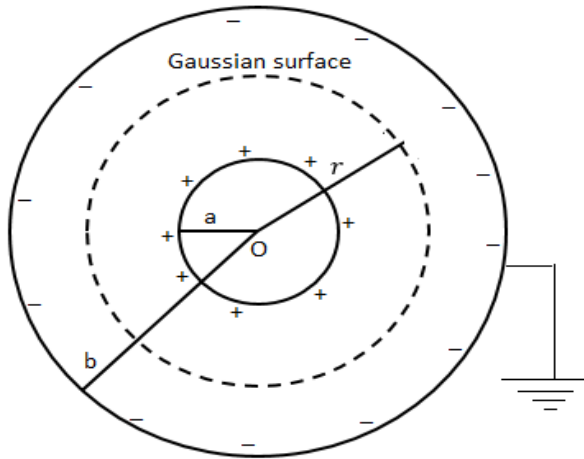


Fig: Spherical Capacitor

A spherical capacitor consists of two spherical metallic spheres having radii a and b . As a Gaussian surface, we draw a sphere of radius r as shown in figure.

From Gauss Law,

$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$

$$\text{or, } E \times 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$\text{or, } E = \frac{q}{4\pi\epsilon_0 r^2}$$

Now the potential difference between the spheres is

$$V = \int_a^b E dr = \int_a^b \frac{q}{4\pi\epsilon_0 r^2} dr = \frac{q}{4\pi\epsilon_0} \int_a^b \frac{1}{r^2} dr = \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_a^b$$

$$\text{or, } V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{b} \right] = \frac{q}{4\pi\epsilon_0} \left[\frac{b-a}{ab} \right]$$

Hence the capacitance of spherical capacitor is given by

$$C = \frac{q}{V} = 4\pi\epsilon_0 \left[\frac{ab}{b-a} \right]$$

$$\text{In the case when } b \rightarrow \infty, C = 4\pi\epsilon_0 \left[\frac{a}{1-\frac{a}{b}} \right] = 4\pi\epsilon_0 a$$

This is the capacitance of isolated spherical capacitor.

Cylindrical capacitor

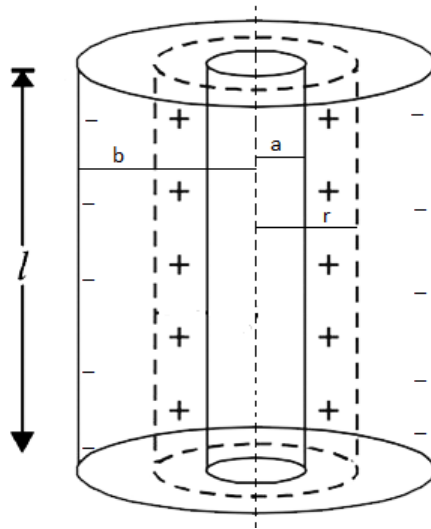


Figure shows a cylindrical capacitor of length l with inner radius a and outer radius b . As a Gaussian surface, we draw a cylinder of radius r and length l as shown in figure.

From Gauss Law,

$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$$

$$\text{or, } E \times 2\pi r l = \frac{q}{\epsilon_0}$$

$$\text{or, } E = \frac{q}{2\pi\epsilon_0 r l}$$

Now the potential difference between the cylinders is

$$V = \int_a^b E dr = \int_a^b \frac{q}{2\pi\epsilon_0 r l} dr = \frac{q}{2\pi\epsilon_0 l} \int_a^b \frac{1}{r} dr = \frac{q}{2\pi\epsilon_0 l} \ln\left(\frac{b}{a}\right)$$

Hence the capacitance of cylindrical capacitor is given by

$$C = \frac{q}{V} = \frac{2\pi\epsilon_0 l}{\ln\left(\frac{b}{a}\right)}$$

Note: Steps to find the capacitance of a capacitor

- i) Use Gauss Law $\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$ and find E .

- ii) Find V using $V = \int E dr$
- iii) Find capacitance using $C = \frac{q}{V}$

Energy stored in a capacitor and Energy Density

Small amount of work done to charge the capacitor by small amount dq is

$$dW = Vdq = \frac{q}{C} dq$$

Where, $V = \frac{q}{C}$ is potential difference between the plates of capacitor.

Now total work done to charge the capacitor from 0 to q is

$$W = \int dW = \int_0^q \frac{q}{C} dq = \frac{1}{C} \int_0^q q dq = \frac{q^2}{2C}$$

This work done is stored as potential energy (U_E) in the capacitor.

$$\therefore U_E = W = \frac{q^2}{2C}$$

$$\text{Hence, } U_E = \frac{q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} qV$$

This is the energy stored in the capacitor.

Now energy stored per unit volume between the plates of capacitor is called electrical energy density and is given by

$$u_b = \frac{U_E}{\text{Volume}} = \frac{\frac{1}{2} CV^2}{Ad} = \frac{CV^2}{2Ad} = \frac{\epsilon_0 A}{d} \frac{V^2}{2Ad} = \frac{1}{2} \epsilon_0 \frac{V^2}{d^2}$$

$$\therefore u_b = \frac{1}{2} \epsilon_0 E^2 \text{ Where, } E = \frac{V}{d} \text{ is the electric field intensity between the plates of capacitor.}$$

This shows that the electrical energy density is proportional to square of electric field intensity.

Gauss Law in Dielectrics

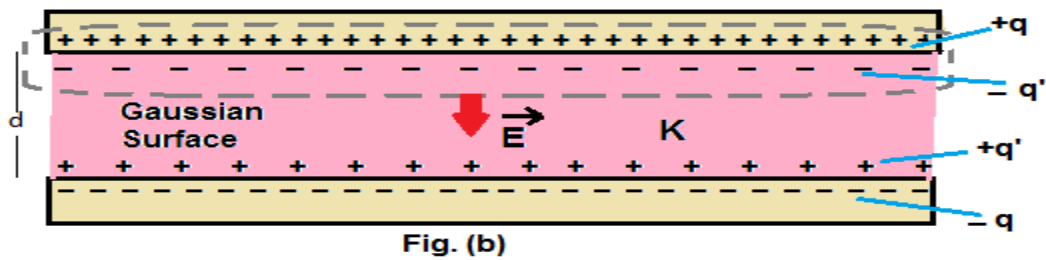
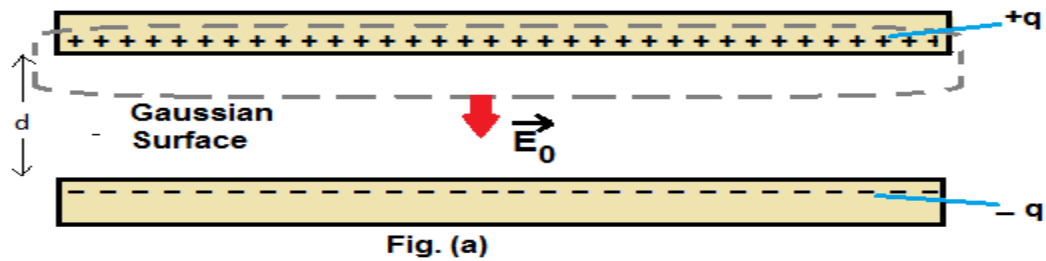


Figure (a) and (b) show the parallel plate capacitor without and with dielectrics.

Applying Gauss law in figure (a)

$$\oint \vec{E}_0 \cdot \vec{da} = \frac{q}{\epsilon_0}$$

$$\text{or, } E_0 A = \frac{q}{\epsilon_0}$$

$$\text{or, } E_0 = \frac{q}{\epsilon_0 A} \dots \dots \dots (i)$$

Here, E_0 is the electric field between the plates of capacitor without dielectrics.

Now consider a dielectric slab having dielectric constant K is inserted between the plates of capacitor then the electric field reduced from E_0 to E due to induced dipole with induced charges q_i .

Now applying Gauss Law in figure (b)

$$\oint \vec{E} \cdot \vec{da} = \frac{q - q_i}{\epsilon_0} \dots \dots \dots (ii)$$

$$\text{or, } EA = \frac{q - q_i}{\epsilon_0}$$

$$\text{or, } E = \frac{q - q_i}{\epsilon_0 A} \dots \dots \dots (iii)$$

$$\text{We have, } K = \frac{E_0}{E}$$

$$\therefore E = \frac{E_0}{K} = \frac{q}{\epsilon_0 AK} \dots \dots \dots (iv)$$

Hence from equation (iii) and (iv),

$$\frac{q}{\epsilon_0 AK} = \frac{q - q_i}{\epsilon_0 A}$$

$$or, q - q_i = \frac{q}{K}$$

$$or, q_i = \left(1 - \frac{1}{K}\right) q$$

Hence the Gauss Law with dielectrics is

$$\oint \vec{E} \cdot d\vec{a} = \frac{q}{K\epsilon_0}$$

Again, from equations (iii) and (iv),

$$\frac{q}{\epsilon_0 AK} = \frac{q - q_i}{\epsilon_0 A}$$

$$or, \frac{q}{\epsilon_0 AK} = \frac{q}{\epsilon_0 A} - \frac{q_i}{\epsilon_0 A}$$

$$or, \frac{q}{A} = \epsilon_0 \frac{q}{\epsilon_0 AK} + \frac{q_i}{A}$$

$$or, D = \epsilon_0 E + P$$

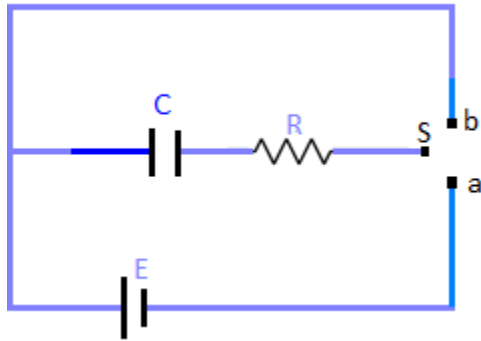
Where, $D = \frac{q}{A} = \frac{\text{Free charge}}{\text{area}}$ is the magnitude of displacement vector.

And $P = \frac{q_i}{A} = \frac{\text{induced charge}}{\text{area}}$ is the magnitude of polarization vector.

In vector form, we can write

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

RC-Circuit (Charging and Discharging of Capacitor)



a) When switch S is connected to a then charging of capacitor takes place.

Using KVL,

$$E = IR + \frac{q}{C}$$

$$\text{or, } CE - q = IRC$$

$$\text{or, } CE - q = RC \frac{dq}{dt}$$

$$\text{or, } \frac{dq}{CE - q} = \frac{1}{RC} dt$$

Integrating both sides,

$$\int \frac{dq}{CE - q} = \frac{1}{RC} \int dt$$

$$\text{or, } -\ln(CE - q) = \frac{1}{RC} t + k_1 \dots \dots \dots (i)$$

Where, k_1 is integration constant.

When $t = 0, q = 0$

$$\therefore k_1 = -\ln CE$$

Hence from equation (i),

$$-\ln(CE - q) = \frac{1}{RC} t - \ln CE$$

$$\text{or, } \ln \left\{ \frac{CE - q}{CE} \right\} = -\frac{1}{RC} t$$

$$\text{or, } 1 - \frac{q}{CE} = e^{-\frac{1}{RC}t}$$

$$\text{or, } 1 - \frac{q}{q_0} = e^{-\frac{1}{RC}t} \quad \text{Where, } q_0 = CE \text{ is maximum charge stored in the capacitor.}$$

$$\text{or, } q = q_0(1 - e^{-\frac{1}{RC}t})$$

This gives the expression for growth of charge.

$$\text{When, } t = RC = \tau_c \text{ then } q = q_0(1 - e^{-1}) = 0.63q_0 = 63\% \text{ of } q_0$$

Hence the time at which charge through the capacitor growth to 63% of the maximum charge is called capacitive time constant.

$$\text{Again charging current is } I = \frac{dq}{dt} = q_0 \times \frac{1}{RC} \times e^{-\frac{1}{RC}t} = I_0 e^{-\frac{1}{RC}t}$$

Where, $I_0 = \frac{q_0}{RC}$ is maximum current.

b) When switch S is connected to b then discharging of capacitor takes place.

Using KVL,

$$IR + \frac{q}{C} = 0$$

$$\text{or, } R \frac{dq}{dt} = -\frac{q}{C}$$

$$\text{or, } \frac{dq}{q} = -\frac{1}{RC} dt$$

Integrating both sides,

$$\int \frac{dq}{q} = -\frac{1}{RC} \int dt$$

$$\text{or, } \ln q = -\frac{1}{RC}t + k_2 \dots \dots \dots (ii)$$

Where, k_2 is integration constant.

$$\text{When } t = 0, q = q_0$$

$$\therefore k_2 = \ln q_0$$

Hence from equation (ii),

$$\ln q = -\frac{1}{RC}t + \ln q_0$$

$$\text{or, } \ln \frac{q}{q_0} = -\frac{1}{RC}t$$

$$\text{or, } \frac{q}{q_0} = e^{-\frac{1}{RC}t}$$

$$\text{or, } q = q_0 e^{-\frac{1}{RC}t}$$

This gives the expression for decay of charge.

When, $t = RC = \tau_c$ then $q = q_0 e^{-1} = 0.37q_0 = 37\% \text{ of } q_0$

Hence the time at which charge through the capacitor decays to 37% of the maximum charge is called capacitive time constant.

Again charging current is $I = \frac{dq}{dt} = -q_0 \times \frac{1}{RC} \times e^{-\frac{1}{RC}t} = -I_0 e^{-\frac{1}{RC}t}$

Where, $I_0 = \frac{q_0}{RC}$ is maximum current.

Here, we see that the magnitude of current is same in the case of both charging and discharging of capacitor.

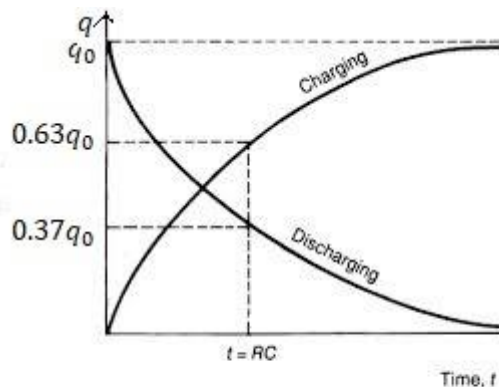


Fig: Variation of charge with time

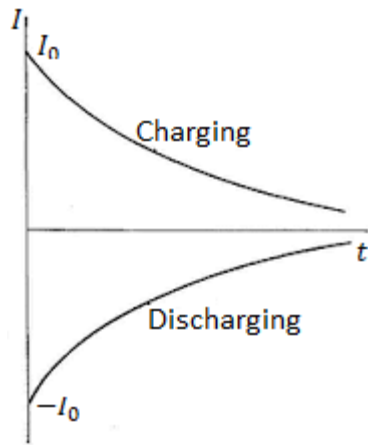


Fig: Variation of current with time

Note: We have the magnitude of current is

$$I = I_0 e^{-\frac{1}{RC}t}$$

$$\text{or, } \frac{I}{I_0} = e^{-\frac{1}{RC}t}$$

$$\text{or, } \frac{1}{RC}t = \ln\left(\frac{I_0}{I}\right)$$

$$\text{or, } C = \frac{t}{R \ln\left(\frac{I_0}{I}\right)}$$

Capacitors

1. A sheet of paper 40mm wide and 1.5×10^{-2} mm thick between metal plates of the same width is used to make a $2 \mu F$ capacitor. If the dielectric constant of the paper is 2.5, what is the length of paper required?

→ Here, width of paper = width of plate = $w = 40\text{mm} = 0.04\text{m}$,

Thickness of paper = distance between plates = $d = 1.5 \times 10^{-2}\text{mm} = 1.5 \times 10^{-5}\text{m}$, capacitance $C' = 2 \mu F = 2 \times 10^{-6} F$

Dielectric constant, $K = 2.5$

Let l be the length of paper required.

We have capacitance of a capacitor with dielectric is

$$C' = \frac{K \epsilon_0 A}{d} = \frac{K \epsilon_0 (l \times w)}{d}$$

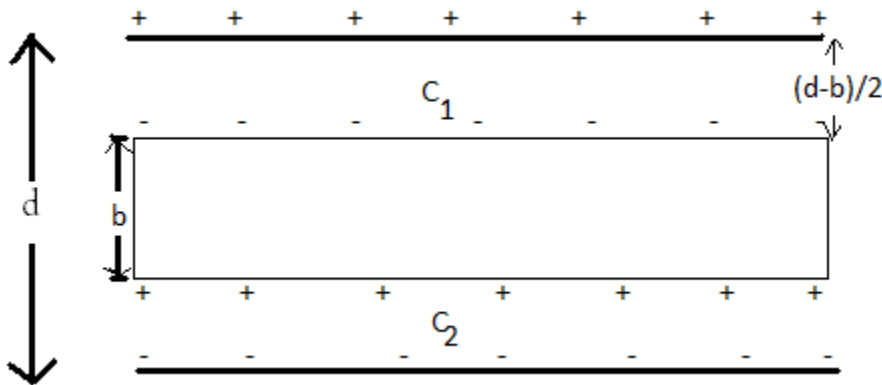
$$\therefore l = \frac{C'd}{K\epsilon_0 W} = \frac{2 \times 10^{-6} \times 1.5 \times 10^{-5}}{2.5 \times 8.85 \times 10^{-12} \times 0.04} = ?$$

2. Into a parallel plate capacitor of separation d and plate area A , an uncharged copper plate of thickness b is inserted, midway between plates. Find the change in capacitance.

→ Here, initial capacitance before inserting the copper plate is $C = \frac{\epsilon_0 A}{d}$

After inserting the copper plate,

$$C_1 = C_2 = \frac{\epsilon_0 A}{(d-b)/2} = \frac{2\epsilon_0 A}{(d-b)}$$



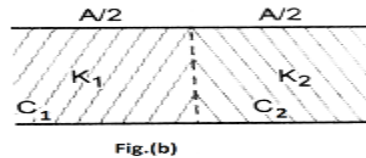
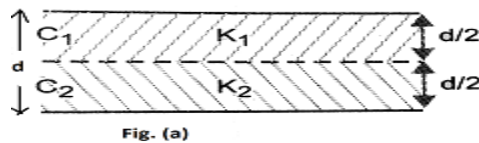
Since, C_1 and C_2 are in series, so net capacitance is

$$C' = \frac{C_1 C_2}{C_1 + C_2} = \frac{\frac{2\epsilon_0 A}{(d-b)} \frac{2\epsilon_0 A}{(d-b)}}{\frac{2\epsilon_0 A}{(d-b)} + \frac{2\epsilon_0 A}{(d-b)}} = \frac{\epsilon_0 A}{(d-b)}$$

$$\therefore \text{Change in capacitance} = C' - C = \frac{\epsilon_0 A}{(d-b)} - \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 A b}{d(d-b)}$$

3. A parallel plate capacitor is filled with two dielectric having dielectric constant k_1 and k_2 in the way as in fig. prove that the capacitance is given by (a)

$$\frac{2\epsilon_0 A}{d} \left(\frac{k_1 k_2}{k_1 + k_2} \right) \text{ and (b) } \frac{\epsilon_0 A}{d} \left(\frac{k_1 + k_2}{2} \right)$$



→ Here, (a) In this case, distance is divided. So

$$C_1 = \frac{K_1 \epsilon_0 A}{d/2} = \frac{2K_1 \epsilon_0 A}{d} \text{ and } C_2 = \frac{K_2 \epsilon_0 A}{d/2} = \frac{2K_2 \epsilon_0 A}{d}$$

Since C_1 and C_2 are in series so net capacitance is

$$C' = \frac{C_1 C_2}{C_1 + C_2} = \frac{\frac{2K_1 \epsilon_0 A}{d} \frac{2K_2 \epsilon_0 A}{d}}{\frac{2K_1 \epsilon_0 A}{d} + \frac{2K_2 \epsilon_0 A}{d}} = \frac{2\epsilon_0 A}{d} \left(\frac{k_1 k_2}{k_1 + k_2} \right)$$

(b) In this case, area is divided. So

$$C_1 = \frac{K_1 \epsilon_0 A}{2d} \text{ and } C_2 = \frac{K_2 \epsilon_0 A}{2d}$$

Since C_1 and C_2 are in parallel so net capacitance is

$$C' = C_1 + C_2 = \frac{K_1 \epsilon_0 A}{2d} + \frac{K_2 \epsilon_0 A}{2d} = \frac{\epsilon_0 A}{d} \left(\frac{k_1 + k_2}{2} \right)$$

4. An air filled parallel plate capacitor has an capacitance of 1.32pF. The separation of the plates is doubled and wax is inserted between them. The new capacitance is 2.57pF. Find the dielectric constant of the wax.

→ Here, Capacitance without dielectric $C=1.32\text{pF}$, Capacitance with dielectric $C'=2.57\text{pF}$

$$\text{We have, } C = \frac{\epsilon_0 A}{d}$$

$$\text{New capacitance with dielectric and distance doubled, } C' = \frac{K \epsilon_0 A}{2d} = \frac{KC}{2}$$

$$\therefore K = \frac{2C'}{C} = ?$$

5. If a parallel plate capacitor is to be designed to operate in an environment of fluctuating temperature, prove that the rate of change of capacitance C with temperature T is given by

$$\frac{dC}{dT} = C \left[\frac{1}{A} \frac{dA}{dT} - \frac{1}{x} \frac{dx}{dT} \right].$$

→ Here, we have the capacitance of parallel plate capacitor with plate area A and plate separation x is

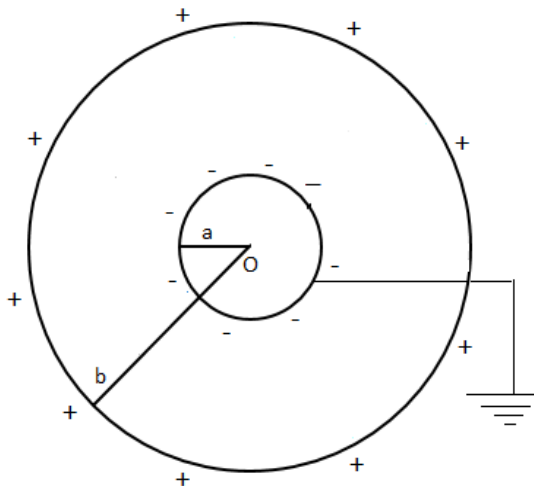
$$C = \frac{\epsilon_0 A}{x}$$

Now differentiating both sides with respect to temperature T ,

$$\frac{dC}{dT} = \epsilon_0 \left[\frac{1}{x} \frac{dA}{dT} - \frac{A}{x^2} \frac{dx}{dT} \right] = \frac{\epsilon_0 A}{x} \left[\frac{1}{A} \frac{dA}{dT} - \frac{1}{x} \frac{dx}{dT} \right] = C \left[\frac{1}{A} \frac{dA}{dT} - \frac{1}{x} \frac{dx}{dT} \right]$$

6. Prove that the capacitance of a concentric spherical capacitor of radii a and b is $C=4\pi\epsilon_0 \left[\frac{b^2}{b-a} \right]$, if outer sphere is charged and inner sphere is earthed.

→ Here, outer sphere behaves as an isolated capacitor whose capacitance is $C_1 = 4\pi\epsilon_0 b$



Again the capacitance of spherical capacitor having inner radius a and outer radius b is $C_2 =$

$$4\pi\epsilon_0 \left(\frac{ab}{b-a} \right)$$

Here, C_1 and C_2 are in parallel, so net capacitance is

$$C = C_1 + C_1 = 4\pi\epsilon_0 \left[b + \frac{ab}{b-a} \right] = 4\pi\epsilon_0 \left[\frac{b^2}{b-a} \right]$$

7. A spherical drop of mercury of radius r has capacitance $C=4\pi\epsilon_0 r$. Calculate the capacitance if two such drops combine to form a single large drop.

→ Here, let R be the radius of large drop. Then capacitance of large drop is $C' = 4\pi\epsilon_0 R$

Since, volume is constant. So

$$2 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$\therefore R = 2^{1/3}r$$

$$\therefore C' = 4\pi\epsilon_0 R = 4\pi\epsilon_0 2^{1/3}r = 2^{1/3}4\pi\epsilon_0 r = 2^{1/3}C$$

8. The space between two concentric conducting spherical shells of radii 1.7cm and 1.2cm is filled with a substance of dielectric constant $K=23.5$. A potential difference $V=73V$ is applied across the inner and outer shells. Determine (a) the capacitance of the device, (b) the free charge q on the inner shell, and (c) the charge q' induced along the surface of inner shell.

→ Here, outer radius, $b=1.7\text{cm}=1.7\times 10^{-2}\text{m}$ inner radius, $a=1.2\text{cm}=1.2\times 10^{-2}\text{m}$ dielectric constant, $K=23.5$

Potential difference, $V = 73V$

a) We have, capacitance $C = 4\pi\epsilon_0 K \left(\frac{ab}{b-a} \right) = \frac{23.5}{9 \times 10^9} \left(\frac{1.7 \times 1.2}{1.7-1.2} \right) \times 10^{-2} = ?$

b) Free charge $q = CV = ?$

c) Induced charge, $q' = \left(1 - \frac{1}{K} \right) q = ?$

9. A parallel plate capacitor has a capacitance of 100 pF, a plate area of 100 cm² and a mica dielectric. At 50 volts potential difference calculate (a) E in the mica (b) the free charge on the plates and (c) the induced surface charge, if dielectric constant for mica, $k = 5.4$

→ Here, Capacitance, $C=100\text{pF}=100\times 10^{-12}\text{F}$, plate area $A=100\text{cm}^2=100\times 10^{-4}\text{m}^2$, potential difference, $V=50V$

Dielectric constant $K = 5.4$

a) We have *electric field intensity*, $E = \frac{q}{\epsilon_0 A} = \frac{CV}{\epsilon_0 A} = ?$

b) Free charge $q = CV = ?$

c) Induced charge, $q' = \left(1 - \frac{1}{K} \right) q = ?$

10. A long cylindrical conductor has length 1m and is surrounded by a co-axial cylindrical conducting shell with inner radius double that of long cylindrical conductor. Calculate the capacitance of this capacitor assuming that there is a vacuum in space between cylinders.

→ Here, length $l = 1\text{m}$, outer radius $b=2a$

we have the capacitance of cylindrical capacitor is $C = \frac{2\pi\epsilon_0 l}{\ln\left(\frac{b}{a}\right)}$

$$= \frac{2 \times \pi \times 8.85 \times 10^{-12} \times 1}{\ln 2} = ?$$

- 11. Two capacitors having capacitance $25 \mu F$ and $5 \mu F$ are connected in parallel and charged with a $100V$ power supply. Calculate the total energy stored in the two capacitors.**

→ Here, $C_1 = 25 \mu F = 25 \times 10^{-6} F$, $C_2 = 5 \mu F = 5 \times 10^{-6} F$, P.D., $V=100V$

Since, C_1 and C_2 are in parallel so net capacitance is

$$C = C_1 + C_2 = 30 \times 10^{-6} F$$

∴ total energy stored in two capacitor is $U = \frac{1}{2} CV^2 = ?$

- 12. A cylindrical capacitor has radii a and b . Show that the half the stored potential energy lies with a cylinder of radius $r = \sqrt{ab}$.**

→ Here, we have the capacitance of cylindrical capacitor having outer radius b and inner radius a is

$$C = \frac{2\pi\epsilon_0 l}{\ln\left(\frac{b}{a}\right)}$$

∴ Energy stored in a cylinder having outer radius b and inner radius a is

$$U = \frac{q^2}{2C} = \frac{q^2}{4\pi\epsilon_0 l} \ln\left(\frac{b}{a}\right)$$

Again Energy stored in a cylinder having outer radius b and inner radius r is

$$U' = \frac{q^2}{4\pi\epsilon_0 l} \ln\left(\frac{b}{r}\right)$$

By question, $U' = \frac{1}{2} U$

$$\frac{q^2}{4\pi\epsilon_0 l} \ln\left(\frac{b}{r}\right) = \frac{1}{2} \frac{q^2}{4\pi\epsilon_0 l} \ln\left(\frac{b}{a}\right)$$

$$\text{or, } \ln\left(\frac{b}{r}\right) = \frac{1}{2} \ln\left(\frac{b}{a}\right)$$

$$\text{or, } \ln\left(\frac{b}{r}\right) = \ln\sqrt{\frac{b}{a}}$$

$$\text{or, } \frac{b}{r} = \sqrt{\frac{b}{a}}$$

$$\text{or, } r = \sqrt{ab}$$

- 13. A fully charged capacitor stores energy W_0 . How much energy remains when its charge has decrease to $\frac{1}{2}$ its original value.**

→ Here, $W_0 = \frac{1}{2} \frac{q_0^2}{C}$

When $q = \frac{q_0}{2}$

$$\text{Then energy } W = \frac{1}{2} \frac{q^2}{C} = \frac{1}{2} \frac{\left(\frac{q_0}{2}\right)^2}{C} = \frac{1}{4} \frac{q_0^2}{C} = \frac{1}{4} W_0$$

- 14. A capacitor of capacitance C is discharged through a resistor of resistance R . After how many time constants is the stored energy $\frac{1}{4}$ of its initial value?**

→ Here, we have in case of discharging, $q = q_0 e^{-\frac{t}{RC}} = q_0 e^{-\frac{t}{\tau_c}}$

$$\text{Initial energy is } U_0 = \frac{1}{2} \frac{q_0^2}{C}$$

$$\text{Now, } U = \frac{1}{2} \frac{q^2}{C} = \frac{1}{2} \frac{q_0^2}{C} e^{-\frac{2t}{\tau_c}} = U_0 e^{-\frac{2t}{\tau_c}}$$

$$\text{or, } \frac{1}{4} U_0 = U_0 e^{-\frac{2t}{\tau_c}}$$

$$\text{or, } \frac{2t}{\tau_c} = \ln 4$$

$$\text{or, } t = \frac{\ln 4}{2} \tau_c = 0.693 \tau_c$$

15. A capacitor C is discharged through a resistor R. After how many time constants does its (i) charge falls to $\frac{1}{2}$ of its initial value (ii) the store energy drop to $\frac{1}{2}$ of its initial value.

→ Here, (i) we have in case of discharging, $q = q_0 e^{-\frac{t}{RC}} = q_0 e^{-\frac{t}{\tau_c}}$

$$\text{or, } \frac{1}{2} q_0 = q_0 e^{-\frac{t}{\tau_c}}$$

$$\text{or, } \frac{t}{\tau_c} = \ln 2$$

$$\text{or, } t = ?$$

$$\text{(ii) Again, } U = \frac{1}{2} \frac{q^2}{C} = \frac{1}{2} \frac{q_0^2}{C} e^{-\frac{2t}{\tau_c}} = U_0 e^{-\frac{2t}{\tau_c}}$$

$$\text{or, } \frac{1}{2} U_0 = U_0 e^{-\frac{2t}{\tau_c}}$$

$$\text{or, } \frac{2t}{\tau_c} = \ln 2$$

$$\text{or, } t = ?$$

16. How many time constants must elapse before a capacitor in an RC circuit is charged for 99% of its maximum charge?

→ Here, we have in case of charging, $q = q_0 \left(1 - e^{-\frac{t}{RC}}\right)$

$$\text{or, } 0.99 q_0 = q_0 \left(1 - e^{-\frac{t}{RC}}\right)$$

$$\text{or, } e^{-\frac{t}{RC}} = 0.01$$

$$\text{or, } -\frac{t}{\tau_c} = \ln 0.01$$

$$\text{or, } t = ?$$

Chapter: Direct Current

- Current $I = \frac{dq}{dt}$
- $I = nev_d A$
- Current density $J = \frac{I}{A} = nev_d$

- Electron density or number of electrons per unit volume $n = \frac{dN_A}{M}$; d is density and M is molar mass
- Resistance $R = \frac{\rho l}{A}$
- Variation of resistance with temperature $R = R_0[1 + \alpha(T_2 - T_1)]$
- Mobility $\mu = \frac{v_d}{E}$ (unit $m^2V^{-1}s^{-1}$)

Ohm's Law

$$V = IR$$

$$\text{or, } El = JA \times \frac{\rho l}{A}$$

$$\text{or, } E = J \times \frac{1}{\sigma}$$

$$\text{or, } J = \sigma E$$

This is modified Ohm's law

$$\text{or, } nev_d = \sigma E$$

$$\text{or, } \sigma = \frac{nev_d}{E} = ne\mu$$

Where, $\mu = \frac{v_d}{E}$ is called mobility.

Microscopic view of resistivity

a) In metal

In the absence of electric field, the free electrons in metals move in random directions so the net current is zero. But when an electric field is applied, the electrons drift in a direction opposite to the electric field with a constant speed called drift velocity.

Consider an electron of mass m is and charge e is placed in an electric field of magnitude E. From Newton's second law of motion, acceleration of the electron is

$$a = \frac{F}{m} = \frac{eE}{m} \dots \dots \dots (i)$$

The drift velocity of the electron is given by

$$v_d = a\tau \text{ where } \tau \text{ is average time between collision or relaxation time.}$$

$$\text{or, } v_d = \frac{eE}{m}\tau \dots \dots \dots (ii)$$

We have the current density is

$$J = nev_d$$

$$\text{or, } v_d = \frac{J}{ne} \dots \dots \dots (iii)$$

From (ii) and (iii)

$$\frac{J}{ne} = \frac{eE}{m}\tau$$

$$\text{or, } J = \frac{ne^2\tau}{m} E$$

We have, $J = \sigma E$ where σ is conductivity

$$\therefore \sigma E = \frac{ne^2\tau}{m} E$$

$$\text{or, } \sigma = \frac{ne^2\tau}{m}$$

This gives the conductivity of metal.

Now the resistivity is given by

$$\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$$

If λ be the mean free path and v_{eff} be the effective speed of electron then

$$\lambda = v_{eff} \times \tau$$

$$\therefore \tau = \frac{\lambda}{v_{eff}}$$

$$\text{Hence, } \rho = \frac{m}{ne^2\tau} = \frac{mv_{eff}}{ne^2\lambda}$$

Also we have, $E = \frac{1}{2}mv_{eff}^2 = \frac{3}{2}kT$ where k is Boltzmann's constant.

$$\therefore m = \frac{3kT}{v_{eff}^2}$$

$$\text{Hence } \rho = \frac{mv_{eff}}{ne^2\lambda} = \frac{3kT}{v_{eff}^2} \times \frac{v_{eff}}{ne^2\lambda} = \frac{3kT}{ne^2\lambda v_{eff}}$$

b) In Semiconductor

If n_e be the density of free electron in the conduction band of semiconductor and v_{de} be the drift speed under the application of electric field then electron current density is

$$J_e = n_e e v_{de} = n_e e \mu_e E \quad (\text{Since, } \mu_e = \frac{v_{de}}{E} \text{ is electron mobility})$$

$$\text{or, } J_e = \sigma_e E$$

Where, $\sigma_e = n_e e \mu_e$ is the electron conductivity in the conduction band.

Similarly for holes, $\sigma_h = n_h e \mu_h$

Now total conductivity $\sigma = \sigma_e + \sigma_h$

$$\text{or, } \sigma = e(n_e \mu_e + n_h \mu_h)$$

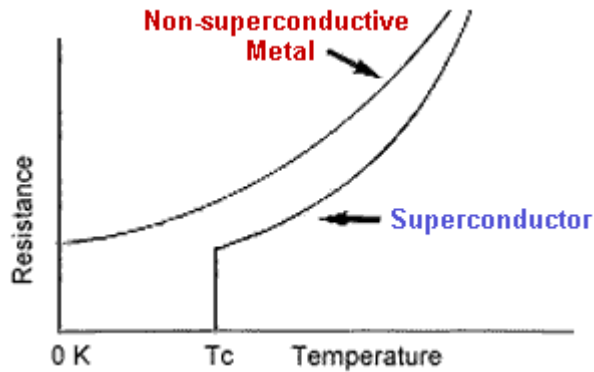
For intrinsic semiconductor, $n_e = n_h = n_i$ then

$$\sigma_i = n_i e (\mu_e + \mu_h)$$

$$\text{And } \rho_i = \frac{1}{\sigma_i} = \frac{1}{n_i e (\mu_e + \mu_h)}$$

Superconductor

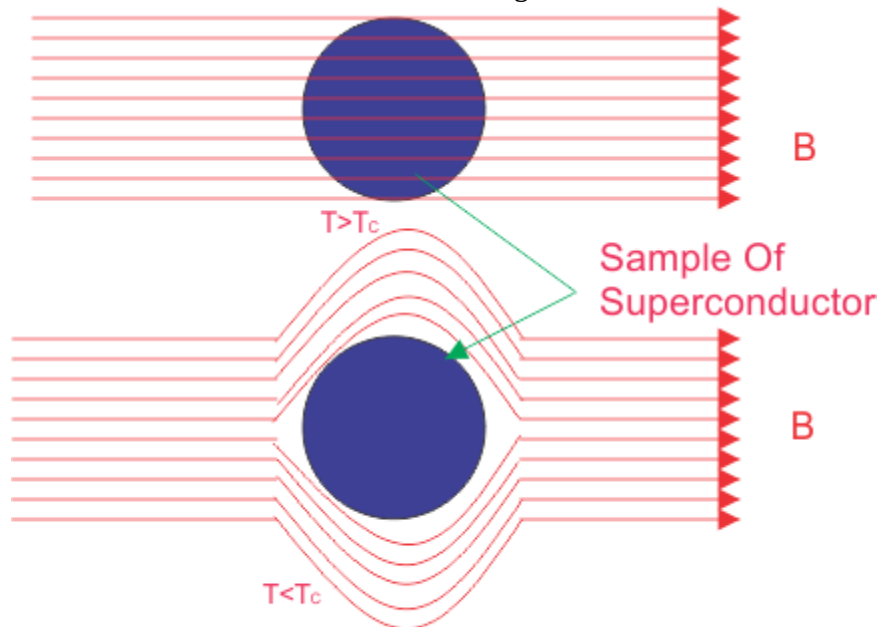
The electrical resistance of metals and alloys decreases as the temperature is lowered. If we study the variation of resistance with temperature, it is found that at very low temperature, the resistance becomes zero. At about 4.2K, the resistance of mercury falls sharply and below this temperature, mercury shows no resistance at all.



This phenomenon in which the electrical resistance suddenly drops to zero when the material is cooled to sufficiently low temperature is called superconductivity and the material is called superconductor. The temperature at which resistance of material suddenly falls to zero is called critical temperature. It is denoted by T_c .

Characteristics of Superconductor

1. If a ring of superconductor is cooled in a magnetic field to a temperature below critical temperature, an induced current is set up in the ring. This current in the superconducting ring flows almost infinitely till the temperature and magnetic field remain unchanged. Such current are called persistent current.
2. A Superconductor, when it is cooled below the critical temperature T_c , expel the magnetic field and doesn't allow the magnetic field to penetrate inside it. This phenomenon in superconductors is called Meissner effect. The meissner effect is shown in figure below.



Meissner Effect

3. If a superconductor is placed in a sufficiently strong magnetic field, the superconductor becomes a normal conductor. The value of magnetic field at which superconductivity is destroyed is called critical magnetic field H_c (function of temperature) which is given by

$$H_c(T) = H_c(0) \left[1 - \frac{T^2}{T_c^2} \right]$$

where $H_c(0)$ is critical magnetic field at 0K.

Direct Current

1. A copper wire of cross-sectional area $5 \times 10^{-6} \text{ m}^2$ carries a steady current of 50A. Assuming one free electron per atoms, calculate (i) Free electron density and (ii) average drift velocity. Density of copper = $8.9 \times 10^3 \text{ kg/m}^3$, Molar mass of copper = 64 and Avogadro's number = $6.02 \times 10^{23} \text{ mol}^{-1}$

→ Here, Cross – sectional area $A = 5 \times 10^{-6} \text{ m}^2$, Current $I = 50 \text{ A}$, Density $d = 8.9 \times 10^3 \text{ kg/m}^3$, Molar mass $M = 64 \text{ gm} = 64 \times 10^{-3} \text{ kg}$, Avogadro's number $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

(i) We have, Free electron density $n = \frac{dN_A}{M} = ?$

(ii) Again, $I = nev_d A$

$$\therefore \text{drift velocity } v_d = \frac{I}{neA} = ?$$

2. A typical copper wire of household wiring has a cross-sectional area $3.31 \times 10^{-6} \text{ m}^2$ and carries a current of 10A. What is the drift speed of the electrons? Density of copper is 8.95 gmcm^{-3} , Avogadro's number $6.02 \times 10^{23} \text{ mol}^{-1}$, molar mass of copper = 63.5gm.

→ Here, Cross – sectional area $A = 3.31 \times 10^{-6} \text{ m}^2$, Current $I = 10 \text{ A}$, Density $d = 8.95 \text{ gmcm}^{-3} = 8.95 \times 10^3 \text{ kg/m}^3$, Molar mass $M = 63.5 \text{ gm} = 63.5 \times 10^{-3} \text{ kg}$, Avogadro's number $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

Same as 1.

3. Calculate the drift speed of electrons when 10A current is supplied through a copper wire of cross-sectional area 1 mm^2 and electron density 10^{28} m^{-3} .

→ Same as 1.

4. If the copper wire connected between car battery to the starting motor has a cross-sectional area 0.21 cm^2 and length 0.85m carries a current of 300 A, calculate the time required for the electrons to get from the battery to the motor. Given, no. of electrons per unit volume is $8.5 \times 10^{28} \text{ m}^{-3}$.

→ Here, $A = 0.21 \text{ cm}^2 = 0.21 \times 10^{-4} \text{ m}^2$, length $l = 0.85 \text{ m}$, $I = 300 \text{ A}$, $n = 8.5 \times 10^{28} \text{ m}^{-3}$

We have $I = nev_d A$

$$\therefore \text{drift velocity } v_d = \frac{I}{neA} = ?$$

Now time required $t = \frac{l}{v_d} = ?$

5. The current density in a cylindrical wire of radius $R = 2 \text{ mm}$ is uniform across a cross-section of the wire and is $2 \times 10^5 \text{ A/m}^2$. What is the current through the outer section of the wire between radial distances 1mm and 2mm?

→ Current $I = J \times A = J \times \pi(R^2 - r^2) = 2 \times 10^5 \times \pi[(2 \times 10^{-3})^2 - (1 \times 10^{-3})^2] = ?$

6. The magnitude J of the current density in a certain wire with a circular cross-section of radius $R = 2 \text{ mm}$ is given by $J = 3 \times 10^8 r^2$, with J in A/m^2 and radial distance r in meters. What is the current through outer section bounded by $r = 0.9R$ and $r = R$?

$$\rightarrow I = \int \vec{j} \cdot d\vec{a} = \int_{0.9R}^R 3 \times 10^8 r^2 (2\pi r dr) = 6 \times 10^8 \times \pi \int_{0.9R}^R r^3 dr = 6 \times 10^8 \times \pi \times \left[\frac{r^4}{4} \right]_{0.9R}^R = ?$$

7. A wire with a resistance of 6Ω is drawn out through a die so that its new length is three times its original length. Find the resistance of longer wire.

→Original resistance $R=6\Omega$

$$\text{We have, } R = \frac{\rho l}{A} \dots \dots \dots (i)$$

$$\text{when length is made three times, } R' = \frac{\rho l'}{A'} = \frac{\rho \times 3l}{A'}$$

Since, volume remains constant,

$$\therefore Al = A'l'$$

$$\text{or, } A' = \frac{Al}{l'} = \frac{Al}{3l} = \frac{A}{3}$$

$$\text{Hence, } R' = \frac{\rho \times 3l}{A'} = \frac{\rho \times 3l}{A/3} = 9 \times \frac{\rho l}{A} = 9R = 9 \times 6 = 54\Omega$$

8. Two conductors are made of same material and have same length. Conductor A is a solid wire of diameter 1mm. Conductor B is a hollow tube of outside diameter 2mm and inside diameter 1mm. What is the resistance ratio R_A/R_B , measured between their ends?

→ Here, $\rho_A = \rho_B$, $l_A = l_B$, $d_A = 1\text{mm}$, $d_{B1} = 2\text{mm}$, $d_{B2} = 1\text{mm}$

$$\text{area of cross – section of conductor A, } A_A = \frac{\pi d_A^2}{4}$$

$$\text{And area of cross-section of conductor B, } A_B = \frac{\pi(d_{B1}^2 - d_{B2}^2)}{4}$$

$$\text{Now, } \frac{R_A}{R_B} = \frac{\frac{\rho_A l_A}{A_A}}{\frac{\rho_B l_B}{A_B}} = \frac{(d_{B1}^2 - d_{B2}^2)}{d_A^2} = ?$$

9. A cylindrical resistor of radius 5mm and length 2cm is made of material that has a resistivity of $3.5 \times 10^{-5} \Omega\text{m}$. What are (a) the magnitude of current density and (b) the potential difference when the energy dissipation rate in the resistor is 1 Watt?

→Here, radius $r=5\text{mm}=0.005\text{m}$, length $l = 2\text{cm} = 0.02\text{m}$, resistivity $\rho = 3.5 \times 10^{-5} \Omega\text{m}$, power $P=1\text{Watt}$

$$\text{a) We have, resistance } R = \frac{\rho l}{A} = \frac{\rho l}{\pi r^2} = ?$$

$$\text{And power } P = i^2 R$$

$$\therefore \text{current } i = \sqrt{\frac{P}{R}} = ?$$

$$\text{Now current density } J = \frac{i}{A} = \frac{i}{\pi r^2} = ?$$

$$\text{b) P.D. } V = iR = ?$$

10. Two copper wires of same length l and cross-sectional areas A and $2A$ are connected to a battery. What will be the ratio of drift velocities when the wires are in (i) series (ii) parallel?

→Here, $l_1 = l_2 = l$, $A_1 = A$, $A_2 = 2A$

$$\text{We have drift velocity } v_d = \frac{I}{neA}$$

$$\therefore \frac{v_{d1}}{v_{d2}} = \frac{I_1}{I_2} \times \frac{A_2}{A_1}$$

i) When the wires are in series, $I_1 = I_2$

$$\therefore \frac{v_{d1}}{v_{d2}} = \frac{I_1}{I_2} \times \frac{A_2}{A_1} = \frac{2A}{A} = 2:1$$

ii) When in parallel, $V_1 = V_2$

$$\text{or, } I_1 R_1 = I_2 R_2$$

$$\text{or, } I_1 \frac{\rho l}{A_1} = I_2 \frac{\rho l}{A_2}$$

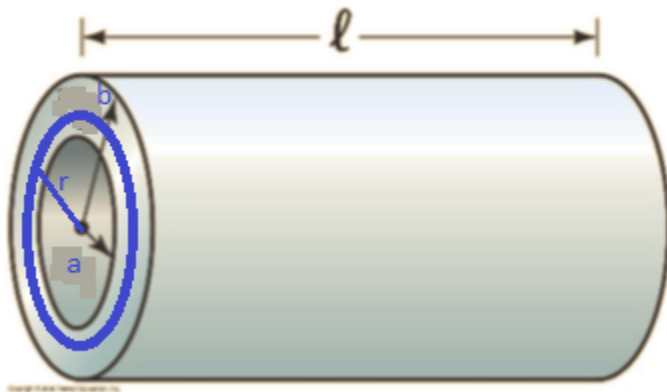
$$\text{or, } \frac{I_1}{I_2} = \frac{A_1}{A_2}$$

$$\therefore \frac{v_{d1}}{v_{d2}} = \frac{I_1}{I_2} \times \frac{A_2}{A_1} = \frac{A_1}{A_2} \times \frac{A_2}{A_1} = 1:1$$

11. The space between two metallic coaxial cylinders of radii a and b is filled with a material of resistivity ρ . If the cylinder has length l , show that the radial resistance between the cylinders is given by $R = \frac{\rho}{2\pi l} \ln \frac{b}{a}$.

→ We have resistance of a cylindrical shell of radius r and thickness dr is

$$dR = \frac{\rho dr}{2\pi r l}$$



∴ total resistance between the cylinder is

$$R = \int_a^b dR = \frac{\rho}{2\pi l} \int_a^b \frac{dr}{r} = \frac{\rho}{2\pi l} \ln \left(\frac{b}{a} \right)$$

12. A conductor of uniform radius 1.2cm carries a current of 3A due to potential gradient of 120V/m. What is the specific resistance of the material?

→ Here, radius $r = 1.2\text{cm} = 0.012\text{m}$, current $I = 3\text{A}$, Potential gradient $E = 120\text{V/m}$

We have $J = \sigma E$

$$\text{or, } \frac{I}{A} = \frac{1}{\rho} E$$

$$\text{or, } \frac{I}{\pi r^2} = \frac{1}{\rho} E$$

$$\therefore \rho = ?$$

13. If the temperature of an iron filament changes by 25°C , what will be the fractional change in resistance? $\alpha = 5 \times 10^{-3} \text{K}^{-1}$.

→ We have, $R = R_0[1 + \alpha(T_2 - T_1)]$

$$\therefore \frac{R - R_0}{R_0} = \alpha (T_2 - T_1) = ?$$

14. The resistance of a wire at 30°C is 3 ohm. At what temperature will the resistance be 5 ohm? The temperature coefficient of the wire is $5.66 \times 10^{-3} \text{ } ^\circ\text{C}^{-1}$.

→ We have, $R = R_0[1 + \alpha(T_2 - T_1)]$

$$\text{or, } 5 = 3[1 + 5.66 \times 10^{-3} \times (T_2 - 30)]$$

or, $T_2 = ?$

15. At what temperature would the resistance of copper conductor be double its resistance at 20°C? $\alpha =$

$$4.3 \times 10^{-3} / \text{K}$$

→ We have, $R = R_0[1 + \alpha(T_2 - T_1)]$

$$\text{or, } 2R_0 = R_0[1 + 4.3 \times 10^{-3} \times (T_2 - 20)]$$

or, $T_2 = ?$

16. Calculate the mean free time and mean free path between collisions for the conduction electrons in copper having electron density $8.5 \times 10^{28} \text{ m}^{-3}$ and resistivity $1.7 \times 10^{-8} \Omega \text{ m}$ Charge of electron is $1.6 \times 10^{-19} \text{ C}$, mass of electron is $9.1 \times 10^{-31} \text{ kg}$, effective speed of electron is $1.6 \times 10^6 \text{ m/s}$.

→ Here, $n = 8.5 \times 10^{28} \text{ m}^{-3}$, $\rho = 1.7 \times 10^{-8} \Omega \text{ m}$, $e = 1.6 \times 10^{-19} \text{ C}$, $m = 9.1 \times 10^{-31} \text{ kg}$, $v_{eff} = 1.6 \times 10^6 \text{ m/s}$

$$\text{We have, } \sigma = \frac{ne^2\tau}{m}$$

$$\therefore \rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$$

$$\text{or, } \tau = \frac{m}{ne^2\rho} = ?$$

Again, mean free path $\lambda = v_{eff}\tau = ?$

17. What will be the conductivity of sodium metal having electron density $2.5 \times 10^{28} \text{ m}^{-3}$ and relaxation time $3 \times 10^{-14} \text{ sec}$?

$$\rightarrow \text{We have, } \sigma = \frac{ne^2\tau}{m} = ?$$

18. A silicon sample has been doped with 10^{17} arsenic atoms/cm³. Calculate the conductivity of the sample at 300K and 400K. The mobilities of electron at 300K and 400 K are $800 \text{ cm}^2/\text{Vs}$ and $420 \text{ cm}^2/\text{Vs}$ respectively.

$$\rightarrow \text{We have, } \sigma_{300} = ne\mu_{300} = ?$$

$$\text{And } \sigma_{400} = ne\mu_{400} = ?$$

19. The intrinsic carrier density of Ge at 27°C is $2.4 \times 10^{17} \text{ m}^{-3}$. Calculate its intrinsic resistivity if the electron and hole mobilities are $0.35 \text{ m}^2 \text{ V}^{-1} \text{ S}^{-1}$ and $0.018 \text{ m}^2 \text{ V}^{-1} \text{ S}^{-1}$.

→ We have, $\rho_i = \frac{1}{n_i e (\mu_e + \mu_h)} = ?$

20. A superconducting state of lead specimen has critical temperature 6.2 K at zero magnetic field and the critical field is 0.064×10^6 A/m at 0 K. Estimate the critical field at 5K.

→ We have, $H_c(T) = H_c(0) \left[1 - \frac{T^2}{T_c^2} \right] = ?$

21. A copper wire and iron wire of equal length l and diameter d joined and a p.d. V is applied between the ends of the composite wire. Calculate, a) the electric field strength in each wire b) the current density in each wire and c) the p.d. across each wire. Assume that $l = 10\text{m}$, $d = 2\text{mm}$ and $V = 100\text{volts}$, resistivity of copper = $1.7 \times 10^{-8}\text{ohm-m}$ and resistivity of iron = $1 \times 10^{-7}\text{ohm-m}$

→ We have resistance of copper wire, $R_1 = \frac{\rho_1 l}{A} = \frac{\rho_1 l}{\pi d^2} \times 4 = ?$

And resistance of iron wire, $R_2 = \frac{\rho_2 l}{A} = \frac{\rho_2 l}{\pi d^2} \times 4 = ?$

c) P.D. across copper wire, $V_1 = \frac{R_1}{R_1 + R_2} V = ?$

P.D. across iron wire, $V_2 = \frac{R_2}{R_1 + R_2} V = ?$

a) Electric field for copper, $E_1 = \frac{V_1}{l} = ?$

Electric field for iron, $E_2 = \frac{V_2}{l} = ?$

b) Current through the composite wire, $I = \frac{V}{R} = \frac{V}{R_1 + R_2} = ?$

So current density, $J = \frac{I}{A} = \frac{I}{\pi d^2} \times 4 = ?$

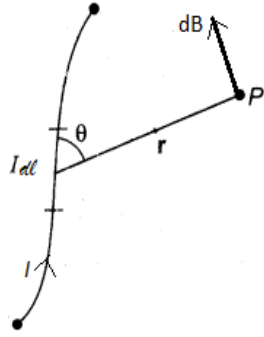
Chapter: Magnetic force & Field

- Magnetic force acting on a moving charge particle is $F = Bqv \sin \theta$, where θ is the angle between B and v .
- Magnetic force on a current carrying conductor is $F = BIl \sin \theta$, where θ is the angle made by conductor with magnetic field.
- Magnetic dipole moment $\mu = NIA$
- Torque on a current loop $\tau = BIN A \sin \theta = \mu B \sin \theta$
where θ is the angle made by the loop with direction of magnetic field.
- Potential energy needed to change the orientation of magnetic dipole $U = -\mu B \cos \theta$

Biot-Savart Law

The magnitude of magnetic field dB at point P due to small current element Idl at a distance r

is $dB \propto \frac{Idl \sin \theta}{r^2}$



$$\text{or, } dB = K \frac{Idl \sin \theta}{r^2}$$

Where, $K = \frac{\mu_0}{4\pi}$ in S.I.

(Here, $\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$ is permeability of free space)

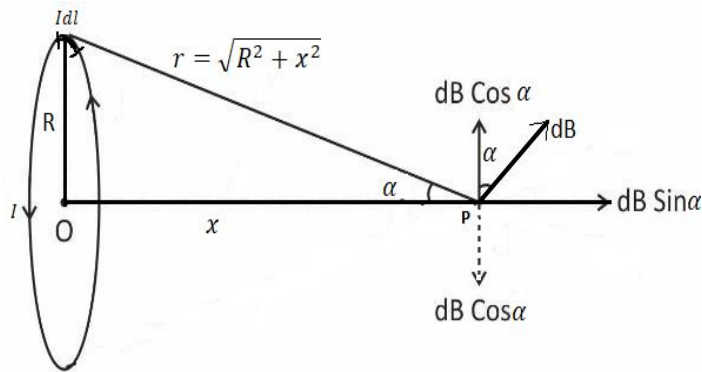
and $K = 1$ in c. g. s.

In vector form $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I(d\vec{l} \times \vec{r})}{r^3}$

Application

1. Magnetic field on the axis of circular current carrying loop

Consider a circular loop of wire of radius R carrying current I . Consider a point P on the axis of circular loop at a distance x from its center at which magnetic field is to be determined.



According to Biot-Savart law, magnetic field at P due to current element Idl is

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin 90^\circ}{r^2} = \frac{\mu_0 Idl}{4\pi r^2}$$

Here, $dB \cos \alpha$ components of magnetic field cancel each other and the effective components are $dB \sin \alpha$.

$\therefore$ net magnetic field at P is

$$B = \int dB \sin \alpha = \int_0^{2\pi R} \frac{\mu_0}{4\pi} \frac{Idl}{r^2} \frac{R}{r} = \frac{\mu_0 IR}{4\pi r^3} \int_0^{2\pi R} dl = \frac{\mu_0 IR}{4\pi r^3} \times 2\pi R = \frac{\mu_0 IR^2}{2 r^3}$$

$$\text{or, } B = \frac{\mu_0 IR^2}{2(R^2 + x^2)^{3/2}}$$

If the coil consists of N number of turns then total magnetic field is

$$B = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}}$$

Case i: At the center of loop, $x = 0$ so $B = \frac{\mu_0 N I}{2R}$.

This gives the maximum value.

Case ii: When $x \gg R$ then $B = \frac{\mu_0 N I R^2}{2x^3}$

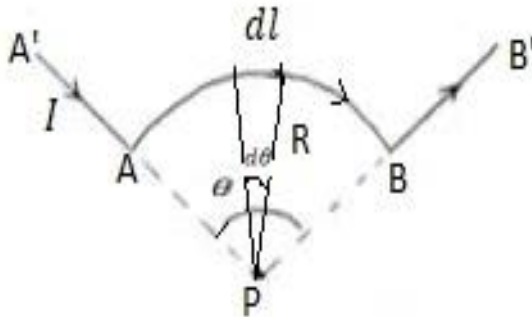
We have the magnetic dipole moment is $\mu = N I A = N I \times \pi R^2$

$$\therefore B = \frac{\mu_0 \mu}{2\pi x^3}$$

This is the magnetic field due to a magnetic dipole. Hence a circular loop behaves as a magnetic dipole for large distance.

2. Magnetic field due to curved wire segment

Consider a curved wire segment carrying current I which consists of three segments $A'A$, AB and BB' . Let B_1 , B_2 and B_3 be the magnetic fields at P (center of arc AB) due to segments $A'A$, AB and BB' respectively.



Here, $B_1 = B_3 = 0$ (Since, $d\vec{l} \times \vec{r} = 0$ for these sections)

Now, magnetic field at P due to current element $I dl$ on the segment AB is

$$dB_2 = \frac{\mu_0 I dl \sin 90^\circ}{4\pi R^2} = \frac{\mu_0 I dl}{4\pi R^2} = \frac{\mu_0 I R d\theta}{4\pi R^2} = \frac{\mu_0 I}{4\pi R} d\theta$$

$$\therefore B_2 = \int dB_2 = \int_0^{\pi/2} \frac{\mu_0 I}{4\pi R} d\theta = \frac{\mu_0 I}{4\pi R} \int_0^{\pi/2} d\theta = \frac{\mu_0 I}{4\pi R} \times \pi/2 = \frac{\mu_0 I}{8R}$$

Hence, net magnetic field at P is $B = B_1 + B_2 + B_3 = \frac{\mu_0 I}{8R}$

Ampere's Law

It states that the line integral of magnetic field around any closed path is equal to μ_0 times the current enclosed by that closed path.

$$i.e. \oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

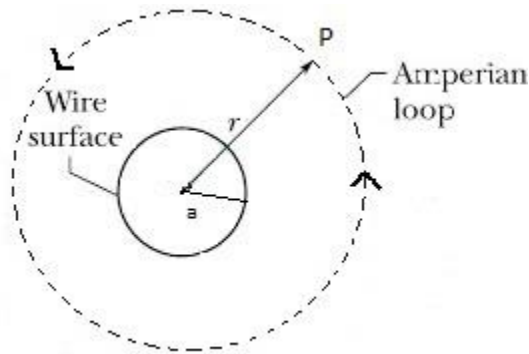
Application

Magnetic Field due to Long Straight Wire

Consider a long straight wire of radius a and carrying current I which is distributed uniformly throughout the conductor.

a) At a point outside the conductor ($r > a$)

Figure shows the cross-section of a long straight wire.



To find the magnetic field for the region $r > a$, we draw a circular Amperian loop of radius r ($r > a$) that encloses the current I .

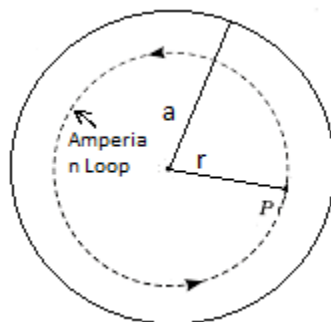
From Ampere's Law,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$or, B \times 2\pi r = \mu_0 I$$

$$or, B = \frac{\mu_0 I}{2\pi r}$$

b) At a point inside the conductor ($r < a$)



To find the magnetic field for the region $r < a$, we draw a circular Amperian loop of radius r ($r < a$) that encloses the current I' .

From Ampere's Law,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I'$$

$$\text{or, } B \times 2\pi r = \mu_0 I'$$

$$\text{or, } B = \frac{\mu_0 I'}{2\pi r}$$

Since, current density is same throughout the conductor,

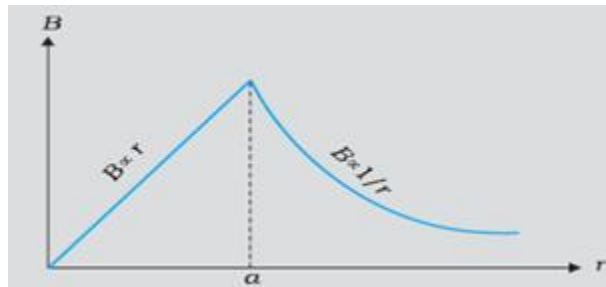
$$\therefore J = \frac{I}{\pi a^2} = \frac{I'}{\pi r^2}$$

$$\text{or, } I' = \frac{r^2}{a^2} I$$

$$\text{Hence, } B = \frac{\mu_0}{2\pi r} \times \frac{r^2}{a^2} I$$

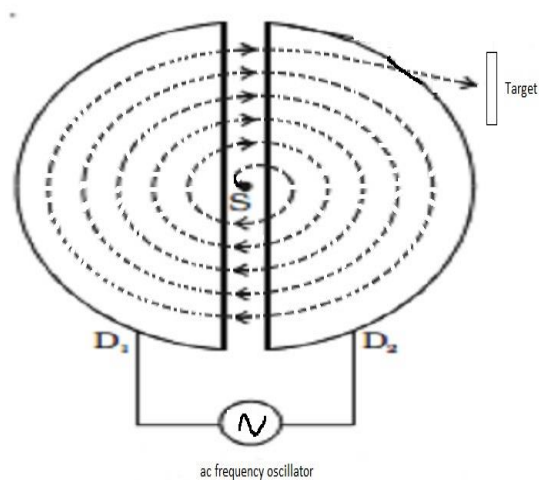
$$\text{or, } B = \left(\frac{\mu_0 I}{2\pi a^2} \right) r$$

At the center, $B = 0$ and on the surface, $B = \frac{\mu_0 I}{2\pi a}$ (maximum value)



Cyclotron

It is a device which is used to accelerate the positively charged particles (proton, deuteron, α – particle etc) to acquire enough energy to carry out nuclear disintegration.



A cyclotron consists of two hollow semicircular metallic surfaces called dee's. These two dee's are connected to a powerful ac frequency oscillator. A source of positively charged particle is placed at the midpoint of two dee's. The whole apparatus is then placed between two poles of a strong electromagnet. The magnetic field due to this magnet is perpendicular to the plane of dee's.

The necessary centripetal force for the charged particle to move in the circular path is provided by the magnetic force.

$$\text{i.e. } Bqv = \frac{mv^2}{r}$$

$$\text{or, } r = \frac{mv}{Bq}$$

Now, time taken by the charged particle to complete the semi-circle inside a dee is

$$t = \frac{\text{distance}}{\text{speed}} = \frac{\pi r}{v} = \frac{\pi}{v} \times \frac{mv}{Bq} = \frac{\pi m}{Bq}$$

which is independent of velocity of charged particle and radius of circular path.

Again, the time period of cyclotron is

$$T = 2t = \frac{2\pi m}{Bq}$$

$$\therefore \text{frequency, } f = \frac{1}{T} = \frac{Bq}{2\pi m}$$

$$\text{Also the angular frequency is } \omega = 2\pi f = \frac{Bq}{m}$$

Now the energy gained by the charged particle is

$$E = \frac{1}{2}mv^2 = \frac{1}{2}m \left\{ \frac{Bqr}{m} \right\}^2 = \frac{1}{2m} B^2 q^2 r^2$$

$$\text{Since, } f = \frac{Bq}{2\pi m}$$

$$\therefore Bq = 2\pi m f$$

$$\text{Hence, } E = \frac{1}{2m} \times 4\pi^2 m^2 f^2 \times r^2$$

$$\text{or, } E = 2\pi^2 f^2 r^2 m$$

Hence, the maximum energy of cyclotron is

$$E_{\max} = \frac{1}{2m} B^2 q^2 r_{\max}^2 = 2\pi^2 f^2 r_{\max}^2 m$$

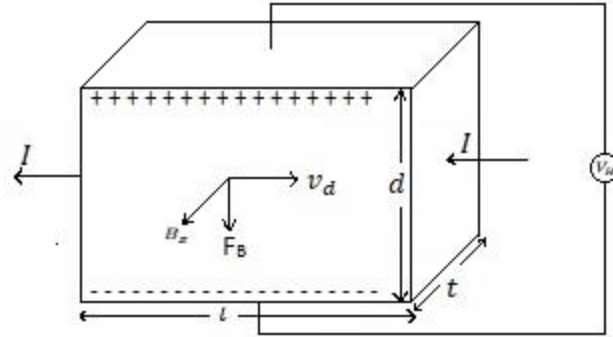
$\therefore$ energy is maximum on the periphery of dee's.

Limitations

1. It cannot accelerate the uncharged particles like neutron.
2. When $v \approx c$, it does not work properly.

Hall Effect

When a magnetic field is applied in a specimen perpendicular to the current, a voltage is developed across the specimen in the direction perpendicular to both current and magnetic field. This effect is called Hall effect and the voltage thus developed is called Hall voltage.



Consider a specimen having rectangular cross-section of length l , width d and thickness t carrying current I along x-axis. If the specimen is placed in a magnetic field B_z along z-axis, then hall voltage V_H is developed along y-axis. At equilibrium condition, the magnetic force directed downward is equal to upward electric force.

$$\text{i.e. } B_z q v_d = q E_H$$

$$\text{or, } E_H = B_z v_d \dots \dots \dots (i)$$

$$\text{we have, } I = -n e v_d A$$

$$\therefore J_x = \frac{I}{A} = -n e v_d$$

$$\text{or, } v_d = -\frac{J_x}{n e}$$

Now from equation (i)

$$E_H = -\frac{J_x B_z}{n e}$$

$$\text{or, } \frac{E_H}{J_x B_z} = -\frac{1}{n e}$$

$$\text{or, } R_H = -\frac{1}{n e}$$

Where, $\frac{E_H}{J_x B_z} = R_H$ is called Hall Coefficient.

Again from equation (i)

$$\frac{V_H}{d} = B_z v_d$$

$$\text{or, } \frac{V_H}{d} = -\frac{IB_z}{neA} \text{ (Since, } I = -nev_d A \text{)}$$

$$\text{or, } \frac{V_H}{d} = -\frac{IB_z}{ne(dt)} \quad (\text{Since, } A = dt)$$

$$\text{or, } V_H = -\frac{IB_z}{net}$$

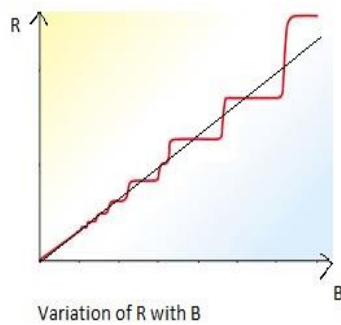
This gives Hall voltage.

Now, the Hall resistance is given by

$$R = \frac{V_H}{I} = \frac{B_z}{net}$$

Also the Hall mobility is given by

$$\mu_H = \frac{v_d}{E_H} = \frac{J_x}{neE_H} = \frac{\sigma}{ne} = \sigma R_H = \frac{R_H}{\rho}$$



Applications

1. It is used for the accurate measurement of magnetic field, Hall mobility etc.
2. It is used to determine the nature of charge carriers.
3. It is used to determine whether the specimen is metal, semiconductor or insulator.

Magnetic Force & Field

Number of electrons per unit volume or electron density $n = \frac{dN_A}{M}$

1. An electron is projected into a magnetic field of flux density 10T with a velocity of $3 \times 10^7 \text{ m/s}$ in a direction at right angles to the field. Compute the magnetic force on the electron and compare with the weight of electron.

→ Here, Magnetic Flux density $B = 10\text{T}$, velocity $v = 3 \times 10^7 \text{ m/s}$, $\theta = 90^\circ$

We have magnetic force, $F = Bqv \sin \theta = Bev = ?$

Weight of electron $F' = mg = ?$ (Take $m = 9.1 \times 10^{-31} \text{ kg}$)

Here, weight of electron is negligible in comparison to magnetic force.

2. A wire of length L and carries a current I . If the wire is formed into a circular coil then prove the maximum torque in a given magnetic field B developed for a single turn is $\tau = L^2 IB / 4\pi$.

→ Here, $L = 2\pi r$, $N = 1$

∴ radius of circular coil $r = \frac{L}{2\pi}$

We have torque $\tau = BIN A \sin \theta$

$$\therefore \tau_{\max} = BINA = BIA = BI \times \pi r^2 = BI \times \pi \times \frac{L^2}{4\pi^2} = \frac{L^2 IB}{4\pi}$$

3. The magnitude of dipole moment of earth is $8 \times 10^{22} \text{ J/T}$. Assume that this produced by charges flowing in the molten outer core of the earth. If the radius of their circular path is 3500km, calculate the current they produce.

→ Here, magnetic dipole moment $\mu = 8 \times 10^{22} \text{ J/T}$, radius $r = 3500 \text{ km} = 3500000 \text{ m}$

We have, $\mu = IA = I \times \pi r^2$

$$\therefore I = \frac{\mu}{\pi r^2} = ?$$

4. A coil of wire has an area of $2 \times 10^{-4} \text{ m}^2$, consists of 100 turns and contains the current of 0.045A. The coil is placed in a uniform magnetic field of magnitude 0.15T. Determine the magnetic moment of the coil. Also calculate the maximum torque that the magnetic field can exert in the coil.

→ Here, area $A = 2 \times 10^{-4} \text{ m}^2$, number of turns $N = 100$, Current $I = 0.045 \text{ A}$,

Magnetic Field $B = 0.15 \text{ T}$

We have magnetic moment $\mu = NIA = ?$

torque $\tau = BIN A \sin \theta$

$$\therefore \tau_{\max} = BINA = ?$$

5. A coil of magnetic moment 1.45 Am^2 is oriented initially with its magnetic moment antiparallel to a uniform 0.835T magnetic field. What is the change in P.E. of the coil when it is rotated 180° so that its magnetic moment is parallel to the field.

→ Here, $\mu = 1.45 \text{ Am}^2$, $B = 0.835 \text{ T}$

We have, initial potential energy $U_1 = -\mu B \cos 180^\circ$

Final potential energy $U_2 = -\mu B \cos 0^\circ$

$\therefore$ Change in P.E. $= U_2 - U_1 = ?$

6. A circular coil of 160 turns has a radius of 1.9cm. (a) Calculate the current that results in a magnetic dipole moment of magnitude 2.3Am^2 . (b) Find the maximum magnitude of torque that the coil carrying this current can experience in a uniform 35mT magnetic field.

$\rightarrow$ Here, $N = 160, r = 1.9\text{cm} = 0.019\text{m}, \mu = 2.3\text{Am}^2$

a) We have, $\mu = NIA = NI \times \pi r^2$

$$\therefore I = \frac{\mu}{N\pi r^2} = ?$$

b) Again,

torque $\tau = BIN A \sin \theta$

$$\therefore \tau_{\max} = BIN A = \mu B = ?$$

7. A long circular coil consisting of 100 turns with diameter 1.2m carries a current of 5A. (a) find the magnetic field at a point along the axis 80cm from the center. (b) At what distance from the center, along the axis, is the field magnitude $1/8$ as great as at the center?

$\rightarrow$ Here, $N = 100, R = 0.6\text{m}, I = 5\text{A}$

a) We have magnetic field at a distance x on the axis is

$$B = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}} = ?$$

b) By question, $B_x = \frac{1}{8} B_0$

$$\text{or, } \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}} = \frac{1}{8} \frac{\mu_0 N I}{2R}$$

$$\text{or, } \frac{R^2}{(R^2 + x^2)^{3/2}} = \frac{1}{8R}$$

$$\text{or, } (R^2 + x^2)^{3/2} = 8R^3$$

$$\text{or, } (R^2 + x^2)^{1/2} = 2R$$

$$\text{or, } R^2 + x^2 = 4R^2$$

$$\text{or, } x = \sqrt{3}R = ?$$

8. What current must be passed through a flat circular coil of 10 turns and radius 5cm to produce a flux density of $2 \times 10^{-4}\text{T}$ at its center?

$\rightarrow$ Here, $N = 10, R = 5\text{cm} = 0.05\text{m}, B = 2 \times 10^{-4}\text{T}$

We have magnetic field at the center of circular coil is

$$B = \frac{\mu_0 N I}{2R}$$

$$\therefore I = ?$$

9. In a hydrogen atom of radius $5.3 \times 10^{-11} \text{m}$, an electron revolve round the nucleus with a speed of $2.2 \times 10^6 \text{m/s}$. Calculate the magnitude of magnetic field at the center of atom and magnetic dipole moment.

→ Here, $R = 5.3 \times 10^{-11} \text{m}$, $v = 2.2 \times 10^6 \text{m/s}$

We have current $I = \frac{q}{t} = \frac{e}{t} = ef = e \frac{\omega}{2\pi} = e \frac{v}{2\pi R} = ?$

Now, magnetic field at the center of circular coil is

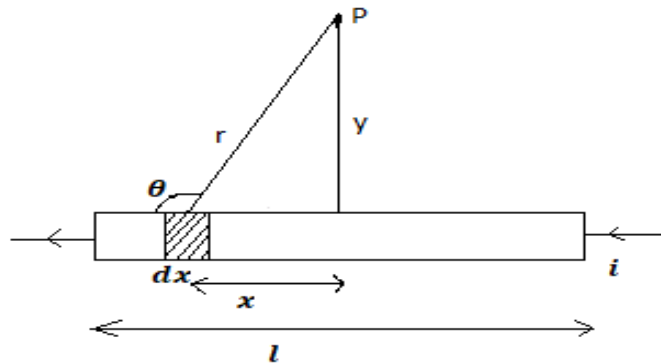
$$B = \frac{\mu_0 I}{2R} = ?$$

And magnetic dipole moment $\mu = IA = I \times \pi R^2 = ?$

10. A straight wire segment of length l carries current i . Show that the magnitude of magnetic field B produced by the segment at a distance y from it along a perpendicular bisector is $B = \frac{\mu_0 i}{2\pi y} \frac{l}{\sqrt{l^2 + 4y^2}}$

→ We have the magnitude of magnetic field at P due to current element $id\mathbf{x}$ is

$$dB = \frac{\mu_0}{4\pi} \frac{id\mathbf{x} \sin\theta}{r^2}$$



∴ net magnetic field at P is

$$B = 2 \int_0^{l/2} dB = 2 \int_0^{l/2} \frac{\mu_0}{4\pi} \frac{id\mathbf{x} \sin\theta}{r^2} = \frac{\mu_0 i}{2\pi} \int_0^{l/2} \frac{dx y}{r^2 r} = \frac{\mu_0 i y}{2\pi} \int_0^{l/2} \frac{dx}{r^3} = \frac{\mu_0 i y}{2\pi} \int_0^{l/2} \frac{dx}{(x^2 + y^2)^{3/2}}$$

$$\text{or, } B = \frac{\mu_0 i y}{2\pi} \frac{1}{y^2} \left[\frac{x}{(x^2 + y^2)^{1/2}} \right]_0^{l/2} = \frac{\mu_0 i}{2\pi y} \left[\frac{l/2}{(\frac{l^2}{4} + y^2)^{1/2}} \right] = \frac{\mu_0 i}{2\pi y} \frac{l}{\sqrt{l^2 + 4y^2}}$$

11. Calculate the magnetic flux density at a perpendicular distance at a point 2cm from a 20 cm long straight wire carrying current of 10A.

→ Here, $y = 2\text{cm} = 0.02\text{m}$, $l = 20\text{cm} = 0.2\text{m}$, $i = 10\text{A}$

We have $B = \frac{\mu_0 i}{2\pi y} \frac{l}{\sqrt{l^2 + 4y^2}} = ?$

12. A long wire is known to have a radius greater than 4mm and to carry a current that is uniformly distributed over its cross-section. The magnitude of magnetic field due to that current is 0.28mT at a point 4mm from the axis of wire and 0.20mT at a point 10mm from the axis of wire. What is the radius of the wire?

→ We have the magnetic field inside the wire is

$$B = \left(\frac{\mu_0 I}{2\pi a^2} \right) r$$

or, $0.28 \times 10^{-3} = \left(\frac{\mu_0 I}{2\pi a^2} \right) \times 4 \times 10^{-3} \dots \dots \dots (i)$

Again the magnetic field outside the wire is

$$B = \frac{\mu_0 I}{2\pi r}$$

or, $0.20 \times 10^{-3} = \frac{\mu_0 I}{2\pi \times 10 \times 10^{-3}} \dots \dots \dots (ii)$

Now, equation (ii)/(i) give

$$a = ?$$

13. In a certain cyclotron, a proton moves in a circle of radius 0.5m. The magnitude of magnetic field is 1.2 T. (a) What is the oscillator frequency? (b) What is the K.E. of the proton, in eV?

→ Here, radius of cyclotron $r = 0.5\text{m}$, magnetic field $B = 1.2\text{T}$, $q = e = 1.6 \times 10^{-19}\text{C}$,

Mass of proton, $m = 1.67 \times 10^{-27}\text{kg}$

a) We have frequency $f = \frac{Bq}{2\pi m} = ?$

b) $K.E = 2\pi^2 f^2 r^2 m = \dots \dots \dots J = \dots \dots \dots \times \frac{1}{1.6 \times 10^{-19}} \text{eV}$

14. A physicist designed a cyclotron to accelerate protons to $1/10^{\text{th}}$ the speed of light. The magnet used will produce a field of 1.4T Calculate (a) the radius of the cyclotron (b) the corresponding oscillator frequency.

→ Here, $v = \frac{c}{10} = 3 \times 10^7 \text{m/s}$, $B=1.4\text{T}$, $q = e = 1.6 \times 10^{-19}\text{C}$, $m = 1.67 \times 10^{-27}\text{kg}$

a) We have $r = \frac{mv}{Bq} = ?$

b) $f = \frac{Bq}{2\pi m} = ?$

15. A cyclotron is operated at an oscillator frequency of 12MHz and has radius 53cm. (a) What is the magnitude of magnetic field needed for deuterons to be accelerated in the cyclotron? (b) What is the resulting K.E. of deuterons?

→ Here, $f = 12\text{MHz} = 12 \times 10^6\text{Hz}$, $r = 53\text{cm} = 0.53\text{m}$, $q = e = 1.6 \times 10^{-19}\text{C}$,

$$m = 2m_p = 2 \times 1.67 \times 10^{-27}\text{kg}$$

a) We have, $f = \frac{Bq}{2\pi m}$

$$\therefore B = ?$$

b) K.E. $= 2\pi^2 f^2 r^2 m = ?$

- 16. Deuterons in a cyclotron describe a circle of radius 0.32m just before emerging from dees. The frequency of the applied emf is 10MHz. Find the flux density of the magnetic field and the velocity of deuterons emerging out of the cyclotron. Mass of deuteron $= 3.32 \times 10^{-27}\text{Kg}$, charge $= 1.6 \times 10^{-19}\text{C}$.**

→ Here, $r = 0.32\text{m}$, $f = 10\text{MHz} = 10 \times 10^6\text{Hz}$, $m = 3.32 \times 10^{-27}\text{kg}$, $q = 1.6 \times 10^{-19}\text{C}$

We have, $f = \frac{Bq}{2\pi m}$

$$\therefore B = ?$$

Again, $r = \frac{mv}{Bq}$

$$\therefore v = \frac{Bqr}{m} = ?$$

- 17. What radius is needed in proton synchrotron to atom particle energy of 10GeV, assuming that a guide field of 1.8wb/m^2 is available?**

→ We have total energy of proton $E = K.E. + \text{rest mass energy}$

or, $E = 10\text{GeV} + m_0 c^2 = 10 \times 10^9 \times 1.6 \times 10^{-19} + 1.67 \times 10^{-27} \times (3 \times 10^8)^2 \text{Joule}$

Again we have $E = mc^2$

$$\therefore m = \frac{E}{c^2} = ?$$

Now, $r = \frac{mv}{Bq} = \frac{mc}{Be} = ?$

- 18. In a Hall effect experiment, a current of 3A sent lengthwise through a conductor 1cm wide, 4cm long, and $10\mu\text{m}$ thick produce a transverse Hall P.D. of $10\mu\text{V}$ when a magnetic field of 1.5T is passed perpendicularly through the thickness of the conductor. From these data, find (a) the drift velocities of the charge carriers and (b) the number density of the charge carrier.**

→ Here, current $I = 3\text{A}$, width $d = 1\text{cm} = 0.01\text{m}$, thickness $t = 10\mu\text{m} = 10 \times 10^{-6}\text{m}$,

Hall voltage $V_H = 10\mu\text{V} = 10 \times 10^{-6}\text{V}$, magnetic field $B_z = 1.5\text{T}$

b) We have, $V_H = \frac{IB_z}{net}$

$$\therefore n = ?$$

a) Again we have $I = nev_d A$

$$\therefore v_d = \frac{I}{neA} = \frac{I}{nedt} = ?$$

- 19. A copper strip 2cm wide and 1mm thick is placed in a magnetic field 1.5T. If a current of 200 A is set up in the strip, calculate (i) hall voltage and (ii) hall mobility if the number of electrons per unit volume is $8.4 \times 10^{28}\text{m}^{-3}$ and resistivity is $1.72 \times 10^{-9}\Omega\text{m}$.**

→ Here, $d = 2\text{cm} = 0.02\text{m}$, $t = 1\text{mm} = 0.001\text{m}$, $B_z = 1.5\text{T}$, $I = 200\text{A}$,

$$\rho = 1.72 \times 10^{-9}\Omega\text{m}, n = 8.4 \times 10^{28}\text{m}^{-3}$$

- a) Hall voltage $V_H = \frac{IB_z}{net} = ?$
 b) Hall mobility $\mu_H = \frac{1}{ne\rho} = ?$

20. A rectangular copper strip 1.5cm wide and 1cm thick carries a current of 5A. Find the hall voltage for a 1.2T magnetic field in a direction perpendicular to the strip. Assume electron density $8.49 \times 10^{28} \text{m}^{-3}$.

→ Here, $d = 1.5 \text{cm} = 0.015 \text{m}$, $t = 1 \text{cm} = 0.01 \text{m}$, $I = 5 \text{A}$, $B_z = 1.2 \text{T}$, $n = 8.49 \times 10^{28} \text{m}^{-3}$

we have Hall voltage $V_H = \frac{IB_z}{net} = ?$

21. A strip of copper $150 \mu\text{m}$ thick is placed in a magnetic field of strength 0.65T perpendicular to the plane of the strip and a current of 23A is set up in the strip. Calculate the hall voltage if $n = 8.5 \times 10^{28} \text{m}^{-3}$ and Hall coefficient.

→ Hall voltage $V_H = \frac{IB_z}{net} = ?$

and $R_H = \frac{1}{ne} = ?$

Chapter: Electromagnetic Induction

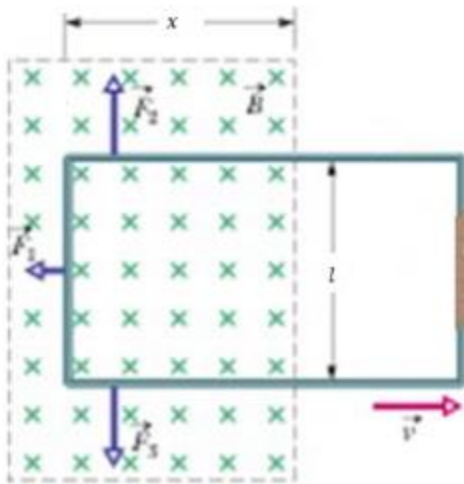
Faraday's Law of electromagnetic induction

It states that the induced emf in a circuit is equal to the negative of the time rate of change of magnetic flux through the circuit.

i.e. induced emf $\mathcal{E} = -\frac{d\Phi_B}{dt}$

Induction and Energy Transfer

Consider a rectangular conducting loop of width l is moving with velocity v under the action of some external force in a uniform magnetic field B . Here, magnetic field is applied perpendicular to the plane of loop.



The rate of doing work is power and is given by

$$P = Fv \dots \dots \dots (i)$$

The force acting on the conductor is

$$F = BIl \sin \theta = BIl$$

We have magnetic flux $\phi_B = BA = Blx$

From Faraday's law,

$$\text{Induced emf is } \varepsilon = \frac{d\phi_B}{dt} = Bl \frac{dx}{dt} = Blv$$

$$\therefore \text{Current } I = \frac{\varepsilon}{R} = \frac{Blv}{R}$$

$$\text{Hence, force on the conductor is } F = B \frac{Blv}{R} l = \frac{B^2 l^2 v}{R}$$

Now, from equation (i)

$$P = \frac{B^2 l^2 v}{R} v = \frac{B^2 l^2 v^2}{R}$$

Again, the rate at which the thermal energy appeared in the loop when the conductor is pulled or pushed is given by

$$P = I^2 R = \frac{B^2 l^2 v^2}{R}$$

$\therefore$ The work that we do in pulling the conductor through the magnetic field appears as thermal energy.

In short, mechanical energy $\rightarrow$ electrical energy $\rightarrow$ thermal energy

Self-induction

The phenomenon in which induced emf is produced due to the change of current passing through a coil is called self-induction.

The change in flux is proportional to the change in current passing through a coil.

$$\text{i.e. } d\phi_B \propto dI$$

$$\text{or, } d\phi_B = LdI$$

Where, L is constant called coefficient of self- induction or self- inductance.

$$\therefore L = \frac{d\phi_B}{dI}$$

According to Faraday's law of e-m induction, $\varepsilon = -\frac{d\phi_B}{dt}$

$$\text{or, } \varepsilon = -\frac{d\phi_B}{dI} \frac{dI}{dt}$$

$$\text{or, } \varepsilon = -L \frac{dI}{dt}$$

This is induced emf due to self- induction.

If $\frac{dI}{dt} = 1 \text{ A/s}$ then $\varepsilon = L$ (numerically)

Hence, when the rate of change of current is 1 A/s, then the induced emf is numerically equal to the self-inductance of the coil.

Self-Inductance of solenoid

The magnetic field inside a solenoid of length l carrying current I is

$$B = \mu_0 nI = \frac{\mu_0 NI}{l} \text{ where, } N \text{ is number turns in the solenoid.}$$

The magnetic flux linkage with the single turn is

$$\phi_B = BA = \frac{\mu_0 NI}{l} A$$

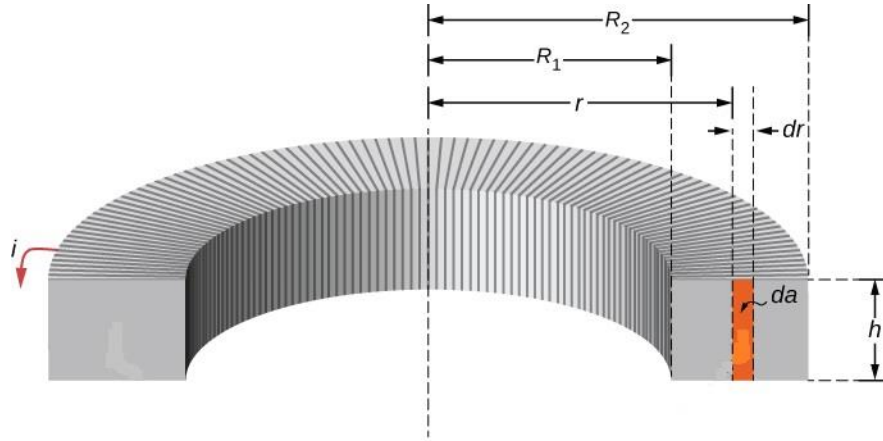
Now self -inductance is given by

$$L = \frac{N\phi_B}{I} = \frac{\mu_0 N^2 A}{l}$$

or, $L = \mu_0 n^2 A l$ which depends only on geometric factors but is independent of current.

Self-Inductance of toroid

Consider a toroid of rectangular cross-section having N number of turns carrying current I . Let R_1 and R_2 be the internal and external radii of the toroid. Consider an elementary section dr at a distance r from its center and h be the width of the strip.



We have the magnetic field inside the toroid is

$$B = \mu_0 n I = \frac{\mu_0 N I}{2\pi r} \text{ where, } N \text{ is number turns in the solenoid.}$$

The magnetic flux through the cross-section of the toroid is

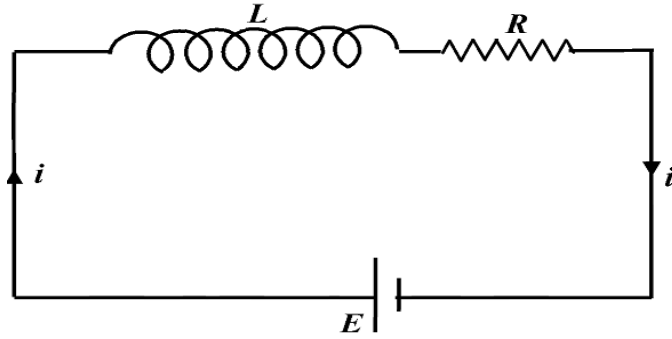
$$\Phi_B = \int \vec{B} \cdot d\vec{A} = \int_{R_1}^{R_2} \frac{\mu_0 N I}{2\pi r} (h dr) = \frac{\mu_0 N I h}{2\pi} \int_{R_1}^{R_2} \frac{dr}{r} = \frac{\mu_0 N I h}{2\pi} \ln\left(\frac{R_2}{R_1}\right)$$

Now self-inductance is given by

$$L = \frac{N \Phi_B}{I} = \frac{\mu_0 N^2 h}{2\pi} \ln\left(\frac{R_2}{R_1}\right) \text{ which depends only on geometric factors but is independent of current.}$$

Energy stored in Magnetic Field

Consider a circuit as shown in figure.



Using KVL,

$$E = IR + L \frac{dI}{dt}$$

$$\text{or, } IE = I^2 R + LI \frac{dI}{dt}$$

Here, the term $LI \frac{dI}{dt}$ represents the rate of energy stored in the magnetic field.

$$\therefore \frac{dU_B}{dt} = LI \frac{dI}{dt}$$

$$\text{or, } dU_B = LI dI$$

$$\therefore U_B = \int dU_B = \int_0^I LI dI = \frac{1}{2} LI^2$$

Hence the energy stored in the magnetic field is

$$U_B = \frac{1}{2} LI^2$$

Now magnetic energy density is given by

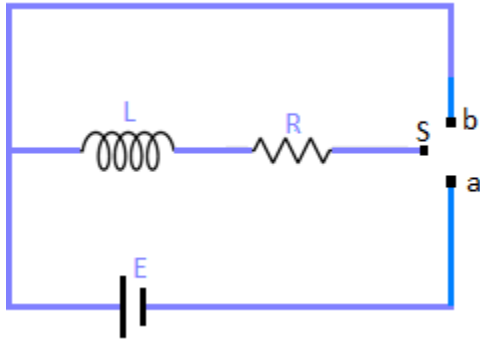
$$u_b = \frac{U_B}{\text{Volume}} = \frac{1}{2} LI^2 \times \frac{1}{Al} = \frac{\mu_0 n^2 Al I^2}{2Al} \quad (\text{Since, } L = \mu_0 n^2 Al)$$

$$\text{or, } u_b = \frac{(\mu_0 n I)^2}{2\mu_0}$$

$$\therefore u_b = \frac{B^2}{2\mu_0}$$

This shows that magnetic energy density is proportional to square of magnetic field.

LR-Circuit (Growth and Decay of Current)



- a) When switch S is connected to a then growth of current takes place.
Using KVL,

$$E = IR + L \frac{dI}{dt}$$

$$\text{or, } E - IR = L \frac{dI}{dt}$$

$$\text{or, } \frac{E}{R} - I = \frac{L}{R} \frac{dI}{dt}$$

$$\text{or, } \frac{dI}{\frac{E}{R} - I} = \frac{R}{L} dt$$

Integrating both sides,

$$\int \frac{dI}{\frac{E}{R} - I} = \frac{R}{L} \int dt$$

$$\text{or, } -\ln\left(\frac{E}{R} - I\right) = \frac{R}{L}t + k_1 \dots \dots \dots (i)$$

Where, k_1 is integration constant.

When $t = 0, I = 0$

$$\therefore k_1 = -\ln \frac{E}{R}$$

Hence from equation (i),

$$-\ln\left(\frac{E}{R} - I\right) = \frac{R}{L}t - \ln \frac{E}{R}$$

$$\text{or, } \ln \left\{ \frac{\frac{E}{R} - I}{\frac{E}{R}} \right\} = -\frac{R}{L}t$$

$$\text{or, } 1 - \frac{I}{E/R} = e^{-\frac{R}{L}t}$$

$\text{or, } 1 - \frac{I}{I_0} = e^{-\frac{R}{L}t}$ Where, $I_0 = E/R$ is maximum current through the circuit.

$$\text{or, } I = I_0(1 - e^{-\frac{R}{L}t})$$

This gives the expression for growth of current.

When, $t = \frac{L}{R} = \tau_L$ then $I = I_0(1 - e^{-1}) = 0.63I_0 = 63\% \text{ of } I_0$

Hence the time at which current through the inductor growth to 63% of the maximum current is called inductive time constant.

b) When switch S is connected to b then decay of current takes place.

Using KVL,

$$IR + L \frac{dI}{dt} = 0$$

$$\text{or, } IR = -L \frac{dI}{dt}$$

$$\text{or, } \frac{dI}{I} = -\frac{R}{L}dt$$

Integrating both sides,

$$\int \frac{dI}{I} = -\frac{R}{L} \int dt$$

$$\text{or, } \ln I = -\frac{R}{L}t + k_2 \dots \dots \dots (ii)$$

Where, k_2 is integration constant.

When $t = 0, I = I_0$

$$\therefore k_2 = \ln I_0$$

Hence from equation (ii),

$$\ln I = \frac{R}{L}t + \ln I_0$$

$$\text{or, } \ln \frac{I}{I_0} = -\frac{R}{L}t$$

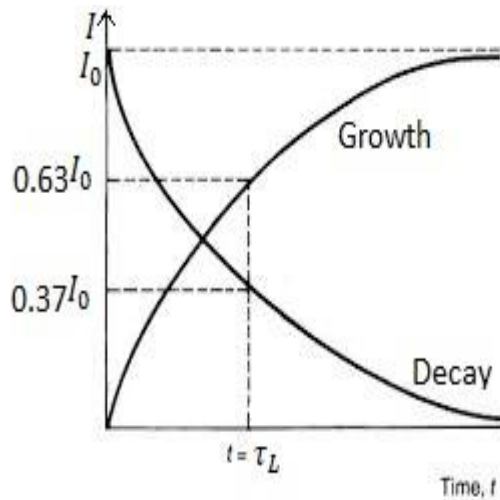
$$\text{or, } \frac{I}{I_0} = e^{-\frac{R}{L}t}$$

$$\text{or, } I = I_0 e^{-\frac{R}{L}t}$$

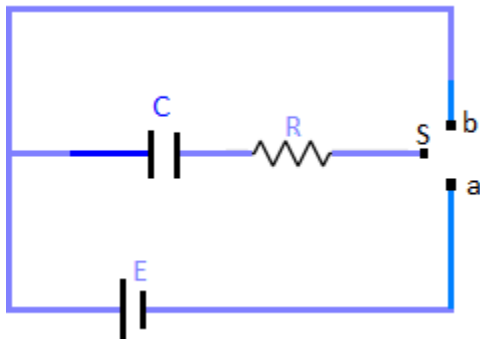
This gives the expression for decay of current.

When, $t = \frac{L}{R} = \tau_L$ then $I = I_0 e^{-1} = 0.37I_0 = 37\% \text{ of } I_0$

Hence the time at which current through the inductor decays to 37% of the maximum current is called inductive time constant.



RC-Circuit (Charging and Discharging of Capacitor)



- c) When switch S is connected to a then charging of capacitor takes place.
Using KVL,

$$E = IR + \frac{q}{C}$$

$$\text{or, } CE - q = IRC$$

$$\text{or, } CE - q = RC \frac{dq}{dt}$$

$$\text{or, } \frac{dq}{CE - q} = \frac{1}{RC} dt$$

Integrating both sides,

$$\int \frac{dq}{CE - q} = \frac{1}{RC} \int dt$$

$$\text{or, } -\ln(CE - q) = \frac{1}{RC} t + k_1 \dots \dots \dots (i)$$

Where, k_1 is integration constant.

When $t = 0, q = 0$

$$\therefore k_1 = -\ln CE$$

Hence from equation (i),

$$-\ln(CE - q) = \frac{1}{RC} t - \ln CE$$

$$\text{or, } \ln \left\{ \frac{CE - q}{CE} \right\} = -\frac{1}{RC} t$$

$$\text{or, } 1 - \frac{q}{CE} = e^{-\frac{1}{RC} t}$$

$$\text{or, } 1 - \frac{q}{q_0} = e^{-\frac{1}{RC} t} \quad \text{Where, } q_0 = CE \text{ is maximum charge stored in the capacitor.}$$

$$\text{or, } q = q_0(1 - e^{-\frac{1}{RC} t})$$

This gives the expression for growth of charge.

$$\text{When, } t = RC = \tau_c \text{ then } q = q_0(1 - e^{-1}) = 0.63q_0 = 63\% \text{ of } q_0$$

Hence the time at which charge through the capacitor growth to 63% of the maximum charge is called capacitive time constant.

Again charging current is $I = \frac{dq}{dt} = q_0 \times \frac{1}{RC} \times e^{-\frac{1}{RC}t} = I_0 e^{-\frac{1}{RC}t}$

Where, $I_0 = \frac{q_0}{RC}$ is maximum current.

d) When switch S is connected to b then discharging of capacitor takes place.

Using KVL,

$$IR + \frac{q}{C} = 0$$

$$\text{or, } R \frac{dq}{dt} = -\frac{q}{C}$$

$$\text{or, } \frac{dq}{q} = -\frac{1}{RC} dt$$

Integrating both sides,

$$\int \frac{dq}{q} = -\frac{1}{RC} \int dt$$

$$\text{or, } \ln q = -\frac{1}{RC} t + k_2 \dots \dots \dots (ii)$$

Where, k_2 is integration constant.

When $t = 0, q = q_0$

$$\therefore k_2 = \ln q_0$$

Hence from equation (ii),

$$\ln q = -\frac{1}{RC} t + \ln q_0$$

$$\text{or, } \ln \frac{q}{q_0} = -\frac{1}{RC} t$$

$$\text{or, } \frac{q}{q_0} = e^{-\frac{1}{RC} t}$$

$$\text{or, } q = q_0 e^{-\frac{1}{RC} t}$$

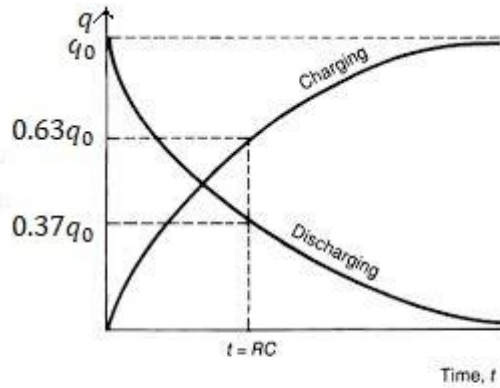
This gives the expression for decay of charge.

When, $t = RC = \tau_c$ then $q = q_0 e^{-1} = 0.37 q_0 = 37\% \text{ of } q_0$

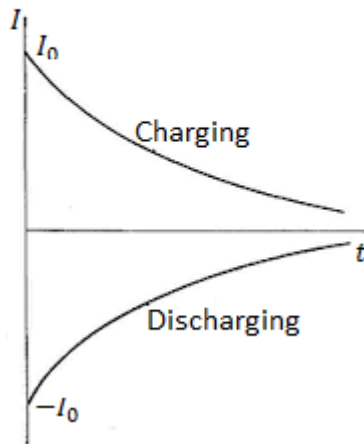
Hence the time at which charge through the capacitor decays to 37% of the maximum charge is called capacitive time constant.

Again charging current is $I = \frac{dq}{dt} = -q_0 \times \frac{1}{RC} \times e^{-\frac{1}{RC}t} = -I_0 e^{-\frac{1}{RC}t}$

Where, $I_0 = \frac{q_0}{RC}$ is maximum current.



Variation of charge with time



Variation of current with time

Note: We have the magnitude of current is

$$I = I_0 e^{-\frac{1}{RC}t}$$

$$\text{or, } \frac{I}{I_0} = e^{-\frac{1}{RC}t}$$

$$\text{or, } \frac{1}{RC} t = \ln\left(\frac{I_0}{I}\right)$$

$$\text{or, } C = \frac{t}{R \ln\left(\frac{I_0}{I}\right)}$$

Displacement Current

Current through the plates of capacitor due to time varying electric field is called displacement current.

$$\text{We have } E = \frac{q}{\epsilon_0 A}$$

$$\text{or, } q = \epsilon_0 A E$$

$$\therefore \text{displacement current } i_d = \frac{dq}{dt} = \epsilon_0 A \frac{dE}{dt}$$

$$\text{or, } i_d = \epsilon_0 A \frac{d}{dt}\left(\frac{V}{d}\right) = \frac{\epsilon_0 A}{d} \frac{dV}{dt} = C \frac{dV}{dt}$$

$$\text{Now the displacement current density is } j_d = \frac{i_d}{A} = \epsilon_0 \frac{dE}{dt}$$

$$\text{Also the induced magnetic field is } B = \frac{1}{2} \mu_0 r j_d$$

Electromagnetic Induction

1. A solenoid 1.3m long 2.6cm in diameter carries a current of 18 A. The magnetic field inside the solenoid is 23mT. Find the length of the wire forming the solenoid and self inductance of the solenoid.

→ Here, length of solenoid $l = 1.3m$, radius of solenoid $r = 1.3cm = 0.013m$,
current $I = 18A$, magnetic field $B = 23mT = 0.023T$

$$\text{We have } B = \mu_0 n I = \frac{\mu_0 N I}{l}$$

$$\therefore N = ?$$

$$\text{Now, length of wire forming the solenoid} = N \times 2\pi r = ?$$

Again, self- inductance

$$L = \frac{\mu_0 N^2 A}{l} = \frac{\mu_0 N^2}{l} \times \pi r^2 = ?$$

2. A toroid has number of turns 1250, internal radius 52 mm, external radius 95 mm and thickness of the ring 13 mm, calculate the inductance.

→ Here, $N = 1250$, $R_1 = 52mm = 0.052m$, $R_2 = 95mm = 0.095m$, $h = 13mm = 0.013m$

$$\text{We have, inductance } L = \frac{\mu_0 N^2 h}{2\pi} \ln\left(\frac{R_2}{R_1}\right) = ?$$

3. A circular loop of wire of radius 5cm carries a current of 100A. What is the energy density at the center of the loop?

→ Here, radius $R = 5\text{m} = 0.05\text{m}$, current $I = 100\text{A}$

$$\text{We have Energy density } u_b = \frac{B^2}{2\mu_0} = \frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2R}\right)^2 = ?$$

4. Calculate the energy required to set up in a cube of 10cm edge (a) a uniform electric field of 10^5V/m (b) a uniform magnetic field of 1T.

→ Here, Side of cube $l = 10\text{cm} = 0.1\text{m}$, $E=10^5\text{V/m}$, $B=1\text{T}$

$$\text{a) } U_E = u_e \times \text{volume} = \frac{1}{2} \epsilon_0 E^2 \times l^3 = ?$$

$$\text{b) } U_B = u_b \times \text{volume} = \frac{B^2}{2\mu_0} \times l^3 = ?$$

5. An inductance L is connected to a battery of emf E through a resistance R. Show that the potential difference across the inductance after time t is $V_L = Ee^{-\frac{Rt}{L}}$. At what time is the p.d. across the inductance equal to that across the resistance such that $i=i_0/2$.

$$\rightarrow \text{We have, } I = I_0 \left(1 - e^{-\frac{R}{L}t}\right)$$

Now, potential difference across the inductor is

$$V_L = L \frac{dI}{dt} = L \times I_0 \times \frac{R}{L} e^{-\frac{R}{L}t} = I_0 R e^{-\frac{R}{L}t} = E e^{-\frac{R}{L}t}$$

$$\text{Where, } E = I_0 R$$

$$\text{For second part, } V_L = V_R$$

$$\text{or, } E e^{-\frac{R}{L}t} = IR$$

$$\text{or, } I_0 R e^{-\frac{R}{L}t} = \frac{I_0}{2} R$$

$$\text{or, } \frac{R}{L}t = \ln 2$$

$$\text{or, } t = \ln 2 \times \frac{L}{R} = 0.693 \tau_L$$

6. A solenoid having an inductance of $6.3\mu\text{H}$ is connected in series with a $1.2\text{ k}\Omega$ resistor. (a) If a 14V battery is connected across the pair, how long will it take for the current through the resistor to reach 80% of its final value? (b) What is the current resistor at time $t=\tau_L$?

→ Here, $L=6.3\mu\text{H}=6.3\times 10^{-6}\text{H}$, $R = 1.2\text{k}\Omega = 1200\Omega$, $E = 14\text{V}$

a) We have, $I = I_0(1 - e^{-\frac{R}{L}t})$

or, $0.8I_0 = I_0(1 - e^{-\frac{R}{L}t})$

or, $e^{-\frac{R}{L}t} = 0.2$

or, $t = ?$

b) At $t = \tau_L$, $I = I_0(1 - e^{-1}) = 0.63I_0 = 0.63 \times \frac{E}{R} = ?$

7. A series circuit has 25Ω resistance and 0.1henry inductance. What will be the initial rate of increase of current if the circuit contains a 12V steady source? What time is required for the current to attain a value of 100mA ?

→ Here, $R=25\Omega$, $L=0.1\text{H}$, $E=12\text{V}$, $I=100\text{mA}=0.1\text{A}$

We have, $I = I_0(1 - e^{-\frac{R}{L}t})$

$\therefore \frac{dI}{dt} = I_0 \times \frac{R}{L} e^{-\frac{R}{L}t} = \frac{E}{L} e^{-\frac{R}{L}t} = ?$

Again, $I = I_0(1 - e^{-\frac{R}{L}t})$

or, $I = \frac{E}{R}(1 - e^{-\frac{R}{L}t})$

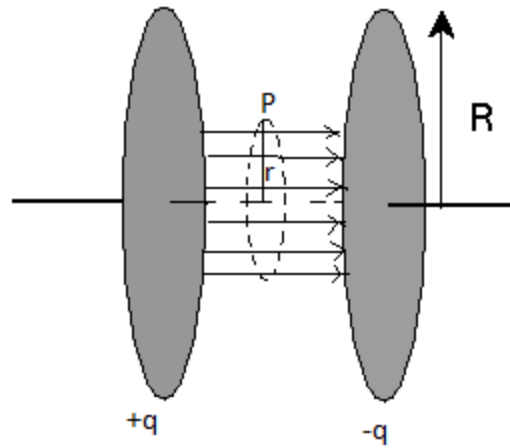
$\therefore t = ?$

8. A parallel plate capacitor with circular plates of radius R be charged, prove that the induced magnetic field at radius r in the region between the plates is

$B = \frac{1}{2} \mu_0 \epsilon_0 r \frac{dE}{dt}$ for $r \leq R$ and

$B = \frac{1}{2r} \mu_0 \epsilon_0 R^2 \frac{dE}{dt}$ for $r \geq R$

→ For $r \leq R$, we have



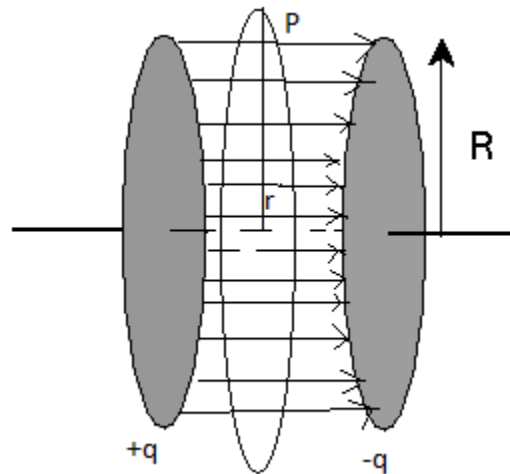
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 A \frac{dE}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 \times \pi r^2 \frac{dE}{dt}$$

$$\text{or, } B = \frac{1}{2} \mu_0 \epsilon_0 r \frac{dE}{dt}$$

For $r \geq R$, we have



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 A \frac{dE}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 \times \pi R^2 \frac{dE}{dt}$$

$$\text{or, } B = \frac{1}{2r} \mu_0 \epsilon_0 R^2 \frac{dE}{dt}$$

9. Calculate the displacement current between the circular plates of area $1.5 \times 10^{-2} \text{ m}^2$ and rate of electric field change is $1.5 \times 10^{12} \text{ V/ms}$. Also calculate the displacement current density and induced magnetic field for $r=R=69\text{mm}$.

→ Here, $A=1.5 \times 10^{-2} \text{ m}^2$, $\frac{dE}{dt}=1.5 \times 10^{12} \text{ V/m-s}$

We have, displacement current $i_d = \epsilon_0 A \frac{dE}{dt} = ?$

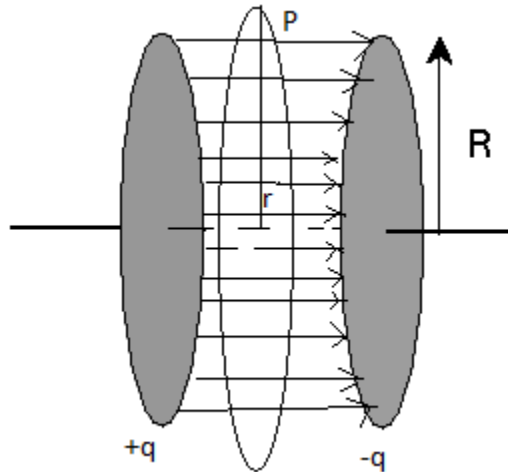
displacement current density $j_d = \epsilon_0 \frac{dE}{dt} = ?$

induced magnetic field $B = \frac{1}{2} \mu_0 r j_d = ?$

10. The induced magnetic field at radial distance 6mm from the central axis of a circular parallel plate capacitor is $2 \times 10^{-7} \text{ T}$. The plates have radius 3mm. At what rate is the electric field between the plates changing?

→ Here, $r = 6\text{mm} = 0.006\text{m}$, $B = 2 \times 10^{-7} \text{ T}$, $R = 3\text{mm} = 0.003\text{m}$

We have for $r > R$



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 A \frac{dE}{dt}$$

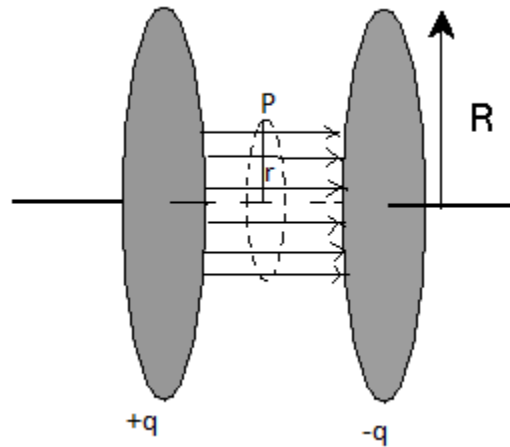
$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 \times \pi R^2 \frac{dE}{dt}$$

$$\text{or, } B = \frac{1}{2r} \mu_0 \epsilon_0 R^2 \frac{dE}{dt}$$

$$\therefore \frac{dE}{dt} = \frac{B \times 2r}{\mu_0 \epsilon_0 R^2} = ?$$

11. As a parallel plate capacitor with circular plates 20cm in diameter is being charged, the current density of the displacement current in the region between the plates is uniform and has a magnitude of 20 A/m^2 . (a) Calculate the magnitude B of the magnetic field at a distance $r=50\text{mm}$ for the axis of symmetry of this region. (b) Calculate dE/dt in this region.

→ Here, $R=10\text{cm}=0.1\text{m}$, $j_d = 20\text{A/m}^2$, $r=50\text{mm}=0.05\text{m}$



For $r < R$,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 A \frac{dE}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 \times \pi r^2 \frac{dE}{dt}$$

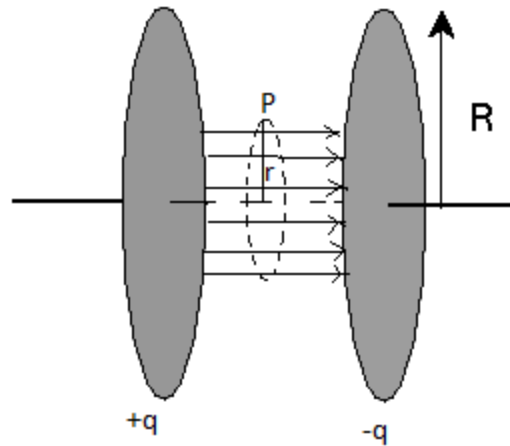
$$\text{or, } B = \frac{1}{2} \mu_0 \epsilon_0 r \frac{dE}{dt} = \frac{1}{2} \mu_0 r j_d = ?$$

$$\text{Again, } j_d = \epsilon_0 \frac{dE}{dt}$$

$$\therefore \frac{dE}{dt} = \frac{j_d}{\epsilon_0} = ?$$

- 12. A parallel plate capacitor with circular plates of 10cm radius is charged by varying electric field of $1.5 \times 10^{12} \text{V/ms}$. Evaluate induced magnetic field if the radius of the plate is 55mm and displacement current in this situation.**

→ Here, $R=10\text{cm}=0.1\text{m}$, $\frac{dE}{dt} = 1.5 \times 10^{12} \text{V/m-s}$, $r=55\text{mm}=5.5\text{cm}=0.055\text{m}$



For $r < R$,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 A \frac{dE}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 \times \pi r^2 \frac{dE}{dt}$$

$$\text{or, } B = \frac{1}{2} \mu_0 \epsilon_0 r \frac{dE}{dt} = ?$$

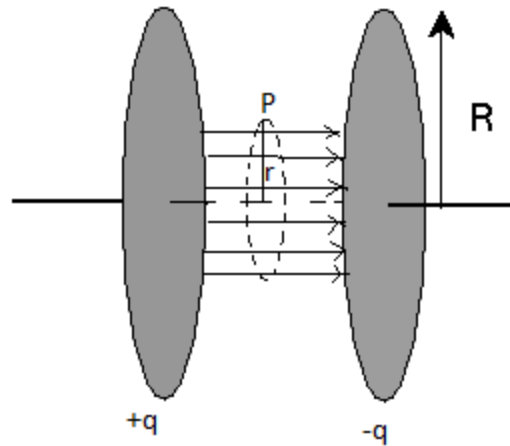
$$\text{Again, } i_d = \epsilon_0 A \frac{dE}{dt} = \epsilon_0 \times \pi \times r^2 \times \frac{dE}{dt} = ?$$

- 13. A circular parallel plate capacitor of area 154cm^2 is being charged has a uniform current density of a displacement current, having magnitude 20 Am^{-2} . Calculate (a) the magnitude of magnetic field at a distance $r=50\text{mm}$ about the central axis between the plates. (b) dE/dt in this region.**

$$\rightarrow \text{Here, } \pi R^2 = 154\text{cm}^2$$

$$\therefore R = 7\text{cm} = 0.07\text{m}, j_d = 20\text{ Am}^{-2}, r = 50\text{mm} = 5\text{cm} = 0.05\text{m}$$

a) For $r < R$,



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 A \frac{dE}{dt}$$

$$\text{or, } B \times 2\pi r = \mu_0 \epsilon_0 \times \pi r^2 \frac{dE}{dt}$$

$$\text{or, } B = \frac{1}{2} \mu_0 \epsilon_0 r \frac{dE}{dt} = ?$$

$$b) j_d = \epsilon_0 \frac{dE}{dt}$$

$$\therefore \frac{dE}{dt} = \frac{j_d}{\epsilon_0} = ?$$

14. A parallel plate capacitor with circular plates is charged by current i . (a) what is the magnitude of $\oint \vec{B} \cdot d\vec{s}$ in terms of μ_0 and i between the plates if $r=a/5$ from the center? (b) What is the magnitude of induced magnetic field for $r=a/5$ in terms of displacement current.

$$\rightarrow \text{(a) We have, } \oint \vec{B} \cdot d\vec{s} = \mu_0 i'$$

$$\text{Since, Current density } J = \frac{i}{\pi a^2} = \frac{i'}{\pi r^2}$$

$$\therefore i' = \frac{r^2}{a^2} \times i = \frac{1}{25} i$$

$$\text{Hence, } \oint \vec{B} \cdot d\vec{s} = \mu_0 \times \frac{1}{25} i = \frac{\mu_0 i}{25}$$

$$\text{(b) Again we have, } \oint \vec{B} \cdot d\vec{s} = \mu_0 i'$$

$$\text{or, } B \times 2\pi r = \mu_0 i'$$

$$\text{or, } B = \frac{\mu_0 i'}{2\pi r} = \frac{\mu_0}{2\pi r} \times \frac{1}{25} i = \frac{\mu_0}{2\pi a} \times \frac{1}{25} i = \frac{\mu_0 i}{10\pi a}$$

15. If a test charge revolves round a circular path of radius 8.5cm where the magnetic field increases at steady rate 0.13T/s, calculate the magnitude of induced electric field at a point 12.5cm and 4cm.

$$\rightarrow \text{Here, } R=8.5\text{cm}=8.5 \times 10^{-2}\text{m}, \frac{dB}{dt} = 0.13\text{T}$$

For $r=12.5\text{cm}=12.5 \times 10^{-2}\text{m}$, we have from Faraday's law

$$\oint \mathbf{E} \cdot d\mathbf{l} = \frac{d\phi_B}{dt}$$

$$\text{or, } E \times 2\pi r = \frac{d(BA)}{dt}$$

$$\text{or, } E \times 2\pi r = \pi R^2 \times \frac{dB}{dt}$$

$$\text{or, } E = \frac{1}{2} \frac{R^2}{r} \frac{dB}{dt} = ?$$

Again for $r=4\text{cm}=4 \times 10^{-2}\text{m}$,

$$\oint \mathbf{E} \cdot d\mathbf{l} = \frac{d\phi_B}{dt}$$

$$\text{or, } E \times 2\pi r = \frac{d(BA)}{dt}$$

$$\text{or, } E \times 2\pi r = \pi r^2 \times \frac{dB}{dt}$$

$$\text{or, } E = \frac{1}{2} r \frac{dB}{dt} = ?$$

16. A long solenoid of radius 2cm has 1000 turns per meter and carries a sinusoidally varying current $I = 5\sin 100\pi t$, where I is in ampere and t is in second. Determine the magnitude of induced electric field at a radius $r=1\text{cm}$ and $r=3\text{cm}$ from its central axis.

$$\rightarrow \text{Here, } R=2\text{cm}=2 \times 10^{-2}\text{m}, n=1000 \text{ turns/m}, I = 5\sin 100\pi t$$

$$\text{We have, } \frac{dB}{dt} = \frac{d(\mu_0 n I)}{dt} = \mu_0 n \frac{d(5\sin 100\pi t)}{dt} = \mu_0 n \times 5 \times 100\pi \cos 100\pi t = \dots \dots \dots \cos 100\pi t \text{ T/s}$$

For $r=1\text{cm}=1\times 10^{-2}\text{m}$,

$$\oint E \cdot dl = \frac{d\phi_B}{dt}$$

$$\text{or, } E \times 2\pi r = \frac{d(BA)}{dt}$$

$$\text{or, } E \times 2\pi r = \pi r^2 \times \frac{dB}{dt}$$

$$\text{or, } E = \frac{1}{2} r \frac{dB}{dt} = \dots \dots \dots \cos 100\pi t \text{ V/m}$$

For $r=3\text{cm}=3\times 10^{-2}\text{m}$, we have from Faraday's law

$$\oint E \cdot dl = \frac{d\phi_B}{dt}$$

$$\text{or, } E \times 2\pi r = \frac{d(BA)}{dt}$$

$$\text{or, } E \times 2\pi r = \pi R^2 \times \frac{dB}{dt}$$

$$\text{or, } E = \frac{1}{2} \frac{R^2}{r} \frac{dB}{dt} = \dots \dots \dots \cos 100\pi t \text{ V/m}$$

17. An air cored solenoid with length 25cm, area of cross-section 30cm^2 and number of turns 490 carries a current of 20A. The current is suddenly switched off in a brief time of 10^{-3} sec. How much is the average back emf induced across the ends of the open switch in the circuit?

→ Here, we have induced emf

$$\varepsilon = L \frac{dI}{dt} = \mu_0 n^2 A l \frac{dI}{dt} = 4\pi \times 10^{-7} \times \left(\frac{490}{0.25}\right)^2 \times 30 \times 10^{-4} \times 0.25 \frac{20}{10^{-3}} = \dots \dots V$$

Chapter: Photon and matter waves

In order to explain the motion of macro particles, classical mechanics is sufficient. To explain the motion of micro particles, scientists developed a new mechanics, known as quantum mechanics or wave mechanics.

De-Broglie Wavelength

When the microparticles are in motion then they show dual nature i.e. particle nature and wave nature. Wave mechanics explains the motion of particles which are in atomic dimension.

According to Planck's law of radiation, the energy of radiation is in the form of packet. The minimum value of energy of radiation is known as quanta and its value is

$$E = hf \dots\dots (i)$$

Where $h = 6.62 \times 10^{-34}$ Js is Planck's constant and f is frequency of radiation.

Also from Einstein's mass energy relation, $E = mc^2 \dots\dots\dots (ii)$

Where, m is mass of the particle and c is velocity of light.

Now from equations (i) and (ii)

$$mc^2 = hf$$

$$\text{or, } \frac{c}{f} = \frac{h}{mc}$$

$$\text{or, } \lambda = \frac{h}{mc} = \frac{h}{P}$$

Where, P is momentum of the particle. This is de-Broglie wavelength.

$$\text{Note: } E = hf = \frac{h}{2\pi} 2\pi f = \hbar \omega$$

$$\text{Also, } P = \frac{h}{\lambda} = \frac{h}{2\pi} \frac{2\pi}{\lambda} = \hbar k$$

$$\text{Phase velocity } u = \frac{\omega}{k} = \lambda f$$

$$\text{Group velocity, } v_g = \frac{d\omega}{dk}$$

Relation between phase velocity and group velocity

If a wave is propagating through a medium, then the rate of displacement of wave is phase velocity or wave velocity. For a wave, the phase velocity is given by

$$u = \frac{\omega}{k} = \lambda f$$

If more than one wave are propagating through a medium, then there is a resultant wave which is formed due to the superposition of component wave. Also, there is a periodically change in velocity. The velocity of resultant wave due to the superposition of component waves is group velocity and is given by

$$v_g = \frac{d\omega}{dk}$$

Since, $\omega = 2\pi f$ so $d\omega = 2\pi f df$

Also, $k = \frac{2\pi}{\lambda}$ so $dk = -\frac{2\pi}{\lambda^2} d\lambda$

$$\therefore v_g = \frac{d\omega}{dk} = \frac{2\pi f df}{-\frac{2\pi}{\lambda^2} d\lambda} = -\lambda^2 \frac{df}{d\lambda} = -\lambda^2 \frac{d\left(\frac{u}{\lambda}\right)}{d\lambda} = -\lambda^2 \left[-\frac{u}{\lambda^2} + \frac{1}{\lambda} \frac{du}{d\lambda} \right] = u - \lambda \frac{du}{d\lambda}$$

This shows that the phase velocity is greater than group velocity.

For free space, $\frac{du}{d\lambda} = 0$

$$\therefore v_g = u$$

i.e. group velocity=phase velocity

Wave Function

It is the mathematical tool which is used to characterize the wave motion of a particle. It may be in the form of real or imaginary. It is represented by ψ (psi). It is given by

$$\psi(x, t) = \text{Re}^{i(kx - \omega t)}$$

$$\text{or, } \psi(x, t) = \text{Re}^{\frac{i}{\hbar}(Px - Et)} \quad (\text{Since, } E = hf = \frac{h}{2\pi} 2\pi f = \hbar\omega \text{ and } P = \frac{h}{\lambda} = \frac{h}{2\pi} \frac{2\pi}{\lambda} = \hbar k)$$

Physical Significance of wave function

The wavefunction is the mathematical tool which characterizes all the properties of micro particles. The product of complex conjugate of a wave function with that wave function gives the probability density of finding the particle in a certain region. So the square of wave function physically represents the probability of finding the particle in the given region.

Normalized wave function

If ψ be any wave function then for ψ to be normalized, $\int \psi^* \psi dx = 1$

In most of the cases, the wave function is not normalized. We can normalize the given wave function by multiplying by a constant, which is known as normalizing constant.

If ψ be unnormalized wave function, it can be normalized by multiplying by a constant A so that A ψ is the normalized wave function.

$$\therefore \int (A\psi)^* (A\psi) dx = 1$$

$$\text{or, } \int A^2 \psi^* \psi dx = 1$$

$$\text{or, } A^2 \int \psi^* \psi dx = 1$$

$$\text{or, } A^2 = \frac{1}{\int \psi^* \psi dx}$$

Schrodinger's Wave Equations

To describe the motion of micro particles, Schrodinger developed two differential equations which are known as Schrodinger's Wave Equations.

Consider a particle of mass m is moving with velocity v in the region of potential energy V then total energy of the particle is

$$E = K.E. + P.E. = \frac{1}{2}mv^2 + V = \frac{p^2}{2m} + V$$

Operating this equation by ψ from right, we get

$$E\psi = \frac{p^2}{2m}\psi + V\psi \dots \dots \dots (i)$$

a) Schrodinger's time dependent wave equation

We have the wave function is $\psi(x, t) = Re^{\frac{i}{\hbar}(Px - Et)} \dots \dots \dots (ii)$

Here, ψ depends on both position and time.

Now, differentiating equation (ii) with respect to t , we get

$$\frac{d\psi}{dt} = -\frac{iE}{\hbar} Re^{\frac{i}{\hbar}(Px - Et)} = -\frac{iE}{\hbar} \psi$$

$$\therefore E\psi = -\frac{\hbar}{i} \frac{d\psi}{dt} = i\hbar \frac{d\psi}{dt} \quad (\text{Since, } i^2 = -1)$$

Again, differentiating equation (ii) with respect to x twice, we get

$$\frac{d\psi}{dx} = \frac{iP}{\hbar} Re^{\frac{i}{\hbar}(Px - Et)}$$

$$\text{And } \frac{d^2\psi}{dx^2} = \left(\frac{iP}{\hbar}\right)^2 Re^{\frac{i}{\hbar}(Px - Et)} = -\frac{P^2}{\hbar^2} \psi$$

$$\text{or, } P^2\psi = -\hbar^2 \frac{d^2\psi}{dx^2}$$

With the values of $E\psi$ and $P^2\psi$, equation (i) becomes

$$i\hbar \frac{d\psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi$$

This is Schrodinger's time dependent wave equation in 1-D.

In 3-D, this becomes

$$i\hbar \frac{d\psi}{dt} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$$

Where, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is square of del operator.

b) Schrodinger's time independent wave equation

We have the wave function is $\psi(x) = \text{Re} e^{\frac{i}{\hbar}(Px-Et)}$ (iii)

Here, ψ depends only on position.

Now, differentiating equation (iii) with respect to x twice, we get

$$\frac{d\psi}{dx} = \frac{iP}{\hbar} \text{Re} e^{\frac{i}{\hbar}(Px-Et)}$$

$$\text{And } \frac{d^2\psi}{dx^2} = \left(\frac{iP}{\hbar}\right)^2 \text{Re} e^{\frac{i}{\hbar}(Px-Et)} = -\frac{P^2}{\hbar^2} \psi$$

$$\text{or, } P^2 \psi = -\hbar^2 \frac{d^2\psi}{dx^2}$$

With the value of $P^2 \psi$, equation (i) becomes

$$E\psi = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi$$

$$\text{or, } \frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + (E - V)\psi = 0$$

$$\text{or, } \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0$$

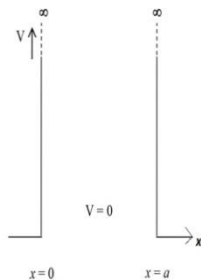
This is Schrodinger's time independent wave equation in 1-D.

In 3-D, this becomes

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V)\psi = 0$$

Where, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is square of del operator.

Particle in 1-D infinite potential well (Particle in 1-D box)



Let us consider a particle of mass m is moving inside the infinite potential well of width a as shown in figure. The potential can be defined as

$$V(x) = 0 ; x > 0, x < a$$

$$V(x) = \infty ; x \leq 0, x \geq a$$

Since the wave function is zero in the region of infinite potential, so outside the well $\psi = 0$. We have to find the wave function only inside the well.

We have Schrodinger's time independent wave equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

But inside the well, $V = 0$

$$\therefore \frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0$$

$$\text{or, } \frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \quad \text{Where, } \alpha = \sqrt{\frac{2mE}{\hbar^2}} \dots \dots \dots (i)$$

Now the solution of this differential equation can be written as

$$\psi(x) = A\sin\alpha x + B\cos\alpha x \dots \dots \dots (ii)$$

Where A and B are constants whose values is to be determined.

At $x = 0, \psi = 0$

$$\therefore A\sin\alpha \cdot 0 + B\cos\alpha \cdot 0 = 0$$

$$\text{or, } B = 0$$

With this value, equation (ii) becomes

$$\psi(x) = A\sin\alpha x \dots \dots \dots (iii)$$

Again, at $x = a, \psi = 0$

$$\therefore A\sin\alpha a = 0$$

This implies that, $\sin\alpha a = 0$ (Since, $A \neq 0$)

$$\text{or, } \sin\alpha a = \sin n\pi$$

$$\text{or, } \alpha a = n\pi$$

$$\therefore \alpha = \frac{n\pi}{a} \dots \dots \dots (iv)$$

Hence the wave function is $\psi(x) = A\sin \frac{n\pi}{a}x$

Here, the value of A can be determined by using normalizing condition as

$$\int_0^a \psi^* \psi dx = 1$$

$$\text{or, } \int_0^a A^2 \sin^2\left(\frac{n\pi x}{a}\right) dx = 1$$

$$\text{or, } \frac{A^2}{2} \int_0^a \left[1 - \cos\left(\frac{2n\pi x}{a}\right)\right] dx = 1$$

$$\text{or, } \frac{A^2}{2} a = 1$$

$$\therefore A = \sqrt{\frac{2}{a}}$$

Hence the wave normalized function is $\psi(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x$; $0 < x < a$

Again from equations (i) and (iv)

$$\sqrt{\frac{2mE}{\hbar^2}} = \frac{n\pi}{a}$$

$$\text{or, } \frac{2mE_n}{\hbar^2} = \frac{n^2\pi^2}{a^2}$$

$$\text{or, } E_n = \frac{n^2\pi^2\hbar^2}{2ma^2} = \frac{n^2 h^2}{8ma^2} ; n = 1, 2, 3, \dots (\neq 0)$$

This shows that energy levels are quantized when a particle is confined in a 1-D infinite potential well.

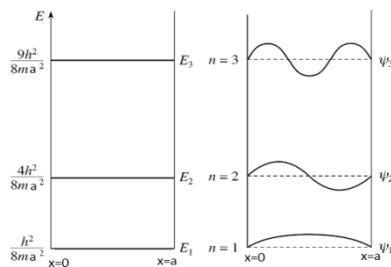
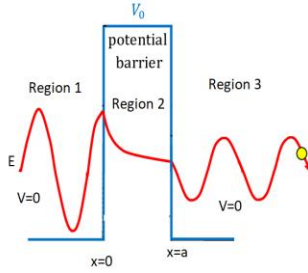


Fig: Wave function and Energy level diagram

Finite Potential Barrier (Tunneling Effect)



Consider a particle of mass m having energy E is incident on a finite potential barrier of width a and potential height V_0 such that $E < V_0$. The potential can be defined as

$$V(x) = 0 \quad ; x < 0, x > a$$

$$\text{And } V(x) = V_0 \quad ; x \geq 0, x \leq a$$

We have Schrodinger's time independent wave equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0 \dots \dots \dots (i)$$

For region 1 this equation becomes,

$$\frac{d^2\psi_1}{dx^2} + \frac{2mE}{\hbar^2} \psi_1 = 0$$

$$\text{or, } \frac{d^2\psi_1}{dx^2} + \alpha^2 \psi_1 = 0 \quad \text{Where, } \alpha = \sqrt{\frac{2mE}{\hbar^2}}$$

The solution of this equation can be written as

$$\psi_1 = Ae^{i\alpha x} + Be^{-i\alpha x} \dots \dots \dots (ii)$$

Where, A is the amplitude of incident wave at $x = 0$ and B is that of reflected wave from $x = 0$.

Again, for region 2, equation (i) becomes

$$\frac{d^2\psi_2}{dx^2} + \frac{2m}{\hbar^2} (E - V_0)\psi_2 = 0$$

For $E < V_0$,

$$\frac{d^2\psi_2}{dx^2} - \frac{2m}{\hbar^2} (V_0 - E)\psi_2 = 0$$

$$\text{or, } \frac{d^2\psi_2}{dx^2} - \beta^2 \psi_2 = 0 \quad \text{Where, } \beta = \sqrt{\frac{2m}{\hbar^2} (V_0 - E)}$$

The solution of this equation can be written as

$$\psi_2 = Ce^{\beta x} + De^{-\beta x} \dots \dots \dots (iii)$$

Where, C is the amplitude of transmitted wave from $x = 0$ and D is that of reflected wave from $x = a$.

Similarly, for region 3, equation (i) becomes

$$\frac{d^2\psi_3}{dx^2} + \frac{2mE}{\hbar^2}\psi_3 = 0$$

$$\text{or, } \frac{d^2\psi_3}{dx^2} + \alpha^2\psi_3 = 0 \quad \text{Where, } \alpha = \sqrt{\frac{2mE}{\hbar^2}}$$

The solution of this equation can be written as

$$\psi_3 = Fe^{i\alpha x} + Ge^{-i\alpha x}$$

Where, F is the amplitude of transmitted wave from $x = a$ and G is that of reflected wave from region 3.

Since, there is no possibility of reflection from region 3, so $G=0$.

$$\therefore \psi_3 = Fe^{i\alpha x} \dots \dots \dots (iv)$$

Here, equations (ii), (iii) and (iv) represent the wave equations for region 1,2 and 3 respectively.

Now the transmission coefficient (T) is given by the ratio of the probability of transmitted amplitude to the incident amplitude.

$$\text{i.e. } T = \left| \frac{F}{A} \right|^2 = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right) e^{-2\beta a}$$

and the reflection coefficient is given by $R=1-T$

Tunneling Effect: When a particle of energy E is incident on a potential barrier of height V_0 such that $E < V_0$, then there is finite probability of finding the particle in the third region. This is called tunneling effect. Tunneling microscopes, tunneling diodes are based on this effect.

Heisenberg's Uncertainty Principle

It states that, it is impossible to measure two variables accurately and precisely same at a time which gives the motion of the atomic system. These two variables are known as canonically conjugate variables. For example, position and momentum, energy and time etc. are canonically conjugate variables.

If Δx be the error in the measurement of position and ΔP be the error in the measurement of momentum then the product of errors is always in the order of Planck's constant.

i.e. $\Delta x. \Delta P \geq \hbar$

Similarly, $\Delta E. \Delta t \geq \hbar$

Photons And Matter Waves

1. Find the de-Broglie wavelength of a 15KV proton.

→ Here, $V=15KV=15000V$

We have, de-Broglie wavelength $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2meV}} = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 1.67 \times 10^{-27} \times 1.6 \times 10^{-19} \times 15000}} = ?$

2. Prove that the group speed of a wave packet is identical to the speed of a particle.

→ We have, group speed

$$v_g = \frac{d\omega}{dk} = \frac{d(\hbar\omega)}{d(\hbar k)} = \frac{dE}{dP} = \frac{d}{dP} \left(\frac{P^2}{2m} \right) = \frac{2P}{2m} = \frac{mv}{m} = v ; \text{the speed of particle}$$

3. An electron and a photon have a wavelength of 0.2nm. What is the momentum of the (a) electron and (b) Photon? What is the energy in eV of (c) electron and (d) Photon?

→ Here, $\lambda_p = \lambda_e = 0.2nm = 0.2 \times 10^{-9}m$,

a) Momentum of electron $P_e = \frac{h}{\lambda_e} = ?$

b) Momentum of photon $P_p = \frac{h}{\lambda_p} = ?$

c) Energy of electron $E_e = \frac{P_e^2}{2m_e} = ?$

d) Energy of Photon $E_p = P_p c = ?$

4. A non-relativistic particle is moving three times as fast as an electron. The ratio of the de-Broglie wavelength of the particle to that of electron is 1.813×10^{-4} . By calculating its mass, identify the particle.

→ Here, $v_p = 3v_e$, $\frac{\lambda_p}{\lambda_e} = 1.813 \times 10^{-4}$

We have, $\frac{\lambda_p}{\lambda_e} = \frac{h/m_p v_p}{h/m_e v_e} = \frac{m_e v_e}{m_p v_p}$

or, $m_p = \frac{v_e \lambda_e}{v_p \lambda_p} m_e = \frac{1}{3} \times \frac{1}{1.813 \times 10^{-4}} \times 9.1 \times 10^{-31} = 1.673 \times 10^{-27} kg$ (neutron)

5. Calculate permitted energy levels of an electron in one dimensional potential well of width $2A^0$.

→ Here, width $a = 2A^0 = 2 \times 10^{-10}m$

We have $E_n = \frac{n^2 h^2}{8ma^2}$; $n = 1, 2, 3, \dots$

or, $E_n = \frac{n^2 h^2}{8ma^2} = \frac{n^2 \times (6.62 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2 \times 10^{-10})^2} = 1.5 \times 10^{-18} \times n^2 \text{ Joules}$

For $n = 1, E_1 = 1.5 \times 10^{-18} \times 1^2 = ?$

For $n = 2, E_2 = 1.5 \times 10^{-18} \times 2^2 = ?$

For $n = 3, E_3 = 1.5 \times 10^{-18} \times 3^2 = ?$ and so on.

6. An electron is confined to an infinite height box of size 0.1nm. Calculate the ground state energy of the electron and radiation frequency. How this electron can be put to the third energy level?

→ Here, width $a = 0.1\text{nm} = 0.1 \times 10^{-9}\text{m}$

We have $E_n = \frac{n^2 h^2}{8ma^2}$

For ground state, $n = 1$

∴ $E_1 = \frac{h^2}{8ma^2} = ?$

So radiation frequency $f = \frac{E_1}{h} = ?$

To put the electron to third energy level, we need to supply energy $= E_3 - E_1 = 9E_1 - E_1 = 8E_1 = ?$

7. An electron is confined to an one dimensional region in which its ground state energy is 2eV.
(a) what is the length of the region? And (b) how much energy is required to promote the electron to its first excited state?

→ Here, Ground state energy $E_1 = 2\text{eV} = 2 \times 1.6 \times 10^{-19}\text{J}$

a) We have $E_n = \frac{n^2 h^2}{8ma^2}$

∴ $E_1 = \frac{h^2}{8ma^2}$

∴ $a = ?$

b) To promote the electron to first excited state, we need to supply energy

$= E_2 - E_1 = 4E_1 - E_1 = 3E_1 = 6\text{eV}$

8. Write down the expression for the energy level of a 1gm particle in the infinite potential well of width 1cm. If the energy of the particle is 1J, what is the value of quantum number at that state?

→ Here, mass $m = 1\text{gm} = 0.001\text{kg}$, width $a = 1\text{cm} = 0.01\text{m}$, $E_n = 1\text{J}$

We have $E_n = \frac{n^2 h^2}{8ma^2}$

∴ $n = ?$

9. A football of 500gm is confined between two impenetrable walls of DasharathRangashala that can be modeled as a box of length 100m. Calculate the minimum speed of the ball.

→ Here, mass $m = 500\text{gm} = 0.5\text{kg}$, width $a = 100\text{m}$

We have $E_n = \frac{n^2 h^2}{8ma^2}$

For minimum speed of ball $n=1$

$$\therefore E_1 = \frac{h^2}{8ma^2}$$

Again, $E_1 = \frac{1}{2}mv^2$

or, $\frac{h^2}{8ma^2} = \frac{1}{2}mv^2$

or, $v = \frac{h}{2ma} = ?$

- 10. A beam of electrons having energy of each 3eV is incident on a potential barrier of height 4eV. If the width of the barrier is $20\text{\AA}$, calculate the percentage transmission of the beam through the barrier.**

→ Here, Energy of electron beam $E = 3\text{eV} = 3 \times 1.6 \times 10^{-19}\text{J}$,

Height of potential barrier $V_0 = 4\text{eV} = 4 \times 1.6 \times 10^{-19}\text{J}$, width $a = 20\text{\AA} = 20 \times 10^{-10}\text{m}$

We have transmission coefficient $T = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0}\right) e^{-2\beta a}$

Where, $\beta = \sqrt{\frac{2m}{\hbar^2} (V_0 - E)} = ?$

$\therefore T = ?$

- 11. For an electron confined to an infinite potential well of width 0.1nm, determine the uncertainty in measuring momentum and kinetic energy.**

→ Here, $\Delta x = 0.1\text{nm} = 0.1 \times 10^{-9}\text{m}$

We have, $\Delta x \cdot \Delta P = \frac{h}{2\pi}$

or, $\Delta P = \frac{h}{2\pi \times \Delta x} = ?$

and $\Delta K.E. = \frac{(\Delta P)^2}{2m} = ?$

- 12. The uncertainty in the location of a particle moving with a velocity $7.28 \times 10^7\text{m/s}$ is double of its de-Broglie wavelength. Find out the uncertainty in measuring the velocity.**

→ Here, velocity $v = 7.28 \times 10^7\text{m/s}$, $\Delta x = 2\lambda = 2 \times \frac{h}{mv}$

we have, $\Delta x \cdot \Delta P = \frac{h}{2\pi}$

or, $\Delta x \cdot m\Delta v = \frac{h}{2\pi}$

or, $\Delta v = \frac{h}{2\pi \times m\Delta x} = \frac{h}{2\pi \times m} \times \frac{mv}{2h} = \frac{v}{4\pi} = ?$

13. Calculate the smallest possible uncertainty in the position of an electron moving with $3 \times 10^7 \text{ m/s}$.

→ Here, $\Delta v = 3 \times 10^7 \text{ m/s}$

$$\text{we have, } \Delta x \cdot \Delta p = \frac{h}{2\pi}$$

$$\text{or, } \Delta x \cdot m\Delta v = \frac{h}{2\pi}$$

$$\text{or, } \Delta x = \frac{h}{2\pi \times m\Delta v} = ?$$

Chapter: Electromagnetic Waves

- Del Operator, $\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$
- $\nabla^2 = \nabla \cdot \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
- Divergence of a vector $\vec{A} = \nabla \cdot \vec{A}$
- Curl of a vector $\vec{A} = \nabla \times \vec{A}$
- Line integral of vector $\vec{A} = \oint \vec{A} \cdot d\vec{l}$
- Surface integral of vector $\vec{A} = \oint \vec{A} \cdot d\vec{a}$
- Volume integral of vector $\vec{A} = \oint \vec{A} dV$
- Gauss Divergence Theorem $\oint \vec{A} \cdot d\vec{a} = \oint (\nabla \cdot \vec{A}) dV$
- Stoke's Theorem $\oint \vec{A} \cdot d\vec{l} = \oint (\nabla \times \vec{A}) \cdot d\vec{a}$
- Electric Flux $\phi_E = \int \vec{E} \cdot d\vec{a}$
- Magnetic Flux $\phi_B = \int \vec{B} \cdot d\vec{a}$
- Current $I = \oint \vec{J} \cdot d\vec{a}$
- Charge $q = \oint \rho dV$

Maxwell's Equations

1) Integral Form

- $\oint \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}$ (Gauss Law in Electrostatics)
- $\oint \vec{B} \cdot d\vec{a} = 0$ (Gauss Law in magnetism)
- $\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$ (Faraday's Law of electromagnetic Induction)
- $\oint \vec{B} \cdot d\vec{l} = \mu_0(I + \epsilon_0 \frac{d\phi_E}{dt})$ (Modified Ampere's Law)

2) Differential Form

- $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$
- $\nabla \cdot \vec{B} = 0$
- $\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$
- $\nabla \times \vec{B} = \mu_0(\vec{J} + \epsilon_0 \frac{d\vec{E}}{dt})$

Physical Significance

- It signifies that the existence of electric monocharges in nature.
- It signifies that the non-existence of magnetic monopoles in nature.
- It signifies that electric field can be generated by the variation of magnetic flux.

- iv) It signifies that magnetic field can be generated by the flow of current in a conductor as well as by the variation of electric flux.

Conversion of Maxwell's equations from integral form to differential form

- i) First Maxwell's equation in integral form is

$$\oint \vec{E} \cdot \vec{da} = \frac{q}{\epsilon_0}$$

If dq be the small amount of charge in elemental volume dV , then $dq = \rho dV$

Where, ρ is volume charge density.

Now, total charge q in the entire volume V is $q = \oint \rho dV$

$$\therefore \oint \vec{E} \cdot \vec{da} = \frac{1}{\epsilon_0} \oint \rho dV$$

Using Gauss Divergence Theorem on L.H.S. we get

$$\oint (\nabla \cdot \vec{E}) dV = \frac{1}{\epsilon_0} \oint \rho dV$$

$$\therefore \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

- ii) Second Maxwell's equation in integral form is

$$\oint \vec{B} \cdot \vec{da} = 0$$

Using Gauss Divergence Theorem on L.H.S. we get

$$\oint (\nabla \cdot \vec{B}) dV = 0$$

$$\therefore \nabla \cdot \vec{B} = 0$$

- iii) Third Maxwell's equation in integral form is

$$\oint \vec{E} \cdot \vec{dl} = -\frac{d\phi_B}{dt}$$

We have, Magnetic Flux $\phi_B = \oint \vec{B} \cdot \vec{da}$

$$\therefore \oint \vec{E} \cdot \vec{dl} = -\frac{d}{dt} \oint \vec{B} \cdot \vec{da}$$

$$\text{or, } \oint \vec{E} \cdot \vec{dl} = -\oint \frac{d\vec{B}}{dt} \cdot \vec{da}$$

Using Stoke's Theorem on L.H.S., we get

$$\oint (\nabla \times \vec{E}) \cdot \vec{da} = -\oint \frac{d\vec{B}}{dt} \cdot \vec{da}$$

$$\therefore \nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$$

- iv) Fourth Maxwell's Equation in integral form is

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 (I + \epsilon_0 \frac{d\phi_E}{dt})$$

We have, Current $I = \oint \vec{J} \cdot \vec{da}$

And Electric Flux $\phi_E = \oint \vec{E} \cdot \vec{da}$

$$\therefore \oint \vec{B} \cdot \vec{dl} = \mu_0 \left(\oint \vec{J} \cdot \vec{da} + \epsilon_0 \frac{d}{dt} \oint \vec{E} \cdot \vec{da} \right)$$

$$\text{or, } \oint \vec{B} \cdot \vec{dl} = \oint \left[\mu_0 \left(\vec{J} + \epsilon_0 \frac{d\vec{E}}{dt} \right) \right] \cdot \vec{da}$$

Using Stoke's Theorem on L.H.S., we get

$$\oint (\nabla \times \vec{B}) \cdot \vec{da} = \oint \left[\mu_0 \left(\vec{J} + \epsilon_0 \frac{d\vec{E}}{dt} \right) \right] \cdot \vec{da}$$

$$\therefore \nabla \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{d\vec{E}}{dt} \right)$$

Electromagnetic Wave Equation in non-conducting and anisotropic medium

We have Maxwell's equation in any medium are

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon} \dots \dots \dots (i)$$

$$\nabla \cdot \vec{B} = 0 \dots \dots \dots (ii)$$

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt} \dots \dots \dots (iii)$$

$$\nabla \times \vec{B} = \mu \left(\sigma \vec{E} + \epsilon \frac{d\vec{E}}{dt} \right) \dots \dots \dots (iv) \quad (\text{Since, } \vec{J} = \sigma \vec{E})$$

For non-conducting and anisotropic medium, $\sigma = 0$ and $\rho = 0$

$\therefore$ Equations (i) and (iv) become

$$\nabla \cdot \vec{E} = 0 \dots \dots \dots (v)$$

$$\nabla \times \vec{B} = \mu \epsilon \frac{d\vec{E}}{dt} \dots \dots \dots (vi)$$

Now taking curl of equation (iii), we get

$$\nabla \times \nabla \times \vec{E} = \nabla \times -\frac{d\vec{B}}{dt}$$

$$\text{or, } (\nabla \cdot \vec{E})\nabla - (\nabla \cdot \nabla)\vec{E} = -\frac{d}{dt}(\nabla \times \vec{B}) \quad [\text{Since, } \vec{A} \times \vec{B} \times \vec{C} = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}]$$

$$\text{or, } -\nabla^2 \vec{E} = -\frac{d}{dt}(\mu \epsilon \frac{d\vec{E}}{dt}) \quad (\text{Since, using equations (v) and (vi)})$$

$$\text{or, } \nabla^2 \vec{E} = \mu\epsilon \frac{d^2 \vec{E}}{dt^2} \dots \dots \dots (vii)$$

Again taking curl of equation (vi) we get

$$\nabla \times \nabla \times \vec{B} = \nabla \times \mu\epsilon \frac{d\vec{B}}{dt}$$

$$\text{or, } (\nabla \cdot \vec{B})\nabla - (\nabla \cdot \nabla)\vec{B} = \mu\epsilon \frac{d}{dt}(\nabla \times \vec{B})$$

$$\text{or, } -\nabla^2 \vec{B} = \mu\epsilon \frac{d}{dt} \left(-\frac{d\vec{B}}{dt} \right) \quad (\text{Since, using equations (ii) and (iii)})$$

$$\text{or, } \nabla^2 \vec{B} = \mu\epsilon \frac{d^2 \vec{B}}{dt^2} \dots \dots \dots (viii)$$

Here, equations (vii) and (viii) represent the electromagnetic wave equations (plane wave equations) for non-conducting and anisotropic medium.

The plane wave solutions of equations (vii) and (viii) can be written as

$$E = E_0 \sin(\omega t - kx)$$

$$\text{and } B = B_0 \sin(\omega t - kx)$$

Where, E_0 and B_0 are maximum values of electric and magnetic fields respectively.

Now comparing equations (vii) and (viii) with differential equation of wave motion ($\nabla^2 y = \frac{1}{v^2} \frac{d^2 y}{dt^2}$), we get the velocity of e-m wave is

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

$$\text{or, } v = \frac{1}{\sqrt{\mu_r \epsilon_r \mu_0 \epsilon_0}} = \frac{c}{\sqrt{\mu_r \epsilon_r}} \quad \text{Where } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \text{ is velocity of light in free space.}$$

This shows that the velocity of e-m wave in a medium is less than velocity of light in free space.

For free space, $\mu_r = 1$ and $\epsilon_r = 1$

$$\therefore v = c$$

Hence, the velocity of e - m wave is equal to the velocity of light in free space.

Ratio of Electric and magnetic Fields

The electric and magnetic fields at a point due to the propagation of e-m wave can be written as

$$E = E_0 \sin(\omega t - kx)$$

$$\text{and } B = B_0 \sin(\omega t - kx)$$

$$\therefore \frac{E}{B} = \frac{E_0}{B_0} \dots \dots \dots (i)$$

We have from Faraday's law of e-m induction,

$$\frac{dE}{dx} = -\frac{dB}{dt}$$

$$\text{or, } -kE_0 \cos(\omega t - kx) = -\omega B_0 \cos(\omega t - kx)$$

$$\text{or, } \frac{E_0}{B_0} = \frac{\omega}{k} = v$$

Now, from equation (i),

$$\frac{E}{B} = \frac{E_0}{B_0} = v$$

For free space,

$$\frac{E}{B} = \frac{E_0}{B_0} = c$$

Note: we have magnetic energy density $u_b = \frac{B^2}{2\mu_0} = \frac{E^2}{2\mu_0 c^2} = \frac{E^2}{2\mu_0} \mu_0 \epsilon_0 = \frac{1}{2} \epsilon_0 E^2 = u_e$; the electric energy density

$$\therefore u_b = u_e$$

Poynting Vector (Poynting Theorem)

The rate of flow of energy per unit area in the case of e-m wave is measured in terms of a vector, called pointing vector and is given by

$$\text{Poynting Vector, } \vec{S} = \frac{1}{\mu} (\vec{E} \times \vec{B})$$

Let us consider an electromagnetic wave is propagating through a medium with velocity v . Let us divide the medium into infinite number of small layers of thickness dx and area of cross-section A .

Now total energy developed in a layer is

$$dU = dU_E + dU_B$$

$$\text{or, } dU = u_e A dx + u_b A dx \quad \left(\text{Since, } u_e = \frac{dU_E}{A dx} \text{ and } u_b = \frac{dU_B}{A dx} \right)$$

$$\text{or, } dU = 2u_b Adx \quad (\text{Since, } u_e = u_b)$$

$$\text{or, } dU = 2 \frac{B^2}{2\mu} Adx \quad (\text{Since, } u_b = \frac{B^2}{2\mu})$$

$$\text{or, } dU = \frac{B}{\mu} \frac{E}{v} Adx \quad (\text{Since, } \frac{E}{B} = v)$$

$$\text{Now, } \frac{dU}{Adt} = \frac{EB}{\mu v} \frac{dx}{dt}$$

$$\text{or, } \frac{dU}{Adt} = \frac{EB}{\mu v} v \quad (\text{Since, } \frac{dx}{dt} = v)$$

$$\text{or, } S = \frac{EB}{\mu}$$

Where, $S = \frac{dU}{Adt}$ is the magnitude of *Poynting Vector*.

$$\text{In vector form, } \vec{S} = \frac{1}{\mu} (\vec{E} \times \vec{B})$$

For free space, $\mu = \mu_0$

$$\therefore \vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

Now the intensity of electromagnetic wave is given by average of the magnitude of $\vec{S}$.

$$\text{We have, } S = \frac{EB}{\mu_0} = \frac{1}{\mu_0} E_0 \sin(\omega t - kx) B_0 \sin(\omega t - kx) = \frac{E_0 B_0}{\mu_0} \sin^2(\omega t - kx)$$

$$\therefore I = \frac{E_0 B_0}{2\mu_0}$$

$$\text{or, } I = \frac{c B_0^2}{2\mu_0} \Rightarrow I \propto B_0^2$$

$$\text{or, } I = cu_b \quad (\text{Since, } u_b = \frac{B^2}{2\mu_0})$$

$$\text{Also, } I = \frac{E_0^2}{2c\mu_0} \Rightarrow I \propto E_0^2$$

$$\text{or, } I = \frac{E_0^2}{2c^2\mu_0} c = \frac{E_0^2}{2\mu_0} \mu_0 \epsilon_0 c = c \times \frac{1}{2} \epsilon_0 E_0^2 = cu_e$$

Continuity Equation (Charge Conservation Theorem)

It states that total charge for an isolated system is conserved. It gives the relation between charge density and current density as

$$\nabla \cdot \vec{J} + \frac{d\rho}{dt} = 0$$

We have the fourth Maxwell's equation is

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{d\vec{E}}{dt} \right)$$

Taking divergence on both sides

$$\nabla \cdot \nabla \times \vec{B} = \mu_0 \left(\nabla \cdot \vec{J} + \epsilon_0 \frac{d(\nabla \cdot \vec{E})}{dt} \right)$$

Since the divergence of curl is always zero, so $\nabla \cdot \nabla \times \vec{B} = 0$

$$\therefore \mu_0 \left(\nabla \cdot \vec{J} + \epsilon_0 \frac{d(\nabla \cdot \vec{E})}{dt} \right) = 0$$

$$\text{or, } \nabla \cdot \vec{J} + \epsilon_0 \frac{d(\nabla \cdot \vec{E})}{dt} = 0$$

From first Maxwell's Equation we have $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

$$\text{or, } \nabla \cdot \vec{J} + \epsilon_0 \frac{d\left(\frac{\rho}{\epsilon_0}\right)}{dt} = 0$$

$$\text{or, } \nabla \cdot \vec{J} + \frac{d\rho}{dt} = 0$$

Electromagnetic Waves

1. The earth's mean radius is $6.37 \times 10^6 \text{m}$ and mean earth-sun distance is $1.5 \times 10^8 \text{km}$. What fraction of the radiation emitted by the sun is intercepted by the disk of the earth?

→ Here, radius of earth $r = 6.37 \times 10^6 \text{m}$, mean earth sun distance $R = 1.5 \times 10^8 \text{km} = 1.5 \times 10^{11} \text{m}$

We have, the intensity of radiation emitted by sun $I = \frac{P}{4\pi R^2}$

And the intensity of radiation received by the disc of earth $I' = \frac{P}{\pi r^2}$

$$\therefore \frac{I}{I'} = ?$$

2. The intensity of direct solar radiation that is not absorbed by the atmosphere on a particular summer day is 100W/m^2 . How close would you stand to a 1KW electric heater to feel the same intensity for uniform distribution of heater radiation?

→ Here, Intensity $I = 100\text{W/m}^2$, Power $P=1\text{KW}=1000\text{W}$

$$\text{We have, } I = \frac{P}{A} = \frac{P}{4\pi r^2}$$

$$\therefore r = ?$$

3. A space probe $2 \times 10^{10}\text{m}$ from a star measures the total intensity of em radiation from the star to be $5 \times 10^3 \text{ W/m}^2$. If the star radiates uniformly in all directions, what is the total average power output?

→ Here, distance $r=2 \times 10^{10}\text{m}$, Intensity $I = 5000\text{W/m}^2$

$$\text{We have, } I = \frac{P}{A} = \frac{P}{4\pi r^2}$$

$$\therefore P = ?$$

4. Sunlight just outside the earth's atmosphere has an intensity of 1.4KW/m^2 . Calculate the magnitude of pointing vector, maximum electric field and maximum magnetic fields for sunlight, assuming it to be a plane wave.

→ Here, Intensity $I = 1.4\text{KW/m}^2 = 1400 \text{ W/m}^2$

Magnitude of pointing vector $S=1400 \text{ W/m}^2$

$$\text{We have, } I = \frac{cB_0^2}{2\mu_0}$$

$$\therefore B_0 = \sqrt{\frac{2\mu_0 I}{c}} = ?$$

$$\text{Again } E_0 = cB_0 = ?$$

5. Sunlight strikes the earth outside its atmosphere with an intensity of $2\text{cal/cm}^2\text{-min}$. Calculate the magnitudes of electric and magnetic fields.

→ Here, Intensity $I = 2\text{cal/cm}^2\text{-min} \times \frac{2 \times 4.2}{10^{-4} \times 60} = 1400 \text{ W/m}^2$

Same as 4.

6. The maximum electric field 10m from an isotropic point source is 2V/m . What are (a) the maximum value of magnetic field (b) average intensity of light and (c) power of the source?

→ Here, distance $r=10\text{m}$, $E_0 = 2\text{V/m}$

a) we have, $B_0 = \frac{E_0}{c} = ?$

b) $I = \frac{cB_0^2}{2\mu_0} = ?$

c) $P = I \times A = I \times 4\pi r^2 = ?$

7. A certain radio station broadcasts at a frequency of 10^{20} Hz . At a point some distance from the transmitter, the maximum magnetic field of the em wave is $1.6 \times 10^{-4}\text{T}$. (a) What is the speed of the propagation of the wave (b) What is the wavelength? (c) What is the maximum electric field?

→ Here, $f = 10^{20}\text{ Hz}$, $B_0 = 1.6 \times 10^{-4}\text{T}$

a) $v = c = 3 \times 10^8\text{ m/s}$

b) $\lambda = \frac{c}{f} = ?$

c) $E_0 = cB_0 = ?$

8. A certain plane wave emitted by a microwave antenna has a wavelength of 3cm and maximum magnitude of electric field $2 \times 10^{-4}\text{V/m}$. (a) What is the frequency of wave? (b) What is the maximum magnetic field? (c) What is maximum energy density? (d) What is intensity of wave?

→ Here, $\lambda = 3\text{cm} = 0.03\text{m}$, $E_0 = 2 \times 10^{-4}\text{V/m}$

a) $f = \frac{c}{\lambda} = ?$

b) $B_0 = \frac{E_0}{c} = ?$

c) $u_b = \frac{B_0^2}{2\mu_0} = ?$

d) $I = \frac{cB_0^2}{2\mu_0} = ?$

9. An isotropic point source emits light at wavelength 500nm at the rate of 200W . A light detector is positioned 400m from the source. What is the maximum rate at which magnetic component of the light changes with time at the detector's location?

→ Here, $\lambda = 500\text{nm} = 500 \times 10^{-9}\text{m}$, $P = 200\text{W}$, $r = 400\text{m}$

We have, $B = B_0 \sin(\omega t - kx)$

$$\therefore \frac{dB}{dt} = \omega B_0 \cos(\omega t - kx)$$

$$\text{or, } \left[\frac{dB}{dt} \right]_{\max} = \omega B_0 \dots \dots \dots (i)$$

$$\text{Again, } I = \frac{cB_0^2}{2\mu_0} = \frac{P}{4\pi r^2}$$

$$\therefore B_0 = ?$$

$$\text{Hence from (i), } \left[\frac{dB}{dt} \right]_{\max} = 2\pi f \times B_0 = 2\pi \times \frac{c}{\lambda} \times B_0 = ?$$

10. What is the radiation pressure 1.5m away from a 500W light bulb? Also calculate the amplitudes of electric and magnetic field there.

→ Here, $r=1.5\text{m}$, $P=500\text{W}$

$$\text{We have radiation pressure } p_r = \frac{I}{c} = \frac{P}{4\pi r^2 c} = ?$$

$$\text{Again, } I = \frac{P}{4\pi r^2} = \frac{cB_0^2}{2\mu_0}$$

$$\therefore B_0 = ?$$

$$\text{Also, } E_0 = cB_0 = ?$$

11. The sun delivers about 10^3W/m^2 of energy to the earth's surface through EM radiation. Calculate (a) the total power incident on a roof of dimensions $8\text{m} \times 20\text{m}$. (b) radiation pressure and force exerted on the roof, assuming roof is perfect absorber.

→ Here, $I=10^3\text{W/m}^2$, Area $A=8\text{m} \times 20\text{m}$

$$\text{a) } P = I \times A = ?$$

$$\text{b) } p_r = \frac{I}{c} = ?$$

$$\text{and Force } F = p_r \times A = ?$$